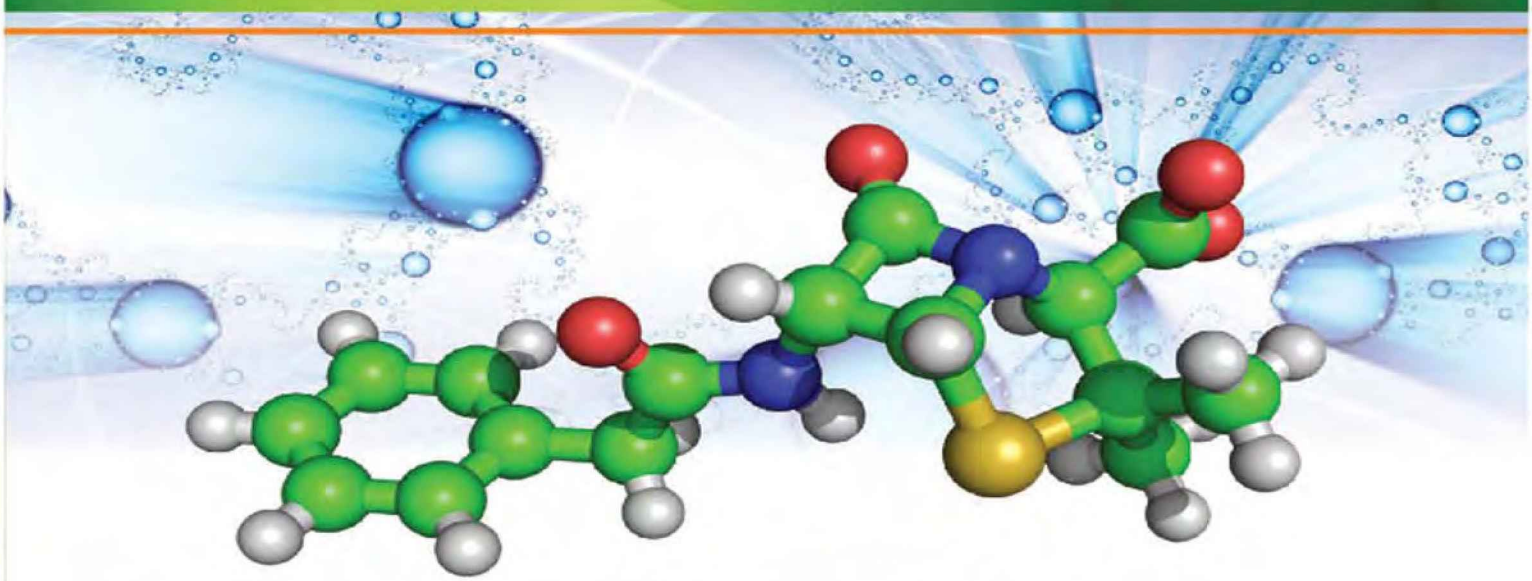
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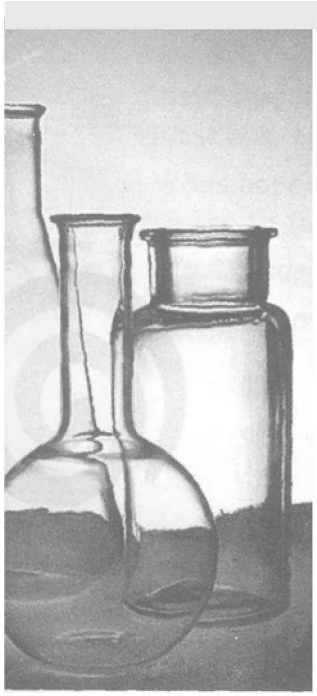
CHEMISTRY

CLASS 9

By

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Preface

In today's competitive world, students need to tackle competitive exams in order to ensure a successful career. This can be achieved only by having a thorough knowledge of the subject which will help them to strengthen the application of the fundamental concepts and enhance their problem solving skills.

The **"IIT Foundation Series—Chemistry"** books for classes 6 to 10 are designed to provide a solid base to the students who aspire to undertake the IIT and other competitive exams for entrance to engineering institutions after Class 12. It will also ensure their excellent performance in all other similar competitive exams with ease.

The distinction features of this series are :

- The syllabus goes beyond the usual school syllabus to cover a wider aspect of the subject.
- **Textbook** and **Practice Material** are present in a single book.
- A **concise description** of each topic with **important concepts, definition** and **words.**
- **Important points** at the end of each chapter giving brief summary of the concepts learned in the chapter.
- **Objective** and **subjective** Questions in each chapter based on the latest examination pattern.
- **Challenging questions/problems solved** for encouraging the students to come up with solutions of more tough questions/problems on their own.
- **Challenging questions/problems solved** for encouraging the students to come up with solutions of more tough questions/problems on their own.
- Exercises for practice with **Simple Problems, Conceptual Problems, Multiple Choice Questions, Numerical Problems, Paragraph Questions, Assertion and Reasoning Questions,** etc. as asked in the IIT entrance exams.

The series hope to serve as a self-learning kit to the students which will give them the confidence to take up any entrance exam as a challenge.

Any queries/suggestions regarding the subject matter can be sent to the publisher.

We will strive our best to solve these queries and incorporate and suggestions given to us.

— Author

Contents

CHAPTER 1

Elements, Compounds and Mixtures 1-13

- Introduction ... 1
- Atom ... 1
- Matter ... 1
- Classification of Matter ... 1
- Pure Substances ... 2
- Elements ... 2
- Symbols of Elements ... 2
- Compounds ... 3
- Naming of Compounds ... 5
- Mixtures ... 6
- Classification of Mixtures ... 6
- Separation of Mixtures ... 7
- Differences between Compounds and Mixtures ... 8

CHAPTER 2

Symbols, Formulae and Equations 14-34

- Review of Important Concepts ... 14
- Chemical Symbols ... 14
- Meaning Conveyed by Symbols ... 14
- Elements are Made up of Atoms ... 15
- Electronic Configuration of Elements (Bohr-Bury Rules) ... 15
- Valence Electrons ... 17
- Ions and their Formulae ... 17
- Modern Concept of Valency ... 20
- Radical ... 20
- Chemical Formula ... 21
- Chemical Equations – Representing Chemical Reactions ... 22
- Writing and Balancing Chemical Equations ... 23
- Special Symbols used in Chemical Equation ... 25
- Meaning of Balanced Chemical Equations ... 25
- Significance of Numbers used in a Chemical Equation ... 26
- Atomic Number and Mass Number ... 28
- Atomic Weight of an Element ... 29
- Molecular Weight of a Compound ... 29
- Simple Calculations based on Chemical Formulae and Equations ... 30

CHAPTER 3

Gaseous State and Gas Laws 35-44

- Introduction ... 35
- Physical Properties of Gases ... 35
- Measurable Properties of Gases ... 35
- Absolute Scale of Temperature or Kelvin Scale ... 36
- Kinetic Theory of Gases ... 37
- Gas Laws ... 37
- Boyle's Law (1662) ... 37
- Aqueous Tension ... 38
- Charles' Law (1787) ... 38
- Ideal Gas Equation ... 38
- Gay Lussac's Law of Combining Volumes ... 40
- Avogadro's Law ... 40

CHAPTER 4

Atomic Structure 45-57

- Introduction ... 45
- Daltons Atomic Model, 1808 ... 45
- The Discharge Tube Experiments – Discovery of Electron ... 46

- Canal Rays – Discovery of Proton ... 47
- Discovery of Neutron ... 47
- Fundamental Particles Present in the Atom ... 49
- Atomic Models ... 49
- Rutherford's Atomic Model, 1911 ... 50
- Bohr's Atomic Model (1913) ... 51
- Some Important Points to Remember ... 51
- Rules of Electronic Configuration ... 52
- Properties of Atom ... 53

CHAPTER 5

Periodic Classification 48-77

- Introduction ... 58
- Importance of Classifying Elements ... 58
- Dobereiner Triads, 1829 ... 58
- Newland's Law of Octaves, 1864 ... 58
- Meyer's Classification (1869, German) ... 59
- Mendeleev's Classification (1869, Russian) ... 59
- Modern Periodic Table ... 60
- Special Names of Some Groups in the Periodic Table ... 62
- Types of Elements ... 62
- Representative Elements ... 62
- Transition Elements ... 64
- Inner Transition Elements ... 65
- Inert Gases (or Noble Gases) ... 65
- Periodicity ... 66
- Periodic Properties ... 67
- Electronic Configuration of Elements ... 67
- Atomic Properties ... 68
- Atomic Radius ... 68
- Ionic Radius ... 69
- Ionisation Energy ... 69
- Electron Affinity (Affinity Energy) ... 70
- Electronegativity ... 71
- Metallic Character ... 72
- Non-metallic Character ... 73
- Summary of General Trends in Atomic Properties ... 74

CHAPTER 6

Chemical Bonding 78-91

- Introduction ... 78
- Chemical Bond ... 78
- Electron Redistribution in Chemical Bonds ... 79
- Ionic bond, Kossel, 1916 ... 80
- Covalent Bond, 1916, Lewis ... 82
- Classification of Covalent Bonds ... 83
- Ionic Compounds and Covalent Compounds ... 84
- Coordinate Covalent Bond (Dative Bond) ... 84
- Van der Waal's Forces ... 85
- Bond Pairs and Lone Pairs ... 87

CHAPTER 7

Chemical Kinetics 92-102

- Introduction ... 92
- Reversible and Irreversible Chemical Reactions ... 92
- Rates of a Chemical Reaction ... 92
- Path of a Chemical Reaction ... 94
- Collision Theory ... 94
- Internal Energy, Activation Energy and Threshold Energy ... 95
- Energy Changes in a Chemical Reaction ... 96
- Effect of Catalyst on Reaction Paths ... 96
- Factors Affecting Reaction Rates ... 97
- Law of Mass Action (Rate Law) ... 97
- Reaction Mechanisms ... 98

CHAPTER 8

Chemical Equilibium 103-111

- Introduction ... 103
- Reversible and Irreversible Changes ... 103
- Forward Reaction and Backward Reactions ... 103
- State of Chemical Equilibrium ... 103
- Dynamic Nature of Chemical Equilibrium ... 104

- Graphical Representation of Irreversible Reactions ... 104
- Graphical Representation of the State of Chemical Equilibrium ... 104
- Characteristics of Chemical Equilibrium ... 105
- Types of Chemical Equilibrium ... 106
- Law of Chemical Equilibrium (Law of Mass Action) . .. 106
- Significance of Equilibrium Constant ... 106
- Le Chatelier's Principle ... 106
- Effects of Changing Physical Conditions of a Chemical Equilibrium ... 106
- Application of Le Chatelier Principle ... 107
- Some Specific Reactions and Favourable Conditions ... 108

CHAPTER 9

Metallurgy 112-125

- Introduction ... 112
- Characteristics of Metals ... 112
- Characteristics of Non-metals ... 112
- Occurrence ... 112
- Metallurgy ... 113
- Steps in Metallurgy ... 113
- Dressing and Concentration of Ore ... 114
- Conversion of Ore into Oxide ... 115
- Reduction of Ore to Pure Metal (Extraction of Metal) ... 115
- Purification and Refining of Metals ... 116
- Metallurgy of Aluminium (Halls Process) ... 116
- Metallurgy of Iron ... 119
- Metallurgy of Zinc ... 121
- Alloys ... 122
- Corrosion – A Special Discussion ... 122

CHAPTER 10

Study of Group IA Element—Hydrogen 126-133

- Introduction to Hydrogen ... 126
- Occurrence ... 126
- Laboratory Preparation of Hydrogen ... 126
- Industrial Manufacture of Hydrogen ... 127
- Physical Properties of Hydrogen ... 128
- Hydrogen ... 128
- Chemical Properties of Hydrogen ... 128
- Test for hydrogen ... 129
- Uses of Hydrogen ... 129

CHAPTER 11

Study of Group IVA Element—Carbon and its Oxides 34-150

- Introduction to Carbon ... 134
- Occurrence ... 134
- Allotropes of Carbon ... 135
- Crystalline Allotropes ... 135
- Amorphous Allotropes ... 137
- Fuels ... 139
- Combustion ... 140
- Oxides of Carbon ... 141
- Carbon Monoxide ... 141
- Carbon Dioxide ... 143
- Fires and Fire Extinguishers ... 145

CHAPTER 12

Study of Group VA Element —Nitrogen and its compounds 151-170

- Introduction to Nitrogen ... 151
- Occurrence of Nitrogen ... 151
- General Methods of Preparation of Nitrogen ... 151
- Laboratory Preparation of Nitrogen ... 152
- Chemical Properties of Nitrogen ... 153
- Test for Nitrogen ... 154
- Uses ... 154
- Compounds of Nitrogen ... 154
- Ammonia and Ammonium Hydroxide ... 155

- Industrial Preparation of Ammonia by Haber's Process ... 156
- Physical Properties of Ammonia Gas ... 157
- Chemical Properties of Ammonia Gas ... 157
- Chemical Properties of Ammonium Hydroxide (NH_4OH) ... 158
- Test for Ammonium Hydroxide ... 159
- Uses of Ammonium Hydroxide ... 159
- Ammonium Salts ... 160
- Uses of Ammonium Salts ... 160
- Nitric Acid ... 161
- Industrial Manufacture of Nitric Acid by Ostwald's Process ... 162

CHAPTER 13

Study of Group VIA Element—Oxygen 171-180

- Introduction to Oxygen ... 171
- Occurrence ... 171
- General Methods of Preparation of Oxygen ... 171
- Laboratory Preparation of Oxygen ... 171
- Industrial Preparation of Oxygen ... 172
- Physical Properties of Oxygen ... 173
- Chemical Properties of Oxygen ... 173
- Test for Oxygen ... 175
- Uses of Oxygen ... 175
- Rusting of Iron ... 176

CHAPTER 14

Study of Group VIA Element—Sulphur and its compounds 181-199

- Introduction to Sulphur ... 181
- Occurrence ... 181
- Industrial Extraction of Sulphur ... 181
- Sulphur Exhibits Allotropy ... 182
- Physical Properties of Sulphur ... 184
- Chemical Properties of Sulphur ... 185
- Uses of Sulphur ... 186
- Introduction Sulphur Dioxide ... 186
- General Methods of Preparation ... 186
- Laboratory Preparation of Sulphur Dioxide ... 186
- Physical Properties of Sulphur Dioxide ... 187
- Chemical Properties of Sulphur Dioxide ... 187
- Test for Sulphur Dioxide ... 188
- Uses of Sulphur Dioxide ... 188
- Introduction to Sulphuric Acid ... 188
- General Methods of Preparation ... 189
- Contact Process – Industrial Manufacture of Sulphuric Acid ... 189
- Physical Properties of Sulphuric Acid ... 190
- Chemical properties of Sulphuric Acid ... 190
- Tests for Sulphuric Acid ... 192
- Uses of Sulphuric Acid ... 192
- Hydrogen Sulphide ... 193

CHAPTER 15

Study of Group VIIA Element—Chlorine 200-208

- Introduction to Chlorine ... 200
- Laboratory Preparation of Chlorine ... 200
- Manufacture of Chlorine ... 201
- Physical Properties of Chlorine ... 201
- Chemical Properties of Chlorine ... 202
- Uses of Chlorine ... 204

Series Content

Class 6

1. Nature of Matter
2. Pure and Mixed Substances
3. Elements, Compounds and Mixtures
4. Symbols and Formulae
5. Changes Around Us
6. Water
7. Air

Class 7

1. Elements Compounds and Mixtures
2. States of Matter
3. Symbols, Formulae and Equations
4. Laws of Chemical Combination
5. Atomic Structure
6. Acids, Bases and Salts
7. Chemistry in Daily Life
8. Air
9. Water
10. Changes Around Us

Class 8

1. Action of Heat on Substances
2. Transformation of Substances
3. Symbols, Formulae and Equations
4. Types of Chemical Changes
5. Atomic Structure
6. Periodic Classification
7. Chemical Bonding
8. Water and Its Constituents
9. Hydrogen
10. Oxygen
11. Sulphur and Its Compounds
12. Nitrogen and Its Compounds
13. Study of Group IVA Element—Carbon and Its Oxides
14. Phosphorous and Its Compounds
15. Common Salt and Its Constituents
16. Metals and Non-Metals

Class 9

1. Elements, Compounds and Mixtures
2. Symbols, Formulae and Equations
3. Gaseous State and Gas Laws
4. Atomic Structure
5. Periodic Classification
6. Chemical Bonding
7. Chemical Kinetics
8. Chemical Equilibrium
9. Metallurgy
10. Study of Group IA Element—Hydrogen
11. Study of Group IVA Element—Carbon and its Oxides
12. Study of Group VA Element—Nitrogen and Its Compounds
13. Study of Group VIA Element—Oxygen
14. Study of Group VIA Element—Sulphur and Its Compounds
15. Study of Group VIIA Element—Chlorine

Class 10

1. Atomic Structure
2. Periodic Table
3. Chemical Bonding
4. Alkali Earth Metals
5. Metals and Metallurgy
6. Acids and Bases
7. Solutions
8. Electrolysis
9. Organic Chemistry
10. Carbohydrates, Proteins, Fats and Oils
11. Stoichiometry
12. Chemistry and Industry
13. Nuclear Chemistry
14. Analytical Chemistry

1 Elements, Compounds and Mixtures

1.0 INTRODUCTION

Any science is a systematic study by experimentation, observation and drawing inferences. Chemistry is a branch of science that deals with a systematic study of the composition of matter and the reactions that take place between matter. A study of chemistry tells us how substances behave with other substances. It also explains how different chemical substances join together to form new substances. For example when wood is burnt it changes to ash with the release of heat and light. Some gases like carbon dioxide are also released.

The burning of wood is a chemical reaction.

In order to understand the nature of different chemical substances and chemical reactions, it is important to know about the fundamental building blocks of all chemicals – ***the atom.***

1.1 ATOM

The word '*atom* 'comes from the Greek word '*atomos'* meaning indivisible. It was John Dalton, an English scientist, who proposed in 1808, that matter consists of very small particles called *atoms*. According to Dalton if matter is divided into smaller and smaller parts we will finally get a very small particle that cannot be divided any further.

Dalton called the smallest particle of matter that was not divisible any further as an atom.

Dalton studied matter and put forth his findings in a theory called ***Dalton's Atomic Theory.***

Dalton's Atomic Theory

1. *Matter is made up of tiny particles called atoms.*
2. *Atoms are indivisible.*
3. *Atoms cannot be created or destroyed.*
4. *Atoms of the same element are identical in mass and other properties.*
5. *Atoms of different elements differ in their masses and properties.*

Further scientific studies showed that atoms could be divided further into still smaller particles.

In 1852, Plucker and Goldstein in 1876 showed that all atoms contained negatively charged particles. Thus they showed that the atom could be divided or split to produce these negatively charged particles.

In 1897 J. J. Thomson called these negatively charged particles electrons. He further put forth the idea that the atom was a positively charged solid ball with the negative electrons scattered inside it.

Only later, Rutherford in 1911 and Neil Bohr in 1912 and Chadwick in 1932 showed that all atoms are made up of three particles – ***electrons, protons and neutrons.***

Atoms are actually made up of negatively charged particles called electrons, positively charged particles called protons and neutrally charged particles called neutrons.

1.2 MATTER

All materials around us are made of matter. Books, tables, chairs, the air we breathe, the water we drink, the food we eat, our body, the sun, the moon, the earth are all made up of matter. Let us define matter.

Matter is anything that occupies space, has mass and can be perceived by our senses.

Thus there are three essential conditions for something to be called matter. They are as follows.

1. It must occupy space.
2. It must have mass.
3. We must be able to perceive it by our senses, i.e., be able to touch, smell, see or taste it.

We see so many different kinds of matter around us. Different kinds of matter differ from each other in their composition and properties. They differ in colour, size, weight, volume, composition, melting point, boiling point and solubility in water. Besides these differences, they differ from each other in many ways.

1.3 CLASSIFICATION OF MATTER

To study all the different kinds of matter accurately and conveniently, matter is classified into two broad categories: ***Pure substances and Mixtures.***

A pure substance consists of particles of only one kind. It has a uniform composition throughout. Such matter is said to be *homogeneous* in nature.

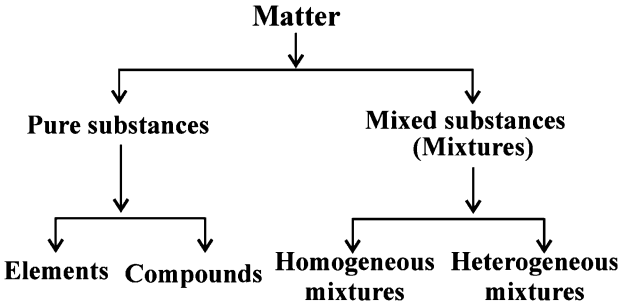
Pure substances are further classified into two sub-groups: ***Elements and Compounds.***

Examples: Pure gold consists of particles of gold only; pure water contains particles of water only.

A mixture consists of particles of more than one kind. It may or may not have a uniform composition throughout. If the matter has a uniform composition throughout then the mixture is said to be a *homogeneous mixture*. If the matter does not have a uniform composition throughout then the mixture is said to be a *heterogeneous mixture*.

Examples: *Homogeneous mixtures:* salt solution contains particles of water and common salt; sugar solution contains particles of water and sugar.

Heterogeneous mixtures: muddy water, sand and pebbles, mixture of iron filings and sulphur.



1.4 PURE SUBSTANCES

Pure substances contain particles of only one kind. Pure substances are further classified into ***Elements and Compounds.***

1.5 ELEMENTS

Elements are simple pure substances made up of only one kind of particles that cannot be decomposed or divided any further into simpler substances by physical or chemical methods.

Examples: Oxygen, nitrogen, carbon, hydrogen, iron, copper, gold, silver, etc.

Characteristics of Elements

An element has the following important characteristics :

(i) An element is made up of atoms of only one kind. It has uniform composition – in other words elements are homogeneous in nature.

(ii) An element cannot be broken by chemical methods or physical methods into simpler substances.

(iii) An element has specific physical properties like physical state, density, colour, odour, taste, melting point and boiling point.

(iv) An element has specific chemical properties like reactivity with other substances.

(v) An element occurs either in the free state as free element or in combination with other elements as a compound.

Known Elements

At present, around 117 elements are known. Out of these around 92 occur naturally in the earth's crust while the rest are artificially created by man.

1. At room temperature, most elements are solids.
2. Mercury is a liquid element at room temperature.
3. There are eleven elements that are gases – Hydrogen, Helium, Oxygen, Nitrogen, Fluorine, Neon, Chlorine, Argon, Krypton, Xenon and Radon.
4. More than 80 elements are metallic elements while around 22 are non-metallic elements.
5. Some elements are radioactive elements as they emit radiation which is often harmful.

Classification of Elements

Based on their properties, the elements are classified into four groups as follows.

(i) Metals (82 are known)

(ii) Non-metals (22 are known)

(iii) Metalloid (7 are known)

(iv) Noble gases (6 are known).

1.6 SYMBOLS OF ELEMENTS

With so many elements existing either in the free state or in the combined state with other elements, it would be more convenient if we could represent these elements – in the free state or in the combined state in a short form.

Illustration: When we use the letters PTO at the end of a page we know that it stands for 'please turn over.' We turn the page and continue to read when we come across these words. Similarly we use the short form etc. at the end of a list of things. This is a short form for etcetera which means that there are other items in the list. Another short form that we often use is e.g., which stands for example.

Elements can be represented using a standard list of symbols or short form representations.

A symbol is a short form representation of an element.

Example: H is the symbol for Hydrogen; O is the symbol for Oxygen, etc.

Symbols for different elements are selected in several ways.

(a) For some elements the first letter of the name of the element is taken.

Examples:

Element	Symbol
Hydrogen	H
Nitrogen	N
Carbon	C
Phosphorus	P
Oxygen	O
Sulphur	S
Fluorine	F

(b) For some elements the first two letters of the name of the element are taken.

Examples:

Element	Symbol
Calcium	Ca
Aluminium	Al
Germanium	Ge
Neon	Ne
Helium	He
Silicon	Si
Argon	Ar
Nickel	Ni

(c) For some elements the symbol is taken based on sound.

Examples:

Element	Symbol
Magnesium	Mg
Manganese	Mn

(d) For some elements the symbols are taken from their Latin names.

Element	Latin name	Symbol
Sodium	Natrium	Na
Potassium	Kalium	K
Lead	Plumbum	Pb
Iron	Ferrum	Fe
Silver	Argentum	Ag
Copper	Cuprum	Cu
Gold	Aurum	Au
Antimony	Stibium	Sb

Symbols convey very useful information about the element they represent.

For example the symbol H used for the element hydrogen conveys the following information:

(i) The symbol H is of the element hydrogen.

(ii) The symbol H represents one atom of hydrogen.

Similarly the symbol O used for the element oxygen conveys the following information:

(i) The symbol O is of the element oxygen.

(ii) The symbol O represents one atom of oxygen.

Elements are made up of atoms

Elements are pure substances that are made of small particles called *atoms*.

Atoms are the smallest particles of matter that may or may not exist independently but always take part in a chemical reaction.

The structure of the atom is as follows.

1. The atoms have electrons, protons and neutrons in them.
   (i) The electrons are negatively charged particles.
   (ii) The protons are positively charged particles.
   (iii) The neutrons are neutral particles.
2. The number of electrons is equal to the number of protons. Since the negatively charged particles (electrons) are equal to the positively charged particles (protons) the atom is electrically neutral.
3. The protons and neutrons are present in a small central region of the atom called the nucleus.
4. The electrons revolve around the nucleus at high speeds.
5. The Atomic Number of an element gives the number of protons in the atom. This number is equal to the number of electrons in the neutral atom. For example, the Atomic Number of sodium is 11. This means that each sodium atom has 11 protons and 11 electrons. Similarly, the Atomic Number of oxygen is 8. This means that each oxygen atom has 8 protons and 8 electrons.
6. The electrons revolve around the nucleus in fixed orbits of increasing radius.

1.7 COMPOUNDS

Elements exist in free state as well as in combined state. When elements exist in the combined state, it is actually the atoms of specific elements that combine to form what Dalton called 'compound

atoms'. We call the combined state of elements as compounds.

A compound is a pure substance formed by the chemical combination of atoms of different elements in a fixed ratio by mass.

Example 1: Chalk is a compound made up of atoms of calcium, carbon and oxygen combined in the ratio 10:3:12 by weight.

Example 2: Water is a compound made up of atoms of hydrogen and oxygen combined in the ratio 1:8 by weight.

Example 3: Carbon dioxide is a compound made up of atoms of carbon and oxygen combined in the ratio 3:8 by weight.

Thus a compound is made up of two or more atoms of different elements combined chemically in a fixed proportion by weight. Compounds have specific physical properties like physical state, density, colour, odour, taste, melting point and boiling point. They also have specific chemical properties like reactivity with other substances. The physical and chemical properties of the compound are different from those of the elements in the compound.

Characteristics of compounds

A compound has the following important characteristics.

(i) Compounds have uniform composition – in other words they are homogeneous in nature.

(ii) The properties of compounds are different from those of its constituent elements.

(iii) Compounds can be broken down into their constituent elements by chemical methods but not by physical methods like boiling, distillation, etc.

(iv) Compounds have a fixed composition of their own.

(v) Energy is either absorbed or released during the formation of a compound.

(vi) A compound can be represented by its formula.

Molecules

Molecules are the smallest particles of a pure substance that can exist independently and are made up of two or more atoms combined chemically in a fixed proportion by weight and exhibit all the properties of that pure substance.

Example 1: Water molecule

(i) Each molecule of water consists of two atoms of hydrogen and one atom of oxygen.

(ii) The two elements hydrogen and oxygen are combined in a ratio of 1:8 by weight.

(iii) A water molecule can exist independently.

Example 2: Sulphur dioxide molecule

(i) Each molecule of sulphur dioxide consists of one atom of sulphur and two atoms of oxygen.

(ii) The two elements sulphur and oxygen are combined in a ratio of 1:1 by weight.

(iii) A sulphur dioxide molecule can exist independently.

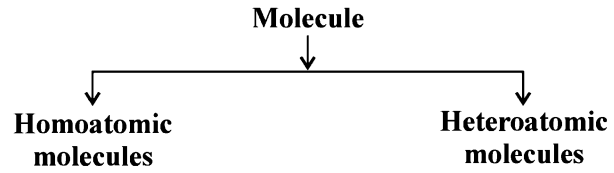
Example 3: Hydrogen molecule

(i) Each molecule of hydrogen consists of two atoms of hydrogen.

(ii) The two atoms are combined chemically.

(iii) A hydrogen molecule can exist independently.

Molecules are of two types.



Homoatomic molecules are formed by atoms belonging to the same element.

Examples :

H_2 – Hydrogen molecule is formed when two atoms of hydrogen combine chemically.

O_2 – Oxygen molecule is formed when two atoms of oxygen combine chemically.

N_2 – Nitrogen molecule is formed when two atoms of nitrogen combine chemically.

S_8 – Sulphur molecule is formed when eight atoms of sulphur combine chemically.

P_4 – Phosphorous molecule is formed when four atoms of phosphorous combine chemically.

Heteroatomic molecules are formed by atoms belonging to different elements.

Examples :

NH_3 – Ammonia molecule is formed when one atom of nitrogen combines chemically with three atoms of hydrogen.

H_2O – Water molecule is formed when two atoms of hydrogen combines chemically with one atom of oxygen.

CO_2 – Carbon dioxide molecule is formed when one atom of carbon combines chemically with two atoms of oxygen.

Atomicity

Homoatomic molecules are made up of atoms of only one element.

Atomicity is the number of atoms of an element present in one molecule of the element.

Depending upon the atomicity, homoatomic elements are called

(a) Monoatomic, if only one atom is present in the molecule of the element.

Examples: Helium (He), Neon (Ne), Copper (Cu)

(b) Diatomic, if two atoms of the element are present in the molecule of the element.

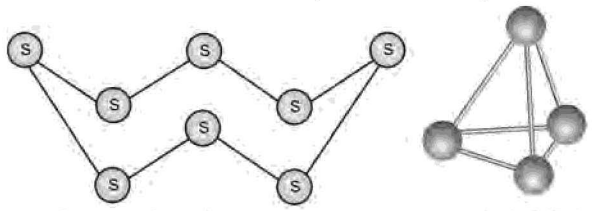
Examples: Hydrogen (H_2), Oxygen (O_2), Chlorine (Cl_2)

(c) Triatomic, if three atoms of the element are present in the molecule of the element.

Example: Ozone (O_3)

(d) Polyatomic if three or more than three atoms of the element are present in the molecule of the element.

Examples: Phosphorous (P_4), Sulphur (S_8)



Structure of a sulphur molecule, S_8

Phosphorous molecule, P_4

Elements that do not combine easily

Atoms of metals and metalloids do not combine with their own types of atoms. Similarly the atoms of inert gases exist free under all conditions. So their atoms are regarded as molecules. The molecules of all these elements are said to be monoatomic molecules.

Representing elements in short form–Formula

Just as symbols are used to represent elements in short form, a chemical formula is used to represent a compound in short form.

A formula is a short form representation of a compound.

To write a formula, the symbols of all the elements are written close together with small subscripts that show the number of atoms of that element that have combined chemically.

Example: CO_2 is the formula of carbon dioxide.

SO_2 is the formula of sulphur dioxide.

H_2O is the formula of water.

$CaCO_3$ is the formula of calcium carbonate.

Formulae of some common compounds

Name of the compound	Formula
Hydrochloric acid	HCl
Nitric acid	HNO_3
Sulphuric acid	H_2SO_4
Sodium hydroxide	NaOH
Calcium oxide	CaO
Calcium carbonate	$CaCO_3$
Sodium chloride	NaCl

Formulae of some common gaseous compounds

Gas	Formula
Carbon dioxide	CO_2
Sulphur dioxide	SO_2
Hydrogen chloride gas	HCl
Methane	CH_4
Carbon monoxide	CO

Formulae of some metallic oxides

Metallic oxides	Formula
Magnesium oxide	MgO
Calcium oxide	CaO
Zinc oxide	ZnO
Copper oxide	CuO

Formulae of some common metallic sulphides

Metallic sulphide	Formula
Magnesium sulphide	MgS
Zinc sulphide	ZnS
Iron sulphide	FeS
Calcium sulphide	CaS

1.8 NAMING OF COMPOUNDS

A few examples of how compounds are named are given below.

A compound in which oxygen is chemically combined with another element is called oxide.

To name the oxide, the name of the element is added to the word oxide.

When the elements zinc and oxygen combine the compound formed is an oxide. The name of the element – in this case zinc, is added to the word oxide to name the compound.

(i) Metal + oxygen → metallic oxide

Zinc + oxygen → zinc oxide

(ii) Non-metal + oxygen → non–metallic oxide

Sulphur + oxygen → sulphur dioxide

A compound in which chlorine is chemically combined with another element is called a chloride.

To name the chloride the name of the element is added to the word chloride.

When the elements zinc and chlorine combine the compound formed is a chloride. The name of the element – in this case zinc, is added to the word chloride to name the compound.

(iii) Element + chlorine → chloride

Zinc + chlorine → zinc chloride

Hydrogen + chlorine → hydrogen chloride

A compound in which sulphur is chemically combined with another element is called a sulphide.

To name the sulphide the name of the element is added to the word sulphide.

When the elements zinc and sulphur combine the compound formed is a sulphide. The name of the element – in this case zinc, is added to the word sulphide to name the compound.

(iv) Element + sulphur → sulphide

Zinc + sulphur → zinc sulphide

1.9 MIXTURES

A mixture is a substance made up of two or more elements or compounds mixed together in a proportion that is not fixed.

Example 1: Salt solution is a mixture of common salt and water. The amount of salt can be increased or decreased. Similarly, the amount of water can be increased or decreased.

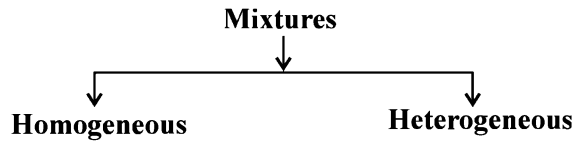
Example 2: Brass is a mixture of copper and zinc.

Example 3: Glucose solution is a mixture of glucose and water.

Example 4: Air is a mixture of gases – mainly nitrogen, oxygen, carbon dioxide and water vapour.

Types of mixtures

Mixtures are classified into two types depending upon their composition.



Homogeneous mixture

A homogeneous mixture has uniform composition throughout the mixture.

Example 1: Sugar solution is a mixture of sugar and water that has a uniform composition throughout the solution.

Example 2: Petrol is a mixture of hydrocarbon fuels that has a uniform composition throughout.

Example 3: Brass is a mixture of copper and zinc that has a uniform composition throughout.

Heterogeneous mixture

A heterogeneous mixture has different composition in different parts of the mixture.

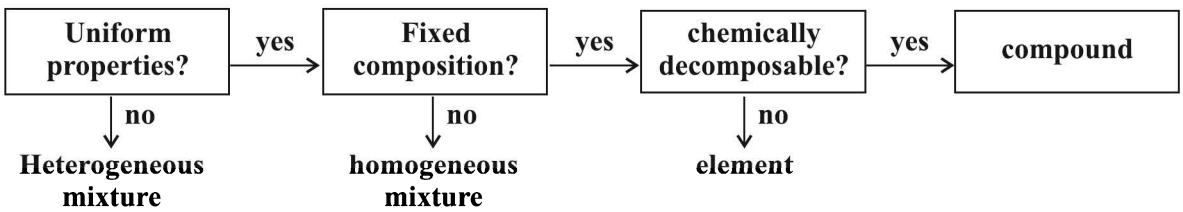
Example 1: A mixture of sand and salt has different composition in different parts of the mixture.

Example 2: A mixture of iron and sulphur has different composition in different parts of the mixture.

A simple flow chart to identify matter

Ask these questions. Depending on the answer you can identify the type of matter.

Classification of matter



1.10 CLASSIFICATION OF MIXTURES

1. **Based on the physical state:** Mixtures are classified based on the physical state of the constituent parts.

(a) Solid-solid mixture

If a mixture contains two solids as components then it is called solid-solid mixture.

Example 1: A mixture of pebbles and pulses – both pebbles and pulse grains are solids.

Example 2: A mixture of iron and sand – both iron and sand are solids.

(b) Solid-liquid mixture

If a mixture contains a solid and liquid as components, it is called a solid-liquid mixture.

Example 1: A mixture of clay and water – clay is a solid mixed in water which is a liquid.

Example 2: A mixture of chalk and water – chalk is a solid mixed in water which is a liquid.

Example 3: A mixture of common salt in sea water – common salt is a solid mixed in sea water which is a liquid.

(c) Liquid-liquid mixture

If a mixture contains two liquids as components, it is called a liquid-liquid mixture.

Example 1: A mixture of oil and water – oil and water are liquids.

Example 2: A mixture of methyl alcohol and ethyl alcohol – both are liquids.

(d) Solid-gas mixture

If a mixture contains a solid mixed with a gaseous substance, it is called a solid-gas mixture.

Example: Smoke is a mixture of soot or dust mixed with air.

(e) Liquid-gas mixture

If a mixture contains a liquid mixed with a gaseous substance, it is called a liquid-gas mixture.

Example: Natural tap water contains oxygen and carbon dioxide (air) dissolved in water.

(f) Gas-gas mixture

If a mixture contains a gas mixed with another gaseous substance, it is called a gas-gas mixture.

Example: Mountain air is a mixture of gases – oxygen, nitrogen, etc.

1.11 SEPARATION OF MIXTURES

Often we need to separate the constituents of a mixture. For example, it is important to separate pebbles from a mixture of pebbles and pulse grains before we use the pulse for cooking.

Separation of the constituents in a mixture is a process that depends on the physical properties of the components.

Need for separation of mixtures

(i) To separate unwanted constituents from the useful components of a mixture.

Example: Separation of cream from milk, separation of husk particles from wheat grains.

(ii) To obtain a pure substance from a mixture of substances that may or may not be impurities.

Example: Separation of salt from sea water, separation of suspended impurities in tap or well water.

Separation of solid-solid mixtures

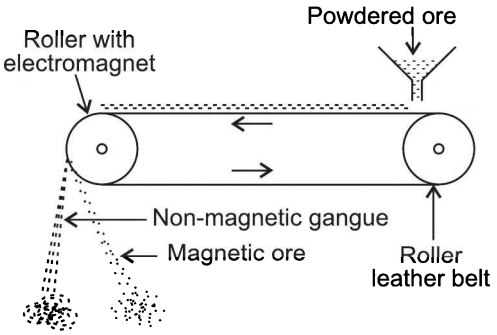
(i) By mechanical picking: This method is used when the solid particles are of different size, shape and colour and visible to the human eye.

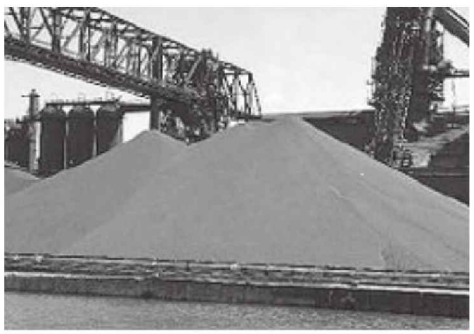
Example: Separating small stones from wheat or rice.

(ii) Magnetic separation: This method is used when one of the components is magnetic and other components are non-magnetic.

Example 1: A mixture of iron filings and powdered sulphur can be easily separated by passing a magnet over the mixture repeatedly so that iron filings are attracted to the magnet leaving the non-magnetic sulphur behind.

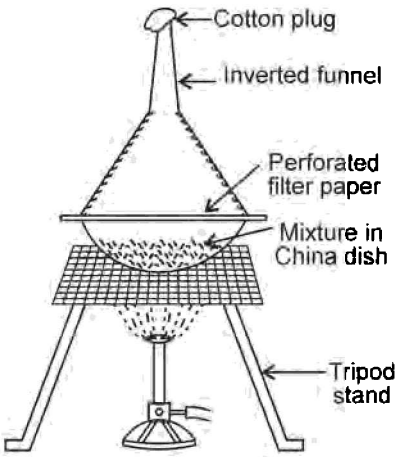
Example 2: This method is used widely by the iron and steel industry. Iron ore is powdered and passed over magnetic rollers when sand and impurities which are non-magnetic fall straight down whereas the iron ore particles which are magnetic fall separately (See sketch below and actual photo of how iron ore is separated from impurities).





(iii) Sublimation: This method is used when one of the components sublimes on heating. (Sublimation is the process of direct change of state from solid to gas).

Example: A mixture of common salt and ammonium chloride.

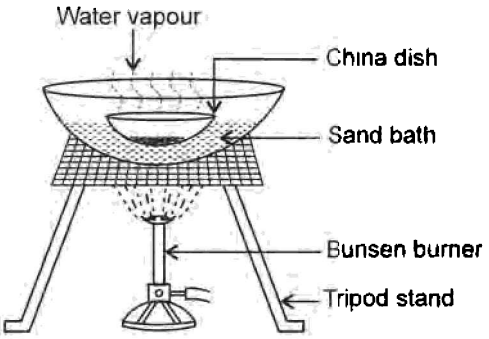


Separation of a mixture by sublimation

Separation of solid-liquid mixture

(i) By evaporation

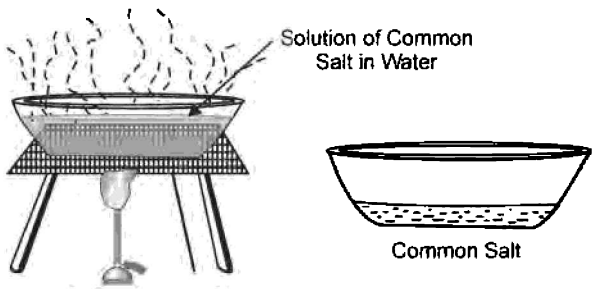
This method is used when the liquid in the mixture has a low boiling point. To separate the liquid, simple evaporation or boiling is used depending on the boiling point of the liquid.



Separation of non-volatile solid by evaporation

Example 1: Sulphur dissolved in carbon disulphide can be separated by placing the solution in the sun. Carbon disulphide has a low boiling point and evaporates after some time leaving sulphur behind.

Example 2: Common salt and water can easily be separated by boiling the solution when water boils off leaving the salt behind.



Separation of Common Salt by evaporation

(ii) Distillation

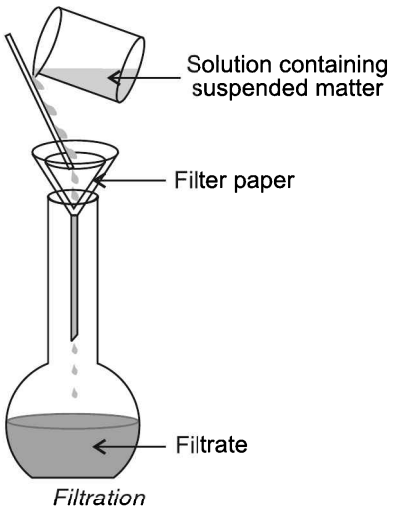
Distillation is the process of converting a liquid into vapour (by heating) and condensation of the vapour to separate the pure liquid. The condensed vapour is called distillate.

Example 1: Mixture of iodine in chloroform can be separated by gentle heating when chloroform vapourises and the vapour condensed to get pure chloroform.

Example 2: A mixture of common salt in water can be separated by strong heating when water evaporates leaving the salt behind. The vapours when condensed give pure water known as distilled water.

(iii) Filtration

Filtration is the process of removing suspended impurities from a solution. This method is used when the solid is insoluble in the liquid and thus forms a heterogeneous mixture. The mixture is poured into a folded filter paper when the liquid passes through the filter paper and is collected as filtrate. The insoluble solid is left behind on the filter paper as residue. The method of folding the filter paper to form a cone which will fit into the mouth of a funnel is shown. The actual method of filtration is also shown.



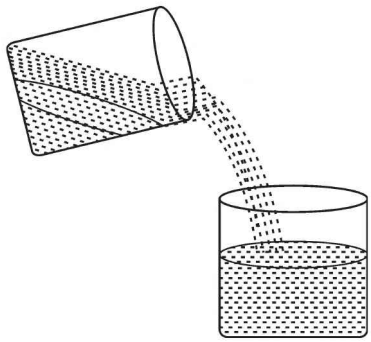
Filtration

Example 1 : Chalk powder and water

Example 2 : Sand and water

(iv) Sedimentation and decantation

This method is used to separate a heterogeneous mixture containing an insoluble solid in a liquid. The mixture is allowed to stand for some time without disturbing the mixture by stirring or in any way. The insoluble solid substances that are heavier settle at the bottom. A clear liquid column forms on the top. This clear liquid column is called supernatant liquid. The solid substance that settles down is called sediment. This process is known as sedimentation. The clear liquid is slowly poured out taking care not to disturb the solid sediment at the bottom too much. This process is known as decantation.

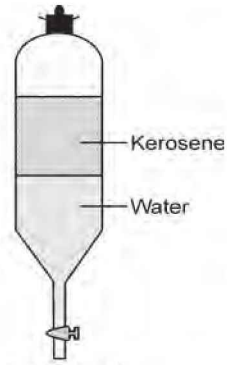


Decantation

Example: Sand and water can be separated by sedimentation and then decantation.

Separation of liquid-liquid mixture

This method can be used for separating two or more liquids that do not mix with each other. The mixture is poured into a **separating funnel**. On standing, such a liquid-liquid mixture separates into **two distinct layers** – the heavier liquid forms the lower layer, while the lighter one forms the upper layer. The tap at the bottom of the funnel is opened and the lower liquid is drained out into a beaker. Now another beaker is placed under the funnel and the lighter liquid is next drained out. Example: A mixture of kerosene and water can be separated in this way. Kerosene, which is lighter than water, will form the upper layer. The lower layer is water (see figure).



Separation by a separating funnel

1.12 DIFFERENCES BETWEEN COMPOUNDS AND MIXTURES

Characteristics	Compound	Mixture
1. Composition	Made up of atoms of elements in a fixed proportion	Made up of elements or compounds or both in any proportion
2. Nature	Particles are of the same kind	Particles are of different kinds
3. Structure	Always homogeneous	May or may not be homogeneous
4. Appearance	Components cannot be seen separately	Components may or may not be seen separately
5. Preparation	Always involves a chemical change	Involves only physical change
6. Properties	Entirely different from those of the constituents	Each component retains its individual properties
7. Separation	Components can be separated by chemical means only	Components can be separated by physical means only
8. Energy changes	Energy is always evolved or absorbed	Generally no energy is evolved or absorbed

EXERCISE

I. Choose the correct answer

1. **The substance which is not a compound**
   (a) Salt (b) Water
   (c) Graphite (d) Sand
2. **A substance that decomposes on heating**
   (a) Oxygen (b) Phosphorous
   (c) Mercuric oxide (d) Sulphur
3. **Elements are represented by**
   (a) Symbol (b) Molecular formula
   (c) Ion (d) Equation
4. **A group of atoms possessing positive charge or negative charge**
   (a) Atom (b) Radical
   (c) Molecule (d) Valency
5. **Substances formed as a result of a chemical reaction**
   (a) atoms (b) reactants
   (c) products (d) molecules
6. **The symbols of gold and silver are respectively**
   (a) Au, S (b) Au, Ag
   (c) Ag, Au (d) G, Ag
7. **A formula tells about**
   (a) an element
   (b) number of atoms of each element in a molecule
   (c) gives a mathematical formula
   (d) the source of the compound
8. **The molecular formula is a short form representation of**
   (a) a compound (b) a mixture
   (c) an element (d) ion
9. **Compounds of oxygen with other elements are called**
   (a) bromides (b) oxides
   (c) chlorides (d) sulphides
10. **The change in which a completely new substance is formed is**
    (a) physical change (b) chemical change
    (c) reversible change (d) temporary change
11. **The substance which is not a compound is**
    (a) marble (b) limestone
    (c) diamond (d) chalk
12. **The substance that is not pure is**
    (a) brass (b) copper wire
    (c) aluminium sheet (d) gold coin
13. **The substance that breaks into simpler substances on simple heating is**
    (a) carbon (b) sulphur
    (c) phosphorous (d) zinc carbonate
14. **The mixture that is not homogeneous is**
    (a) sugar solution (b) brine
    (c) soil (d) petrol
15. **The substance that has only one kind of atom in it is**
    (a) marsh gas
    (b) dry ice (carbon dioxide in solid state)
    (c) well water d. ozone
16. **The method that can be used to separate a homogeneous mixture of two liquids with differing densities is**
    (a) decantation using a funnel with a tap
    (b) filtration
    (c) fractional crystallisation
    (d) fractional distillation
17. **Chromatography can be used to separate a _____ mixture.**
    (a) solid-solid b. complex mixture
    (c) liquid-gas d. solid-liquid
18. **Fractional crystallization uses the difference in _____of the components of a mixture.**
    (a) rates of diffusion (b) density
    (c) melting point (d) solubility
19. **The common element in all amalgams is**
    (a) iron (b) mercury
    (c) copper (d) oxygen
20. **An alloy is a homogeneous mixture of**
    (a) non-metals (b) inert gases
    (c) metals (d) metalloids

II. Give one word answers

1. A mixture in which the composition is uniform
   _____
2. The name of the scientist who gave the name electron
   _____
3. Electrons, protons and neutrons are called
   _____
4. The instrument used for separating two immiscible liquids with difference in densities
   _____
5. Elements with properties between that of metals and non-metals
   _____

III. Match the following.

1. **Match the mixture in Set A with the type of mixture in Set B**

Set A	Set B
i. salt in water	a. solid in solid
ii. soda water	b. gas in gas
iii. air	c. solid in liquid
iv. bronze	d. gas in liquid

2. Match the examples of changes in Set A with their type in Set B.

Set A	Set B
i. CO_2 gas to CO_2 solid	a. Chemical change
ii. Hydrogen to Helium	b. Nuclear fission
iii. C to CO_2	c. Nuclear fusion
iv. Uranium to Lead	d. Physical change

3. Match the separation method in Set A that can be used for separating mixtures given in Set B.

Set A	Set B
i. Sedimentation	a. mixture of CO_2 and O_2 gas
ii. Fractional distillation	b. mixture of sand in water
iii. Fractional crystallisation	c. mixture of kerosene and petrol
iv. Diffusion	d. mixture of NaCl and KNO_3

IV. State whether the following statements are True or False

1. A homogeneous mixture is a pure substance.
   _____
2. All pure substances are not compounds but all compounds are pure substances.
   _____
3. Separation of mixtures depends on chemical properties.
   _____
4. The symbols of sodium, mercury and gold, silver are derived from their Latin names.
   _____
5. Homogeneous mixture has uniform but not fixed composition.
   _____

V. Find the odd one out

1. Boron, Germanium, Arsenic, Calcium, Antimony
2. Chlorine, Oxygen, Hydrogen, Steam, Nitrogen

VI. Select the suitable word from the words given below and fill in the blanks. A word can be used more than once.

metalloid, electrons, hydrogen, metallic, proton, oxygen, xenon, neutron, atomicity, graphite

1. The non-metal which is a good conductor is _____.
2. The particle(s) that is / are not negatively charged in the atom is / are _____.
3. Polonium is a _____.
4. The gaseous elements in the given list are _____.
5. Platinum is a _____ element.
6. The most abundant free element that is not very abundant in the earth is _____.
7. The particle that was discovered the last _____.
8. As sulphur has eight atoms to a molecule its _____ is eight.
9. The particle that balances the charge of the electron in the atom is _____
10. Water molecule has double the number of _____ atoms as the number of _____ atoms.

VII. Fill in the blanks

1. Kalium is the Latin name of _____
2. The name of the element based on the planet Uranus is _____ and its symbol is _____
3. _____ is the combining capacity of elements.
4. Respiration is an example of _____
5. An _____ is the smallest unit of matter.
6. Protons carry _____ charge.
7. The particles of matter which take part in a chemical reaction are called _____
8. The atomicity of Helium is _____
9. _____ are malleable and ductile.
10. Non-metals have _____ valency
11. In a chemical reaction_____are rearranged.

VIII. Tick mark (✓) against the most suitable answer from the alternatives suggested in the following.

1. Boiling Point and Melting Point are (physical / chemical) properties.
2. Reaction of an element with other elements are (physical / chemical) properties.
3. To identify pure substances the (physical / chemical) properties are compared with those of known substances.
4. The most abundant element in the universe is (hydrogen / helium).
5. At room temperature, the liquid non-metal is (bromine / mercury) and liquid metal is (bromine / mercury).

IX. Each question contains two statements. Statement A is an Assertion Statement and Statement R is the Reason Statement. Each question has four choices out of which only one is correct. Choose the correct one.

Choose A if Statement A is true, Statement R is true; Statement R is correct explanation for statement A

Choose B if Statement A is true, statement R is true; Statement R is not correct explanation for statement A

Choose C if Statement A is true, Statement R is false

Choose D if Statement A is false, statement R is true

1. **Statement A:** A compound is a pure substance.

   Statement R: Atoms of different elements in a compound combine in a fixed ratio by mass.

2. **Statement A:** An atom is electrically neutral.

   Statement R: The number of protons is equal to the number of electrons in an atom.

3. **Statement A:** An ion is not electrically neutral.

   Statement R: The number of electrons and protons are not equal in an ion.

4. **Statement A:** Pure compounds are homogeneous substances.

   Statement R: Compounds cannot be separated into constituents by physical methods.

5. **Statement A:** Each component in a mixture retains its individual properties.

   Statement R: Components can be separated by physical means only.

6. **Statement A:** A mixture is not a pure substance.

   Statement R: The simple components of a mixture taken separately are impure.

7. **Statement A:** A compound is broken into simpler substances by physical methods.

   Statement R: A compound is made up of two or more atoms combined chemically.

8. **Statement A:** Separation of the components of a mixture is done by chemical methods.

   Statement R: Components of a mixture have different physical properties.

9. **Statement A:** A compound will have the properties of the elements present in it.

   Statement R: The elements in a compound are combined chemically.

10. **Statement A:** A substance that has uniform composition is an element or a compound.

    Statement R: A heterogeneous mixture does not have uniform composition.

X. Answer the following questions.

1. Write the symbols for the following elements :

   sodium, potassium, lead, phosphorus, carbon, fluorine, bromine.

2. Write the elements represented by the following symbols.

   Ca, Zn, Fe, Cu, Hg, Ag, C, N

3. Classify the following into elements and compounds :

   sodium, water, oxygen, chlorine, carbon dioxide, sulphur, hydrochloric acid.

4. Classify the following elements into metals and non-metals.

   Sodium, Chlorine, Magnesium, Oxygen, Aluminium, Sulphur.

5. What does each of the following represent?
   (a) $4O_2$ (b) $4O$
   (c) O_3 (d) H_2O

6. What are pure and impure substances?

7. Name a technique used to separate:
   (a) Iron and sulphur
   (b) Salt in sea water
   (c) Ammonium chloride and sand
   (d) Coconut oil and water.
   (e) Chalk powder and water
   (f) Sand and water
   (g) Sand and iodine
   (h) Alcohol and water

8. Name the change of state in which naphthalene changes into vapour.

9. What is sublimation? Give three examples of substances that sublime.

10. Differentiate between: Evaporation and Boiling. Give one similarity between the two.

11. Name three different methods in each case for separation of
    (a) Solid-solid mixture (b) Solid-liquid mixture
    (c) Liquid-liquid mixture

12. Give an example of
    (a) Homogeneous mixture
    (b) Heterogeneous mixture
    (c) Monoatomic molecule
    (d) Diatomic molecule
    (e) Triatomic molecule
    (f) Polyatomic molecule
    (g) A solid non-metal
    (h) A liquid metal

13. Classify the following as pure substances and mixtures :
    (a) Common salt (b) Milk
    (c) Bronze (d) Sea water
    (e) Copper wire (f) Carbon dioxide

14. Name the four categories in which elements have been classified.
15. Name the two most abundant elements in the earth's crust.
16. Name the two most abundant metals in the earth's crust.
17. Name the three most common elements present in the human body.
18. What are compounds? Give any two characteristics of compounds.

KEY TO EXERCISE

I. 1. c 2. c 3. a 4. b 5. c 6. b 7. b 8. a
9. b 10. b 11. c 12. a 13. d 14. c 15. d 16. a
17. b 18. d 19. b 20. c

II. 1. homogeneous 2. G J Stoney 3. Fundamental particles
4. separating funnel 5. metalloids

III. 1. i–c ii–d iii–b iv–a 2. i–d ii–c iii–a iv–b
3. i–b ii–c iii–d iv–a

IV. 1. False 2. True 3. False 4. True 5. True

V. 1. calcium 2. steam

VI. 1. graphite 2. neutron, proton 2. metalloid 4. hydrogen, oxygen and xenon 5. metallic
6. hydrogen 7. neutron 8. atomicity 9. proton 10. hydrogen, oxygen

VII. 1. potassium 2. Uranium, U 3. Valency 4. oxidation 5. atom
6. positive 7. atom 8. one 9. Metals 10. negative 11. atoms

VIII. 1. physical 2. chemical 3. physical 4. hydrogen 5. bromine, mercury

IX. 1. B 2. A 3. A 4. B 5. B 6. C 7. D
8. D 9. D 10. B

2 Symbols, Formulae and Equations

2.0 REVIEW OF IMPORTANT CONCEPTS

We have seen that matter is classified based on chemical composition into pure and impure substances. Pure substances are further classified into Elements and Compounds. Elements cannot be broken into simpler substances by simple physical or chemical methods. Compounds are made up two or more elements combined chemically in a fixed proportion by weight. Compounds can be broken into simpler substances by chemical methods. Elements combine chemically in fixed proportion by weight to form compounds.

2.1 CHEMICAL SYMBOLS

Representing Elements

There are more than hundred known elements. Some examples of elements are hydrogen, sodium, oxygen, carbon and nitrogen, etc. A convenient way of representing elements is by using symbols.

A symbol is used to represent an element in short form.

Generally a symbol comprises of the first letter or first two letters of the name of the element or its Latin equivalent. Symbols of first thirty elements are given below. The Latin equivalent names, where present, are mentioned in brackets. The symbol is underlined in the name.

	Element	Symbol
1.	**H**ydrogen	H
2.	**He**lium	He
3.	**Li**thium	Li
4.	**Be**ryllium	Be
5.	**B**oron	B
6.	**C**arbon	C
7.	**N**itrogen	N
8.	**O**xygen	O
9.	**F**luorine	F
10.	**Ne**on	Ne
11.	Sodium (Latin name is **Na**trium)	Na
12.	**Mg**nesium	Mg
13.	**Al**uminium	Al
14.	**Si**licon	Si
15.	**P**hosporous	P
16.	**S**ulphur	S
17.	**Cl**orine	Cl
18.	**Ar**gon	Ar
19.	Potassium (Latin name is **K**alium)	K
20.	**Ca**lcium	Ca
21.	**Sc**andium	Sc
22.	**Ti**tanium	Ti
23.	**V**anadium	V
24.	**Cr**omium	Cr
25.	**Mn**ganese	Mn
26.	Iron (Latin name is (**Fe**rrum)	Fe
27.	**Co**balt	Co
28.	**Ni**ckel	Ni
29.	Copper (Latin name is **Cu**prum)	Cu
30.	**Zn**c	Zn

Some more symbols

Some more symbols taken from Latin names are given below.

Element	Latin name	Symbol
Silver	**Ag**entum	Ag
Tin	**Sn**nnum	Sn
Lead	**Pb**um	Pb
Mercury	**Hg**rargyrum	Hg
Antimony	**Sb**ium	Sb
Gold	**Au**rum	Au

2.2 MEANING CONVEYED BY SYMBOLS

Symbols convey very important information about an element.

For example the symbol H used for the element hydrogen conveys the following information:

(i) The symbol H is of the element hydrogen

(ii) The symbol H represents one atom of hydrogen

(iii) If we look for this element in the Periodic Table of Elements, we will get further information about its atomic number and atomic weight or mass number as follows:

Atomic Number = 1
Mass Number = 1

Similarly the symbol O used for the element oxygen conveys the following information:

(i) The symbol O is of the element oxygen.

(ii) The symbol O represents one atom of oxygen.

(iii) From the Periodic Table of elements, we know that its atomic number is 8, mass number is 16.

2.3 ELEMENTS ARE MADE UP OF ATOMS

Elements are pure substances that are made up of small particles called *atoms*.

An atom is the smallest unit of matter that may or may not exist independently, but takes part in chemical reactions.

It has been found that atoms are made up of certain particles that are inside the atom as follows.

(i) The atoms have *electrons*, *protons* and *neutrons* in them.

  (a) The *electrons are negatively charged particles.*

  (b) The *protons are positively charged particles.*

  (c) The *neutrons are neutrally charged particles.*

(ii) The number of electrons in an atom is equal to the number of protons in it. Since the negatively charged particles (electrons) are equal to the positively charged particles (protons) the *atom is electrically neutral.*

(iii) The protons and neutrons are present in a small central region within the atom called the *nucleus*. Since protons and neutrons are found inside the nucleus of the atom they are called *nucleons*.

(iv) The electrons revolve around the nucleus at fixed distances from the nucleus at large speeds.

(v) The **atomic number** of an element gives the number of protons in an atom. This number is also equal to the number of electrons in the neutral atom. For example the atomic number of sodium is 11. This means that each sodium atom has 11 protons and 11 electrons. Similarly, the atomic number of oxygen is 8. This means that each oxygen atom has 8 protons and 8 electrons.

(vi) The *mass number* of an element gives the sum of the number of protons and neutrons in an atom. For example the mass number of sodium is 23. This means that the sum of the protons and neutrons in the nucleus of sodium atom is 23.

(vii) The difference between mass number and atomic number gives the number of neutrons in the atom of the element. For example the difference between mass number of sodium (23) and its atomic number (11) is 23 – 11 = 12. There are 12 neutrons in sodium atom.

(viii) The electrons revolve around the nucleus in fixed orbits of increasing radius.

(ix) Each orbit can hold *a maximum of* $2n^2$ *electrons*, where n is the number of the orbit.

(x) The *electronic configuration* of an element gives the distribution of electrons in each orbit of the atom.

2.4 ELECTRONIC CONFIGURATION OF ELEMENTS (BOHR-BURY RULES)

The electrons in an atom are arranged according to certain rules known as Bohr-Bury rules.

(i) The centre of the atom consists of the nucleus with the nucleons (protons and neutron) in it.

(ii) The electrons revolve in fixed orbits around the nucleus of the atom.

(iii) The orbits are numbered from the centre outwards. The first orbit closest to the nucleus is numbered as 1 and is denoted as K-shell. The next is numbered as 2 and is denoted as L-shell. The next with number 3 is denoted as M-shell. The next with number 4 is denoted as N-shell and so on.

(iv) The maximum number of electrons in an orbit is given by $2n^2$, where n is the number of the orbit.

The first orbit can have a maximum of 2 electrons.

The second orbit can have a maximum of 8 electrons.

The third can have a maximum of 18 electrons and so on.

The last orbit however irrespective of its number cannot have more than 8 electrons.

(v) The maximum number of electrons in the outermost shell cannot exceed 8 and in the

second last shell 18, irrespective of the shell number.

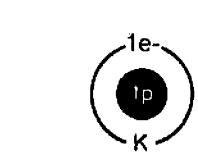
(vi) It is not necessary that the last orbit should be filled with electrons for the next orbit to start filling – a new orbit starts filling as soon as the last orbit gets 8 electrons. (For example the maximum number of electrons in the 3rd shell is 18. But the fourth shell starts filling as soon as the 3rd shell has 8 electrons.

Orbit	Number	Maximum number of electrons ($2n^2$)
K-shell	n = 1	2
L-shell	n = 2	8
M-shell	n = 3	18
N-shell	n = 4	32

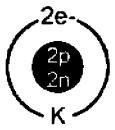
The electronic configuration of the first twenty elements is tabulated below.

Element Symbol	Atomic Number (Z)	Mass Number (A)	Protons (= Z)	Neutrons (A – Z)	Electrons (= Z)	Electronic Configuration			
						K-shell	L-shell	M-shell	N-shell
Hydrogen–H	1	1	1	0	1	1			
Helium–He	2	4	2	2	2	2			
Lithium–Li	3	7	3	4	3	2	1		
Beryllium–Be	4	9	4	5	4	2	2		
Boron–B	5	11	5	6	5	2	3		
Carbon–C	6	12	6	6	6	2	4		
Nitrogen–N	7	14	7	7	7	2	5		
Oxygen–O	8	16	8	8	8	2	6		
Fluorine–F	9	19	9	10	9	2	7		
Neon–Ne	10	20	10	10	10	2	8		
Sodium–Na	11	23	11	12	11	2	8	1	
Magnesium–Mg	12	24	12	12	12	2	8	2	
Aluminium–Al	13	27	13	14	13	2	8	3	
Silicon–Si	14	28	14	14	14	2	8	4	
Phosphorous–P	15	31	15	16	15	2	8	5	
Sulphur–S	16	32	16	16	16	2	8	6	
Chlorine–Cl	17	35	17	18	17	2	8	7	
Argon–Ar	18	40	18	22	18	2	8	8	
Potassium–K	19	39	19	20	19	2	8	8	1
Calcium–Ca	20	40	20	20	20	2	8	8	2

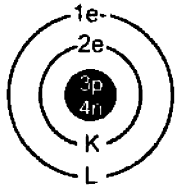
Atomic structures of the first twenty elements are given below.



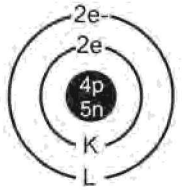
Hydrogen atom : A=1; Z=1



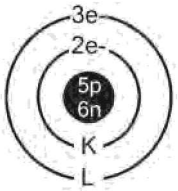
Helium atom : A=4; Z=2



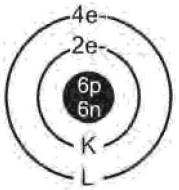
Lithium atom : A=7; Z=3



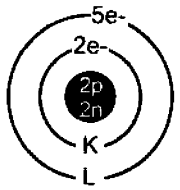
Beryllium atom : A=9; Z=4



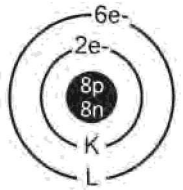
Boron atom : A=11; Z=5



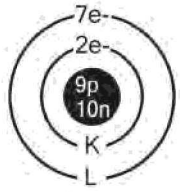
Carbon atom : A=12; Z=6



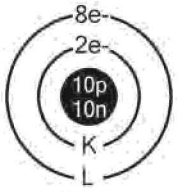
Nitrogen atom : A=14; Z=7



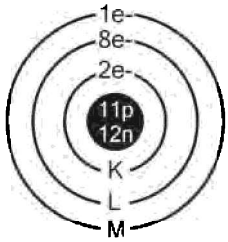
Oxygen atom : A=16; Z=8



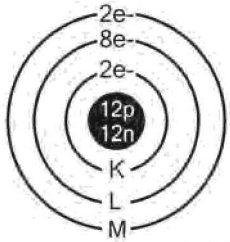
Fluorine atom : A=19; Z=9



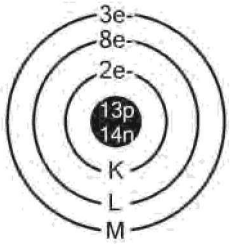
Neon atom : A=20; Z=10



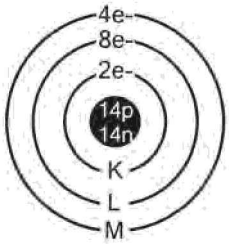
Sodium atom A=23; Z=11



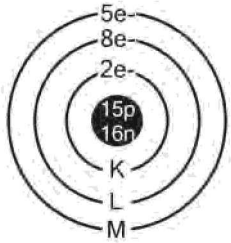
Magnesium atom : A=24; Z=12



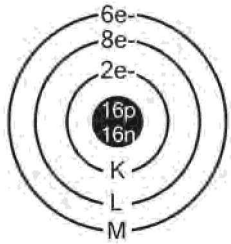
Aluminum atom : A=27; Z=13



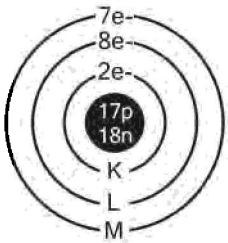
Silicone atom : A=28; Z=14



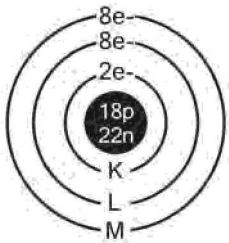
Phosphorus atom : A=31; Z=15



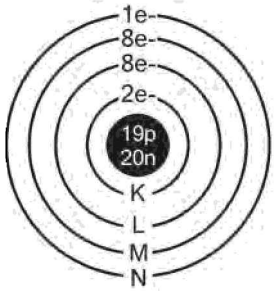
Sulphur atom : A=32; Z=16



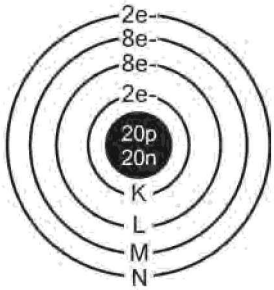
Chlorine atom : A=35; Z=17



Argon atom : A=40; Z=18



Potassium atom : A=39; Z=19



Calcium atom : A=40; Z=20

2.5 VALENCE ELECTRONS

The valence electrons are the electrons that are present in the last or outermost shell of an atom.

The number of electrons in the last orbit of an atom in general helps in determining the valency of the element.

To find the valency of the element, the atomic number must be known. Then the electronic configuration needs to be known. The number of electrons in the last shell gives its valence electrons.

Significance of valence electrons

The valence electrons of an element decide the chemical properties of the element.

The valence electrons of an element are of extreme importance. They decide

(i) which element can combine with another elements and

(ii) how easily or with how much difficulty the element will combine.

Valency – the traditional concept

Valency is the combining capacity of an element with another element.

Illustration

Let us take the example of the combining capacity of a few elements *with hydrogen.*

Each atom of carbon combines with 4 atoms of hydrogen. So, valency of carbon is 4.

Each atom of nitrogen combines with 3 atoms of hydrogen. So, valency of nitrogen is 3.

Each atom of oxygen combines with 2 atoms of hydrogen. So, valency of oxygen is 2.

Each atom of chlorine combines with 1 atom of hydrogen. So, valency of chlorine is 1.

2.6 IONS AND THEIR FORMULAE

Formation of ions

When an atom loses or gains electrons it gets a charge.

On losing electrons, the positive charge is more than the negative charge in an atom. Thus the atom *acquires a positive charge.*

On gaining electrons, the negative charge is

more than the positive charge in the atom. The atom thus *acquires negative charge.*

An atom that acquires an electrical charge – either positive (on losing electrons) or negative (on gaining electrons) is called an *ion.*

Anions and Cations

If an atom gains electrons, the atom gets a negative charge. Such a negatively charged 'atom' is called a negative ion or *anion.*

If an atom loses electrons, the atom gets a positive charge. Such a positively charged 'atom' is called a positive ion or *cation.*

Why do atoms lose or gain electrons?

Inert gas elements like neon, argon, krypton, xenon and radon have 8 electrons in the last shell. They are chemically inactive as they do not react with other elements. Hence these elements are considered to be very stable.

When an atom has 8 electrons in the last shell they are highly stable. The stability can be seen in the fact that such elements are chemically inactive. All atoms lose or gain electrons to get 8 electrons in the last shell.

Importance of 8 electrons in the last shell

It is seen that elements with 8 electrons in the last shell do not combine with other elements. They are said to be non-reactive, inert and **stable**.

In fact, the reason for some elements combining with other elements is to get 8 electrons in the valence shell, as 8 valence electrons gives **extra stability**.

They get eight electrons by **losing, gaining or sharing electrons** with other atoms of the same or different elements.

All chemical reactions involve transfer or sharing of valence electrons so that each atom gets eight electrons.

Single shell atoms

In single shell atoms, the maximum number of electrons in the shell is 2. Hence when there is only one shell, atoms try to get 2 electrons in the shell to get stability. This they do in the same way - by losing, gaining or sharing electrons.

Consider sodium atom. It has an electronic configuration of (2, 8, 1). Thus, sodium atom can get a stable configuration of 8 electrons in the last shell, *if one electron is removed from the valence shell.*

Sodium loses one electron from the M-shell and gets 8 electrons in the L-shell that is now the *new* last shell.

Consider chlorine atom. It has an electronic configuration of (2, 8, 7). Thus, chlorine atom can get a stable configuration of 8 electrons in the last shell, *if one electron is added in the valence shell.*

Chlorine gains one electron in the M-shell and gets 8 electrons in this shell and thus becomes more stable.

Elements with 1, 2 or 3 electrons in the valence shell lose these electrons so that the new valence shell has 8 electrons.

Elements with 5, 6 or 7 electrons in the valence shell gain 3, 2 or 1 electron respectively to get 8 electrons in the valence shell.

Elements with 4 electrons cannot *lose or gain* 4 electrons with equal ease. Such elements with 4 electrons in the valence shell *share 4 electrons with other elements to get 8 electrons in the valence shell*

We will study how elements lose, gain or share electrons to form chemical bonds with other elements in subsequent chapters.

Formation of cations

Element	Symbol	Atomic Number	Electronic Configuration	Ion Formation	Valence electrons after losing/ gaining electrons
Hydrogen	H	1	(1)	$H - 1e^- \rightarrow H^+$	0
Helium	He	2	(2)	Does not lose electrons as the atom has only one shell and the shell has 2 electrons.	–

Lithium	Li	3	(2, 1)	$Li - 1e^- \rightarrow Li^{1+}$	2
Beryllium	Be	4	(2, 2)	$Be - 2e^- \rightarrow Be^{2+}$	2
Boron	B	5	(2, 3)	$B - 3e^- \rightarrow B^{3+}$	2
Neon	Ne	10	(2, 8)	Does not lose electrons as the atom has 8 electrons in the valence shell	–
Sodium	Na	11	(2, 8, 1)	$Na - 1e^- \rightarrow Na^+$	8
Magnesium	Mg	12	(2, 8, 2)	$Mg - 2e^- \rightarrow Mg^{2+}$	8
Aluminium	Al	13	(2, 8, 3)	$Al - 3e^- \rightarrow Al^{3+}$	8
Silicon	Si	14	(2, 8, 4)	$Si - 4e^- \rightarrow Si^{4+}$	8
Argon	Ar	18	(2, 8, 8)	Does not lose electrons as the atom has 8 electrons in the valence shell	–
Potassium	K	19	(2, 8, 8, 1)	$K - 1e^- \rightarrow K^+$	8
Calcium	Ca	20	(2, 8, 8, 2)	$Ca - 2e^- \rightarrow Ca^{2+}$	8

Formation of anions

Element	Symbol	Atomic Number	Electronic Configuration	Ion Formation	Valence electrons after losing/ gaining electrons
Hydrogen	H	1	(1)	$H + 1e^- \rightarrow H^{1-}$	2
Helium	He	2	(2)	Does not gain electrons as the atom has 2 electrons in the valence shell	–
Carbon	C	6	(2, 4)	$C + 4e^- \rightarrow C^{4-}$	8
Nitrogen	N	7	(2, 5)	$N + 3e^- \rightarrow N^{3-}$	8
Oxygen	O	8	(2, 6)	$O + 2e^- \rightarrow O^{2-}$	8
Fluorine	F	9	(2, 7)	$F + 1e^- \rightarrow F^-$	8
Neon	Ne	10	(2, 8)	Does not gain electrons as the atom has 8 electrons in the valence shell	–
Phosphorous	P	15	(2, 8, 5)	$P + 3e^- \rightarrow P^{3-}$	8
Sulphur	S	16	(2, 8, 6)	$S + 2e^- \rightarrow S^{2-}$	8
Chlorine	Cl	17	(2, 8, 7)	$Cl + 1e^- \rightarrow Cl^-$	8
Argon	Ar	18	(2, 8, 8)	Does not gain electrons as the atom has 8 electrons in the valence shell	–

2.7 MODERN CONCEPT OF VALENCY

Valency is the number of electrons lost or gained by an atom of an element in its valence shell.

Illustration

Atoms lose electrons to become cations. Referring to the table of cation formation we know that :

Sodium loses one electron. So valency of sodium is +1.

Magnesium loses two electrons. So valency of magnesium is +2.

Aluminium loses three electrons. So valency of aluminium is +3.

Atoms gain electrons to become anions. Referring to the table of anion formation we know that :

Chlorine gains one electron. So valency of chlorine is –1.

Oxygen gains two electrons. So valency of oxygen is –2.

Nitrogen gains three electrons. So valency of nitrogen is –3.

Variable valency

Some metals show variable valency. In different compounds they can lose different number of electrons to exhibit *variable valency.*

Example: Iron shows two valencies in $FeCl_2$ and $FeCl_3$. In $FeCl_2$, the valency of iron is +2. In $FeCl_3$, the valency of iron is +3.

$Fe - 2e^- \longrightarrow Fe^{2+}$ (two electrons lost by iron atom)

$Fe - 3e^- \quad Fe^{3+}$ (three electrons lost by iron atom)

The lower valency is suffixed with –ous in its name. The higher valency is suffixed with –ic in its name. Thus Fe^{2+} is called *ferrous ion* and Fe^{3+} is called *ferric ion.*

Other examples:

Cu^+ - Cuprous	Cu^{2+} - Cupric
Pb^{2+} - Plumbous	Pb^{4+} - Plumbic
Cr^{2+} - Chromous	Cr^{3+} - Chromic
Au^+ - Aurous	Au^{3+} - Auric
Sn^{2+} - Stannous	Sn^{4+} - Stannic
Ag^+ - Argentous	Ag^{2+} - Argentic

2.8 RADICAL

Radical is a group of atoms that carries a charge (positive or negative) and behaves as a single unit in a chemical reaction.

Example: Sulphate ion (SO_4^{2-}) is a radical.

$H_2\underline{SO_4} + 2NaCl \qquad Na_2\underline{SO_4} + 2HCl$

The sulphate part in H_2SO_4 moves as a unit and combines with sodium.

Other examples of radicals:

Positive radicals:	NH_4^+	– ammonium
	ZnO_2^{2+}	– zincate
	AlO_2^{2+}	– aluminate
Negative radicals:	OH^-	– hydroxide radical
	SO_4^{2-}	– sulphate radical
	NO_3^-	– nitrate

Given below is an alphabetical list of cations and anions. Some of these ions will be used more as you go into higher classes. But, you can keep this list as a reference.

List of cations – Given below is a list of cations in the alphabetical order.

List of Cations

Name of ion	Ion	Name of ion	Ion
Aluminium	Al^{3+}	Hydrogen	H^+
Ammonium	NH_4^+	Lithium	Li^+
Antimonic	Sb^{5+}	Manganous	Mn^{2+}
Antimonous	Sb^{3+}	Manganic	Mn^{3+}
Arsenic	As^{5+}	Magnesium	Mg^{2+}
Argentic	Ag^{2+}	Mercuric	Hg^{2+}
Arsenous	As^{3+}	Mercurous	Hg_2^{2+}
Auric	Au^{3+}	Nickel	Ni^{2+}
Argentous	Ag^+	Plumbous	Pb^{2+}
Aurous	Au^+	Potassium	K^+
Beryllium	Be^{2+}	Platinum	Pt^{4+}
Boron	B^{3+}	Phosphonium	PH_4^+
Barium	Ba^{2+}	Plumbic	Pb^{4+}
Calcium	Ca^{2+}	Radium	Ra^{2+}
Chromic	Cr^{3+}	Rubidium	Rb^{2+}
Cobaltic	Co^{3+}	Silver	Ag^+
Cobaltous	Co^{2+}	Sodium	Na^+
Cupric	Cu^{2+}	Stannic	Sn^{4+}
Cuprous	Cu^+	Stannous	Sn^{2+}
Ferrous	Fe^{2+}	Strontium	Sr^{2+}
Ferric	Fe^{3+}	Zinc	Zn^{2+}

List of Anions

Name of Ion	Ion	Name of Ion	Ion
Acetate	CH_3COO^-	Bromate	BrO_3^-
Aluminate	AlO_2^-	Bisulphite	HSO_3^-
Arsenite	AsO_3^{3-}	Cyanide	CN^-
Arsenate	AsO_4^{3-}	Cyanate	CNO^-
Bicarbonate	HCO_3^-	Carbonate	CO_3^{2-}
Bromide	Br^-	Chromate	CrO_4^{2-}
Bromite	BrO_2^-	Carbide	C^{4-}
Borate	BO_3^{3-}	Chlorate	ClO_3^-
Bisulphate	HSO_4^-	Chlorite	ClO_2^-
Boride	B^{3-}	Chloride	Cl^-

Dichromate	$Cr_2O_7^{2-}$	Oxalate	$C_2O_4^{2-}$
Dichromate	$Cr_2O_7^{2-}$	Phosphate	PO_4^{3-}
Hypobromite	BrO^-	Perchlorate	ClO_4^-
Hypoiodite	IO^-	Permanganate	MnO_4^-
Hypochlorite	ClO^-	Peroxide	O_2^{2-}
Hydride	H^-	Plumbate	PbO_3^{2-}
Iodide	I^-	Phosphide	P^{3-}
Formate	$HCOO^-$	Phosphite	PO_3^{3-}
Fluoride	F^-	Superoxide	O_2^-
Ferricyanide	$[Fe(CN)_6]^{3-}$	Sulphide	S^{2-}
Ferrocyanide	$[Fe(CN)_6]^{4-}$	Sulphite	SO_3^{2-}
Formate	$HCOO^-$	Sulphate	SO_4^{2-}
Fluoride	F^-	Stannite	SnO_2^{2-}
Manganate	MnO_4^{2-}	Stannate	SnO_4^{2-}
Nitrite	NO_2^-	Silicate	SiO_3^{2-}
Nitride (Azide)	N_3^{3-}	Thiocyanate	SCN^-
Nitrate	NO_3^-	Thiosulphate	$S_2O_3^{2-}$
Oxide	O^{2-}	Zincate	ZnO_2^{2-}

2.9 CHEMICAL FORMULA

A chemical formula is a short form representation of a chemical compound.

Writing formulae

A formula is a collection of symbols written with small *positive whole numbers* as subscripts.

The following guidelines will help you write the formula of a compound.

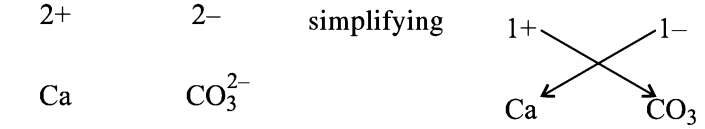
(a) Write the symbol of the positive ion. Leave a gap and in the same line write the symbol of the negative ion. (You can refer to the ions list if you do not know the formula or symbol of the ions.

(b) Write the charges of the respective ions on top of the ions.

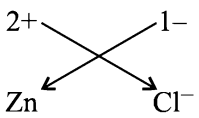
(c) If the two numbers can be simplified by dividing by a common factor divide and write the two numbers in simpler terms.

(d) Bring the two symbols close together. Write the charge on the negative ion as the subscript of the positive ion and vice-versa. Using the crisscross method as shown below write the formula.

Example 1: Writing the formula of calcium carbonate. Look for the symbol/ formula of calcium and carbonate in the list of cations and anions respectively.



Example 2: Writing the formula of zinc chloride.



Formula: $ZnCl_2$

EXERCISE

1. Write the formulae of the following compounds.

Sodium sulphite: _____

Calcium sulphate: _____

Potassium sulphide: _____

Zinc carbonate: _____

Sodium bicarbonate: _____

Potassium nitrate: _____

Ammonium nitrite: _____

Aluminium nitride: _____

Ammonium hydroxide: _____

Chromic sulphate: _____

Ammonium carbonate: _____

Lithium dichromate: _____

2. Write the names of the following compounds.

NH_4NO_3: _____

KI: _____

$Cu(OH)_2$: _____

$ZnCl_2$: _____

CH_3COONa: _____

PbS:	_____	$CaSiO_3$:	_____
Na_2O_2:	_____	Na_2ZnO_2:	_____
$NiCl_2$:	_____	$PbSO_3$:	_____
$Mg(HCO_3)_2$	_____	$KHSO_3$:	_____

Information conveyed by a chemical formula

As we become familiar with the formulae we will see that a formula conveys very important information about a compound.

For example the formula of $CaCO_3$ conveys the following information:

(i) The formula $CaCO_3$ represents the compound calcium carbonate.

(ii) The formula represents one molecule of calcium carbonate.

(iii) Each molecule of calcium carbonate has 1 atom of calcium, 1 atom of carbon and 3 atoms of oxygen.

(iv) If the atomic weights of the three elements are added, the molecular weight of calcium carbonate is found to be $40 + 12 + (3 \times 16) = 100$.

(v) We also get information of the composition by weight of the elements in the compound as follows.

(vi) If 100g of calcium carbonate is taken, it has 40g of calcium, 12g of carbon and 48g of oxygen.

2.10 CHEMICAL EQUATIONS – REPRESENTING CHEMICAL REACTIONS

Chemical Reaction

A chemical reaction takes place when one or more substances (elements / compounds) undergo a chemical change forming one or more *new* substances with or without energy changes.

The substances that undergo the change are called *reactants*.

The substances that are formed as a result of the chemical change are called *products*.

Example: Carbon burns in air to form carbon dioxide gas releasing heat in the process.

Carbon is the *reactant*. Carbon dioxide is the *product*. Heat energy is released in the conversion.

A chemical equation is a short form representation of a chemical reaction using symbols and formulae.

Illustration

The chemical equation for the above chemical reaction is

$$C + O_2 \xrightarrow{\Delta} CO_2 + \text{Heat released}$$

The reactants and products are separated by a $\longrightarrow$ mark. The reactants are to the left of the arrow and products to the right. The symbol above the arrow mark means that the reactants must be heated for the reaction to take place.

Some more examples of chemical reactions and their equations are given below.

(a) Hydrogen reacting with oxygen to give water.

$$2H_2 + O_2 \qquad 2H_2O$$

(b) Potassium chlorate undergoing decomposition on heating to form potassium chloride and oxygen.

$$2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$$

(c) Silver nitrate reacting with hydrogen chloride to give silver chloride and nitric acid.

$$AgNO_3 + HCl \longrightarrow AgCl + HNO_3$$

(d) Zinc reacts with dilute hydrochloric acid to form zinc chloride and liberate hydrogen.

$$Zn + 2HCl \longrightarrow ZnCl_2 + H_2$$

Significance of numbers in a chemical equation

The numbers before the formula in a chemical equation are used for *balancing the chemical equation.*

A balanced chemical equation is a direct consequence of the *Law of Conservation of Mass.*

Law of Conservation of Mass states that matter can neither be created nor destroyed but can be changed from one form to another.

Now consider the chemical reaction hydrogen reacting with oxygen to give water.

The *unbalanced* equation is written as

$$H_2 + O_2 \longrightarrow H_2O$$

On the reactants side (left hand side) of the equation, there are two H-atoms and two O-atoms.

On the products side (right hand side) of the equation, there is one molecule of water – each water molecule having two H-atoms and one O-atom.

It can be seen that *there is one O-atom extra on the reactant side*. If the equation is written like this then it shows that one O-atom has been destroyed. But matter can neither be created nor destroyed.

It is therefore necessary to ensure that atoms of each element are equal on both sides. This is done by writing numbers before the reactants and the products such that the atoms of each element on both sides are equal. This is called ***balancing the chemical equation.***

To balance the chemical equation, we take two hydrogen molecules (i.e., four H-atoms) on the reactants side. On the products side we form 2 water molecules. The unbalanced equation has now been balanced and is rewritten as

$$2H_2 + O_2 \longrightarrow 2H_2O$$

Now the atoms of each element are equal on both sides.

This represents a *balanced chemical equation.*

> If there is no number before the reactant or product in a chemical equation it is understood that the number is one. In the given example, there is no number before oxygen. This implies that oxygen and hydrogen react in the ratio of 1: 2 atoms.

2.11 WRITING AND BALANCING CHEMICAL EQUATIONS

Chemical equations may be balanced by using any one of two methods given below.

(i) Trial and error method

(ii) Sally's method

(i) Trial and error method of balancing equations

In the trial and error method the following steps are to be taken.

(a) Balance the metallic elements first on both sides.

(b) Next balance the non-metallic elements.

(c) Balance gaseous elements last.

Illustrative example on balancing a equation :

Potassium permanganate reacts with sulphuric acid as follows

$$KMnO_4 + H_2SO_4 \longrightarrow K_2SO_4 + MnSO_4 + H_2O + O_2$$

Step 1: Balance the potassium and manganese atoms first (highlighted in grey).

$$\mathbf{2}KMnO_4 + H_2SO_4 \longrightarrow K_2SO_4 + \mathbf{2}MnSO_4 + H_2O + O_2$$

Step 2: Balance non-metallic element next – sulphur (highlighted in grey).

$$2KMnO_4 + \mathbf{3}H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + H_2O + O_2$$

Step 3: Balance hydrogen next – gaseous element (highlighted in grey).

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + \mathbf{3}H_2O + O_2$$

Step 4: Balance oxygen.

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + \mathbf{2½}O_2$$

Step 5: Multiply the full equation to convert fractional coefficient to whole numbers.

$$\mathbf{4}KMnO_4 + \mathbf{6}H_2SO_4 \longrightarrow \mathbf{2}K_2SO_4 + \mathbf{4}MnSO_4 + \mathbf{6}H_2O + \mathbf{5}O_2$$

This represents a balanced chemical equation.

(ii) Sally's method

Sally's method of balancing chemical equations is a more scientific method.

Before we illustrate the use of this method let us define a term – *frequency number*. We will use this for the specific purpose of balancing equations as the success with this method is quite high.

Frequency number of an element participating in a chemical reaction is the number of times the element ***occurs on both sides of the equation***.

For example in the *unbalanced* equation

$$K_2Cr_2O_7 + HCl \longrightarrow KCl + CrCl_3 + H_2O + Cl_2$$

Potassium occurs two times in the equation – on the reaction side once and on the products side once. Hence the frequency number of potassium is 2.

Similarly, the frequency number of chromium (Cr) is 2, of oxygen (O) is 2, of hydrogen (H) is 2 and of chlorine (Cl) is 4.

After finding the frequency number of each element in the equation, the following steps are taken to balance the chemical equation.

Step 1 – Write the frequency numbers of all the elements in the given chemical equation.

Step 2 – *Start balancing the equation from that element which has the least frequency number.*

Step 3 – *Other elements should be balanced in the order of increasing frequency numbers.*

Step 4 – *If two or more elements have same frequency number, then balance the metallic element first.*

Step 5 – *If there are two or more metallic elements with same frequency number, balance the metals in decreasing order of atomic numbers* – the highest atomic number first and the smallest atomic number last.

Step 6 - *If there are two or more non-metallic elements, with same frequency number, balance the non metals in decreasing order of atomic numbers* – the highest atomic number first and the smallest atomic number last.

Illustrative examples

Example 1

Aluminium carbonate reacts with dilute nitric acid to form aluminium nitrate, water and carbon dioxide. This reaction can be written in equation form as under.

$Al_2(CO_3)_3 + HNO_3 \rightarrow Al(NO_3)_3 + CO_2 + H_2O$

Let us now write the frequency numbers and order of balancing each element.

Elements	Al	C	O	H	N
Frequency number	2	2	5	2	2
Order of balancing the elements	1st	3rd	5th	4th	2nd

Explanation for the order in which the elements are balanced

Aluminium, carbon, hydrogen and nitrogen have same frequency number.

However, aluminium being a metal gets 1st place in balancing.

Amongst the non-metals with frequency number 2, nitrogen has the highest atomic number followed by carbon and hydrogen.

Thus, order of balancing nitrogen is 2nd, carbon is 3rd and hydrogen is 4th.

The element oxygen, which has frequency number 5 gets 5th place in the order of balancing.

Balancing the equation

(i) ***Balancing aluminium***: The number of atoms of aluminium towards the reactants side is 2 in $Al_2(CO_3)_3$, but in the products side it is one in $Al(NO_3)_3$. Thus, in order to balance aluminium we multiply $Al(NO_3)_3$ by 2 as shown in the equation (a) below.

$Al_2(CO_3)_3 + HNO_3 \rightarrow 2Al(NO_3)_3 + CO_2 + H_2O$ (a)

(ii) ***Balancing nitrogen***: In equation (a) number of nitrogen atoms in HNO_3 towards the reactants side is 1. The number of nitrogen atoms in 2 $Al(NO_3)_2$ towards the products side is 6. Thus, in order to balance nitrogen we multiply HNO_3 by 6 as shown in equation (b) below.

$Al_2(CO_3)_3 + 6HNO_3 \rightarrow 2Al(NO_3)_3 + CO_2 + H_2O$ (b)

(iii) ***Balancing carbon***: In equation (b) number of carbon atoms in $Al_2(CO_3)_3$ towards the reactants side is 3. The number of carbon atoms in CO_2 towards the products side is 1. Thus, in order to balance carbon we multiply CO_2 by 3 as shown in equation (c) below.

$Al_2(CO_3)_3 + 6HNO_3 \rightarrow 2Al(NO_3)_3 + 3CO_2 + H_2O$ (c)

(iv) ***Balancing hydrogen***: In equation (c) number of hydrogen atoms in $6HNO_3$ towards the reactants side is 6. The number of hydrogen atoms in H_2O towards the products side is 2. Thus, in order to balance hydrogen, we multiply H_2O by 3 as shown in equation (d) below.

$Al_2(CO_3)_3 + 6HNO_3 \rightarrow 2Al(NO_3)_3 + 3CO_2 + 3H_2O$ (d)

(v) ***Balancing oxygen***: In equation (d) total number of oxygen atoms on the side of reactants is 27 and the total number of atoms on the side of products is also 27.

Thus, the equation is fully balanced.

Example 2

Lead nitrate crystals on strong heating decompose to form lead monoxide, nitrogen dioxide gas and oxygen gas. The reaction can be written in the form of chemical equation as under

$Pb(NO_3)_2 \rightarrow PbO + NO_2 + O_2$

Let us now write the frequency numbers and order of balancing each element.

Element	Pb	N	O
Frequency number	2	2	4
Order of balancing order the elements	1st	2nd	3rd

Balancing the equation

(i) **Balancing lead**: The number of atoms of lead is one on reactants side and products side.

Thus, it needs no balancing.

(ii) **Balancing of nitrogen**: The reactants side has 2 nitrogen atoms and the products side one nitrogen atom. Thus, NO_2 in the products side is multiplied by 2 so as to balance nitrogen as shown in equation (a)

$$Pb(NO_3)_2 \rightarrow PbO + 2NO_2 + O_2 \quad (a)$$

(iii) **Balancing of oxygen**: The number of oxygen atoms towards the reactants side is 6 (an even number). The number of oxygen atoms towards the side of products is 7 (an odd number)

While balancing, if any of the following two situations exist, then check whether the condition given below is met and THEN apply the rule that follows.

Situation 1 – There is a difference of one atom between reactants and products

Situation 2 – The number of atoms towards the reactants are even and towards the products are odd or vice versa

Condition: The element under consideration should be present in its elementary form in the equation either on the reactants side or products side.

Rule: If the above situation is present and also the condition is met then the rule that is to be applied is as follows.

Multiply the whole equation by 2 excluding the elementary state of the element (do not multiply elementary state of the element by 2).

In the above equation, since the element oxygen is present in the elementary form (as O_2) on the products side we multiply equation (a) with 2.

Thus, multiplying equation (a) with 2 but taking care not to multiply O_2, we get equation (b) below.

$$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2 \quad (b)$$

It can now be seen that the number of oxygen atoms towards the reactants side is 12.

Similarly, number of atoms towards the products side is 12.

Thus, the equation is fully balanced.

2.12 SPECIAL SYMBOLS USED IN CHEMICAL EQUATION

A chemical equation conveys important information about a chemical reaction. The reactants and products are shown using their symbols and formulae. Besides this, special symbols are used to give important information. The special symbols used in a chemical equation are given below.

Special symbols used in chemical equations

Symbol	Meaning
$X_{(s)}$	X is a solid
$Y_{(l)}$	Y is a liquid
$Z_{(g)}$	Z is a gas
$C_{(aq)}$	C is in the form of an aqueous solution, i.e, dissolved in water.
A ↑ or $A_{(g)}$	A is a gas which is evolved
B ↓	B is an insoluble product and is precipitated
→	The reaction takes place in only one direction – the forward direction.
⇌	The reaction is reversible. It can take place in both directions – forward and backward.
Δ	Heat energy supplied
Examples:	$2Pb(NO_3)_2$ (s) $\xrightarrow{\Delta}$ $2PbO(s) + 4NO_2(g) + O_2(g)$
	$AgNO_3$ (aq) + NaCl(aq) → $NaNO_3$ (aq) + AgCl↓

2.13 MEANING OF BALANCED CHEMICAL EQUATIONS

A chemical equation using special symbols given above conveys information as follows.

(a) Reactants – a chemical equation gives the symbols or formulae of the substances that are reacting and their physical states.

(b) Products – a chemical equation gives the symbols or formulae of the substances formed and their physical states.

Example – $C(s) + O_2(g) \xrightarrow{\Delta} CO_2(g) + \Delta$

Meaning – Carbon and oxygen are the reactants. Heat is required to start the reaction. The reaction as a whole is exothermic. Carbon is in the solid state while oxygen is in the gaseous state. Carbon dioxide is the product. It is formed in the gaseous state.

(c) A chemical equation gives information about the *nature of a chemical reaction* – whether the reaction is reversible or irreversible as follows.

Irreversible chemical reaction is a chemical reaction that takes place in such a way that the reactants change to products before the reaction ends.

Example – $2H_2(g) + O_2(g) \xrightarrow[\text{or flame}]{\text{electric spark}} 2H_2O(l) + \Delta$

Meaning – Hydrogen reacts with oxygen giving water. The reaction is *irreversible*.

Reversible chemical reaction is a chemical reaction that takes place in two directions – the reactants change to products and simultaneously the products change to the reactants again.

Example – $N_2 + 3H_2 \rightleftharpoons 2NH_3$

Meaning – Nitrogen reacts with hydrogen giving ammonia. The reaction is *reversible* – ammonia decomposes to give nitrogen and hydrogen simultaneously.

(d) A chemical equation gives information about the heat energy ***released or absorbed***.

Example: $S(s) + O_2(g) \quad SO_2(g) + \Delta$

$CaCO_3(s) \quad CaO(s) + CO_2(g) - \Delta$

(e) A chemical equation gives information about the ***reaction conditions***. The reaction conditions include the following.

(i) The temperature at which the reaction takes place.

(ii) The pressure of the gaseous reactants.

(iii) The catalyst used for the reaction.

Example: $N_2 + 3H_2 \xrightarrow[200-900\ atm]{450^\circ-500^\circ C,\ Fe\ Mo} 2NH_3 + \Delta$

Meaning: Nitrogen reacts with hydrogen at 450^0 – 500^0 C in the presence of Iron + Molybdenum as catalyst at a pressure of 200 to 900 atmosphere pressure to form ammonia in an exothermic reaction.

(f) A chemical equation gives information about the ***ratio of the weights*** of the reactants and products formed.

Example: Carbon reacts with oxygen to give carbon dioxide. Whenever this reaction takes place, the ratio of weights of the reactants and products is as given below.

Weights of carbon **:** oxygen **::** carbon dioxide is 3 **:** 8 **::** 11.

Thus 12 gm of carbon will react with 32 gm of oxygen. The two react to give 44gm of carbon dioxide.

(g) A chemical equation gives information about the ***ratio of the volumes*** of gaseous reactants and products formed.

Example: Hydrogen reacts with chlorine to give hydrogen chloride. Both reactants and the product are gaseous. The volumes of hydrogen and chlorine that are taken and the hydrogen chloride that is formed are all measured *ensuring that conditions of temperature and pressure are same in all the three cases*. It is found that whenever this reaction takes place, the ratio of volumes of the reactants and products is as given below.

Volumes of hydrogen : chlorine :: hydrogen chloride is 1 **:** 1 **::** 2.

Thus, ½ *litre* of hydrogen will react with ½ *litre* of chlorine. The two react to give 1 *litre* of hydrogen chloride – all volumes being measured at the same temperature and pressure.

Limitations of a chemical reaction

A chemical equation conveys useful information. But, it also has some limitations as follows.

(a) While a chemical equation tells us about the nature of reactants and products, *it does not tell about the concentration of both.*

(b) While a chemical equation tells us how the different atoms in the reactants get rearranged in the products, *it does not tell about the speed of the reaction.*

(c) While a chemical equation tells us about the direction of the reaction and its reversible or irreversible nature, *it does not tell us about the change in colour that takes place if any.*

(d) While a chemical equation tells us about the physical state of the reactants and products, *it does not tell us about the completion of the reaction.*

2.14 SIGNIFICANCE OF NUMBERS USED IN A CHEMICAL EQUATION

We have seen above that numbers are used before symbols or formuale in a chemical reaction to balance a chemical equation.

Conversely, in a balanced chemical equation the numbers before the reactants and products give us information about :

(i) how many atoms or molecules of the reactants are taking part.

(ii) the number of atoms or molecules of the products that form.

For example, the equation, $C + O_2 \quad CO_2$ conveys that one atom of carbon reacts with one molecule of oxygen to give one molecule of carbon dioxide.

The equation, $2H_2 + O_2 \longrightarrow 2H_2O$ conveys

that two molecules of hydrogen react with one molecule of oxygen to give two molecules of water.

The equation, $2KClO_3 \longrightarrow 2KCl + 3O_2$ conveys that two molecules of potassium chlorate decompose to give two molecules of potassium chloride and 3 molecules of oxygen.

EXERCISE

1. Assign the formulae for the following compounds:

(a) potassium bisulphate
(b) ammonium bisulphide
(c) calcium bisulphate
(d) magnesium acetate
(e) barium monohydrogen phosphate
(f) mercurous nitrate
(g) potassium oxalate
(h) copper (II) chloride
(i) tin (IV) oxide
(j) calcium bicarbonate
(k) sodium dichromate
(l) aluminium chlorate
(m) sodium dihydrogen phosphate
(n) silver (I) phosphate
(o) sodium hydrogen sulphate
(p) aluminium carbonate
(q) ammonium phosphate
(r) aluminium oxide
(s) sodium zincate
(t) sodium thiosulphate
(u) calcium oxalate
(v) calcium dihydrogen phosphate
(w) magnesium chlorite
(x) sodium phosphate
(y) iron (III) oxalate
(z) ammonium oxalate

2. In the following table, name the compounds formed between the cations and anions by matching the horizontal row with the vertical column. You can use the number given in the box for writing your answers. (The table after completion will help you to remember the formulae and names of some of the *most commonly used compounds*. There are one hundred and ten compounds in this table).

Cation → / Anion ↓	NH_4^+	Mg^{2+}	Na^+	Ba^{2+}	Pb^{2+}	Fe^{3+}	Cu^{2+}	Cr^{3+}	Ca^{2+}	Zn^{2+}
HSO_3^-	1	2	3	4	5	6	7	8	9	10
CO_3^{2-}	11	12	13	14	15	16	17	18	19	20
HCO_3^-	21	22	23	24	25	26	27	28	29	30
OH^-	31	32	33	34	35	36	37	38	39	40
NO_2^-	41	42	43	44	45	46	47	48	49	50
Cl^-	51	52	53	54	55	56	57	58	59	60
HCO_3^-	61	62	63	64	65	66	67	68	69	70
HCO_3^-	71	72	73	74	75	76	77	78	79	80
SO_3^{2-}	81	82	83	84	85	86	87	88	89	90
NO_3^-	91	92	93	94	95	96	97	98	99	100
SO_4^{2-}	101	102	103	104	105	106	107	108	109	110

3. Balance the following equations

(a) $Fe + H_2O \rightarrow Fe_3O_4 + H_2$
(b) $Al + NaOH + H_2O \rightarrow NaAlO_2 + H_2$
(c) $Mg(HCO_3)_2 \rightarrow MgCO_3 + CO_2 + H_2O$
(d) $NH_3 + Cl_2 \rightarrow NCl_3 + HCl$
(e) $C + HNO_3 \rightarrow CO_2 + NO_2 + H_2O$
(f) $I_2 + HNO_3 \rightarrow HIO_3 + NO_2 + H_2O$
(g) $BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + HCl$
(h) $C_4H_{10} + O_2 \rightarrow CO_2 + H_2O$
(i) $(NH_4)_2Cr_2O_7 \rightarrow N_2 + Cr_2O_3 + H_2O$
(j) $C + H_2O \rightarrow CO + H_2$
(k) $CuO + NH_3 \rightarrow Cu + H_2O + N_2$
(l) $Mg + N_2 \rightarrow Mg_3N_2$

4. **Write the equations and balance the following chemical reactions.**
   (a) Sulphur trioxide reacts with water to give sulphuric acid (H_2SO_4)
   (b) Zinc displaces copper from copper sulphate to give copper and zinc sulphate.
   (c) Potassium chlorate on heating gives potassium chloride and oxygen gas.
   (d) Sodium reacts with water to give sodium hydroxide and hydrogen gas.
   (e) Silver nitrate reacts with sodium chloride to give silver chloride precipitate and sodium nitrate.
   (f) Ammonium hydroxide reacts with sulphuric acid to give ammonium sulphate and water.
   (g) Lime water turns milky when carbon dioxide gas is bubbled through limewater.
   (h) Hydrogen chloride reacts with manganese dioxide to give chlorine gas and manganese chloride and water.
   (i) Hydrogen peroxide decomposes to water and oxygen on slight warming.
   (j) Ammonia gas decomposes to give nitrogen and oxygen gases.
   (k) Ammonium nitrate when heated decomposes to nitrous oxide (N_2O) and water.
   (l) Sodium nitrate when heated decomposes to sodium nitrite and oxygen.
5. **If the nitride of a metal M has the formula MN, what is the formula of its sulphide, chloride and carbonate?**
6. **The symbol of a divalent metal is M. What is the formula of the metal sulphate?**
7. **By looking up at the elements in the Periodic Table of Elements write the symbols of the following elements:**
   (a) all gases (there are 11 gases in the periodic table)
   (b) all inert gas elements (there are 6 inert gas elements)
   (c) any ten metallic elements
   (d) any ten non-metallic elements which are not gases.
   (e) any five metalloids.
8. **Write the formulae of any two compounds with the following complex ions:**
   (a) zincate (b) oxalate
   (c) permanganate (d) thiosulphate
   (e) stannate

2.15 ATOMIC NUMBER AND MASS NUMBER

An atom is the basic unit of matter in elements. A molecule is the basic unit of a compound. Each element has an unique *Atomic Number, Mass Number and Atomic Weight.*

Atomic Number of an element

Atomic number of an element is the number of protons or positive charge present in the nucleus of the atom. It is denoted by Z.

Example : Atomic number of carbon is 6. This means that carbon atom has 6 protons in its nucleus and 6 electrons revolving around it. Two electrons are in the first orbit and remaining 4 electrons are in the second orbit.

Mass Number of an element

Mass number of an element is the sum of the number of protons and neutrons in the nucleus of an atom. It is denoted by A.

An element X with atomic number, Z and mass number, A is written as $_zX^A$ or .

Example: Mass Number of carbon is 12. This means that the sum of the number of protons and neutrons in the nucleus of carbon atom is 12. Since, carbon atom has 6 protons (atomic number 6), the number of neutrons is 12 – 6 = 6.

An atom is made up of protons, neutron and electrons. *The actual mass of the atom* is equal to the mass of electrons, protons and neutrons.

Mass of light particles like electrons, protons and neutrons is measured in a unit of mass called amu (atomic mass unit).

1amu = 1.66×10^{-24}gm

Mass of protons = 1amu

Mass of neutron = 1 amu

Mass of electron = 1/ 1836 amu.

The mass of an electron is considered to be negligible.

Take the example of hydrogen atom.

Each atom has 1 proton and 1 electron. As the mass of electron is negligible, the mass of H-atom is 1amu.

Now take the example of sodium atom. Each sodium atom has 11 electrons, 11 protons and 12 neutrons (See electronic configuration of elements). Thus, the mass of each sodium atom is 23 amu. When this mass is compared with the mass of H-atom (1 amu), the ratio or Atomic Weight of sodium is 23. The Mass Number (neutrons + protons) is also 23.

As weight of H-atom is 1 amu, the ratio of weight of an atom of an element to that of H-atom (Atomic Weight) will almost be equal to the Mass Number (neutrons + protons).

Example: Oxygen is represented as $_8O^{16}$. This means the element oxygen has atomic number 8 and mass number 16.

2.16 ATOMIC WEIGHT OF AN ELEMENT

Atomic Weight of an element is the ratio of the mass of one atom of the element to the mass of one hydrogen atom.

Mathematically, this is written as

$$\text{Atomic Weight} = \frac{\text{Mass of one atom of element}}{\text{Mass of one atom of hydrogen}}$$

Thus, Atomic Weight

(a) is a ratio (b) has no units

(c) is not the true weight of the atom.

(d) tells how many times each atom of the element is heavier than a hydrogen atom.

Example : Atomic weight of nitrogen is 14. It means that each atom of nitrogen is 14 times heavier than hydrogen atom.

The atomic number, mass number and atomic weight of the first twenty elements are given below.

Atomic weights are given to the nearest whole number.

Element Symbol	Atomic Number (Z)	Mass Number (A)	Atomic Weight
Hydrogen – H	1	1	1
Helium – He	2	4	4
Lithium – Li	3	7	7
Beryllium – Be	4	9	9
Boron – B	5	11	11
Carbon – C	6	12	12
Nitrogen – N	7	14	14
Oxygen – O	8	16	16
Fluorine – F	9	19	19
Neon – Ne	10	20	20
Sodium – Na	11	23	23
Magnesium – Mg	12	24	24
Aluminium – Al	13	27	27
Silicon – Si	14	28	28
Phosphorous – P	15	31	31
Sulphur – S	16	32	32
Chlorine – Cl	17	35	35
Argon – Ar	18	40	40
Potassium – K	19	39	39
Calcium – Ca	20	40	40

2.17 MOLECULAR WEIGHT OF A COMPOUND

Molecular Weight of a compound is the ratio of the mass of one molecule of the compound to the mass of one hydrogen atom.

Mathematically, this is written as

Molecular Weight

$$= \frac{\text{Mass of one molecule of compound}}{\text{Mass of one hydrogen atom}}$$

Thus, Molecular Weight

(a) is a ratio

(b) has no units

(c) is not the true weight of the molecule.

(d) tells how many times each molecule of the compound is heavier than hydrogen atom.

Example: Molecular weight of CO_2 is 44. It means that each molecule of CO_2 is 44 times heavier than hydrogen atom.

Finding the Molecular Weight of a compound from its formula

Consider the compound carbon dioxide. The formula of this compound is CO_2. The formula rep-

resents one molecule of the compound consisting of one carbon atom and two oxygen atoms. The molecular weight of carbon dioxide is found by adding the atomic weights of all atoms present in one molecule.

Thus, molecular weight of CO_2 is $12 + (2 \times 16) = 44$.

The molecular weight of H_2O is $(2 \times 1) + 16 = 18$.

The molecular weight of SO_2 is $32 + (2 \times 16) = 64$.

The molecular weight of $CaCO_3$ is $40 + 12 + (3 \times 16) = 100$.

The molecular weight of H_2SO_4 is $(2 \times 1) + 32 + (4 \times 16) = 98$.

EXERCISE

1. Tabulate the atomic weight of chlorine, iron, fluorine, potassium, nitrogen, oxygen and carbon.
2. Find molecular weights of
   (a) Sodium carbonate [$Na_2CO_3.10H_2O$]
   (b) Mohr's salt [$FeSO_4.(NH_4)_2\ SO_4.6H_2O$]
   (c) Carnalite [$KCl.MgCl_2.6H_2O$]
   (d) Microcosmic salt - $NaNH_4HPO_4.4H_2O$
3. Given the formula of chlorophyll is $C_{55}H_{72}MgN_2O_5$, find its molecular weight.

2.18 SIMPLE CALCULATIONS BASED ON CHEMICAL FORMULAE AND EQUATIONS

1. Calculating percentage composition

Finding the percentage composition of an element in a compound is shown using illustrative examples.

Example 1: Consider water molecule (H_2O). The molecular weight of water is $2 + 16 = 18$.

Percentage composition of oxygen

$$= \frac{16}{18} \times 100 = \frac{800}{9} = 88.89\%$$

$\therefore$ Percentage composition of hydrogen

$= 100 - 88.89 = 11.11\%$

Example 2: Calculate the percentage composition of each element present in sulphur dioxide.

Sulphur dioxide has the formula SO_2. The molecular weight is $32 +$ $= 64$.

Percentage composition of sulphur

$$= \frac{\text{the proportion of sulphur in the molecule}}{\text{molecular weight}} \times 100$$

$$= \frac{32}{64} \times 100 = 50\%$$

Percentage composition of oxygen

$= 100 - 50 = 50\%$.

Example 3: Calculate the percentage composition of water in washing soda.

Washing soda has the formula $Na_2CO_3.10H_2O$.

The molecular weight

$$= \overline{2\times23} + 12 + \overline{3\times16} + 10(1\times2+16) = 46+12+48+180 = 286$$

Percentage composition of water

$$= \frac{\text{the proportion of water in the molecule}}{\text{molecular weight}} \times 100$$

$$= \frac{186}{286} \times 100 = 62.94\%$$

2. Calculating weights of reactants / products

Finding the weights of reactants required or products formed in a reaction is shown using illustrative examples.

Example 1: Find the weight of oxygen required to completely burn 100gm of carbon. The *balanced equation* for the complete burning of carbon is $C + O_2 \rightarrow CO_2$.

The atomic weight of carbon is 12 and that of oxygen is 16.

Thus, 12 units of carbon react with $\overline{2\times16} = 32$ units of oxygen (two atoms) to give

units of carbon dioxide.

Using unitary method, 12g carbon requires 32g oxygen.

100g carbon requires ? g oxygen

$12x = 32 \times 100$

$$x = \frac{32\times100}{12} = 266.67\,gm$$

Example 2: In the above example, what is the weight of carbon dioxide formed?

Using unitary method, 12 g carbon on complete burning gives 44 g of carbon dioxide.

100 g of carbon on complete burning gives ? g carbon dioxide.

$12y = 44 \times 100$

$$y = \frac{44\times100}{12} = 366.67\,gm$$

EXERCISE

I. Choose the correct answer

1. A chemical equation does not tell about
- (a) the rate at which a reaction takes place
- (b) the colour of the reactants or products
- (c) the actual concentration of the reactants and products.
- (d) all the three (a, b and c)

2. If 12g of carbon burns completely in 40g oxygen in a closed container, the product has
- (a) 44g of carbon dioxide
- (b) 8 g of oxygen
- (c) both (a) and (b)
- (d) cannot say

3. If the formula of the oxide of a metal M is M_2O, then the formula of its chloride is
- (a) MCl_2
- (b) MCl
- (c) MCl_3
- (d) MCl_4

4. If the balanced equation is $aPb(NO_3)_{2\ (S)} \rightarrow bPbO_{(S)} + cNO_{2(g)} + dO_{2(g)}$, the value of the coefficients a, b, c and d is
- (a) 2, 4, 1, 2 respectively.
- (b) 2, 4, 2, 1 respectively.
- (c) 2, 2, 4, 1 respectively.
- (d) 2, 2, 2, 4 respectively.

5. If the formula of a metal nitride is MN, the formula of the metal sulphate is
- (a) $M_2(SO_4)_3$
- (b) MSO_4
- (c) $M_3(SO_4)_2$
- (d) $M(SO_4)_2$

6. The *unbalanced* equation for the complete burning of a hydrocarbon is given by

$$C_xH_y + O_2 \rightarrow CO_2 + H_2O$$

The coefficients in the balanced chemical equation for the complete burning of the hydrocarbon from left to right will be
- (a) 1, (x + y/4), x, y/2
- (b) 2, (x + y/4), x, y/2
- (c) 1, (x + y/4), x, y
- (d) 1, (x + y), x, y/2

7. Qualitative meaning of a symbol
- (a) tells about the name of the element
- (b) represents one atom of the element
- (c) both (a) and (b)
- (d) tells about the quality of the element

8. Quantitative meaning of a symbol
- (a) tells about the atomic number of the element
- (b) tells about the mass number of the element
- (c) both (a) and (b)
- (d) tells about the quantity of the element

9. Qualitative meaning of a formula
- (a) tells about the name of the compound
- (b) represents one molecule of the compound
- (c) both (a) and (b)
- (d) tells about the quality of the compound

10. Quantitative meaning of a formula
- (a) tells about the molecular weight of the compound
- (b) tells about the number of atoms of each element of the molecule in the compound
- (c) both (a) and (b)
- (d) tells about the quantity of the compound

11. The symbols of mercury is _____ and lead is _____respectively.
- (a) Hg, Pb
- (b) Pb, Hg
- (c) Le, Me
- (d) Ld, My

12. A symbol of an element does not convey information about
- (a) atomic number.
- (b) atomicity.
- (c) atomic weight.
- (d) mass number.

13. The atomicity of an element tells about
- (a) physical properties.
- (b) number of electrons.
- (c) number of atoms in one molecule of the element.
- (d) its combining capacity.

14. A formula tells about
- (a) number of atoms of each element present in one molecule of the compound
- (b) a mathematical formula.
- (c) an element.
- (d) the source of the compound.

15. The molecular formula is a short form representation of
- (a) a compound
- (b) an element.
- (c) a mixture.
- (d) an alloy.

16. The formula of ammonium dichromate is
- (a) $Am_4Cr_2O_7$
- (b) $(NH_4)_2Cr_2O_7$
- (c) $(NH_4)_3Cr_2O_7$
- (d) $NH_4(Cr_2O_7)_2$

17. A balanced chemical equation supports

(a) Law of chemical equilibrium.
(b) Law of conservation of energy.
(c) Law of conservation of mass.
(d) none of these.

18. $Na_2C_2O_4$ is the formula of sodium

(a) carbonate. (b) oxalate.
(c) formate. (d) acetate.

19. The ion that is not monovalent is

(a) ammonium. (b) sodium.
(c) bicarbonate. (d) sulphite.

20. The trivial name of $Na_2CO_3.10H_2O$ is

(a) soda ash. (b) washing soda.
(c) soda. (d) baking soda.

II. Give one word answers

1. What is an atom called when it loses or gains electrons?
2. The sub atomic particle not present in hydrogen atom.
3. The formula used to calculate the maximum number of electrons in an orbit of an atom.
4. The region in an atom in which nucleons are present.
5. The other name of a negative ion
6. The term used when the valence shell has helium configuration
7. The maximum number of electrons that can be present in the second last shell of an atom
8. The term used for a group of atoms that carries a charge and behaves as a unit in a chemical reaction.
9. The difference between mass number and atomic number gives the number of which sub-atomic particle in an atom?
10. What is the term given to the ratio of the weight of one molecule of a compound to the weight of one hydrogen atom?

III. This section contains three questions. Each question contains statements given in two columns, which have to be matched. Statements (i, ii, iii, iv) in Set A have to be matched with Statements (a, b, c, d) in Set B. More than one match or no match between options in the Sets is possible.

1. Match the ions in Set A with typical examples given in Set B.

Set A	Set B
(i) divalent cation	(a) Al^{3+}
(ii) divalent anion	(b) Ca^{2+}
(iii) a trivalent radical	(c) ZnO_2^{2-}
(iv) a trivalent ion	(d) PO_4^{3-}

2. Match the formula of the compounds in Set A with their names in Set B.

Set A	Set B
(i) $(NH_4)_2CO_3$	(a) Ammonium oxalate
(ii) $HCOONH_4$	(b) Ammonium acetate
(iii) $(NH_4)_2C_2O_4$	(c) Ammonium formate
(iv) CH_3COONH_4	(d) Ammonium carbonate

3. Match the electronic configuration of the element in Set A with the number of electrons lost or gained in Set B.

Set A	Set B
(i) Li (2, 1)	(a) one electron gained
(ii) Cl (2, 8, 7)	(b) two electrons lost
(iii) Ca (2, 8, 8, 2)	(c) two electrons gained
iv. S (2, 8, 6)	d. one electron lost

4. Match the trivial names in Set A with the chemical names in Set B.

Set A	Set B
i. Tear gas	a. N_2O
ii. Marsh gas	b. $CCl_3.NO_2$
iii. Laughing gas	c. $CO.Cl_2$
iv. War gas	d. CH_4

IV. State whether the following statements are True or False

1. The atomic weight of an element is the actual weight of the atom of the element.
2. A cation has more electrons than protons.
3. Neon has octet configuration
4. In $_{20}Ca^{40}$ the ratio of number of neutrons to number of protons is 1.
5. If the number of protons in an ion is three more than the number of electrons, the charge that the ion will carry is –3.
6. The symbol of zinc ion is Zi^{3+}.
7. The symbol of nitride ion is NO_3^-.
8. The compound formed between Al^{3+} and N^{3-} has the formula Al_3N_3.
9. In a balanced chemical equation the number of times each element occurs on each side will be same.
10. A balanced chemical equation is a true reflection of the Law of conservation of mass.

V. Find the odd one out

1. ferrous ion, ammonium ion, sodium ion, sulphate ion, calcium ion.
2. chloride ion, oxide ion, bromide ion, iodide ion, hydride ion
3. $2H_2 + O_2 \rightarrow 2H_2O$; $H_2 + Cl_2 \rightarrow 2HCl$; $N_2 + H_2 \rightarrow$

NH_3; $2Ca + O_2 \rightarrow 2CaO$; $2Zn + O_2 \rightarrow 2ZnO$

4. hydrogen, calcium, oxygen, fluorine, Cl
5. atomic number, mass number, atomic weight, number of electrons, number of neutrons

VI. Fill in the blanks

1. When an atom gains electrons it forms a / an _____.
2. An atom is electrically neutral as the _____ = _____
3. The number of atoms of each element on the reactants side will be _____to that on the product side.
4. Symbols convey both _____ and _____meanings.
5. The formula of nitride ion is _____, that of nitrate ion is _____ and that of nitrite ion is _____.
6. The formula of _____ ion is SO_4^{2-}, while that of _____ ion is SO_3^{2-} and that of _____ ion is S^{2-}.
7. The symbol is used in a chemical equation to denote a _____ reaction.
8. The symbol used to indicate an insoluble product formed in a chemical reaction is _____
9. The maximum of number of electrons in the second last shell of a two shell atom cannot exceed _____.
10. If the number of shells is one, then the stable configuration is known as _____.

VII. Each question contains two statements. Statement A is an Assertion Statement and Statement R is the Reason Statement. Each question has four choices out of which only one is correct. Choose the correct one.

Choose A if Statement A is true, Statement R is true; Statement R is correct explanation for statement A

Choose B if Statement A is true, statement R is true; Statement R is not correct explanation for statement A

Choose C if Statement A is true, Statement R is false

Choose D if Statement A is false, statement R is true

1. **Statement –A**: Ferrous ion is a divalent ion.

   Statement –R: Iron atom loses two electrons to carry two positive charges.

2. **Statement –A:** A balanced and complete equation tells about energy changes that take place.

   Statement –R: Balancing a chemical equation is important to study a chemical reaction.

3. **Statement –A:** As both calcium and sulphate carry two positive charges the formula of calcium sulphate is $Ca_2(SO_4)_2$

   Statement –R: Calcium sulphate is a compound formed by combination of calcium and sulphate ions.

4. **Statement –A:** The formula of a metal nitride is MN.

   Statement –R: The metal ion carries three positive charges and nitrogen ion carries three negative charges.

5. **Statement –A**: The atomic weight has no units.

   Statement –R: The atomic weight is a ratio of two weights.

6. **Statement –A**: The molecular weight is the ratio of weights of two molecules.

   Statement –R: The molecular weight is the true weight of the molecule.

7. **Statement –A**: The percentage of sulphur by weight in sulphur dioxide is 50%.

   Statement –R: The percentage of oxygen by weight in sulphur dioxide is 50%.

8. **Statement –A**: Each molecule of water is 18 times heavier than one atom of hydrogen.

   Statement –R: The ratio by weight of hydrogen **:** oxygen in water is 1:8.

9. **Statement –A**: Each atom of sodium is 23 times heavier than one atom of hydrogen.

   Statement –R: The atomic weight of sodium is 23.

10. **Statement –A**: Oxygen forms a di-negative ion as it gains two electrons to get a stable configuration.

    Statement –R: The electronic configuration of oxygen is (2, 6).

VIII. Answer the following questions.

1. What is the quantitative meaning conveyed by a symbol?
2. What is the quantitative meaning conveyed by a formula?
3. What are the qualitative and quantitative meanings conveyed by the symbol N?
4. How do cations differ from anions?
5. What are the rules of electronic configuration?
6. If YCl_2 is the chloride of a metal Y, write the formula of its sulphate and nitrate.
7. Find the percentage composition of hydrogen in microcosmic salt ($NaNH_4HPO_4.4H_2O$).
8. Find the percentage of water of crystallization in blue vitriol ($CuSO_4.5H_2O$).

9. What weight of sulphur will be required to completely consume 96g of oxygen to convert it to sulphur dioxide (SO_2)?
10. What weight of water needs to be decomposed (electrolysed) to get 100g of hydrogen?
11. Complete and balance the following reactions.
    a. $Fe + ? \rightarrow FeCl_3$
    b. $? + H_2SO_4 \rightarrow BaSO_4\downarrow + HCl$
    c. $Ca(HCO_3)_2 \xrightarrow{\Delta} ? + ?$
    d. $Fe + CuSO_4 \rightarrow ? + Cu$
12. If the formula of the chloride of a metal X is XCl_3 write the formula of the metal phosphate, oxide, and nitride.
13. If a metal M shows variable valency of 1+ and 2+, write the formulae of oxide, sulphate and chloride of both valencies.
14. What is the valency of nitrogen in NO, NO_2 and N_2O?
15. Write the formulae of the given compounds: aluminium sulphate, ammonium sulphate, magnesium nitride.
16. Name the given compounds: CH_3COONa, $Fe_2(SO_4)_3$, $(NH_4)_3PO_4$.
17. What are the limitations of a chemical reaction?
18. Justify the statement - *A balanced chemical reaction is a direct application of Law of conservation of mass.*
19. (a) State the modern concept of valency.
    (b) State the traditional concept of valency.
20. Effect the following conversions: (a) $Fe^{2+} \rightarrow Fe^{3+}$
    (b) $Cl_2 \rightarrow 2Cl^-$
21. If the formula of the oxide of a metal X is X_2O_3 and the formula of a sodium compound is Na_3Y what is the formula of the compound formed by X and Y?
22. A metal, M loses two electrons to form a cation and a non-metal, N gains one electron to form an anion. Write the formula of the compound formed between them.
23. Name the element that has no neutrons.
24. An element is represented both as $_{17}X^{37}$ and $_{17}X^{35}$. What is your observation? What can you say about the properties of the two?
25. Write the names of any three elements that have Latin names with their symbols.

KEY TO EXERCISE

I. 1. d 2. c 3. b 4. c 5. a 6. a 7. c 8. c
9. c 10. c 11. a 12. b 13. c 14. a 15. a 16. b
17. c 18. b 19. d 20. b

II. 1. ion 2. neutron 3. $2n^2$ 4. nucleus 5. anion 6. duplet configuration
7. 18 8. radical 9. neutrons 10. molecular weight

III. 1. i. b ii. c iii. d iv. a, d 2. i. d ii. c iii. a iv. b
3. i. d ii. a iii. b iv. c 4. (i) b (ii) d (iii) a (iv) c

IV. 1. false 2. false 3. true 4. true 5. false 6. false 7. false 8. false
9. true 10. true

V. 1. sulphate ion 2. oxide ion 3. $N_2 + H_2 \rightarrow NH_3$
4. Cl 5. atomic weight

VI. 1. anion 2. number of electrons, number of protons
3. equal 4. quantitative, qualitative
5. N^{3-}, NO_3^-, NO_2^- 6.sulphate, sulphite, sulphide
7. reversible 8. $\downarrow$
9. 2 10. duplet configuration

VII. 1. A 2. D 3. D 4. A 5. A 6. C 7. A
8. B 9. A 10. A

3 Gaseous State and Gas Laws



Robert Boyle



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Jacques Charles

3.0 INTRODUCTION

The title of this chapter raises another question why only gas laws? Why not solid laws or liquid laws? In reality there are many laws for both solids and liquids. Only their applicability depends on several factors. Gases have the interesting property that the *physical* behaviour of one gas is pretty much identical to the behaviour of any other gas. (The *chemical* behaviour, describing possible reactions, of all gases will not be the same). This chapter deals with the relations between the four measurable quantities – volume, pressure, temperature and the number of moles of a gas.

A *gas* is a collection of molecules which move about randomly in all directions. In a gas, the distance a molecule travels between one collision (either with another gas molecule or the wall of the container) is large *compared to the size of the molecules*. The molecules of a gas are relatively far apart and due to this the *physical* properties of a gas do not depend much on the particular element or compound whose molecules make up the gas. The molecules can be compared to very small marbles which move about rapidly, colliding occasionally with the walls of the container or each other. A gas in a container will spread out (change its volume and shape) to fill the entire container. This behaviour is in sharp contrast to that of a solid, whose shape and volume is independent of the container, and a liquid, whose shape changes to that of the container but whose volume remains constant.

3.1 PHYSICAL PROPERTIES OF GASES

The physical properties of a gas are listed below.

1. Gases have low density, melting point and boiling point.
2. Gases do not have any specific shape. Gases take the shape of the container in which they are stored.
3. Gases do not have any specific volume. Gases occupy the entire volume available to them.
4. The gas molecules constantly collide with each other and with the walls of the container.
5. The collisions made by molecules of gas on the walls of the container are responsible for the pressure exerted by the gas.
6. *The collisions of the gas particles are elastic* – there is no change in total kinetic energy during collision with each other or with the walls of the container.
7. Gases have expandability, compressibility and diffusibility. They move in all directions against gravity colliding with each other and the walls of the container in which they are kept in all possible directions.

3.2 MEASURABLE PROPERTIES OF GASES

A measurable physical quantity is the physical quantity that can be measured directly. All physical properties cannot be measured directly. Some physical quantities are derived (see box).

The ***measurable physical properties*** of a gas are ***volume, pressure, temperature and mass.***

Volume of a gas

As gas particles occupy the entire free space available to them, the volume of a gas is the volume of the container in which it is stored. The volume of a gas is typically measured in litres. The actual space in which the gas particles move is the volume of the gas.

Volume of a gas is measured in litres, millilitres or cubic centimetres. ($1l$ = 1 dm^3 = 1000ml = 1000cc)

Example : 1 l of oxygen = 1 dm^3 of oxygen = 1000 ml of oxygen = 1000 cc of oxygen.

Pressure of a gas

Pressure of a gas depends on the force exerted by the gas particles when they collide with the walls of the container. Pressure is the force exerted by the gas per unit area of the container.

Pressure is measured in atmospheres. (1 atm = 76cm of Hg = 760 mm of Hg = 760 torr, where 1 torr = 1 mm of Hg).

Example: 1 atm pressure = 76 cm of Hg = 760 mm of Hg = 760 torr.

Mass of a gas

Mass of a gas is measured as the *number of molecules* of the gas. Each molecule has a certain mass, hence more the number of molecules more the mass.

Mass of a gas is measured in g in CGS system or in kg in MKS system.

Measured and derived physical quantities

A measurable physical quantity is measured **directly** by using instruments. For example we can measure mass by using a balance, pressure using a barometer, temperature using a thermometer.

A derived physical quantity is **derived** from measured physical quantity. For example density is derived by dividing the mass by volume.

Example: 1 kg of oxygen gas = 1000 g of oxygen.

Temperature of a gas

Temperature of a gas is proportional to the average kinetic energy of the moving molecules. Temperature is normally measured in Celsius scale. *While dealing with gases, it is important to measure or convert all temperature into Kelvin scale.*

3.3 ABSOLUTE SCALE OF TEMPERATURE OR KELVIN SCALE

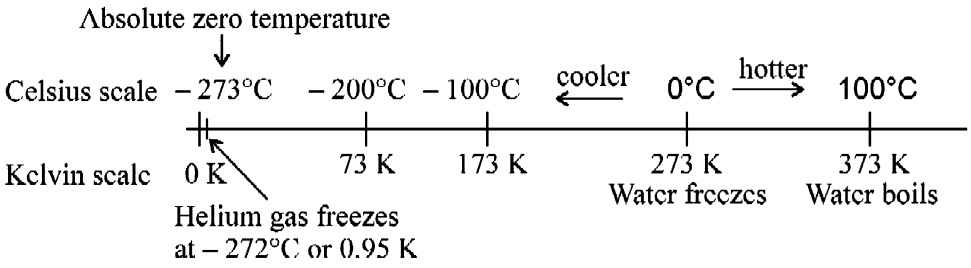
Gases can have temperatures below 0°C. Such values will be negative. To express all temperatures as positive values, Kelvin proposed a temperature scale called *Absolute scale of temperature* also known as *Kelvin scale*.

Temperature expressed in Celsius scale can be converted to Kelvin scale by using the formula:

K = °C + 273

Example: The freezing point of water = 0°C = 0 + 273 = 273 K.

The boiling point of water = 100°C = 100 + 273 = 373 K.



Absolute scale of temperature versus Celsius scale readings

Absolute zero temperature

As we have just seen, the freezing point of water on the Kelvin scale is 273 K. Gases can be cooled to still lower temperatures.

Absolute zero temperature is a hypothetical temperature at which the volume of a gas becomes zero.

Absolute zero temperature = 0 K = – 273°C.

Standard Temperature and Pressure

Standard Temperature and Pressure also known as ***STP*** is equal to 0°C and 1 atm respectively.

This is also sometimes referred to as *Normal Temperature and Pressure* ***(NTP)***.

Normal or standard temperature = 0°C = 273 K

Normal or standard pressure = 1 atm or 76 cm of Hg

3.4 KINETIC THEORY OF GASES

There are significant differences in the Kinetic theory of matter as applied to gases in comparison to solids or liquids. The differences are as follows.

(i) Gases contain large number of tiny, discrete particles (the particles may be atoms, molecules or ions) that are in a state of *high* random motion when compared to solids or liquids.

(ii) The inter-particle distances are large when compared to solids and liquids.

(iii) The forces of attraction between the particles of the gas are very weak when compared to solids and liquids.

(iv) The particles of the gas continuously collide with one another and with the walls of the container. On collision, they change direction without loss of energy. The number of such collisions are very large in gases when compared to liquids or solids.

3.5 GAS LAWS

The ***gas laws*** are a set of laws that give the relationship between temperature in Kelvin scale (T), pressure (P) and volume (V) of a gas and mass. As already seen, the condition in which a gas is present can be described in terms of four variables: *pressure* (P), *volume* (V), *temperature* (T), and the mass of gas. *The Gas laws are the relationship between a pair of the variable while the other two are kept constant (as shown).*

Gas law	Relationship between	Constants
1. *Boyle's law*	P and V	T and mass are constant
2. *Amonton's law*	P and T	V and mass are constant
3. *Charles' law*	V and T	P and mass are constant
4. *Avogadro's hypothesis*	V and mass	P and T are constant

3.6 BOYLE'S LAW (1662)

According to Boyle's Law for *a given mass of a dry gas at constant temperature*, volume is inversely proportional to pressure.

$$V \propto 1 / P$$

Mathematically Boyle's Law is written as

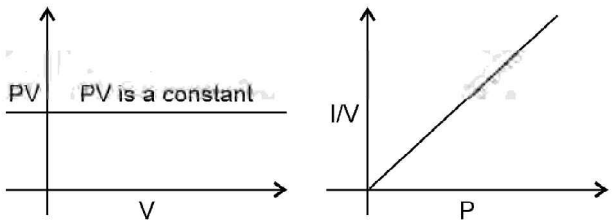
$P_1V_1 = P_2V_2$ = constant

or $\frac{P_1V_1}{P_2V_2} = 1$

where P_1, V_1 are the respective initial values and P_2, V_2 are the respective final values.

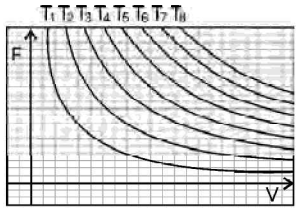
Graphical representation of Boyle's Law

The graphical representation of Boyle's Law are called *Isotherms*. Boyles Law can be represented in several ways.



PV–V *isotherms of a gas at different volumes*

1/V–P *isotherms of a gas at different pressures*



P-V *isotherms of a gas at different temperatures*

Illustration

Consider a given mass of gas kept at constant temperature in a cylinder with a movable piston.

If volume is decreased by plunging the piston inside, the pressure increases as the molecules are pushed together and more collisions take place.

If volume is increased by pulling the piston outward the pressure decreases as the molecules spread out to occupy the increased volume and the number of collisions also decreases. If the pressure and volume are measured and the data compiled, it will be seen that the product of P and V is a constant.

Consider a dry gas kept in a cylinder fitted with a movable piston. Let the volume of the cylinder be 1 *l* and the pressure of the gas be 1 atm. If the volume of the gas is changed to 0.25 atm, 0.5 atm, 0.75 atm, then the new pressures in each case will be as given in the table. In each of these, the value of the product of pressure and volume will be constant as per Boyle's Law. It is to be noted that in each of these cases, the mass of the gas and its temperature is constant.

Volume	Pressure	PV (value)
1. 1*l*	1 atm	1 × 1 = 1
2. 0.25*l*	4 atm	0.25 × 4 = 1
3. 0.5*l*	2 atm	0.5 × 2 = 1
4. 0.75*l*	4/3 atm	0.75 × 4/3 = 1

3.7 AQUEOUS TENSION

When a gas is collected over water, some amount of water vapour will be present in the gas. The water vapour exerts its own pressure.

So, actual pressure of dry gas =

total pressure – the pressure of water vapour.

It can also be seen that as the temperature increases water vapour content will be more.

At a given temperature, the pressure exerted by saturated water vapour is called Aqueous Tension.

Thus, pressure of dry gas =

total pressure – aqueous tension.

3.8 CHARLES' LAW (1787)

According to Charles' Law for *a given mass of a dry gas at constant pressure*, volume is directly proportional to absolute temperature.

$$V \propto T$$

Mathematically Charles' Law is written

as, $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ = constant

where V_1, T_1 are the respective initial values and V_2, T_2 are the respective final values.

Illustration

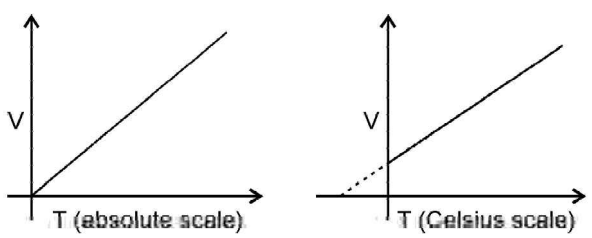
Consider a given mass of gas kept in a cylinder with a movable piston at constant pressure.

If volume is decreased by slowly plunging the piston inside, the temperature increases as the molecules are pushed together and more collisions take place.

If volume is increased by pulling the piston outward the pressure decreases as the molecules spread out to occupy the increased volume and the number of collisions also decreases. If the absolute temperature and volume are measured and the data compiled it will be seen that the data is in accordance with Charles' Law.

Graphical representation of Charles Law

The graphical representations of Charles' Law are called *Isobars*. Charles Law can be represented in two ways.



Some conditions of Charles' Law

1. The gas should be dry.
2. Wet gases have water vapour molecules which will add to the pressure exerted by the gas. Thus the readings will not be accurate.
3. The measured temperature should be in or converted to Kelvin scale.
4. Since the relationship is between volume and absolute temperature, the mass should be kept constant. If the mass changes, the number of molecules will change. This changes the number of collisions which will have an effect on other variables like temperature and volume.
5. The volume and absolute temperature are measured at the same pressure.

3.9 IDEAL GAS EQUATION

The Ideal Gas Equation is obtained by combining the three gas laws.

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} = \text{constant}$$

This equation can be used to derive all the three laws –

Boyle's Law keeping $T_1 = T_2$,

Charles' Law keeping $P_1 = P_2$,

Amonton–Gay Lussac's' Law keeping $V_1 = V_2$.

The Ideal gas equation has the following important consequences:

(i) If temperature and pressure are kept constant, then the volume of the gas is directly proportional to the number of molecules of the gas.

(ii) If the temperature and volume remain constant, then the pressure of the gas is directly proportional to the number of molecules of the gas.

(iii) If the number of gas molecules and the temperature remain constant, then the pressure is inversely proportional to the volume of the gas.

(iv) If the temperature changes and the number of gas molecules are kept constant, then either pressure or volume (or both) will change in direct proportion to the temperature.

Important guidelines while solving Gas law problems

1. Tabulate initial and final data given in the problem. Ensure that all corresponding units are in the same system of units.
2. Till you are fairly confident, always write the formula. Make the unknown as the subject of the equation.
3. Always ensure that you convert temperature from °C to K scale before you start calculating.

SOLVED PROBLEMS IN BOYLE'S LAW CHARLE'S LAW AND IDEAL GAS EQUATION

1. **A gas occupies 1500cc at a pressure of 720 mm of Hg. Find at what pressure will its volume be 1000 cc, given temperature is a constant.**

Ans: P_1 = 720 mm; V_1 = 1500 ml, temperature constant
P_2 = ? V_2 = 1000 ml
Applying Boyle's Law Equation
$P_1V_1 = P_2V_2$ P_2 = 1080 mm

2. **The volume of a gas was found to be 561 cm³, when pressure was 600 mm of Hg. If pressure is decreased by 25% find the new volume of gas.**

Ans: P_1 = 600 mm; V_1 = 561 cc, temperature constant
P_2 = 600 – 25% of 600 = 450 cc
V_2 = ?
Applying Boyle's Law Equation
$P_1V_1 = P_2V_2$ V_2 = 748 cc

3. **279ml of gas at 87°C is cooled to standard temperature at constant pressure. Find the volume of gas at standard temperature.**

Ans: V_1 = 279 ml; T_1 = 87°C = 360K
V_2 =? T_2 = 0°C = 273K
Applying Charles Law Equation

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \quad \text{or} \quad V_2 = 211.58 \text{ ml}$$

4. **To what temperature must a gas at 127°C be cooled, so that its volume is reduced to one-fifth of its initial volume at same pressure?**

Ans: V_1 = x units; T_1 = 127°C = 400K
V_2 = x/5 T_2 = ?°C = ?K
Applying Charles' Law Equation

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \quad T_2 = 80\text{K} = -193°\text{C}$$

5. **1.57dm³ of dry hydrogen gas is at a pressure of 750 mm of Hg when the temperature is 37.5°C. Find the volume of gas at STP.**

Ans: P_1 = 750 mm; V_1 = 1.57 dm³;
T_1 = 37.5°C = 310.5K
P_2 = 760 mm; V_2 = ? dm³; T_2 = 0°C = 273K
(STP conditions)
Using the Ideal gas Equation

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \quad \text{or } V_2 = 1.36 \text{ dm}^3$$

EXERCISE

1. A sample of neon gas occupies a volume of 2.8 *l* at 1.8 atm. What would its volume be at 1.2 atm at same temperature?
2. A balloon full of air has a volume of 2.75 *l* at a temperature of 18°C. What is the balloon's volume at 45°C at same pressure?
3. If 3.0 *l* of a gas at 20°C is heated to 30°C what is the new volume of the gas at same pressure?
4. A sample of argon has a volume of 0.43 ml at 24°C. At what temperature in degrees Celsius will it have a volume of 0.57 ml at same pressure?
5. To what pressure would you have to compress 48.0 *l* of oxygen gas at 99.3 kPa in order to reduce its volume to 16.0 *l* at same temperature?
6. If a barometer at your home reads 768.2 mm of Hg, what is the atmospheric pressure in kPa?
7. What is the starting volume of a 24.7 *l* gas sample that exerts a pressure of 0.999 atm at same temperature? The original pressure is 1.011 atm.
8. What is the starting temperature of 150 ml of gas at same pressure when cooled to 33°C if final volume is 120 ml?
9. What is the volume occupied by 20.4 litres of CO_2 at 1200 torr and at same temperature if new pressure is 760 torr?

Answers

1. 4.2 *l*	2. 3.01 *l*	3. 3.1 *l*
4. 121°C	5. 298 kPa	6. 102.1 kPa
7. 24.4 l	8. 109.5°C	9. 12.92 *l*

3.10 GAY LUSSAC'S LAW OF COMBINING VOLUMES

According to Gay Lussac's Law of combining volumes

'Whenever gases react chemically, they do so in volumes, which bear a simple ratio to each other and also to the products if they are also gaseous, provided the volumes are measured at same temperature and pressure for all the gases'.

Example 1

H_2 + Cl_2 → 2HCl

(1 vol) (1 vol) (2 vol)

In this example, ratio of hydrogen *gas* that reacts with chlorine *gas* forming hydrogen chloride *gas* is **1 : 1 : 2.**

Illustration: 100 ml of hydrogen *gas* will react with 100 ml of chlorine *gas* to give 200 ml of hydrogen chloride *gas*.

Example 2

N_2 + $3H_2$ → $2NH_3$

(1 vol) (3 vol) (2 vol)

In this example, ratio of nitrogen *gas* that reacts with hydrogen *gas* forming ammonia *gas* is 1 : 3 : 2.

Illustration: 100 ml of nitrogen *gas* will react with 300 ml of hydrogen *gas* to give 200 ml of ammonia *gas*.

Some conditions of Gay-Lussac's Law of combining volumes

1. The law is applicable to reacting gases.
2. All volumes are measured under same conditions of temperature and pressure.
3. The volumes of the product are taken only if they are gaseous.

3.11 AVOGADRO'S LAW

'Equal volumes of all gases under similar conditions of temperature and pressure contain equal number of molecules'.

Some conditions of Avogadro's Law

1. The law is applicable to individual gases.
2. All volumes are measured under same conditions of temperature and pressure.

Illustration

If 1*l* of hydrogen *gas* at STP contains say, 'n' molecules, then according to Avogadro's Law's 1*l* of oxygen *gas* at STP has the same number of molecules, i.e., 'n' molecules. 1*l* of nitrogen *gas* at STP also has the same number of molecules, i.e., 'n' molecules and 1*l* of CO_2 at STP will also have the same number of molecules, i.e., 'n' molecules.

Limiting and excess reactants

A limiting reactant is exhausted at the end of a reaction. An *excess reactant* is available to the extent of the excess amount at the end of the reaction.

SOLVED PROBLEMS IN GAY-LUSSAC'S LAW AND AVOGADRO'S LAW

1. **80cc of methane is mixed with 200cc of pure oxygen at room temperature and pressure. The mixture is ignited, when it burns as given below. Find composition of the resulting mixture, if it is cooled to room temperature. The equation of the reaction is $CH_{4(g)}+2O_{2(g)} \rightarrow CO_{2(g)}+2H_2O_{(l)}$**

Ans: Applying Gay Lussac's law,

$CH_{4(g)}$ + $2O_{2(g)}$ → $CO_{2(g)}$ + $2H_2O_{(l)}$

1 vol 2 vol 1 vol (*liquid*)

Ratio of CH_4 and O_2 reacting and CO_2 formed is 1 : 2 : 1.

(Water is not considered as it is a liquid and not a gas.) 80 ml of CH_4 will require 160 ml of O_2.

As the volumes that are available are 80 ml of CH_4 and 200ml of O_2 hence CH_4 is the *limiting reactant* while O_2 is the *excess reactant*. Amount of excess O_2 is 200 – 160 = 40 ml.

1 vol of CH_4 forms 1 vol of CO_2

80 ml of CH_4 will form = 80 ml of CO_2

Composition of the resulting mixture is 40 ml of excess O_2 and 80 ml of CO_2.

2. **A mixture of C_2H_4 and O_2 is ignited. 200cc of CO_2 is collected at STP. Find amount of C_2H_4 and O_2 at STP in the original mixture. The equation of the reaction is $C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$.**

Ans: Applying Gay Lussac's law,

C_2H_4 + $3O_2$ → $2CO_2$ + $2H_2O$

1 vol 3 vol 2 vol (*liquid*)

Ratio of C_2H_4 and O_2 reacting and CO_2 formed is 1 : 3 : 2

So, if vol of CO_2 formed is 200 ml,

Volume of CH_4 required is 100 ml and vol of O_2 required is 300 ml.

This is the composition of the reactant mixture (at the start of the reaction).

3. **500cc of H_2 at STP contains 'n' molecules. How many molecules are present in 25cc of oxygen at STP?**

Ans: Applying Avogadro Law,
500 cc of H_2 at STP has 'n' molecules
500 cc of O_2 at STP will also have 'n' molecules
25 cc of O_2 will have = 0.05 n molecules

4. **6×10^{20} molecules are present in H_2 having volume V at STP. Find the volume of oxygen, which has 3×10^{24} molecules at STP.**

Ans: 6×10^{20} molecules of H_2 occupies 'V' volume at STP
3×10^{24} molecules of H_2 occupies = 5000V at STP
Applying Avogadro Law, at STP
3×10^{24} molecules of O_2 will also occupy 5000V

Simple applications of Avogadro's Law

1. Finding molecular formula of a gas

Experimentally it is found that one volume of hydrogen reacts with one volume of chlorine to give two volumes of hydrogen chloride all volumes being measured under similar conditions.

1 vol of Hydrogen + 1 volume of chlorine → 2 volumes of hydrogen chloride,
Applying Avogadro's law,
1 molecule of hydrogen + 1 molecule of chlorine → 2 molecules of hydrogen chloride

Since we know that each molecule of hydrogen has two atoms of hydrogen and each molecule of chlorine has two atoms of chlorine,

2 atoms of hydrogen + 2 atoms of chlorine → 2 molecules of chlorine

Thus, each molecule of hydrogen chloride must have at least one atom of hydrogen and one atom of chlorine.

∴ the formula of hydrogen chloride is HCl.

2. Finding atomicity of a gas

Experimentally it is found that 2 volumes of ammonia decompose to give one volume of nitrogen and 3 volumes of hydrogen.

2 volumes of ammonia → 1 volume of nitrogen + 3 volumes of hydrogen

Applying Avogadro's Law,

2 molecules of ammonia → 1 molecule of nitrogen + 3 molecules of hydrogen

Since we know the formula of ammonia as NH_3, two molecules of ammonia have 2 atoms of nitrogen and 6 atoms of hydrogen.

These two atoms of nitrogen must be present in one molecule of nitrogen.

Thus the atomicity of nitrogen is 2 and its formula is N_2.

These six atoms of hydrogen must be present in three molecules of hydrogen.

Each molecule of hydrogen thus has two atoms.

Thus the atomicity of hydrogen is 2 and its formula is H_2.

3. Finding vapour density

We can use Avogadro's Law to derive a formula for Vapour Density of a gas.

Vapour Density *is the ratio of the mass of a given volume of gas or vapour to the weight of same volume of hydrogen under the same conditions of temperature and pressure.*

Vapour Density

$$= \frac{\text{mass of given volume of gas}}{\text{mass of same volume of hydrogen}}$$

{under similar conditions}

Since equal volumes of gases have same molecules by Avogadro's Law,

Vapour Density

$$= \frac{\text{mass of 'x' molecules of gas}}{\text{mass of 'x' molecules of hydrogen}}$$

$$= \frac{\text{mass of 1 molecule of gas}}{\text{mass of 1 molecule of hydrogen}}$$

$$= \frac{\text{mass of 1 molecule of gas}}{\text{mass of 2 atoms of hydrogen}}$$

$$= \frac{\text{mass of 1 molecule of gas}}{2 \times \text{mass of 1 atom of hydrogen}}$$

$$= \frac{1}{2} \times \frac{\text{mass of 1 molecule of gas}}{\text{mass of 1 atom of hydrogen}}$$

$$= \frac{1}{2} \times \text{molecular weight of gas}$$

∴ Mol. Wt. = 2 × VD

EXERCISE

I.Choose the correct answer

1. **For a given mass of a gas, if pressure increases:**
   (a) volume and temperature remain constant
   (b) volume increases provided temperature remains constant
   (c) temperature increases provided volume remains constant
   (d) temperature decreases provided volume remains constant

2. **For a given mass of a gas at constant temperature, if the volume V becomes four times, the pressure P will become:**
   (a) 4P (b) P/4
   (c) 2P (d) 4P/T
3. **Measurement of the amount of dry gas collected over water from volume of moist gas is based on**
   (a) Gay Lussac's law (b) Boyle's law
   (c) Charle's law
   (d) Dalton's law of partial pressure
4. **At constant temperature, the pressure of V ml of a dry gas was increased from 1 atm to 3 atm. The new volume will be**
   (a) 3V (b) V/3
   (c) V^3 (d) $2V^3$
5. **For a given mass of a gas, if pressure is reduced to half and its temperature is doubled, then volume V will become**
   (a) 4V (b) $2V^2$
   (c) V/4 (d) 8V
6. **At absolute zero temperature, the total kinetic energy of the molecules is**
   (a) maximum (b) minimum
   (c) zero (d) cannot be predicted
7. **Two flasks A and B of 1 *l* each are filled with O_2 and SO_2 gas at 27°C and 1 atm pressure. The flasks will contain**
   (a) same number of atoms
   (b) same number of moles
   (c) more number of moles in flask B as compared to flask A
   (d) same amount of gases in gram
8. **According to combined gas law**
   (a) $PV \alpha \frac{1}{T}$ (b) $\frac{P}{T} \alpha V$
   (c) $PV \alpha T$ (d) $\frac{V}{T} \alpha P$
9. **The gram molar volume of a gas is the volume occupied at S.T.P by**
   (a) one gram of the gas
   (b) 6.02×10^{23} grams of the gas
   (c) 22.4 g of the gas
   (d) one gram mole of the gas
10. **The kinetic energy of one mole of any gas depends upon**
   (a) pressure of the gas
   (b) nature of the gas
   (c) volume of the gas
   (d) absolute temperature of the gas
11. **If a gas expands at constant temperature**
   (a) the pressure increases
   (b) the kinetic energy of the molecules remains same
   (c) the kinetic energy of the molecules decreases
   (d) the kinetic energy of the molecules of the gas increases
12. **At the same temperature and pressure, which of the following gases will have highest kinetic energy per mole?**
   (a) hydrogen (b) oxygen
   (c) methane (d) all have same
13. **A gas at 10°C occupies a volume of 283 ml. If it is heated to 20°C keeping the pressure constant, the new volume will be**
   (a) 293 ml (b) 283 ml
   (c) 566 ml (d) 586 ml
14. **The volume of a gas measured at 27°C and 1 atm pressure is 10 *l*. What final temperature would be required to reduce the volume to 5.0 *l* at 1 atm pressure?**
   (a) 450K (b) 327K
   (c) 150K (d) 100°C
15. **The temperature of a certain mass of a gas is increased from 50°C to 51°C at same pressure. The volume of gas**
   (a) remains same
   (b) will increase by $\frac{1}{273}$ th of its volume at 273 K.
   (c) will increase by $\frac{1}{273}$ th of its volume at 50°C
   (d) will increase by an amount $\frac{50}{51} \times \frac{1}{273}$
16. **At what temperature in the celsius scale, V (volume)of a certain mass of gas at 27°C will be doubled keeping the pressure constamt?**
   (a) 54°C (b) 327°C
   (c) 427°C (d) 527°C
17. **At constant temperature if pressure increases by 1% the percentage decrease of volume is**
   (a) 1% (b) 100/ 101%
   (c) 1/101% (d) 1/100%

II. Match the following.

Column I	Column II
1. Boyle's Law	a. $P_1V_1/T_1 = P_2V_2/T_2$
2. Charles' Law	b. $P_1V_1=P_2V_2$
3. Ideal Gas Equation	c. $V_1T_1=V_2/T_2$
4. Unit of pressure	d. dm^3
5. Unit of volume	e. K
6. Unit of temperature	f. kPa

III. State whether the following statements are True or False

1. Volume of a dry gas is inversely proportional to pressure only if temperature is constant.
2. Volume of a dry gas is directly proportional to Celsius temperature only if mass and pressure of gas are constant.

3. 0° C is a hypothetical temperature at which the volume of all gases becomes equal to zero.

IV. Fill in the blanks

1. Gases possess_____ or _____.
2. Variables for gases are volume, _____ and _____
3. Temperature on Celsius scale can be converted to temperature on _____ scale by adding 273.
4. STP means a temperature of _____ and pressure of _____
5. Boyle's and Charles' Laws can be explained on the basis of _____ theory of gases.

V. Solve the following problems. (Boyle's Law, Charles' Law and Ideal Gas Equation)

1. An enclosed gas at constant temperature occupies 75 *l* at a pressure of 700 mm of Hg. Find pressure of gas if its volume is increased to 1000 *l* given temperature is a constant.
2. Find pressure of a gas, when its volume is 750 ml, if initial volume of gas is 1250 ml and pressure is 0.8 atm given temperature is a constant.
3. A gas occupies 800 cc at a pressure 'x'. If pressure is altered to 2.5 atm, volume of gas becomes 900 cc. Find the value of 'x' given temperature is a constant.
4. Calculate initial pressure of 4.5 *l* of dry nitrogen gas, if it occupies 3.3 *l* at 1.2 atm, given temperature is a constant.
5. 5 dm^3 of dry oxygen is allowed to expand to 7dm^3 when its pressure, is 700 mm of Hg. Find initial pressure given temperature is a constant.
6. At constant temperature, a gas is at a pressure of 1080 mm of Hg. At what pressure will its volume decrease by 40%?
7. At constant temperature, a gas at a pressure of 1200 mm of Hg occupies a volume of 1500 ml. If its volume is decreased by 40%, find pressure.
8. A dry gas occupies 224 cm^3 at normal pressure. If its volume is increased by 25%, find the new pressure of gas given temperature is a constant.
9. A 12 dm^3 vessel with nitrogen at 152 cm of Hg is connected to another empty 6 dm^3 vessel. Find new pressure given temperature is a constant.
10. A gas occupies 200 ml at a temperature of 27°C and 760 mm of pressure. Find the volume of gas at –3°C and 760 mm of Hg
11. A gas at constant pressure occupies a volume of 300 ml at a temperature of –73°C. Find the volume at 127°C, at same pressure.
12. A dry gas occupies a volume of 1054 dm^3 at a temperature of –73°C. At what temperature will its volume be 4216 dm^3 if the pressure remains constant?
13. A gas occupies 150 ml at 57°C. Find the temperature to which the gas should be heated so that the volume triples at same pressure.
14. A gas occupies a volume of 400 ml. On heating to 127°C its volume becomes 1600 ml. Find initial temperature of gas assuming pressure is constant.
15. A gas is enclosed in a vessel at standard temperature. At what temperature will the volume of enclosed gas be one-eighth of initial volume, pressure remaining constant?
16. At constant pressure a gas at –33°C is heated to 127°C. Find the % increase in the volume of gas.
17. 5 dm^3 of dry hydrogen is collected at a temperature of 37°C and 720 mm of Hg pressure. Find volume of gas at STP.
18. A fixed volume of gas occupies 760 ml at 27°C and 700 mm of Hg. What is its volume at STP?
19. A gas occupies a volume of 512 ml at STP. Find its volume at 27°C and at a pressure of 720 mm of Hg.
20. 225 ml of hydrogen at STP is heated to a temperature of 97°C when its pressure is 810 mm of Hg. Find the volume of gas.
21. 224 ml of dry CO_2 is at temperature of 27°C and pressure of 720 mm of Hg. Find volume of gas at temperature of –73°C and pressure of 3600 mm of Hg.
22. A gas occupies a volume of 100 ml at 0°C and 760 mm of Hg pressure. If the Kelvin temperature of the gas is increased by one-fifth and its pressure is increased by half, find the final volume of gas.
23. The pressure of 1 mole of a gas at STP is doubled and the temperature is raised to 546 K. What is the final volume of gas?
24. A fixed mass of gas has a volume of 750 ml at –23°C and 800 mm of Hg. Find the pressure for which its volume will be 720 ml at a temperature of –3°C.
25. A given mass of dry gas is kept at constant volume while doubling the pressure. Then, at constant pressure Kelvin temperature is doubled. Then keeping temperature constant, the volume is doubled. Express final volume, pressure and temperature in terms of initial values (V_1, P_1 and T_1 respectively).

VI. Solve the following problems. (Gay Lussac's Law and Avogadro's Law)

1. Find the volume of O_2 required and volume of NO formed when 112 cc of NH_3 gas is oxidized catalytically at room temperature and pressure.
   $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$
2. 400 cc of methane is burnt in just sufficient air containing 78% nitrogen to form CO_2 and steam. If all

measurements are made at 100°C find the composition of the resultant mixture.

3. Find volume of Cl_2 gas required and volume of HCl gas formed at STP when 40 ml of CH_4 reacts completely with Cl_2 as given below :
$CH_{4(g)} + 2Cl_{2(g)} \rightarrow CH_2Cl_{2(g)} + 2HCl_{(g)}$
4. 5000 molecules are present in volume V of H_2 at STP. Find the volume of N_2 gas that has 2.5×10^{98} molecules at STP.
5. 6×10^{22} molecules are present in 2.24 dm^3 of Cl_2 at STP. Find volume of SO_2 gas at STP, which has 3×10^{19} molecules.
6. 2.24 dm^3 of dry NH_3 gas contains 6×10^{22} molecules at STP. Find the number of molecules in 5.6 cc of dry N_2 at STP.
7. 10 cc of gas has 10^5 molecules at certain temperature and pressure. Find the number of molecules of O_2 present in 200 cc at same temperature / pressure.

VII. Solve the following problems.

1. At 17°C the volume of a gas is 800 ml. At what temperature will the volume of the gas double?
2. The volume of a gas with some marble is 100 cc at 760 mm pressure. On increasing the pressure to 1000 mm the total volume becomes 80 cc. What is the volume occupied by marble?
3. 400 ml of O_2 at 27°C is cooled to –15°C at constant pressure. Find the contraction in volume.
4. An open vessel at 27°C is heated until three fifths of the air is expelled. Assuming the volumes of the vessel remains constant; find the temperature to which the vessel has to be heated.
5. A ball is inflated to a volume of 5 litre at 25°C and 1 atm. At what temperature will the volume of the ball reach 6 litres, if pressure is not changed?
6. A 2.5 *l* flask can withstand pressure of 14 atm. It is filled with CO_2 at 27°C and 750 mm. At what temperature will it explode?
7. 250 ml oxygen at 27°C, 740 mm was collected over water. What volume will the gas occupy at STP? (Aqueous tension of water at 27°C = 26.5 mm).
8. At 27°C and 5 atm. H_2 gas is present in a 15*l* cylinder. How many balloons of volume 1500 ml can be filled at 27°C and 1 atm?
9. Represent three isotherms at temperatures of T_1, T_2 and T_3 in a single P-V graph if $T_3 > T_2 > T_1$.
10. When a gas is allowed to expand from a region of high pressure to a region of low pressure there is a drop of temperature. What is this called?
11. The equation of decomposition of a gaseous hydride is $4A_xH_{y(g)} \rightarrow 2A_{2(g)} + 6H_{2(g)}$. What is the simplest formula of A_xH_Y?
12. What volume of NO is formed when 90 *l* of O_2 reacts with NH_3 at STP as in the given equation?
$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$

KEY TO EXERCISE

I. 1. c 2. b 3. d 4. b 5. a 6. c 7. b
8. c 9. d 10. d 11. b 12. d 13. a 14. c
15. b 16. b 17. c

II. 1. b 2. c 3. a 4. f 5. d 6. e

III. 1. false 2. false 3. false

IV. 1. shape, volume 2. pressure, temperature 3. Kelvin 4. 273K, 1 atm
5. kinetic

V. 1. 52.5mm 2. 1.33atm 3. 2.81atm 4. 0.88atm 5. 980mm 6. 1800mm 7. 2000mm
8. 608mm 9. 101.33 10. 180ml 11. 600ml 12. 527°C 13. 717°C 14. –173°C
15. –238.875°C 16. 66.67% 17. 4.17dm^3 18. 637ml 19. 593.89ml
20. 286.12ml 21. 29.87ml 22. 80ml 23. 22.4dm^3 24. 900mm 25. $4V_1, P_1, 4T_1$

VI. 1. 140ml, 112 ml 2. 400 ml of CO_2; 800 ml of steam; 2836.37 ml of N_2 from air
3. Vol of Cl_2 required is 80 ml; vol of HCl formed is 80 ml
4. 5×10^{94} V 5. 1.12 ml 6. 1.5×10^{20} molecules 7. 2×10^6 molecules

VII. 1. 307°C 2. 16.67 cc 3. 56ml 4. 750K 5. 84.6°C 6. 3983°C
7. 213.60 ml 8. 50 9. (See isotherms in text) 10. Joule Thomson effect
11. AH_3 12. 72 *l*

4 Atomic Structure

4.0 INTRODUCTION

Matter is anything that occupies space, has mass and can be perceived by our senses. What is matter made up of ? What is its composition? That matter is made up of small particles called *atoms* was recognized quite early by man.

(i) **Democritus**, (400BC), a Greek philosopher proposed that matter is made up of extremely small particles – the '*atoms*', meaning indivisible. However he did not base his observation on any verifiable scientific experiment.

(ii) **Dalton**, an English scientist, in 1808, proposed that matter is made up of very small indivisible particles. He called these particles ***atoms*** – meaning indivisible. He carried out several experiments on how elements combine and put forth his inferences in what is known as Dalton's Atomic Model.

(iii) **William Crookes** (1878), **J J Thomson** (1879) and **Goldstein** (1886) later proved that the atom was divisible. They proved that an atom is made up of smaller particles–***electrons*** and ***protons***.

(iv) **Rutherford** in 1911 discovered that protons are present in a very small region inside the atom called the *nucleus*.

(v) **James Chadwick** in 1932 proved that the atom has another type of particle called ***neutrons***.

(vi) The works of **Rutherford** (1911), **Neil Bohr** (1912), **Moseley** (early 1910s), **Sommerfield** (1916), **de Broglie** (1924), **Schrodinger** (1926), **Heisenberg** (1927) and several other scientists helped in knowing about the arrangement of these particles inside the atom.

Modern atomic theory states that :

(i) All matter is made up of atoms.

(ii) All atoms have three types of subatomic particles–protons, electrons, and neutrons.

(iii) All protons are positively charged particles carrying unit positive charge.

(iv) All electrons are negatively charged particles carrying unit negative charge.

(v) All neutrons are electrically neutral particles and carry no charge.

(vi) Protons and neutrons have almost exactly the same mass.

(vii) Electrons have a mass that is about 1/1836 of the mass of a proton.

(viii) The neutrons and protons are in the centre of the atom in the nucleus of the atom.

(ix) The mass of the atom is due to the nucleus which has protons and neutrons.

(x) The maximum number of electrons in a shell is $2n^2$.

(xi) The electrons revolve around the nucleus in fixed orbits called shells.

4.1 DALTONS ATOMIC MODEL, 1808

Dalton postulates are as follows.



Dalton

(a) Matter consists of small indivisible particles called *atoms*.

(b) Matter can neither be created nor destroyed.

(c) Atoms of same element are similar in all respects. Conversely, atoms of different elements are different in all respects.

Do you know that Dalton's idea about the atom being indivisible was not contested till almost 50 years.

(d) Atoms combine in small whole numbers to form *compound atoms* (today called molecules).

(e) Atom is the smallest unit of matter that takes part in chemical reactions.

Dalton's Atomic Model became the basis of further research on the structure of the atom. Today Dalton's theory is almost discarded.

Defects of Dalton's Atomic Model

(i) Atoms can be divided further into sub-atomic particles like electrons, protons and neutrons and also other particles like mesons, positrons, etc.

(ii) Nuclear fusion and fission reactions show that atoms can be created and destroyed.

(iii) Discovery of *isotopes* (atoms with same atomic number but different mass number) proves that atoms of same element need not be similar in all respects.

Example: Chlorine exists in two isotopic forms - $_{17}Cl^{37}$ and $_{17}Cl^{35}$. This shows that atoms of the same element can be different in some respects. In this example, mass number of chlorine in the two isotopes is different.

(iv) Discovery of *isobars* (atoms with different atomic number but same mass number) proves that atoms of different elements need not be different in all respects.

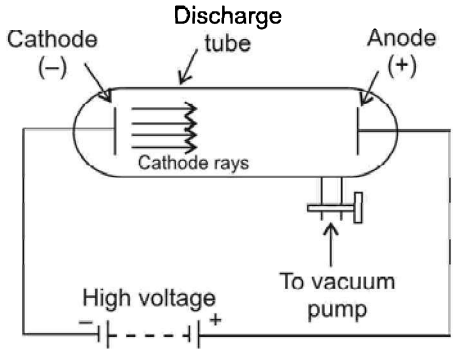
Example: Consider atoms of Argon and Calcium - $_{18}Ar^{40}$ and $_{20}Ca^{40}$. The mass numbers of Argon and Calcium are same. This shows that atoms of different elements can be same in some respects.

(v) Discovery of polymers and macromolecules proves that atoms *can combine in large numbers.*

4.2 THE DISCHARGE TUBE EXPERIMENTS – DISCOVERY OF ELECTRON

The Discharge Tube

The discharge tube consists of a cylindrical glass tube 50 cm long and 4 cm in diameter with two electrodes of aluminium at the two ends. The pressure of the gas can be decreased with the help of a pump and can be controlled by a gauge. A high potential difference of 10,000 – 15,000 V is set up between the electrodes (see figure below).



At atmospheric pressure, all gases behave like insulators. They do not normally allow electricity to flow between the electrodes. At normal pressure inside the tube, a potential difference of about 30,000 V is required to produce a spark between two points in air 1cm apart!!

Thus, at very low pressures, by applying a high potential difference between the electrodes an electric spark can be produced between the plates. As pressure is lowered further a luminous glow of light can be seen all along the length of the tube.

Cathode Rays

As pressure in the cylinder was decreased the following observations were made.

(i) At pressures above 100 mm no electricity flows between the plates as the gas in between the plates acts as an insulator.

(ii) As pressure decreases to 10 mm of Hg, electricity begins to flow between the plates. This flow is called ***discharge***. Luminous rays can be seen going from one plate to the other accompanied by crackling sound.

(iii) As pressure falls to 5 mm of Hg, the discharge becomes steady and the tube is filled with a luminous glow, the colour depending on the gas inside.

For example air gives a green discharge, neon gives bright red, carbon dioxide gives pale blue, hydrogen gives blue, nitrogen gives red discharge.

(iv) As a the pressure is reduced further to 1mm of Hg, the luminous glow separates from the cathode. A separate glow is seen on the cathode.

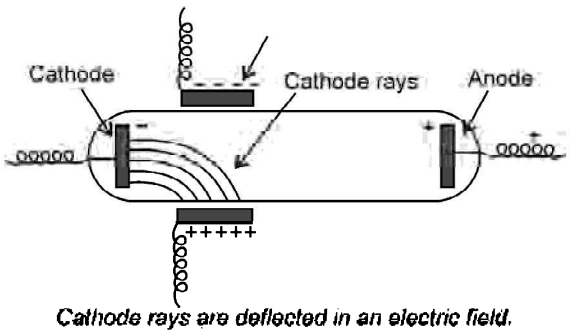
(v) As pressure falls to 0.5 mm of Hg, the glow detaches from the cathode. The luminous glow inside the tube starts decreasing as pressure falls further.

(vi) At a pressure of 0.01 mm of Hg, there is no visible discharge. Now, the cathode emits ***invisible radiation*** that moves towards the anode. If the anode is coated with zinc sulphide, ***scintillations*** or flashes of light can be seen and heard.

Important conclusions

(i) The scintillations show that some *radiation coming* ***from the cathode is hitting the anode***.

(ii) The sound produced proves that the radiations ***consists of particles with mass***.



Cathode rays are deflected in an electric field.

(iii) Since they come from cathode (negative plate) and travel towards anode (positive plate) *these particles must be negatively charged.*

(iv) The radiation travels in a straight line. When an electric field is applied perpendicular to the path of the rays, the rays get deflected towards the positive plate of the electric field (see figure). This further proves that the *radiation consists of negatively charged particles.*

(v) **Charge of the cathode ray particles:** When the charge of the particles in the cathode rays hitting the anode was measured, it was found that all the particles have the same charge. When the gas was changed and the charge measured, it was found that all the particles have the same charge. Even when the electrodes were changed it was found that *all the cathode ray particles have the same charge.*

(vi) **Mass of the cathode ray particles:** When the mass of the particles in the cathode rays hitting the anode was measured, it was found that all the particles have the same mass. When the gas was changed and the mass measured, it was found that all the particles had the same mass. Even when the electrodes were changed it was found that *all the cathode ray particles had the same mass.*

(vii) **e/m ratio:** Thus the ratio of charge (e) and mass (m) called *e/m ratio of the cathode ray particles was a constant* irrespective of the gas used or electrode used.

The following inferences were drawn out of these observations.

1. From where do the cathode ray particles originate?

The particles originate from the cathode.

2. What is the cathode made of?

The cathode is a metal that is a solid – it is made up of matter. The matter is made up of small particles called atoms (Dalton's Atomic Model).

3. So what is the conclusion?

The particles must be coming from inside the atoms of the metal.

4. What if the metal plate is changed?

Exactly the same kind of particles can be seen coming from the changed plate.

5. How can it be said beyond doubt that the particles are exactly same?

The e/m ratio of the particles is found to be same for all cathode ray particles irrespective of the cathode used.

6. What does this prove?

The particles in the cathode rays *are present inside all atoms.* As the particles are present inside all atoms they are the ***fundamental particles*** present in all atoms. (Just as the eyes, ears, nose, etc. are fundamental parts of every human body)

Discovery of electron



J.J. Thomson

Plucker in 1852 and Goldstein in 1876 discovered the cathode rays. The Discharge Tube experiments showed that these rays contain negatively charged particles. The 'e/m' ratio of these particles was found to be a constant even if the electrodes were changed in the discharge tube. So it was concluded that these particles are present in atoms of all elements.

J J Thomson (1887) called the cathode ray particles as electrons.

4.3 CANAL RAYS – DISCOVERY OF PROTON

An indirect inference

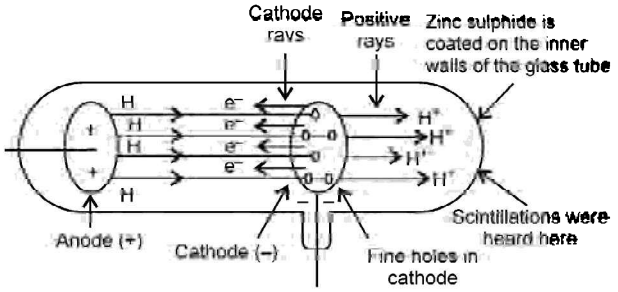
If all atoms have negatively charged particles inside them, then all of them should carry negative charge. But *atoms* are electrically neutral.

What does this mean? The atoms must contain an equal amount of positive charge inside them!

Canal Rays

Goldstein in 1886 made some holes on the cathode plate and found that when a high potential is applied across the electrodes some rays travelled in the direction opposite to the cathode rays (see figure).

These rays travelled away from the anode in a straight line and passed through the holes in the cathode. When the glass tube was coated with ZnS, scintillations were seen and heard there. They were called ***canal rays*** as they passed through the canals or holes in the cathode plate. But the major difference was that the positive rays produced when the gas was changed ***did not have the same charge or mass***.



Canal rays travel in the direction opposite to the cathode rays

Important conclusions

(i) The sound produced proves that the canal rays *consist of particles with mass.*

(ii) The canal rays have particles with positive charge as they move away from the anode.

(iii) The radiation travels in a straight line. When an electric field is applied perpendicular to the path of the rays, they get deflected towards the negative plate of the electric field. This further proves that the radiation consists of *positively charged particles*.

(iv) **Charge of the canal ray particles**: When the charge of the particles in the canal rays hitting the glass walls was measured, it was found that all particles have the same charge if the gas inside was the same. But *when the gas was changed, it was found that the charge was same for some gases and different for other gases. The charge was not a constant and depended on the gas used.*

(v) **Mass of the canal ray particles**: When the mass of the particles in the canal rays was measured, it was found that all the particles had the same mass if the gas was not changed. But *when the gas was changed and the mass measured, it was found that mass of the particles also changed. The mass of the canal ray particles was not a constant. Further when hydrogen gas was used the mass of the canal ray particles was found to be the least i.e., 1 amu*. When other gases were used the positive ray particles had mass greater than 1 amu.

(vi) **e/m ratio**: *Thus the e/m ratio of the canal ray particles was not a constant for all gases. However, when hydrogen gas was used it was found that the e/m ratio was 1. All other gases gave e/m values that were smaller than 1.*

Hydrogen atom

The H-atom has 1 proton and 1 electron.

Using the e/m ratio of hydrogen and charge 'e' of an electron, (found from cathode rays) the mass of the proton was found.

e/m of H^+ was found to be 95710000 C kg^{-1}.

Mass of electron is 9.1×10^{-31} kg

Charge of electron, e = 1.601×10^{-19} C

Mass of hydrogen atom or proton,

m_p = (e / e/m)

Thus, $m_p = 1.674 \times 10^{-27}$ kg

The ratio of mass of proton to electron

m_p / m_e =1840.

Thus the proton was found to be 1840 times (approximate) the mass of the electron. It can be seen that the mass of electron is almost negligible when compared to that of proton.

These conclusions can be arrived at by answering appropriate questions as follows.

What are the observations?

When the charge of the particles in the positive rays was measured, it was found that **all the particles** have the same charge as long as the gas was not changed. But when the gas was changed and the charge measured, it was found that the particles do not always have the same charge.

What is the conclusion?

All the particles in the positive rays are different particles but carry the same charge. There exist different positive particles with different masses and different charges.

What is the origin of these positive particles?

When an electric discharge takes place across the electrodes, one or more electrons in the atoms of the gas are knocked out. The remaining part of the atom acquires a positive charge. It is these positive ions that travel in the form of positive rays.

What do the e/m ratios tell us?

The mass of the positive ray particles is very large when compared to the cathode ray particles. But, the mass of the particles in the positive rays produced when hydrogen gas is used is the smallest. The charge also is the smallest.

What about positive rays formed with other gases?

It has been found that other gases also produce positive rays. But the charge on the particles is same or two times the charge present on the particle produced when hydrogen is used. Also, the mass is several times that of the particle produced when hydrogen is used.

What is the inference?

The e/m ratio when hydrogen gas is used is the smallest. This leads to the inference that the positive particle present in hydrogen atom is the smallest positive particle.

Discovery of proton

Goldstein in 1886 discovered the canal rays. He experimented with various gases and electrodes in the discharge tube. Later Thomson calculated 'e/m' ratios of the canal rays. He observed that 'e/m ratio' was highest when he used hydrogen gas. So, he concluded that the lightest positively charged particle is present in hydrogen atom.

The positive particle present in the canal rays when hydrogen gas was used in the discharge tube was considered as a unit positive charge and was called proton.

4.4 DISCOVERY OF NEUTRON

The presence of the neutron was not conclusively established till 1932. Doubts about the presence of another particle were expressed as early as 1911 by Rutherford. Rutherford could not explain the atomic weight of some atoms like helium and carbon.

Weight of helium atom

If the atom contain only electrons and protons then the weight of helium atom with atomic number 2, should be the sum of the masses of 2 electrons and 2 protons. If the mass of electrons is considered negligible then the atomic weight of helium should be the mass of the two protons present in the atom of helium. However, the actual atomic mass of helium was found to be 4 amu (2 amu more than what was expected).

Rutherford in 1920 predicted that the atom must contain another particle, *which must be neutral* and *have a mass equal to that of a proton.*

It was only in 1932 that **Chadwick** discovered these neutral particles in the atom by bombarding beryllium with α-particles.

$_{4}Be^{9} + {}_{2}He^{4} \rightarrow {}_{6}C^{12} + {}_{0}n^{1}$ + Energy (= 5.5 Mev)

He called the particles **neutrons** (Chadwick was given the Nobel Prize in Physics in 1935 for this discovery).

4.5 FUNDAMENTAL PARTICLES PRESENT IN THE ATOM

The electron, proton and neutron are the **fundamental particles present in all atoms** with the notable exception of hydrogen (which contains only 1 proton and 1 electron). Subsequent research has shown the presence of many other sub-atomic particles.

Properties of fundamental particles

The mass and specific charge of the fundamental particles is as follows.

Particle	Position	Symbol	Charge	Charge (amu)	Mass Kg
Electron	Revolves around the nucleus	$_{-1}e^{0}$	−1 unit	1/1837	9.1×10^{-31}
Proton	In the nucleus	$_{+1}p^{1}$	+1 unit	1	1.67×10^{-27}
Neutron	In the nucleus	$_{0}n^{1}$	Zero	1	1.67×10^{-27}

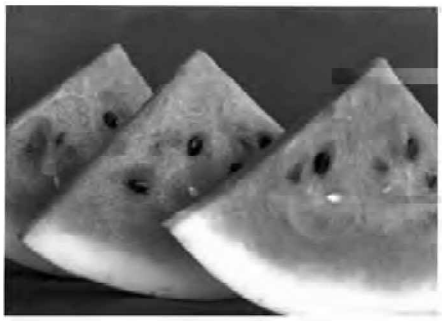
One unit of electrical charge = 1.6×10^{-19} coulombs and mass of 1 amu = 1.66×10^{-24} gm

4.6 ATOMIC MODELS

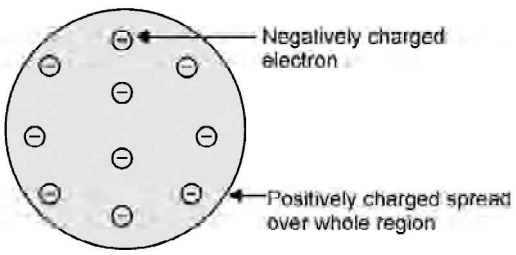
An atomic model gives the arrangement of the fundamental particles inside the atom.

Thomson's Atomic Model

This was the earliest atomic model proposed by Thomson. He stated that the electrons and protons are distributed evenly inside the atom, which is spherical. However the model could not explain why the electrons and protons do not attract each other. If



A water melon – the seeds are dispersed in the red mass.



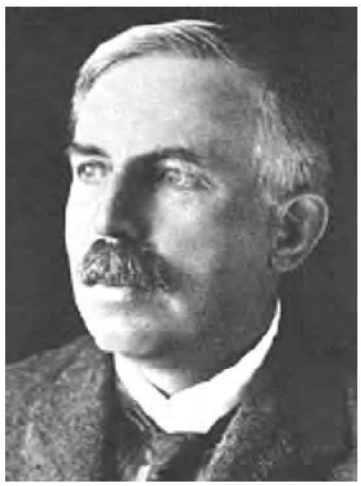
Thomson's Atomic Model – the electrons are dispersed in the positive mass

they attract each other the atom would collapse – this obviously does not happen as atoms are stable and the electrons and protons are separate from each other as shown by the Discharge Tube experiments.

4.7 RUTHERFORD'S ATOMIC MODEL, 1911

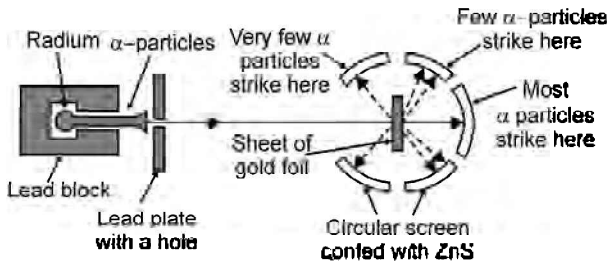
The next attempt at proposing an atomic model was by Rutherford. Rutherford based his findings from the observations of the α-ray scattering experiments.

The experiment: Radium is a radioactive material that emits alpha particles. Alpha particles are denoted by the symbol α. Alpha particles are He^{2+} ions.



Rutherford

A beam of α-particles from a radium source is passed through a lead plate with a very small pin sized hole, to make the beam as thin as possible. The thin beam of α-particles is made to fall on a very thin gold foil. The foil is surrounded by a circular screen coated with ZnS (see figure).

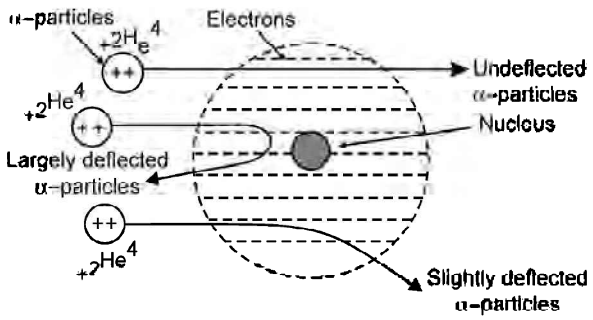


Rutherford's Alpha scattering experiment

Observations

1. Most of the α-particles passed through the gold foil and hit the screen in a straight line.
2. A few α-particles suffered small deflections from their original paths.
3. A still fewer alpha particles suffered large deflections from their original paths.

Rutherford tried to explain these observations using a magnified atom.



The cause of deflection of the alpha particles was explained by Rutherford using a magnified atom

(a) Most α-particles pass through the gold foil *because the atom is largely hollow.*

(b) A few α-particles suffer some deflection from their original path because *they pass close to a positive region that he called* ***'nucleus.'***

(c) A very few α-particles suffer large deflections because *they come on a head collision course with the nucleus.*

Postulates of Rutherford's Atomic Model

Based on these observations Rutherford came to some inferences that he presented in his model known as Rutherford Atomic Model.

According to his model

(i) An atom is a hollow spherical body.

(ii) The entire positive charge or protons is present in a central region called nucleus.

(iii) The nucleus is a very small part of the atom.

(Reason: Most alpha particles pass through the gold foil / atom. If the nucleus was very large then many more alpha particles would have suffered deflection).

(iv) Electrons revolve around the nucleus at high speeds in a circular path.

(Reason: The electrons cannot be stationary as they would be pulled into the nucleus).

(v) The positive charge is in the nucleus. The mass of the atom is almost entirely due to the nucleus.

(vi) There are two types of forces in the atom.

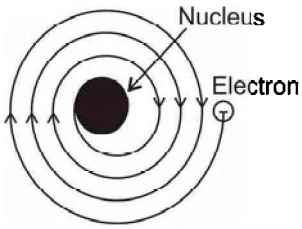
(a) a centripetal force due to the electrostatic force of attraction between electrons and protons. This force pulls the electrons towards the nucleus.

(b) a centrifugal force due to the high speed with which the electrons revolve around the nucleus. This force is directed away from the nucleus.

Due to these two opposing but equal forces, the atom is stable.

Defects of Rutherford Atomic Model

As per laws of electrodynamics any moving charged particle moving at a high speed in a circular path will radiate energy continuously.



The electron when it moves around the nucleus should thus radiate energy. If this is true then it should lose energy and get increasingly pulled towards the nucleus with greater force. Thus the electron must spiral into the nucleus. The atom should collapse if this happens. But the atom is stable.

4.8 BOHR'S ATOMIC MODEL (1913)

To overcome the defects of Rutherford's atomic model, Neil Bohr gave his atomic model.

He agreed with Rutherford that the protons are present in a small region at the centre of the atom in the nucleus and also that the electrons revolve around the nucleus with large speeds.



Neil Bohr

Postulates of Bohr's Atomic Model

(a) Electrons revolve around the nucleus with definite energy in concentric circular orbits called *stationary orbits*. These orbits are denoted as K, L, M, N....

(b) The electrons have the least energy in the K-shell of an atom followed by the L-shell, M-shell, etc.

(c) As long as the electron revolves in the same circular orbit, energy is neither radiated nor absorbed.

(d) Energy of the electron changes only when it moves from one orbit to another orbit. Energy is absorbed when an electron jumps from an inner orbit to an outer orbit and vice-versa.

4.9 SOME IMPORTANT POINTS TO REMEMBER

(a) ***Atomic Number*** (denoted as Z) is the number of protons present in an atom. It will be equal to the number of electrons in a neutral atom.

(b) ***Mass Number*** (denoted as A) is the number of neutrons and protons present in the nucleus.

(c) ***Nucleons*** are the particles present in the nucleus of an atom. Example: protons and neutrons.

(d) The expression (A – Z) gives the number of neutrons in the atom.

(e) **Representing an element:** An element is represented using its symbol with atomic number as a subscript and mass number as superscript.

$_{Z}X^{A}$ where X is any element, Z is its atomic number and A is its mass number.

Example: $_{17}Cl^{35}$

(f) Valence electrons are the electrons present in the outermost shell. These electrons take part in chemical reactions. The number of valence electrons determines the chemical reactivity of the element.

(g) ***Duplet configuration*****:** In the case of single shell atoms like hydrogen and helium, the maximum number of electrons that are possible in the valence shell is 2. When the valence shell in a single shell atom has 2 electrons the configuration is called Duplet Configuration.

(h) ***Octet configuration*:** In the case of multi-shell atoms all elements excluding hydrogen and helium, the maximum number of electrons that are possible in the valence shell is 8. When the valence shell in a multi-shell atom has 8 electrons the configuration is called Octet Configuration.

4.10 RULES OF ELECTRONIC CONFIGURATION

(a) The maximum number of electrons in a given orbit, n is $2n^2$.

The 1st orbit can thus hold a maximum of 2 electrons.

The 2nd orbit can hold a maximum of 8 electrons.

The 3rd orbit can hold a maximum of 18 electrons and so on.

(b) The orbitals are successively filled with electrons in increasing order of energy levels. The lower energy orbits are filled with electrons first, then the next higher energy orbit and so on. K-orbit is filled with electrons followed by L, M, N, etc.

(c) The outermost orbit of an atom (valence orbit/ shell) cannot have more than 8 electrons.

Note – The only exception being a single shell atom in which case the last and only shell cannot hold more than 2 electrons.

(d) The second last shell in an atom cannot have more than 18 electrons.

(e) A new shell starts filling as the outermost shell gets 8 electrons.

(f) A given shell need not be completely filled with electrons for a new shell to start.

Electronic Configuration of the first 20 elements

Atomic Number	Element	Electrons in the shell			
		K-shell	L-shell	M-shell	N-shell
1	H	1			
2	He	2			
3	Li	2	1		
4	Be	2	2		
5	B	2	3		
6	C	2	4		
7	N	2	5		
8	O	2	6		
9	F	2	7		
10	Ne	2	8		
11	Na	2	8	1	
12	Mg	2	8	2	
13	Al	2	8	3	
14	Si	2	8	4	
15	P	2	8	5	
16	S	2	8	6	
17	Cl	2	8	7	
18	Ar	2	8	8	
19	K	2	8	8	1
20	Ca	2	8	8	2

4.11 PROPERTIES OF ATOM

Some important properties of an atom are

(a) Ionisation Energy

(b) Electron Affinity

(c) Electronegativity

(d) Atomic and Ionic radius

(a) Ionisation Energy (IE) is the energy required to remove the valence electron from a neutral atom in its gaseous and ground state. It is measured in electron volts (eV) or kJoules / mole.

Explanation of the terms

(i) Only the valence electron can be removed from an atom. Hence the term valence electron is used.

(ii) The valence electron in an atom moves in its orbit with certain energy. To remove this electron from the valence shell, certain energy has to be absorbed by the electron so that it can overcome the pull of the nucleus and break free from this pull. ***It is an endothermic process.*** Energy can be supplied to the atom by heating it.

(iii) If the element is in its solid or liquid state the energy is used to change its physical state. For example, if heat is supplied to an element in the solid state then the heat energy is used to convert it to the liquid state. The energy supplied is therefore not used for removing the electron from the valence orbit but to change the physical state. Hence the term 'in the gaseous state' is used.

(iv) The atom must be present in its ground state and not in its excited state. In the excited state the electron may not be in its valence orbit. It may be present in a larger orbit. As it has more energy it will be present in a larger orbit. So the term in its ground state is used.

Example: IE of Hydrogen is 496 kJ / mole. It means that 496 kJ of energy must be supplied to remove one electron each from one mole of H-atoms in gaseous and ground state.

Some important points to remember:

(a) Ionisation energy is low for metals and high for non-metals and highest for inert gas elements.

(b) This is because metals have 1, 2 or 3 electrons in the valence shell. If the metal atoms lose these valence electrons then they get a stable inert gas configuration. Thus, metals lose these electrons easily. So energy required to pull out these electrons will be low.

(c) Non-metal atoms have 5, 6, or 7 electrons in the valence shell. They have a tendency to gain electrons so that they can get inert gas configuration. Hence energy required to pull out an electron will be large.

(d) Inert gas elements have a stable octet or duplet configuration. It is difficult to pull out an electron. So they will have the highest IE values.

(e) The energy required to pull out the first electron will be smaller than the energy required to pull out the second electron from uni-positive ion. This is because the pull of the nucleus on the lesser number of electrons after an atom has lost the first electron, is more.

(b) Electron Affinity (EA) is the energy released when an electron is added to the valence shell of a neutral atom in its gaseous and ground state. It is measured in electron volts (eV) or kJoules / mole.

Explanation of the terms:

(i) An electron can be added only to the valence shell. Hence the term valence electron is used.

(ii) When an electron is added to the valence shell of non-metals it gets inert gas configuration. The atom releases some energy as it gets more stability. ***It is an exothermic process***. Energy is released in the form of heat.

(iii) The element must be present in its gaseous state so that whatever energy is released is not used for state conversion.

(iv) The atom must be present in its ground state and not in its excited state. In the excited state the electron may enter an orbit other than the valence orbit. So the term in its ground state is used.

Example: EA of Cl atom is 348 kJ / mole. It means 348 kJ of energy is released when an electron is added to one mole of each chlorine atoms.

Some important points to remember:

(a) Electron affinity is low for metals and high for non-metals and zero for inert gas elements.

(b) This is because metals have 1, 2 or 3 electrons

in the valence shell. These atoms have a tendency to lose electrons but not to gain them. So energy released when an electron is added will be low.

(c) Non-metal atoms have 5, 6, or 7 electrons in the valence shell. They have a tendency to gain electrons so that they can get inert gas configuration. Hence energy released when an electron is added will be large.

(d) Inert gas elements have a stable octet or duplet configuration. It is difficult to add an electron. So they will have zero EA values.

(e) Energy is released when the first electron is added due to extra stability of the ion. But energy is absorbed when the second electron is added to the uni-negative ion as the incoming electron has to overcome the repulsive forces of the uni-negative ion. Hence second EA value is endothermic while first EA value is exothermic.

(c) Electronegativity (EN) is the tendency of a bonded atom to pull a shared pair of electrons towards itself. It is measured on a scale of 0 to 4 called the Pauling Scale.

Example: On this scale Fluorine has the highest EN of 4.0 while Cesium and Francium have the least EN of 0.7. It means that tendency of F atom in say H-F to pull the shared pair of electrons is very high.

(d) Atomic radius is the distance from the centre of the nucleus to the outermost orbit. It is measured in Angstrom units. ($1A° = 10^{-8}$ cm). For example atomic radius of H-atom is $0.59A^0$.

(e) Ionic radius is the distance from the centre of the nucleus to the outermost orbit of an ion. It is measured in Angstrom units.

(i) Cation radius is smaller than the parent atom as a smaller number of electrons are attracted by the same number of protons. Hence the ion is smaller than the parent atom from which it is formed.

(ii) Anion radius is larger than the parent atom as more number of electrons are attracted by the same number of protons. Hence the ion is larger than the parent atom from which it is formed.

EXERCISE

I. Choose the correct answer

1. Cathode rays are
(a) stream of protons
(b) stream of alpha particles
(c) stream of electrons
(d) high energy photons

2. Cathode rays have
(a) mass only without any charge
(b) negative charge with negligible mass
(c) neither mass nor charge
(d) unit mass and unit charge

3. Specific charge of an electron =
(a) 1.602×10^{-19} C (b) 4.8×10^{-10} esu
(c) 5.28×10^{17} esu/g (d) 9.1×10^{-28}gm

4. Unit charge =
(a) 1.6×10^{-19}C (b) 1.6×10^{-10} C
(c) 1.6×10^{10} C (d) 1.6×10^{19} C

5. The ratio of specific charge of a proton and an alpha particle is
(a) 2 : 1 (b) 1 : 2
(c) 1 : 4 (d) 1 : 1

6. The mass of neutron is of the order of
(a) 10^{-25} kg (b) 10^{-24} kg
(c) 10^{-27} kg (d) 10^{-31} kg

7. The heaviest particle among the following is
(a) meson (b) neutron
(c) proton (d) electron

8. Which one of the following is not a fundamental particle?
(a) electron (b) proton
(c) neutron (d) alpha particle

9. The particle having the biggest specific charge is
(a) electron (b) proton
(c) neutron (d) alpha particle

10. Neutrons are obtained by
(a) bombarding radium with alpha particles
(b) bombarding of Be with alpha particles
(c) radioactive disintegration of Uranium
(d) none of these

11. Which of the following subatomic particle is not present in an atom?
(a) neutron (b) proton
(c) electron (d) positron

12. Which of the following has the same atomic number and the same mass number?
(a) protium (b) deuterium
(c) tritium (d) helium

13. **The difference between mass number and atomic number is numerically equal to**
(a) number of electrons (b) number of protons
(c) number of neutrons
(d) sum of neutrons and protons

14. **e/m ratio of canal rays when hydrogen gas is used is _____amongst all gases.**
(a) smallest (b) largest
(c) equal to infinity (d) cannot say

15. **Which of the following properties of an element is a whole number?**
(a) atomic weight (b) atomic volume
(c) atomic number (d) atomic radius

16. **The atomic number and mass number of an element are 26 and 56 respectively. The number of neutrons in the nucleus will be**
(a) 26 (b) 30
(c) 36 (d) 56

17. **Rutherford's alpha particle scattering experiment showed for the first time that the atom has**
(a) electrons (b) protons
(c) neutrons (d) nucleus

18. **Rutherford's alpha rays scattering experiment is related to the size of the**
(a) nucleus (b) electrons
(c) proton (d) neutron

19. **Sodium atom and sodium ion differ by _____**
(a) 1 electron (b) 1 proton
(c) 1 neutron (d) one electron and 1 proton

20. **Element with electronic configuration (2, 8, 2) is**
(a) metal (b) non-metal
(c) metalloid (d) inert gas

21. **The ratio of number of neutrons to protons in $_{82}Pb^{208}$ and $_{83}Bi^{209}$ called n / p ratio is**
(a) higher in Pb than Bi (b) lower in Pb than Bi
(c) equal in both
(d) Ratio is not dependent on given data

22. **The electronic configuration of elements A, B, C and D are (2, 8, 1), (2, 8, 2), (2, 8, 6) and (2, 8,7) respectively. Which forms a divalent anion?**
(a) A (b) B
(c) C (d) D

23. **A neutral atom has a nucleus with a nuclear charge 13 times and mass 27 times that of hydrogen nucleus. How many electrons would be present in the stable positively charged ion?**
(a) 13 (b) 27
(c) 14 (d) 10

24. **Assuming the nucleus and atom to be spheres such that the diameter of the nucleus is of the order of 10^{-11} cm and that of atom is of the order of 10^{-8} cm, the ratio of volumes of atom to nucleus is (Hint: volume of sphere = $\frac{4}{3}\pi r^3$, where r is radius of sphere)**
(a) $10^9 : 1$ (b) $1 : 10^9$
(c) $10^3 : 1$ (d) $1 : 10^3$

25. **An element X has the same mass number and atomic number. This element is**
(a) inert gas (b) hydrogen
(c) alpha particle (d) helium

26. **Energy is absorbed when an electron jumps from**
(a) K-shell to L-shell (b) L-shell to K-shell
(c) both a and b (d) cannot say

27. **The increasing order of e / m ratios of electron, proton, neutron and alpha particle is**
(a) neutron < alpha particle < proton < electron
(b) electron < proton < alpha particle < neutron
(c) neutron < alpha particle < proton = electron
(d) alpha particle > neutron > electron > proton

28. **The weight of 1 hydrogen atom is**
(a) 1.66×10^{-24} kg (b) 1.66×10^{-27}gm
(c) 1.66×10^{-24} gm (d) 1.66×10^{-24} kg

II. Give one word answers

1. The distance from the nucleus to the outermost shell of an atom.
2. The electrons in the last shell of an atom.
3. The term used when the last shell has 8 electrons.
4. Is energy absorbed or released to remove an electron from the valence shell of a neutral gaseous atom?
5. The discovery of what proved that atoms of the same element may be different from each other?

III. Match the following.

1. **Match the property in Column I with the elements Column II**

Set A	Set B
(i) Fractional atomic weight	(a) Hydrogen
(ii) A = Z for	(b) Chlorine
(iii) Variable valency	(c) A – Z
(iv) Number of neutrons =	(d) Fe^{2+}, Fe^{3+}

2. **Match the valence electrons in Column I with their nature in Column II**

Set A	Set B
(i) atoms with 1, 2 or 3 valence electrons	(a) non-metal
(ii) atoms with 8 valence electrons	(b) metal
(iii) atoms with 5, 6 or 7 valence electrons	(c) ions
(iv) atoms in which number of electrons is not equal to protons	(d) noble gas element

3. Match the e/m ratios in Column I with the particles in Column II

	Set A		Set B
(i)	0	(a)	electron
(ii)	1	(b)	proton
(iii)	½	(c)	neutron
(iv)	1836	(d)	Alpha particle

IV. State whether the following statements are True or False

1. The cathode rays move from anode to cathode.
2. The canal rays are deflected towards the negative plate when an electric field is applied perpendicular to the path of the rays.
3. The nucleus is a very small part of the atom.
4. The mass of the atom is due to the electrons and protons.
5. The electrons are outside the nucleus of the atom.

V. Read the paragraphs below and answer the questions that immediately follow the paragraph. Each question has four options out of which only one option is correct. Choose the correct one.

Energy required to pull out the first electron from the valence shell (first Ionisation Energy value) will be quite less when compared to energy required to pull out the second electron from the valence shell (second Ionisation Energy value). After the first electron has been pulled out, the atom acquires a positive charge. The same number of protons attracts a lesser number of electrons. So the pull per electron is increased in the ion. Hence to pull out the second electron is more difficult. Atoms that get eight electrons in the valence shell on losing electrons will lose electrons easily and hence will have low Ionisation Energy values.

1. The ionisation energy value of nitrogen is _____ oxygen.
   (a) more than (b) smaller than
   (c) same as (d) can be more or less
2. Ionisation energy values of inert gas elements are quite
   (a) low (b) high
   (c) equal to other elements (d) cannot say
3. Ionisation energy values of a halogen will be _____that of the next succeeding inert gas element.
   (a) more than (b) smaller than
   (c) same as (d) can be more or less

VI. Fill in the blanks

1. Experiments with the cathode rays showed for the first time that atoms are _____.
2. The alpha scattering experiments showed for the first time the existence of a _____ in the atom.
3. Atoms with same mass number but different atomic number are called _____.
4. Atoms with different mass number but same atomic number are called _____.
5. Cl^{37} and Cl^{35} are _____.
6. Ar^{40} and Ca^{40} are _____.
7. Electrons in the orbit closest to the nucleus have the _____ energy.
8. The mass of the canal ray particles when different gases are used is _____.
9. The ratio of charge to mass of neutrons is _____.
10. The configuration of helium is called _____.

VII. Each question contains two statements. Statement A is an Assertion Statement and Statement R is the Reason Statement. Each question has four choices out of which only one is correct. Choose the correct one.

Choose A if Statement A is true, Statement R is true; Statement R is correct explanation for statement A

Choose B if Statement A is true, statement R is true; Statement R is not correct explanation for statement A

Choose C if Statement A is true, Statement R is false

Choose D if Statement A is false, statement R is true

1. **Statement –A:** The cathode ray particles in the Discharge Tube experiments are fundamental particles present in all atoms

   Statement –R: The e/m ratio of the cathode ray particles is a constant even if the electrode or gas in the tube is changed.
2. **Statement –A:** The e/m ratio of canal ray particles is not a constant.

   Statement –R: The canal ray particles are not fundamental particles.
3. **Statement –A:** The mass of Helium nucleus is more than the mass of two protons present in it.

   Statement –R: Helium nucleus contains protons that are heavier than protons present in other atoms.
4. **Statement –A:** Some elements have fractional atomic weight.

   Statement –R: Isotopes of an element are responsible for fractional atomic weight.

VIII. Answer the following questions.

1. How did Dalton say that an atom is indivisible?
2. What was Dalton's Atomic Model famous for?
3. What were the important postulates of Dalton apart from the indivisible atom?
4. Why was Dalton wrong about atoms of same element being similar and vice-versa?
5. 'Atoms combine in small whole numbers.' Why is this postulate of Dalton wrong?
6. How did Rutherford use a magnified atom to explain the following observations of the alpha scattering experiments?

   i. Only a very few alpha particles suffered large deflections from their original paths.

   ii. Most alpha particles pass through the gold foil in a straight line.
7. What did Rutherford postulate regarding the electrons in the atom?
8. According to Rutherford what are the forces experienced by an electron that prevents it from collapsing into the nucleus?
9. State the postulates of Bohr atomic model.
10. What are the defects of Thomson's atomic model?
11. Define ionization energy, electron affinity, atomic radius and electronegativity of an atom.
12. Compare the radius of an atom with a cation formed by it. Justify.

KEY TO EXERCISE

I. 1. c 2. b 3. a 4. a 5. b 6. c 7. b 8. d
9. d 10. b 11. d 12. a 13. c 14. b 15. c 16. b
17. d 18. a 19. a 20. a 21. a 22. c 23. a 24. a
25. b 26. a 27. a 28. c

II. 1. atomic radius 2. valence electrons 3. octet configuration 4. absorbed
5. isotopes

III. 1. i. b ii. a iii. d iv. c 2. i. b ii. d iii. a iv. c 3. i. c ii. b iii. d iv. a

IV. 1. false 2. true 3. true 4. false 5. true

V. 1. a 2. b 3. c

VI. 1. divisible 2. nucleus with positive charge 3. isobars 4. isotopes 5. isotopes
6. isobars 7. least 8. different 9. 0 10. duplet configuration

VII. 1. A 2. B 3. C 4. A

5 Periodic Classification


J W Dobereiner


Lothar Meyer


Dmitri Mendeleev


H G J Moseley

5.0 INTRODUCTION

Before the 19th century, the number of elements known was very small. As more elements were discovered their study became difficult. Today more than 110 elements are known. A scientific study of all these elements becomes easy if they are classified based on some common features. The Periodic Table is one such classification that is accepted widely. The term '*periodic*' implies that some property of the elements gets repeated after a certain interval or period. As the elements are arranged in the form of a table the word table is used. The Periodic Table is a table of elements in which the elements are arranged horizontally in increasing order of Atomic Numbers. Every successive element differs by one.

5.1 IMPORTANCE OF CLASSIFYING ELEMENTS

Classification of elements helps us to relate the physical and chemical properties and predict the existence of other elements. Thus,

(a) classification of elements helps us to study them better.

(b) classification of elements helps us to correlate their properties.

(c) classification of elements helps us to establish relationships between them.

5.2 DOBEREINER'S TRIADS, 1829

Dobereiner found that there were several elements which could be grouped into threes based on similarities of chemical properties. He was also successful in establishing that *the atomic weight of the middle element was the average of the atomic weight of the other two elements in each triad.*

Based on these two rules, he classified elements into ***groups of three (triad)***.

Examples: Some Dobereiner Triads with the atomic weights in brackets are given below.

Ca(40), Sr(88), Ba(137) is a triad;
Similarly, Li(7), Na(23), K(39) is another triad;
Cl(35.5), Br(80). I(127) is another triad.

It can be seen that the atomic weight of the middle element is approximately the average of the atomic weights of the other two elements. It was also seen that each element in a triad had chemical properties that were similar to the other elements in the triad.

Merits: The main achievement of this classification is that it recognised a relationship between properties of elements and their atomic weights.

Drawback of this classification

As more elements were discovered, it was found that there are many elements that cannot be grouped into triads as proposed by Dobereiner. Also, when elements were placed in a triad based on the average of atomic weight rule, some elements that were dissimilar were being placed in the triad. This and further discoveries led to the discard of this classification.

5.3 NEWLAND'S LAW OF OCTAVES, 1864

Newland arranged the elements *known till then in increasing order of their atomic weights.*

He found that *when the elements were arranged in increasing order of atomic weight, the eighth element starting from a given one anywhere in the list, had some properties that were similar to the first.*

Newland compared this to the eight notes of an octave of a music, and hence he called this classification as *Law of Octaves.*

Example: H, Li, Be, B, C, N, O
followed by F, Na, Mg, Al, Si, P, S

Here, H, F are similar in some properties. Similarly, Li and Na are similar in some properties.

Merits: The main achievement of this classification was that it improved upon the earlier attempts by Dobereiner and used atomic weight as a basis of classification. Further the classification recognised *the repetition of certain properties periodically*.

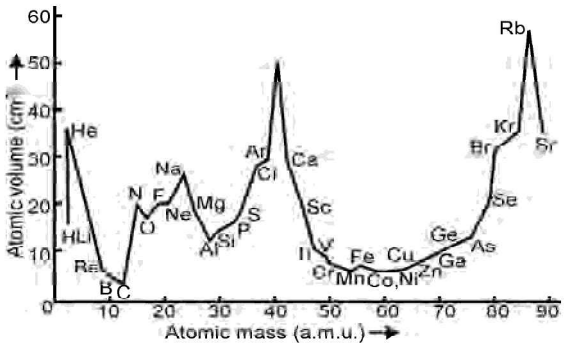
Drawback of this classification

As more elements continued to be discovered, it was found that this classification could not be used for elements beyond atomic weight 40 (Calcium). Besides this, the classification could not place the transition elements in the list.

5.4 MEYER'S CLASSIFICATION (1869, GERMAN)

A German chemist Julius Lothar Meyer in 1869 made a table of 28 elements and determined the atomic volumes of these elements. When the atomic volumes of the elements were plotted against the atomic weight, a series of peaks were produced.

Observations: The peaks had alkali metals sodium, potassium, rubidium, and cesium. Each fall and rise to a peak, corresponded to a ***period*** like in a wave. In each period a number of physical properties other than atomic volume also fell and rose, such as valence electrons and melting point.



Hydrogen, the first in the list of elements is a special case and can be considered as making up the first period all by itself. The second and third period in Meyer's table included seven elements each, and duplicated Newland's law of octaves.

Li	Be	B	C	N	O	F
Na	Mg	Al	Si	P	S	Cl

However, the next wave had more than seven elements. The third wave had about 17 to 18 elements. *This clearly showed where Newland's Law had failed. One could not enforce the law of octaves throughout the table of elements, with seven elements in each row. After the first two periods, the length of the period had to be longer.*

If the curve is studied more closely it will be seen that metals such as Na, K and Rb that have similar properties occur as peaks of the curve. Non-metals like F, Cl and Br that have similar properties occur at the rising part of the curve. Noble gases such as Ne, Ar and Kr that have similar properties occur just before the metals mentioned above.

H and He seem to be the only exception to the rule as they do not have any similarity in chemical properties.

The credit for classifying elements on the basis of atomic weights in this way does not go to Meyer but to Mendeleev who published similar findings a few months earlier.

5.5 MENDELEEV'S CLASSIFICATION (1869, RUSSIAN)

In 1869 working on the research done by Newland, a 35 year old Russian chemist named Dmitri Mendeleev presented a more scientific table of elements. He managed to point out a clear relationship between chemical groups.

Mendeleev saw that the physical and chemical properties recur *periodically*, if the elements were arranged in order of their atomic weights. The elements were arranged in the form of a table based on the *periodic recurrence of the properties* and so the tabular arrangement was called *Periodic Table*. In Mendeleev's time there were about 60 to 70 known elements.

Mendeleev's periodic law states *that the properties of elements are a periodic function of their atomic masses*.

Figure below shows the chart of elements initially made by Mendeleev. The elements are arranged in such a manner that the vertical columns are called *groups* and horizontal columns are called *periods*.

1 H Hydrogen						
7 Li Lithium	9 Be Beryllium	11 B Boron	12 C Carbon	14 N Nitrogen	16 O Oxygen	19 F Fluorine
23 Na Sodium	24 Mg Magnesium	27 Al Aluminium	28 Si Silicon	31 P Phosphorus	32 S Sulphur	35.5 Cl Chlorine
39 K Potassium	40 Ca Calcium		48 Ti Titanium			

In this table, the elements in each group have similar physical and chemical properties (for example valence electrons and melting point). The *periods* or horizontal rows are made with elements arranged in increasing order of atomic masses. As one goes vertically down a *group*, the elements show increase in atomic volume. *The first two periods are similar to Newland's law of octaves.*

Next Mendeleev did something very clever. *He placed undiscovered elements that had similar properties under vertical columns.* For example he placed titanium (Ti) under silicon (Si) as he saw that Ti and Si had similar properties. Thus there was a gap below aluminum (Al) in the group and after calcium (Ca) in the horizontal period. *He also found it necessary to leave some gaps in his table as the elements were not known then.* Rather than considering these gaps as a failure of his classification, Mendeleev said that *they represented elements undiscovered.*

In 1871, he pointed to three gaps in particular, those falling next to the elements boron, aluminum, and silicon in the table. He named the unknown elements as *eka-aluminum, and eka-silicon* ("eka" is the Sanskrit word for "one"). He also predicted various properties of these missing elements, such as density, boiling point, making use of the properties of the elements above and below the gaps in his table. Much later on, Gallium (Ga) and Germanium (Ge) were found, which had same properties as *eka-aluminum, and eka-silicon* respectively. *This demonstrated the success of Mendeleev's periodic table of elements.*

A modified version of Mendeleev's Periodic Table based on atomic weights giving due importance to similarity of chemical properties was *in use for quite some time.*

Merits of Mendeleev's periodic table

1. When Mendeleev presented his periodic table, inert gas elements like He, Ne, Ar were not discovered. When they were discovered, they could be neatly put as the last group of elements, without disturbing the rest of the table.
2. Mendeleev could successfully predict many unknown elements. He left gaps for these elements predicting that these elements existed. As many as eight out of ten gaps left by Mendeleev were filled later showing the predictive nature of his classification. Although the periodic table is now complete with about 117 different elements (naturally occurring and artificially made), *scientists still use Mendeleev's method to predict properties of elements by looking vertically across groups and horizontally across periods of elements.*

Drawbacks of Mendeleev's classification

Although very successful, Mendeleev's periodic table had the following drawbacks.

1. The positions of isotopes could not be accommodated within the table. If one obeys Mendeleev's periodic law then the variation in chemical properties vertically in a group is not followed strictly. For example, consider an isotope of carbon, C^{14}. This would have to be accommodated along with nitrogen (see table above). But C^{14} shows properties same as normal carbon (C^{12}). Mendeleev was thus unable to place isotopes in his table.
2. In order to make the elements fit the requirements, Mendeleev was forced in one or two cases to put an element of slightly higher atomic weight ahead of one of slightly lower atomic weight. Thus, tellurium (Te) (atomic weight 127.6, valency 2) had to be put ahead of iodine (I) (atomic weight 126.9, valency 1) in order to keep tellurium in the valency-2 column and iodine in the valency-1 column.
3. Hydrogen was placed both in IA and VIIA.
4. Position of rare earths was not clear. They were placed in Group IIIA.
5. Transition elements were placed without reasonable justification into Group VIII.
6. Some elements with hardly any similarities were placed together. For example alkali metals, Cu, Ag, Au are kept in Group I.
7. Mendeleev's classification does not give importance to the electronic configuration of elements which is a very important factor in deciding chemical property.

5.6 MODERN PERIODIC TABLE (LONG FORM OF PERIODIC TABLE)

To overcome the drawbacks of Mendeleev's classification, several scientists started working on making the classification error free. In 1913, a scientist named Mosley found that the properties of an element depend on the atomic number.

Based on this, Mendeleev's Periodic Table was modified to arrange the elements in increasing order of atomic number. The Modern Periodic Table is based on the Modern Periodic law which states that

Today there are no gaps in the Periodic Table as elements with atomic number 1 to about 117 have been discovered or artificially created in the laboratory

'*The physical and chemical properties of the elements are a periodic function of their atomic numbers*'. The elements down a group have similar valence configuration which is mainly responsible for similarity in chemical properties.

The main advantage of arranging elements in increasing order of atomic number is that the number of gaps can be predicted exactly. Since atomic number is a whole number we can arrange the elements in successive serial order. If elements are arranged in increasing order of atomic weight new elements as and when discovered will necessitate a rearrangement. In the Modern Periodic Table for new elements not yet discovered a place is kept blank for that atomic number.

Building the Modern Periodic Table

In the previous chapter we had seen how the electronic configuration of the first twenty elements is written. The configuration is now reproduced below.

Element symbol	Atomic number	Electronic configuration			
		K-shell	L-shell	M-shell	N-shell
Hydrogen – H	1	1			
Helium – He	2	2			
Lithium – Li	3	2	1		
Beryllium – Be	4	2	2		
Boron – B	5	2	3		
Carbon – C	6	2	4		
Nitrogen – N	7	2	5		
Oxygen – O	8	2	6		
Fluorine – F	9	2	7		
Neon – Ne	10	2	8		
Sodium – Na	11	2	8	1	
Magnesium – Mg	12	2	8	2	
Aluminium – Al	13	2	8	3	
Silicon – Si	14	2	8	4	
Phosphorous – P	15	2	8	5	
Sulphur – S	16	2	8	6	
Chlorine – Cl	17	2	8	7	
Argon – Ar	18	2	8	8	
Potassium – K	19	2	8	8	1
Calcium – Ca	20	2	8	8	2

Now we will construct the Periodic Table starting from atomic number 1. These elements are arranged and their electronic configurations are also shown. Try to follow the comments given in the left and right side.

Before starting let us remember that :

(i) the maximum number of electrons in *the last shell* cannot be more than 8 electrons *even if it can hold more electrons.*

(ii) the maximum number of electrons in *each shell* is given by $2n^2$ where n is the shell number.

K-shell can hold of a maximum 2 electrons
L-shell can hold of a maximum 8 electrons
M-shell can hold of a maximum 18 electrons and so on

The K-shell starts filling.	H_1 (1)							He_2 (2)	The K-shell is completely filled.
The L-shell starts filling.	Li_3 (2,1)	Be_4 (2,2)	B_5 (2,3)	C_6 (2,4)	N_7 (2,5)	O_8 (2,6)	F_9 (2,7)	Ne_{10} (2,8)	The L-shell is completely filled.
The M-shell starts filling.	Na_{11} (2,8,1)	Mg_{12} (2,8,2)	Al_{13} (2,8,3)	Si_{14} (2,8,4)	P_{15} (2,8,5)	S_{16} (2,8,6)	Cl_{17} (2,8,7)	Ar_{18} (2,8,8)	The M-shell is completely filled.

The other atomic numbers are also filled on the basis of their electronic configuration. The table is now reproduced below without the comments.

H_1 (1)							He_2 (2)
Li_3 (2,1)	Be_4 (2,2)	B_5 (2,3)	C_6 (2,4)	N_7 (2,5)	O_8 (2,6)	F_9 (2,7)	Ne_{10} (2,8)
Na_{11} (2,8,1)	Mg_{12} (2,8,2)	Al_{13} (2,8,3)	Si_{14} (2,8,4)	P_{15} (2,8,5)	S_{16} (2,8,6)	Cl_{17} (2,8,7)	Ar_{18} (2,8,8)

Main features of the Modern Periodic Table

1. In the Modern periodic table, the elements are arranged in increasing order of their atomic number.
2. The vertical columns are called *groups*.
3. The horizontal columns are called *periods*.
4. There are 18 groups and 7 periods.
5. From left to right the *atomic number increases by 1 for each successive element*.
6. The number of elements in each period is as follows.

Period	1	2	3	4	5	6	7
Number of elements	2	8	8	18	18	32	20 plus
Atomic Numbers	1-2	3-10	11-18	19-36	37-54	55-86	87-118

7. The first period contains only 2 elements.
8. The second and third period contains 8 elements.
9. The 4th, 5th and 6th periods contain the *Transition Elements* – each series containing 10 transition elements.
10. The 6th period is the one of the longest periods. This period includes the first set of *fourteen Inner Transition Elements called The Lanthanides.*
11. The 7th period is also one of the longest periods. This period includes the second set of *fourteen Inner Transition Elements called The Actinides*.
12. The two sets of inner transition elements are placed separately at the bottom of the table.

5.7 SPECIAL NAMES OF SOME GROUPS IN THE PERIODIC TABLE

Some of the groups in the Periodic table are known by special names as follows.

Group IA elements are called *alkali metals* as they form oxides / hydroxides with a strong alkaline nature.

Group IIA elements are called *alkaline earth metals* as their oxides were known earlier than the metals, their oxides are alkaline and exist in the earth's crust.

Groups VIII (Fe, Co, Ni / Ru, Rh, Pd / Os, Ir, Pt). This group consists of three triads. The first group is called *ferrous metal group*. The second and third metals are called *platinum metal group*. The group having Cu, Ag and Au are called *coinage metals* as they are used in the manufacture of currency coins.

Group VA elements (N, P, As, Sb and Bi) are called *picogens* (Greek for 'suffocation') as the hydrides of this group (NH_3, PH_3, AsH_3) have pungent odour and cause suffocation.

Group VIA are called *chalcogens* (Greek for ore-forming). Oxygen and sulphur are two important elements of this group associated with ores of many metals.

Group VIIA elements are known as *halogens* (Greek for salt producer) as they form salts.

Group 0 elements are known as *inert gas or noble gas elements* as they occur in very small quantities and are non-reactive.

5.8 TYPES OF ELEMENTS

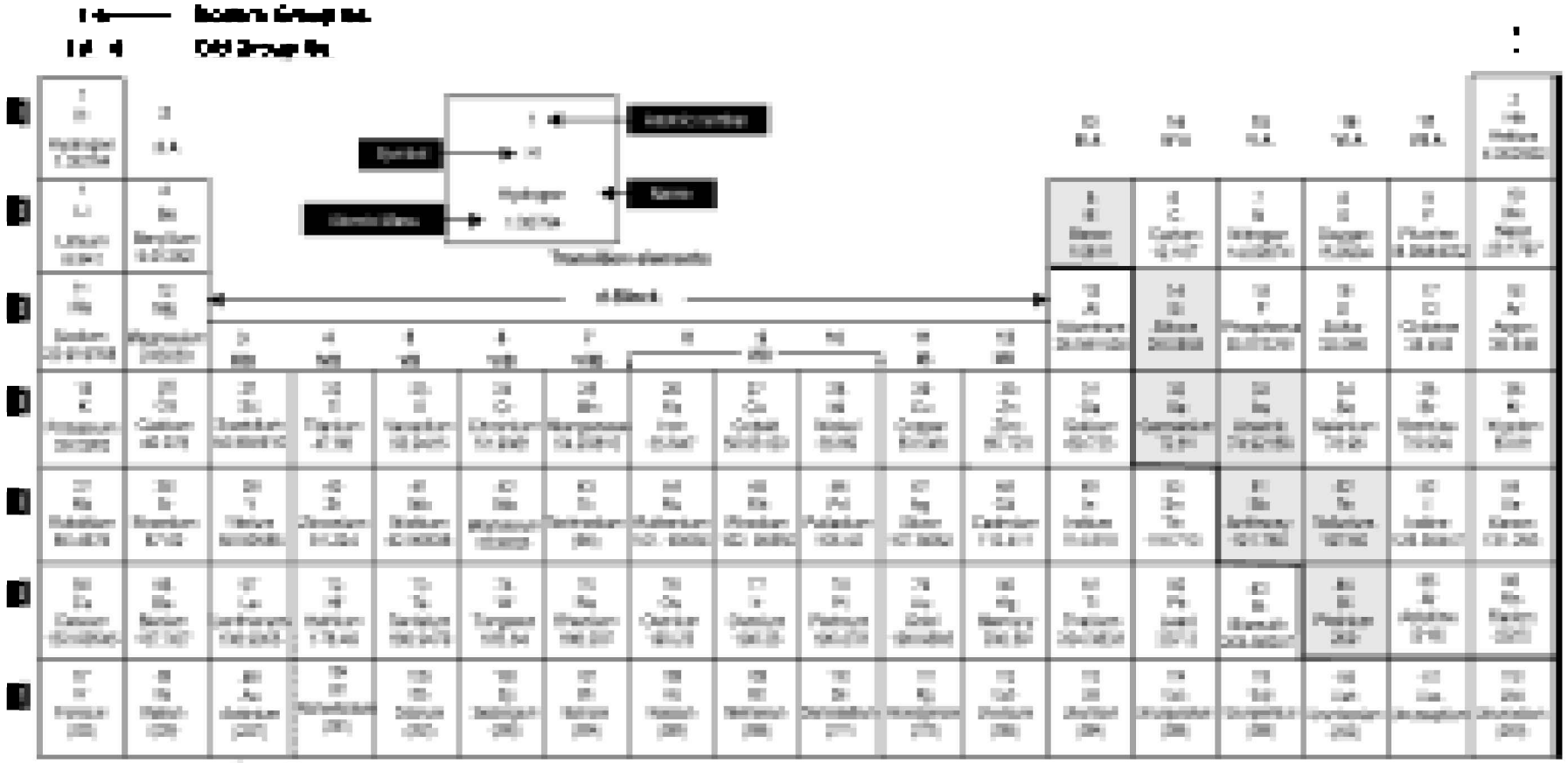
Elements can be divided into four types.

1. Representative elements
2. Transition elements
3. Inner transition elements
4. Inert Gas elements or Noble gases

5.9 REPRESENTATIVE ELEMENTS

The elements belonging to Groups IA, IIA, IIIA, IVA, VA, VIA, and VIIA (Group numbers 1,

The Modern Periodic Table



2, 13, 14, 15, 16, 17 respectively) are called *representative elements*. The representative elements are further sub-divided into *s-block* and *p-block*.

Representative elements

s-block elements	**p-block elements**
Elements of Groups IA and IIA	Elements of Groups IIIA, IVA, VA, VIA, and VIIA

Electronic configuration of representative elements

(i) Representative elements of the second period

The electronic configuration of the representative elements of the second period is as follows.

	At.No.	Electronic Configuration	
Li	3	2, 1	The last shell is filling with electrons
Be	4	2, 2	
B	5	2, 3	
C	6	2, 4	
N	7	2, 5	
O	8	2, 6	
F	9	2, 7	

(ii) Representative elements of the third period

The electronic configuration of the representative elements of the third period is as follows.

	At.No.	Electronic Configuration	
Na	11	2, 8, 1	The last shell is filling with electrons
Mg	12	2, 8, 2	
Al	13	2, 8, 3	
Si	14	2, 8, 4	
P	15	2, 8, 5	
S	16	2, 8, 6	
Cl	17	2, 8, 7	

Main characteristics of Representative elements

(a) In the representative elements the valence shell is incomplete.

(b) The representative elements include metals, non-metals and metalloid elements.

(c) All s-block elements excluding hydrogen are metals.

(d) In the p-block, all the metalloid elements, all the non-metal elements excluding hydrogen and some metal elements are present.

(e) Amongst the representative elements, the non-metals form covalent compounds with other non-metals and electrovalent compounds with metals.

(f) The representative elements that are metals are good conductors of heat and electricity.

5.10 TRANSITION ELEMENTS

The *transition elements* are elements belonging to Groups IB, IIB, IIIB, IVB, VB, VIB, VIIB and VIII. The transition elements are also known as *d-block* elements.

Electronic configuration of transition elements

(i) Transition elements of the fourth period

The electronic configuration of the transition elements of the fourth period, after calcium (2, 8, 8, 2) is as follows.

	At.No.	Electronic configuration	
Sc	21	2, 8, 9, 2	The second last shell is filling with electrons
Ti	22	2, 8, 10, 2	
V	23	2, 8, 11, 2	
Cr	24	2, 8, 13, 1	
Mn	25	2, 8, 13, 2	
Fe	26	2, 8, 14, 2	
Co	27	2, 8, 15, 2	
Ni	28	2, 8, 16, 2	
Cu	29	2, 8, 18, 1	
Zn	30	2, 8, 18, 2	

(ii) Transition elements of the fifth period

The electronic configuration of the transition elements of the fifth period after strontium (2, 8, 18, 8, 2) is as follows.

	At.No.	Electronic configuration	
Y	39	2, 8, 18, 9, 2	The second last shell is filling with electrons
Zr	40	2, 8, 18, 10, 2	
Nb	41	2, 8, 18, 11, 2	
Mo	42	2, 8, 18, 13, 1	
Tc	43	2, 8, 18, 13, 2	
Ru	44	2, 8, 18, 14, 2	
Rh	45	2, 8, 18, 15, 2	
Pd	46	2, 8, 18, 16, 2	
Ag	47	2, 8, 18, 18, 1	
Cd	48	2, 8, 18, 18, 2	

Main characteristics of Transition elements

(a) In the transition elements the second last shell is filling with electrons.

(b) All transition elements are metals with high melting and boiling points.

(c) All transition elements are good conductors of heat and electricity.

(d) Some of the transition elements are attracted towards a magnet.

(e) Many transition elements are good catalysts.

(f) Most transition elements exhibit variable valencies.

5.11 INNER TRANSITION ELEMENTS

The elements of the sixth and the seventh periods of the Group IIIB, i.e., the lanthanides and the actinides, are collectively known as the inner transition elements.

Electronic configuration of Inner transition elements

(i) Inner transition elements of the sixth period

The electronic configuration of the inner transition elements of sixth period, after Lanthanum (2, 8, 18, 18, 9, 2) is as follows.

	At.No.	Electronic configuration	
Ce	58	2, 8, 18, 20, 8, 2	The third last shell is filling with electrons
Pr	59	2, 8, 18, 21, 8, 2	
Nd	60	2, 8, 18, 22, 8, 2	
Pm	61	2, 8, 18, 23, 8, 2	
Sm	62	2, 8, 18, 24, 8, 2	
Eu	63	2, 8, 18, 25, 8, 2	
Gd	64	2, 8, 18, 25, 9, 2	
Tb	65	2, 8, 18, 27, 8, 2	
Dy	66	2, 8, 18, 28, 8, 2	
Ho	67	2, 8, 18, 29, 8, 2	
Er	68	2, 8, 18, 30, 8, 2	
Tm	69	2, 8, 18, 31, 8, 2	
Yb	70	2, 8, 18, 32, 8, 2	
Lu	71	2, 8, 18, 32, 9, 2	

(ii) Inner transition elements of the seventh period

The electronic configuration of the transition elements of seventh period after Actinium (2, 8, 18, 32, 18, 9, 2) is as follows.

	At.No.	Electronic configuration	
Th	90	2, 8, 18, 32, 19, 9, 2	The third last shell is filling with electrons
Pa	91	2, 8, 18, 32, 20, 9, 2	
Np	93	2, 8, 18, 32, 23, 8, 2	
Pu	94	2, 8, 18, 32, 24, 8, 2	
Am	95	2, 8, 18, 32, 25, 8, 2	
Cm	96	2, 8, 18, 32, 25, 9, 2	
Bk	97	2, 8, 18, 32, 27, 8, 2	
Cf	98	2, 8, 18, 32, 28, 8, 2	
Es	99	2, 8, 18, 32, 29, 8, 2	
Fm	100	2, 8, 18, 32, 30, 8, 2	
Md	101	2, 8, 18, 32, 31, 8, 2	
No	102	2, 8, 18, 32, 32, 8, 2	
Lr	103	2, 8, 18, 32, 32, 9, 2	

Main characteristics of Inner transition elements

(a) In the Inner transition elements the third last shell is filling with electrons.

(b) The Inner transition elements form coloured ions.

(c) The Inner transition elements show variable valencies.

(d) Actinides are radioactive by nature.

(e) The Inner transition elements beyond atomic number 92 up to 103 are synthetic as well as radioactive. They are not found in nature in the earth's crust.

5.12 INERT GASES (OR NOBLE GASES)

The elements of the Group 0, which is the 18th vertical column, are known as inert gases or noble gases.

Electronic configuration of inert gas elements

	At.No.	Electronic configuration
He	2	2
Ne	10	2, 8
Ar	18	2, 8, 8
Kr	36	2, 8, 18, 8
Xe	54	2, 8, 18, 18, 8
Rn	86	2, 8, 18, 32, 18, 8
Uno	118	2, 8, 18, 32, 32, 18, 8

The last inert gas element is a synthetic element and does not occur naturally on earth.

Main characteristics of Inert Gas elements

(a) All the inert gas elements have 8 electrons in their valence shell excluding helium, which has 2 electrons.

(b) The valency or combining capacity of inert gas elements is zero.

(c) The inert gas elements do not react with other elements.

(d) All the inert gas elements are gases at room temperature.

5.13 PERIODICITY

'Periodicity' is the repetition of some properties of the elements after specific intervals in the Modern Periodic table.

Reasons for periodicity (repetition of properties)

The main reason for periodicity in the elements or repetition of some properties after specific intervals is that the valence configuration repeats after these specified intervals of elements.

Illustration 1

Consider the electronic configuration and valence configuration of elements of Group IA.

Period		At.No.	Electronic configuration	Interval (difference in atomic numbers)
1	H	1	1	–
2	Li	3	2, 1	2
3	Na	11	2, 8, 1	8
4	K	19	2, 8, 8, 1	8
5	Rb	37	2, 8, 18, 8, 1	18
6	Cs	55	2, 8, 18, 18, 8, 1	18
7	Fr	87	2, 8, 18, 32, 18, 8, 1	32

The chemical properties of the elements are dependent on the valence configuration. The chemical properties of the elements with the same valence configuration will be similar. All the elements in Group IA have 1 electron in the valence shell. They show several similarities of chemical properties. The chemical properties repeat after specific intervals of 2, 8, 8, 18, 18, 32 elements. In other words the elements of Group IA exhibit *periodicity*.

Group IA elements show the following similar properties.

(i) Group IA elements are metals.

(ii) Group IA elements form positive ions.

(iii) Group IA elements are very reactive and are therefore found mostly in the combined state.

(iv) Group IA elements are colourless elements.

(v) Group IA elements combine with hydrogen forming metallic hydrides.

(vi) Group IA elements form basic oxides with oxygen.

(vii) Group IA elements are good reducing agents

(viii) Group IA elements react vigorously with water liberating hydrogen.

Illustration 2

Consider the electronic configuration and valence configuration of elements of Group VIIA.

		At.No.	Electronic configuration	Interval (difference in atomic numbers)
1	–	–	–	–
2	F	9	2, 7	–
3	Cl	17	2, 8, 7	8
4	Br	35	2, 8, 18, 7	18
5	I	53	2, 8, 18, 18, 7	18
6	As	85	2, 8, 18, 32, 18, 7	32

All the elements in Group VIIA have 7 electrons in the valence shell. They show several similarities of chemical properties. The chemical properties repeat after specific intervals of 8, 18, 18, 32 elements. In other words, the elements of Group VIIA exhibit *periodicity*.

Group VIIA elements show the following similar properties.

(i) Group VIIA elements are non–metals.

(ii) Group VIIA elements form negative ions.

(iii) Group VIIA elements are very reactive and are therefore found mostly in the combined state.

(iv) Group VIIA elements are coloured elements.

(v) Group VIIA elements form hydrides with hydrogen.

(vi) Group VIIA elements form acidic oxides with oxygen.

(vii) Group VIIA elements are good oxidising agents.

(viii) Group VIIA elements are only slightly soluble in water, but they dissolve much better in organic solvents like carbon disulphide, chloroform, alcohol, etc.

These illustrations show that elements having same valence configuration have the same properties.

The cause of periodicity is the recurrence of similar electronic configuration.

5.14 PERIODIC PROPERTIES

'Periodic properties' are the properties that show gradual variation (i.e. increase or decrease) in a period or a group in the Modern Periodic Table.

Reasons for periodic properties (graded variation of properties)

(i) Periodic properties in a group

In a group as we go down, a shell is added with every consecutive element. Thus the nuclear charge and atomic size increases in a graded manner. This is mainly responsible for the graded variation in properties like atomic radius, density, melting point, boiling point, metallic character, basic character of oxides and reactivity with other elements – all of these increase as we move down a group.

(ii) Periodic properties in a period

In a period as we go from left to right, one electron is added to the valence shell with every consecutive element. Thus in a period the nuclear charge increases by one and an electron is added to the same valence shell. Thus as we move from left to right in a period, the valence electrons increases by one. The valency also increases by one from 1 to 4 and then decreases by one from 4 to 1. This is responsible for the graded variation of properties.

5.15 ELECTRONIC CONFIGURATION OF ELEMENTS

Electronic configuration of elements in a Group

In a group from top to bottom the following features are significant.

1. **Number of shells**: The number of shells increases by one with every consecutive member of the group. As a rule the number of shells an element has is *equal to the number of the period* to which that element belongs. For example, in halogens (Group VIIA) :

Elements	Electronic	Number of shells configuration	Period to which the element belongs
F(9)	2, 7	2	Second
Cl(17)	2,8,7	3	Third
Br(35)	2,8,18,7	4	Fourth
I(53)	2,8,18,18,7	5	Fifth
At(85)	2,8,18,32,18,7	6	Sixth

2. **Electrons in the outermost shell**: The valence configuration or the number of electrons in the valence shell is the same. Therefore, in a group **valency** is the same.

(i) Group IA elements have one electron each in their valence shells and a valency of 1.

(ii) Group IIA elements have two electrons in their valence shells and a valency of 2.

(iii) Group VIIA elements have seven electrons in their valence shells and a valency of 1.

Valence electrons and Valency

Valency depends on the number of valence electrons. If 1, 2, 3 or 4 electrons are present in the outermost shell, valency will be 1, 2, 3 or 4 respectively.

But if 5, 6 or 7 electrons are present in the valence shell, then valency is given by subtracting the number of valence electrons from 8.

Thus, if 5, 6 or 7 electrons are present in the valence shell, then valency is given by 8 – 5 = 3, 8 – 6 = 2, or 8 – 7 = 1 respectively.

Electronic configuration of elements in a Period

In a period from left to right the following features are significant.

1. **Number of shells**: On moving from left to right in a given period, the electron gets added to the same shell. The number of shells in the atom remains constant. The number of shells in the atom is equal to the number of the period of that element. In the 3rd period, the number of shells is three, i.e., equal to the number of the period.

Elements of the 3rd period	Na	Mg	Al	Si	P	S	Cl	Ar
Atomic number	11	12	13	14	15	16	17	18
Electronic configuration	2,8,1	2,8,2	2,8,3	2,8,4	2,8,5	2,8,6	2,8,7	2,8,8
No. of shells	3	3	3	3	3	3	3	3

2. **Electrons in the outermost shell**: In a given period, the number of electrons in the valence shell increases from left to right. But the valency increases from 1 to 4 up to Group IV, and then it decreases from 4 to 1 in Group VIIA.

5.16 ATOMIC PROPERTIES

The atoms have specific properties. The student should frequently refer to the Periodic Table given above while studying this section for better understanding.

(i) **Atomic radius**
(ii) **Ionisation energy (ionisation potential)**
(iii) **Electron affinity**
(iv) **Electronegativity**
(v) **Metallic character**
(vi) **Non-metallic character**

5.17 ATOMIC RADIUS

Atomic radius is the distance between the centre of the nucleus of an atom and its outermost shell.

Atomic radius is measured in Angstrom units $\left(1\ \overset{0}{A} = 10^{-8}\text{cm}\right)$

Factors affecting Atomic Radius

Atomic size depends upon (i) *number of shells*
(ii) *nuclear charge*

(i) Number of shells

As the number of shells increases the size of the atom increases.

(ii) Nuclear charge

Nuclear charge is the positive charge present in the nucleus of an atom. It is equal to the number of protons in the nucleus, i.e., the atomic number. An increase in nuclear charge decreases the size of the atom, because the electrons are attracted towards the nucleus with a greater force, thereby bringing the outermost shell closer to the nucleus.

Variation of atomic radius

(a) In a group

In a group, the size of an atom increases as one proceeds from top to bottom. This is due to the successive addition of shells with every consecutive element in a Group.

Example: The atomic radius of Group IA elements is tabulated below.

Elements Group IA	Atomic Number	Atomic Radius $\left(\overset{0}{A}\right)$	Electronic configuration	Shells
Hydrogen	1	0.59	1	1
Lithium	3	1.52	2,1	2
Sodium	11	1.86	2,8,1	3
Potassium	19	2.31	2,8.8.1	4
Rubidium	37	2.44	2,8,18,8,1	5
Cesium	55	2.62	2,8,18,18,8,1	6

(b) In a period

As we move from left to right in a period, the atomic number increases by one with every successive element. The number of protons increases by one and *an electron is added to the same shell.* As the electrons are added to the same shell, the pull of the increased nuclear charge is felt more by the valence electrons. The valence shell size therefore shrinks.

The atomic radius decreases with every successive element in a period from Group IA to Group VIIA and then increases slightly with the inert gas element (inert gas). Thus, in a period, the halogen (group VIIA) has the smallest size and the alkali metal (Group IA) has the largest size.

Example: The atomic radius of 2nd Period elements is tabulated below.

Elements of 2nd period	Li	Be	B	C	N	O	F	Ne
Atomic Number	3	4	5	6	7	8	9	10
Atomic radius (A^0)	1.55 (Alkali metal)	1.12	0.88	0.77	0.70	0.66	0.64 (Halogen)	1.12

5.18 IONIC RADIUS

The cation is smaller than the parent atom as *the same number of protons attracts a fewer number of electrons*. The *'pull'* per electron in the cation increases in comparison to the parent atom. Hence there is a contraction in size and the *cation is smaller than the parent atom.*

Example: The atomic size of sodium atom (Na) is 1.86 A^0, but sodium ion (Na^+) is 1.02 A^0.

Similarly, atomic size of Fe is 1.17 A^0, but, Fe^{2+} is 0.76 A^0 and Fe^{3+} is 0.64 A^0.

The anion is larger than the parent atom as *the same number of protons attracts a greater number of electrons*. The *'pull'* per electron in the anion decreases in comparison to the parent atom. Hence there is an increase in size and the *anion is larger than the parent atom.*

Example: The size of chlorine atom (Cl) is 0.99 A^0, but chlorine ion (Cl^-) is 1.81 A^0.

5.19 IONISATION ENERGY

Ionisation Energy is the *energy required* for removing an electron from the valence shell of a gaseous atom or ion. (Ionisation energy is sometimes referred to as Ionisation Potential).

Ionisation Energy is measured in eV or kJ/mole. (1eV = 96.45kJ/mole).

Reason for absorption of energy

Ionisation energy is the energy absorbed by the valence electron due to which it escapes from the pull of the nucleus. *This is an endothermic process.*

Factors affecting Ionisation Energy

Ionisation energy is dependent on

(i) *Atomic radius* (ii) *Nuclear charge.*

(i) **Atomic radius** – Larger the atom, lesser the attractive force felt by the valence electron. Hence, lesser energy is required to pull out an electron. Thus ionisation energy is small. For smaller atoms the reverse is true. Smaller atoms have higher ionisation energy values.

(ii) **Nuclear charge** – If the nuclear charge is more, the attractive force on the valence electrons is more. The energy required to pull out the valence electron will be high – Ionisation Energy will be high.

Thus, actual ionisation energy is a net sum of both these two factors.

Variation in Ionisation Energy

a. In a group

As we go down a group the number of shells and the nuclear charge increase. As the shells increase, the valence electron is farther away from the nucleus thereby reducing the *'pull'* of the nucleus. At the same time the increase in nuclear charge increases the *'pull'* of the nucleus on the valence electron. Thus, these two *'pulls'* have opposite effects.

The decreased pull due to increasing distance from the nucleus *has a larger effect. Thus, as we go down a given group Ionisation Energy decreases.*

Example: The ionisation energy of group IA elements is tabulated below.

Group IA	H	Li	Na	K	Rb	Cs
Ionisation Energy(kJ/mole)	1312	520	498	419	403	375

b. In a Period

As we move across a period, the atomic size becomes smaller. The valence electrons are thus closer to the nucleus and are held more strongly. Hence *the ionisation energy in general increases across a period though there is no definite and uniform trend.*

Example: The ionisation energy of 2nd period elements is tabulated below.

Elements of 2nd period	Li	Be	B	C	N	O	F	Ne
Ionisation Energy (kJ/mole)	520	899*	801	1088	1402	1314	1681	2080**

The increase is not uniform across the period. Two anomalies can be noticed.

1. There is a sudden jump from 1681 to 2080 between F and Ne. (Neon is an inert gas element and hence very stable making it the most difficult to remove electrons in the period).
2. There is a sudden jump from 520 to 899 kJ/mole between Li and Be.
3. There is a *decrease* from 1402 to 1314 kJ/mole between N and O.

The last two anomalies will be discussed in the next class.

Some important points to note :

(i) Small atoms have more ionisation energy.

(ii) In general, ionisation energy increases across a period and decreases down a group.

(iii) Ionisation potential is inversely proportional to the atomic radius.

(iv) In a period ionisation energy is the highest for inert gas elements.

(v) In a group ionisation energy is the highest for the first element in the group.

(vi) On losing the first electron, the atom becomes an ion with one positive charge. To remove another electron from the ion requires more energy than removing the second electron because the same number of protons is attracting a lesser number of electrons. If IE_1 is the ionisation energy required to pull out the first electron, and IE_2 is the ionisation energy required to pull out the second electron, then $IE_2 > IE_1$.

$Mg - 1e^- \rightarrow Mg^+$, IE_1

(12 protons attract 12 electrons)

$Mg^+ - 1e^- \rightarrow Mg^{2+}$, IE_2

(12 protons attract 11 electrons)

$IE_2 > IE_1$

5.20 ELECTRON AFFINITY (AFFINITY ENERGY)

Electron Affinity is the ***energy released*** when an electron is added to the valence shell of the neutral gaseous atom of an element.

Electron Affinity is measured in eV or kJ/mole.

Reason for release of energy

Energy is released due to extra stability (in terms of valence configuration), of the atom when an electron is added particularly to non-metals. *This is an exothermic process when the first electron is added and an endothermic process when the second electron is added to the anion thus formed (due to repulsion).*

Factors affecting Electron Affinity

Electron affinity is dependent on

(i) *Atomic radius* (ii) *Nuclear charge.*

(1) **Atomic radius** – Electron affinity measures energy released when an electron is added to the valence shell. Larger the atom, farther the valence shell, lesser the tendency to attract the incoming electron. Hence for large atoms the electron affinity value will be small.

(2) **Nuclear charge** – Larger the atom more the nuclear charge, stronger the attraction for the incoming electron.

Thus, actual electron affinity is a net sum of both these two factors.

Variation in Electron Affinity

(a) In a group

As we go down a group the number of shells increases. As the shells increase, the inner shell elec-

trons repel the incoming electron making it more difficult to add an electron. Hence, *as we go down a group electron affinity decreases.*

Example: The electron affinity of group IA elements is tabulated below.

Group IA	H	Li	Na	K	Rb	Cs
Electron Affinity (kJ/mole)	72.3	58.9	79.1	66.5	–	–

(b) In a Period

As we move across a period, the atomic size becomes smaller. The incoming electron is pulled more strongly into the atom. Thus *electron affinity increases across a period.*

Example: The electron affinity of the 2nd period elements is tabulated below.

Elements of 2nd period	Li	Be	B	C	N	O	F	Ne
Electron Affinity (kJ/mole)	58.9	0	83.0	127.5	0	140.9	327.9	0

Some important points to note

(i) Electron affinity is measured in eV or kJ/mole.

(ii) Non-metals have higher electron affinity than metals as non-metals have a greater tendency to gain electrons.

(iii) Electron affinity values of very few elements are known as most elements are metals. Metals have a greater tendency to lose electrons not gain them.

(iv) Electron affinity is zero for inert gas elements as they show no tendency to gain electrons.

(v) In general, the second element *in a group* has a higher electron affinity value than the first after which it decreases. Example: Electron Affinity of Cl > F > Br > I.

(vi) *Energy is released (exothermic process) only when the first electron is added.*

(vii) *Energy is absorbed (endothermic process) when the second electron is added since it overcomes the repulsion of the negative ion.*

(viii) In general, electron affinity increases across a period and decreases down a group.

(ix) In a given period, the halogens have the highest electron affinity values. Chlorine has the highest electron affinity value amongst all elements.

5.21 ELECTRONEGATIVITY

Electronegativity is the tendency of an atom to attract the shared pair of electrons *in a bond* towards itself.

Electronegativity has no units as it is indicates only a tendency to attract a shared pair of electrons in a bond. Electronegativity is measured using Pauling Scale.

Electronegativity is dependent on

(i) *Atomic radius* (ii) *Nuclear charge.*

1. **Atomic radius** – Larger the atom farther the valence shell which is involved in electron sharing. Thus the pull of the nucleus is less and the bond pair of electrons is pulled less. Hence for large atoms the electronegativity will be small.
2. **Nuclear charge** – Larger the atom more the nuclear charge, stronger the attraction for the bond pair in a bond.

Thus, actual electronegativity is a net sum of both these two factors.

Variation in Electron Affinity

(i) In a group

As we go down a group, the inner electrons shield the nucleus from the bond pair of electrons in a bond. Thus electronegativity is less. *Electronegativity decreases down a group as the atomic size increases.*

Example:

Group IA	H	Li	Na	K	Rb	Cs	Fr
Electro negativity(Pauling Scale)	2.1	1.0	0.9	0.8	0.8	0.7	0.7

(ii) In a period

As we go across a period the nuclear charge increases with a decrease in the size of the atom. The bond pair is thus closer to the nucleus which pulls it more strongly. Thus *as we move across a period electronegativity increases*.

Example:

Elements of 2nd period	Li	Be	B	C	N	O	F	Ne
Electro negativity(Pauling Scale)	1.0	1.5	2.0	2.5	3.0	3.5	4.0	0

Some important points to note

(i) On the Pauling scale fluorine has the highest electronegativity value of 4.0.

(ii) All other elements have electronegativity values that are lesser than 4.0.

(iii) The least electronegativity amongst all metals is that of cesium. In fact cesium is the most electropositive amongst all elements as it has the easiest tendency to lose an electron to become a cation.

(iv) In a given period, the alkali metals have the least electronegative values.

(v) Higher the ionisation energy and electron affinity of an element, more the electronegativity.

(vi) *More the electronegativity value of an element more the tendency of the element to form an ionic bond rather than a covalent bond with other elements*.

Predicting nature of bond using electronegativity

The difference in electronegativities of the combining atoms can be used to predict the nature of the bond between them.

Example:

(i) **Bond in NaCl**: The electronegativity of sodium is 0.9 whereas that of chlorine is 3.0. The difference is 2.1. Consider a pair of electrons one belonging to sodium and another belonging to chlorine atoms. This pair of electrons between Na and Cl atoms will be pulled strongly by chlorine as it is more electronegative than sodium. *The pull is so strong that the electron belonging to sodium is transferred to the valence orbit of chlorine. This results in an ionic bond forming between Na and Cl atoms.*

(ii) **Bond in HCl**: The electronegativity of hydrogen is 2.1 whereas that of chlorine is 3.0. The difference is 0.9. Consider a pair of electrons one belonging to hydrogen and another belonging to chlorine atom. This pair of electrons between the H and Cl atoms will be pulled by both hydrogen and chlorine atom as the electronegativity difference is not large. However, *the pull of the chlorine atom is slightly more than that of hydrogen atom*. This results in the electron belonging to hydrogen to be closer to chlorine atom. This results in a small negative charge developing on chlorine due to the extra electron. *This results in a polar covalent bond between H and Cl.*

(iii) **Bond in Cl-Cl**: As both atoms in the molecule of chlorine belong to the same element the shared pair is pulled equally by both atoms and the pair is in the middle. Thus, *the sharing is equal and the bond between Cl atoms in chlorine molecule is pure covalent*.

Nature of bond and electronegativity differences

If the electronegativity difference between combining atoms is

(i) more than 1.7, the bond is ionic.

(ii) equal to 1.7, the bond is polar covalent (partially ionic, partially covalent).

(iii) less than 1.7, the bond is covalent.

5.22 METALLIC CHARACTER

Metals are the elements that have a tendency to lose their valence electrons to form a positive ion.

$$Na - e^- \rightarrow Na^+$$
$$(2, 8, 1) \quad (2, 8)$$
$$Mg - 2e^- \rightarrow Mg^{2+}$$
$$(2, 8, 2) \quad (2, 8)$$

Exception: Hydrogen is the only element that has a tendency to lose an electron to form a proton and yet is not a metal.

$$H - e^- \rightarrow H^+$$

Factors affecting metallic character

The metallic character of elements depends on the ease with which the atoms of the element lose electrons. The more the ease with which electrons

are removed from the atom, more the metallic character. Atoms of metal have 1, 2 or 3 electrons in the valence shell. They have a tendency to lose these valence electrons to get octet configuration.

The ease which electrons can be removed from the atom depends on

(i) *atomic size* (ii) *nuclear charge.*

(i) **Atomic size**: Larger the size of the atom, the farther the outermost orbit, and thus, lesser the nuclear pull exerted on the valence electrons. Due to this, the valence electron(s) can be removed more easily. Thus such elements are more metallic by nature.

(ii) **Nuclear charge:** If the nuclear charge is greater, the force exerted by the nucleus on the electron(s) is also greater. This makes it difficult to remove the electron(s) from the outermost orbit. Thus such elements are less metallic by nature.

Variation in metallic character down a group

On moving down a group, atomic size increases, and though nuclear charge also increases, *yet the effect of increased atomic size is greater compared to the increased nuclear charge*. Hence it is easier to remove electrons as we go down a group. Therefore, *metallic nature increases down a group.*

Example: In group IA, the increasing order of metallic character is

Li < Na < K < Rb < Cs < Fr

Variation in metallic character across a period

On moving across a period, as atomic size decreases and nuclear charge increases. Thus it becomes more difficult to remove electrons. Therefore, *the metallic nature decreases across a period, moving from left to right.*

Example: In the 3rd period, sodium is the most metallic element and chlorine the least.

Group IA	Group IA	Group IIA	Group IIIA	Group IVA	Group VA	Group VIA	Group VIIA	Group O
Element	Na	Mg	Al	Si	P	S	Cl	Ar
Atomic number	11	12	13	14	15	16	17	18
Electronic Configuration	2, 8, 1	2, 8, 2	2, 8, 3	2, 8, 4	2, 8, 5	2, 8, 6	2, 8, 7	2, 8, 8
Metallic property	Metal	Metal	Metal	Metalloid	Non-metal	Non-metal	Non-metal	Noble gas
Valency	1	2	3	4	3	2	1	0

5.23 NON-METALLIC CHARACTER

Non-metals are the elements that have a tendency to gain electrons to form a negative ion.

$$Cl + e^- \rightarrow Cl^-$$
$$(2, 8, 7) \qquad (2, 8, 8)$$
$$O + 2e^- \rightarrow O^{2-}$$
$$(2, 6) \qquad (2, 8)$$

Factors affecting non-metallic character

The non-metallic character of elements depends on the ease with which the atoms of the element gain electrons. If it gains electrons easily, the non-metallic character is more. Atoms of non-metal have 5, 6, or 7 electrons in the valence shell. They tend to gain 3, 2 or 1 electrons respectively to get octet configuration.

The ease which electrons can be gained by the atom depends on

(i) *atomic size* (ii) *nuclear charge*

(i) **Atomic size:** The smaller the atomic size, the greater the nuclear pull, since the outermost shell is nearer. It is easier to gain electrons. Therefore, the element is more non-metallic by nature.

(ii) **Nuclear charge:** The greater the nuclear charge, the greater the tendency to gain electron (s), hence the more non-metallic the element.

Variation in non-metallic character down a group

On moving down a group the size of the atom increases as shells are added. The influence of the nucleus on the outer electrons is less as distance increases. Besides this, the inner electrons also shield the nucleus from the outer electrons. Due to this the elements tend to lose electrons easily as we go down a group. Thus, metallic character increases

and non-metallic decreases. The last element in a group has the highest metallic character and least non-metallic character when compared to other elements in the group.

Example: Non-metallic character decreases down Group VIIA as given below:

F > Cl > Br > I

Variation in non-metallic character across a period

Across a period the atom size shrinks with increase in nuclear charge. Due to this an incoming electron is pulled more strongly into the atom. Thus it becomes easy to add electrons. As we move across a period the number of valence electrons increases progressively. Elements with 5, 6 or 7 electrons in a period have high electron affinity values. The non-metallic character increases as it becomes easier across a period to add electrons.

Example: Non-metallic character increases across a period as given below for elements of the 3rd period.

Na < Mg < Al < Si < P < S < Cl

5.24 SUMMARY OF GENERAL TRENDS IN ATOMIC PROPERTIES

Atomic Property	In a group	In a period
1. Atomic Radius	↑	↓
2. Ionisation Potential	↓	↑
3. Electron Affinity	↓	↑
4. Electronegativity	↓	↑
5. Electropositivity	↑	↓
6. Metallic Character	↑	↓
7. Non-metallic Character	↓	↑

EXERCISE

I. Choose the correct answer

1. Metals belonging to the same group in the periodic table are
(a) Mg, Na (b) Mg, Cu
(c) Mg, Ba (d) Mg, K

2. In the periodic table, the elements with atomic number 38 belongs to
(a) period 4 and group IIA
(b) period 4 and group IVA
(c) period 3 and group IVA
(d) period 5 and group IIA

3. The atomic numbers of four elements are given below. Which one is an alkaline earth metal?
(a) 10 (b) 20
(c) 30 (d) 40

4. Which of the following is a defect in the Mendeleev's periodic table?
(a) The position of alkali metals
(b) The position of transition metals
(c) The position of isotopes
(d) All of these

5. The elements of a group of the periodic table have
(a) the same number of protons
(b) the same valence shell
(c) the same valence electrons
(d) the same electron affinity

6. The tenth element in the periodic table resembles the element with atomic number
(a) 2 as well as 30 (b) 2 as well as 54
(c) 8 as well as 18 (d) 8 only

7. Without looking at the periodic table, select from each of the following sets, the elements belonging to the same group by writing their configuration.
(a) Z = 12, 38, 4, 88 (b) Z = 9, 16, 3, 35
(c) Z = 5, 11, 27, 19 (d) Z = 24, 47, 42, 55

8. Third period of periodic table contains
(a) 8 elements (b) 32 elements
(c) 3 elements (d) 18 elements

9. Group number of elements in the periodic table indicates:
(a) valency with respect to hydrogen
(b) the atomicity
(c) the number of electrons in the outermost shell
(d) none of these

10. The atomic numbers of some elements are given. Without looking at the periodic table, select the elements of group III A of the periodic table.
(a) 3, 11, 19, 37 (b) 5, 13, 21, 39
(c) 7, 15, 31, 49 (d) 5, 13, 31, 49

11. The elements which occupy positions in the periodic table just after the zero group of elements are
(a) non–metals (b) gases
(c) radioactive elements (d) alkali metals

12. The statement that is not true for long form of periodic table is
(a) It reflects the sequence of filling the electrons in the order of the K, L, M, N shell....etc

(b) It helps to predict the stable valency states of the elements
(c) It reflects tends in physical and chemical properties of the elements
(d) It helps to predict the relative ionic nature of the bond between any two elements

13. Elements belonging to the same group of the periodic table generally have the same:
(a) electronic configuration
(b) number of electrons in the outermost shell
(c) chemical properties
(d) physical properties

14. The basis of periodic law presented by Mendeleev was
(a) valency (b) atomic mass
(c) atomic number (d) atomic volume

15. The halogens are
(a) normal elements
(b) noble elements
(c) transition elements
(d) inner transition elements

16. In the modern periodic table one of the following does not have appropriate position:
(a) inert gases
(b) inner-transition elements
(c) transition elements (d) fluorine

17. Which of the following is not a representative element?
(a) Fe (b) K
(c) Ba (d) N

18. Elements of 2nd and 3rd periods of periodic table are called
(a) transition elements (b) typical elements
(c) metalloid (d) bridge elements

19. Which of the following statements is correct?
(a) X^- ion is larger in size than X atom
(b) X^+ ions is larger in size than X atom
(c) X^+ ion is larger in size than X^- ion
(d) X^+ and X^- ions are equal in size

20. The electro negativity of F, Cl, Br, and I are 4.0, 3.0, 2.8, and 2.5 respectively. The hydrogen halide with the highest degree of ionic character is
(a) HF (b) HCl
(c) HBr (d) HI

21. The attraction that an atom exerts on a pair of electrons that is being shared with another atom for forming covalent bond is referred to as its
(a) electron affinity (b) electronegativity
(c) ionisation energy (d) valency

22. The difference between ions and atoms is of
(a) relative size (b) configuration
(c) presence of charge (d) all of the above

23. Na^+ is smaller than Na atom because
(a) nucleus in each case contains different nucleons
(b) sodium atom has an electron lesser than sodium ion
(c) sodium atom has 11 electrons and sodium ion has 10 electrons
(d) the force of attraction is less in Na^+ than in Na^-

24. Lowest ionization potential in periods is shown by
(a) inert gases (b) halogens
(c) alkali metals (d) alkaline earth metals

25. The elements on the right side of the periodic table are
(a) non-metals (b) metals
(c) transition elements (d) metalloids

26. The statement that is not correct for the periodic classification of elements is:
(a) The properties of elements are periodic functions of their atomic number.
(b) Non-metallic elements are lesser in number than metallic elements.
(c) The first ionization energies of the elements along the period do not vary in a regular manner with the increase in atomic number.
(d) In the transition elements the second last shell is filled with electrons monotonically with increase in atomic number.

27. The most electropositive element is
(a) Cs (b) Ga
(c) Li (d) Pb

28. The process requiring the absorption of energy is
(a) $F \rightarrow F^-$ (b) $Cl \rightarrow Cl^-$
(c) $O \rightarrow O^{2-}$ (d) $H \rightarrow H^-$

29. The maximum ionization potential in a period is shown by
(a) alkali metals (b) inert gases
(c) representative elements (d) halogens

30. Ionisation energy of nitrogen is more than oxygen because
(a) attraction of electrons by the nucleus is more
(b) half filled 'p' orbitals are more stable
(c) nitrogen atom is small
(d) more penetration effect

II. Give one word answers

1. A group of three elements where the average atomic weights of the first and last elements is the equal to the atomic weight of the middle element.
2. The name of the periodic law that states that properties of the elements are a periodic function of their atomic masses.
3. The elements in which the second last shell is filling.
4. The elements in which the third last shell is filling.

5. The shell that is filling in the representative elements.
6. The repetition of properties after specific intervals of elements in the periodic table.

III. Match the following

This section contains three questions. Each question contains statements given in two columns, which have to be matched. Statements (i, ii, iii, iv) in Set A have to be matched with Statements (a, b, c, d) in Set B. More than one match or no match between options in the Sets is possible.

1. Match the elements in Set A with their Group number in Set B.

Set A	Set B
(i) O, S, Se, Te, Po	(a) 11
(ii) Fe, Ru, Os	(b) 15
(iii) N, P, As, Sb, Bi	(c) 16
(iv) Cu, Ag, Au	(d) 8

2. Match the elements in Set A with their nature in Set B.

Set A	Set B
(i) inert gas elements	(a) metals
(ii) elements with atomic number > 92	(b) non-reactive
(iii) halogens	(c) radioactive elements
(iv) transition elements	(d) non-metals

3. Match the element that belongs to the Group mentioned in Set A with the characteristic given in Set B.

Set A	Set B
(i) Group 1	(a) inert gas element
(ii) Group 17	(b) halogen element
(iii) Group 18	(c) chalcogen
(iv) Group 16	(d) alkali metal

4. Look at the Periodic Table and match the elements in Set A with the Group and period given in Set B.

Set A	Set B
(i) Z = 32	(a) Group 3, 7th period
(ii) Z = 38	(b) Group 3, 6th period
(iii) Z = 89	(c) Group 14, 4th period
(iv) Z = 58	(d) Group 2, 5th period

5. Match the element in Set A with the other name for it in Set B.

Column I	Column II
(i) Sulphur (S)	(a) picogen
(ii) Cobalt (Co)	(b) chalcogen
(iii) Nitrogen (N)	(c) coinage metal
(iv) Gold (Au)	(d) ferrous metal

6. Match the element in Set A with the number of valence electrons in Set B.

Set A	Set B
(i) inert gas element	(a) 1
(ii) alkali metal	(b) 7
(iii) alkali earth metal	(c) 2
(iv) halogen element	(d) 8

IV. State whether the following statements are True or False

1. Hydrogen is the only non-metal in Group IA.
2. There are 11 elementary gases in the Periodic table.
3. All inert gas elements have 8 electrons in the valence shell.
4. The group number of inert gas elements is 8 as the number of valence electrons is 8.
5. There are 28 inner transition elements.

V. Find the odd one out

1. hydrogen, lithium, sodium, potassium, calcium
2. (2), (2, 8), (2, 8, 8), (2, 8, 18, 8, 1), (2, 8, 18, 8)
3. carbon, sulphur, nitrogen, oxygen, fluorine
4. valency, valence electrons, metallic nature, atomic radius, acidic nature of oxide
5. atomic radius, ionisation energy, number of protons, electron affinity, electronegativity,

VI. Fill in the blanks

1. The gradual change in a property as we move across a period or go down a group is called _____.
2. Group IIA and Group IA elements are called _____ and _____ respectively.
3. The period number is also numerically equal to the _____ present in one atom of the each element of that period.
4. The group number is numerically equal to the _____.
5. The shortest and longest period of naturally occurring elements is _____and _____ respectively.

VII. Each question contains two statements. Statement A is an Assertion Statement and Statement R is the Reason Statement. Each question has four choices out of which only one is correct. Choose the correct one.

Choose A if Statement A is true, Statement R is true; Statement R is correct explanation for statement A

Choose B if Statement A is true, statement R is true; Statement R is not correct explanation for statement A

Choose C if Statement A is true, Statement R is false

Choose D if Statement A is false, statement R is true

1. **Statement –A**: In a given period the halogen element has the largest atomic radius.
   Statement –R: The halogens accept an electron easily.
2. **Statement –A**: All inert gas elements are gases.
   Statement –R: All inert gas elements have the lowest ionisation energy values in a given period.
3. **Statement –A**: The Modern Periodic Table uses increasing atomic weights for classification.
   Statement –R: The Modern Periodic Table can explain similarity of chemical properties based on valence electrons.
4. **Statement –A**: Dobereiner Triads cannot be used for most elements.
   Statement –R: Most elements do not satisfy the requirements of Dobereiner's classification.
5. **Statement –A**: Mendeleev's classification could predict the existence of many elements.
   Statement –R: Mendeleev's classification could predict even the place for all isotopes of the elements.
6. **Statement –A**: There are more metals than non-metals in the periodic table.
   Statement –R: Metalloids are few in number.
7. **Statement –A**: Metalloids are found to the right of the Periodic Table.
   Statement –R: Metalloids separate the metals from the non metals.
8. **Statement –A**: All inert gas elements excluding helium have 8 electrons in the valence shell.
   Statement –R: The atom of helium has a single shell.
9. **Statement –A**: Newland's Law of Octaves recognizes for the first time the repetition of chemical properties periodically.
   Statement –R: Newland arranged the elements in increasing order of atomic weight.
10. **Statement –A**: Group 1 elements are called alkali metals.
    Statement –R: Group 1 elements form oxides that dissolve in water to give alkalis.

VIII. Answer the following questions.

1. Give the benefits of classifying elements.
2. Give some of the earlier basis for classification.
3. List out the demerits of earlier attempts at classification.
4. Differentiate between periodicity of properties and periodic property with three examples each.
5. Give the main characteristics of
   (i) representative elements
   (ii) transition elements
   (iii) inner transition elements
   (iv) inert gas elements
6. One of the demerits of Mendeleev's classification was that there was no place for all isotopes of the elements. How is this defect invalid in the Modern Periodic table?
7. Name all elements that have a single shell in the periodic table.
8. If the atomic weights of the first two elements of a Dobereiner triad are 7 and 23, what is the atomic weight of the remaining element of the triad?
9. Name the other name of Group 0, Group VA, Group VIA and Group VIIA elements.
10. What is the other name of Lanthanides and Actinides?

KEY TO EXERCISE

I 1. c 2. d 3. b 4. c 5. c 6. b 7. a 8. a
9. c 10. d 11. d 12. b 13. b 14. b 15. a 16. b
17. a 18. b 19. a 20. a 21. b 22. d 23. c 24. c
25. a 26. c 27. a 28. c 29. b 30. b

II. 1. Dobereiner's Triads 2. Mendeleev's periodic Law 3. Transition elements
4. Inner transition elements 5. valence shell 6. periodicity

III. 1. i. c ii. d iii. b iv. a 2. i. b, d ii. a, c iii. d iv. a
3. i. d ii. b iii. a iv. c 4. i. c ii. d iii. a iv. b
5. i. b ii. d iii. a iv. c 6. i. d ii. a iii. c iv. b

IV. 1. true 2. true 3. false 4. false 5. true

V. 1. calcium 2. (2, 8, 18, 8, 1) 3. sulphur
4. atomic radius (is not periodicity of properties)
5. number of protons (is not periodic property)

VI. 1. periodic property 2. Alkali earth metals, alkali metals 3. number of shells
4. number of valence electrons 5. period 1 and period 6

VII. 1. D 2. B 3. D 4. A 5. C 6. B 7. B 8. B 9. A 10. A

6 Chemical Bonding



G N Lewis



Walther Kossel

6.0 INTRODUCTION

Today we know more than 90 elements that occur naturally. Most of these elements do not occur free but as compounds. What is the reason that most elements do not occur free? Why do elements combine with other elements to form compounds? The answer to both these questions lies in the reactivity of the elements. Elements that occur free are relatively non-reactive. *They do not react easily with other elements and so occur free*. Elements that do not occur free are relatively more reactive. *They react easily with other elements and so occur as compounds*. When elements react with other elements they form chemical bonds.

In 1704, Isaac Newton proposed that atoms attach to each other by some force. In 1819, Berzelius proposed a theory of elements combining due to the electronegative and electropositive character of the combining atoms. By the mid 19th century, various scientists developed the theory of valency. According to this theory, compounds were joined owing to an attraction of positive and negative poles or charges.

It was in 1916, G N Lewis developed a concept in which two atoms share one to six electrons, thus forming a bond. That same year, W Kossel put forward a theory similar to that of Lewis. He stated that bonds form by transfer of electrons between atoms. The bonds were due to attractive forces.

6.1 CHEMICAL BOND

In the chapters on Symbols, Formulae and Equations and Types of chemical changes, we have seen that elements combine to form compounds with the help of chemical bonds. Some important points need to be noted about combining atoms.

(i) Elements do not combine randomly with any or every other element.

(ii) Whether an atom combines with another atom depends on several factors. One of the most important factors is the electronic configuration of the atoms.

(iii) The maximum number of electrons in a shell with number, n, is given by the formula $2n^2$.

(iv) The number of electrons in the valence shell gives the valency of the atom.

(v) Atoms combine with other atoms in a fixed ratio by weight to form molecules.

(vi) *Inert gas elements are very stable and do not form compounds easily*. Before we examine the reason for this let us tabulate the electronic configuration of the inert gas elements.

Observations

A careful study of the tabulation below shows that :

(a) except helium all other inert gas elements have more than one shell with electrons present in them.

(b) except helium all other inert gas elements *have eight electrons in the last shell.*

(c) helium has only one shell with 2 electrons in this shell.

Inert gas element	Atomic Number	Configuration	Number of electrons in the valence shell	Remarks
Helium, He	2	2	2	Atom has only one shell.
Neon, Ne	8	2, 8	8	Atom has more than one shell.
Argon, Ar	18	2, 8, 8	8	Atom has more than one shell.
Krypton, Kr	36	2,8,18,8	8	Atom has more than one shell.
Xenon, Xe	54	2,8,18, 18, 8	8	Atom has more than one shell.
Radon, Rn	86	2, 8, 18, 32, 18, 8	8	Atom has more than one shell.

Duplet configuration – When the *only shell* in an atom or ion has *two electrons* the configuration is called *duplet configuration*.

Examples: – Helium atom with configuration (2) has duplet configuration.
– Lithium atom with configuration (2, 1), looses an electron to form lithium ion with configuration (2). Thus lithium ion (Li^+) has duplet configuration.

Octet configuration – When the last shell in an atom or ion with more than one shell has *eight electrons* the configuration is called *octet configuration*.

Examples: – Neon with configuration (2, 8) has octet configuration.
– Sodium atom with configuration (2, 8, 1), loses an electron to form sodium ion with configuration (2, 8). Thus sodium ion (Na^+) has octet configuration.

Reason for inert gas elements not forming compounds easily

Inert gases have either 2 electrons or 8 electrons in the valence shell. This is known as *inert gas configuration*. The inert gas elements are *very stable* and hence are *non reactive*. They do not form bonds easily.

Why do other elements combine to form compounds?

(a) Elements combine to get more stability in terms of electronic configuration

Atoms combine to get a duplet or octet configuration, which is more stable.

(b) Elements combine to get more stability in terms of energy

Every atom has a certain amount of energy known as internal energy. Atoms combine to form compounds as the internal energy of the molecule formed is less than the sum of the internal energies of the individual atoms. The compound has less energy and hence is more stable.

Example: $2Na + Cl_2 \rightarrow 2NaCl$

The compound NaCl has less internal energy than the sum of the internal energies of Na and Cl_2.

Exceptions: Some compounds do have more energy than the sum of the energies of the individual atoms. Such compounds are unstable and *usually react further to form new compounds with less energy*.

Example: $CaCO_3 \rightarrow CaO + CO_2$

Here the compound CaO has more energy and is unstable. It usually combines with atmospheric CO_2 to form $CaCO_3$ again which is more stable.

$$CaO + CO_2 \rightarrow CaCO_3$$
(from air)

Reactivity of elements

An element is said to be *reactive* when it reacts with other elements to form compounds. Elements that are reactive *will not be found free in nature* and exist as compounds.

The inert gas elements as seen above are found to be *less reactive*. Very few compounds of these elements are known. They occur free in nature.

The elements that do not have inert gas configuration are generally not found free in nature. They are *reactive* combining with elements to form compounds.

6.2 ELECTRON REDISTRIBUTION IN CHEMICAL BONDS

A chemical bond is formed by redistribution of electrons.

When atoms combine chemically there is a

redistribution of electrons in the valence shell of the atoms taking part.

The redistribution can be by way of

(a) transfer of electrons from one atom to another or

(b) sharing of electrons between the two atoms.

Due to the transfer or sharing of electrons both the participating atoms get a stable inert gas configuration – *either duplet or octet configuration.*

When two atoms form a bond, there is a redistribution of electrons either by transfer or sharing of electrons between atoms.

Chemical bonds can be classified in a very broad way based on whether electrons are transferred or shared into *ionic bond* and *covalent bond.*

Every element that does not have an inert gas configuartion tries to get an inert gas configuration by one of the following methods.

(i) *sharing of electrons with other atoms*

(ii) *mutual transfer of electrons (by donating or gaining electrons) from other atoms.*

The type of bond that forms between atoms depends on how the redistributuion takes place. There are two basic types of bonds: *Covalent Bond* and *Ionic Bond.*

Covalent bonds are formed by sharing of electrons between atoms.

Ionic Bonds are formed by transfer of electrons between atoms.

6.3 IONIC BOND, KOSSEL, 1916

Chemical bonds form due to redistribution of electrons. One type of redistribution is by mutual transfer of electrons.

An ionic bond is a chemical bond formed between two atoms (usually a metal and a non-metal) by transfer of electrons from valence shell of one atom to valence shell of another atom.

The transfer of electron takes place in such a way that both the atoms attain inert gas configuration.

The actual bond is formed due to electrostatic force of attraction between the resulting two oppositely charged ions.

Example: Ionic bond formation in NaCl

Sodium atom

Sodium (Na) has a configuration (2, 8, 1). *Sodium atom* has a *tendency to lose one electron (oxidation) to form Na^+ ion.* On losing the electron the Na^+ ion that is formed has the stable configuration of neon – an inert gas. Hence the ion of sodium is more stable than the atom of sodium. This is also the reason why sodium does not occur in the free state as sodium but occurs as compounds of sodium only.

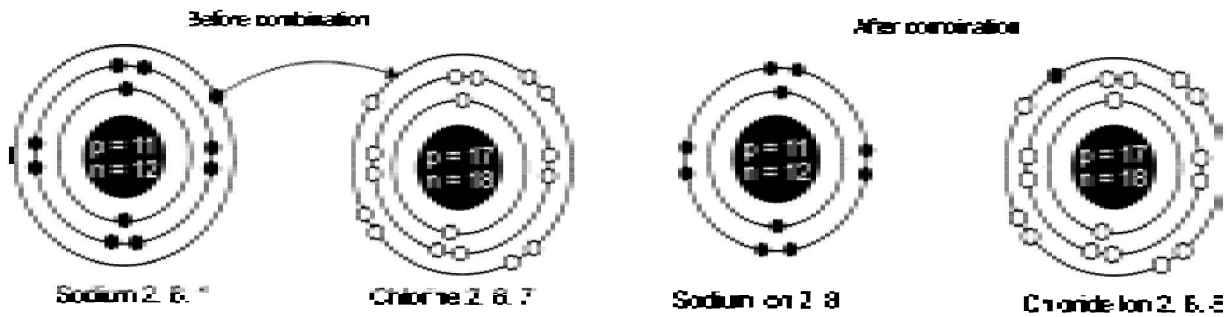
Chlorine atom

Chlorine (Cl) has a configuration (2, 8, 7). *Chlorine atom* has a *tendency to accept one electron (reduction) to form Cl^- ion.* On gaining one electron the Cl^- ion that is formed has the stable configuration of argon – an inert gas. Hence the ion of chlorine is more stable than the atom of chlorine. This is also the reason why chlorine will not occur in the free state but occurs as compounds of chlorine only.

Ionic bond formation between Na^+ and Cl^- ions

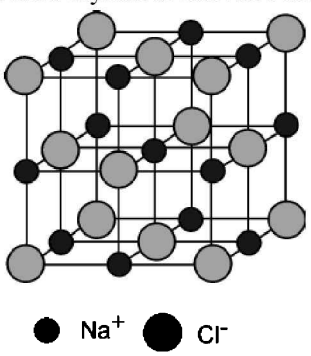
The actual bond in NaCl is formed by attraction of the oppositely charged ions – Na^+ ion and Cl^- ions.

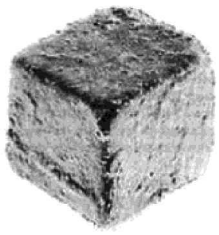
The transfer of an electron from the valence shell of sodium atom to the valence shell of chlorine atom is shown using electron dot diagram (before transfer and after transfer) below.



Electron dot diagram of sodium chloride

Crystal lattice of NaCl crystal is shown below using big dots for chlorine and small dots for sodium ions.





The image of the unit cell of NaCl seen under microscope.

Unit cell: The unit cell is the smallest unit of the crystal that gets repeated to form the larger crystal. The unit cell of NaCl is in the shape of a cube in which the corners are occupied by alternating Na^+ and Cl^- ions.

Other examples:

Compound	Cation formation and configuration changes	Anion formation and configuration changes	Ionic bond formation
1. KCl	$K - 1e^- \rightarrow K^+$ (2, 8, 8, 1) → (2, 8, 8) K^+ ion gets argon configuration	$Cl + 1e^- \rightarrow Cl^-$ (2, 8, 7) → (2, 8, 8) Cl^- ion gets argon configuration	$K^+ + Cl^- \rightarrow KCl$
2. MgO	$Mg - 2e^- \rightarrow Mg^{2+}$ (2, 8, 2) → (2, 8) Mg ion gets neon configuration	$O + 2e^- \rightarrow O^{2-}$ (2, 6) → (2, 8) O^{2-} ion gets neon configuration	$Mg^{2+} + O^{2-} \rightarrow MgO$
3. Al_2O_3	$2Al - 6e^- \rightarrow 2Al^{3+}$ (2, 8, 3) → (2, 8)Al^{3+} Al ion gets neon configuration	$3O + 6e^- \rightarrow 3O^{2-}$ (2, 6) → (2, 8) O^{2-} ion gets neon configuration	$2Al^{3+} + 3O^{2-} \rightarrow Al_2O_3$
4. $CaCl_2$	$Ca - 2e^- \rightarrow Ca^{2+}$ (2, 8, 8, 2) → (2, 8, 8) Ca^{2+} ion gets argon configuration	$2Cl + 2e^- \rightarrow 2Cl^-$ (2, 8, 7) → (2, 8, 8) Cl^- ion gets argon configuration	$Ca^{2+} + 2Cl^- \rightarrow CaCl_2$

Factors favouring ionic bond formation

Two atoms X and Y will form an ionic bond if:

(a) A has 1, 2 or 3 electrons in the valence shell.
(b) B has 5, 6 or 7 electrons in the valence shell.
(c) A loses electrons easily.
(d) B gains electrons easily.
(e) there is a net release of energy when A and B form a compound.

Properties of ionic compounds

Ionic compounds exhibit some common and characteristic properties. They are given below.

(a) Ionic compounds are generally crystalline solids at room temperature. They are never liquids or gases at ordinary temperature and pressure as the ions lack freedom of movement present in gases and liquids. Thus they are rigidly held in the crystal lattice.
(b) Ionic compounds are generally poor conductors of electricity in the solid state.
(c) In molten (fused) state or in aqueous solution the ions conduct electricity.
(d) Ionic compounds are hard with high melting point and boiling point.



A large mass in a small volume

(e) Ionic compounds are soluble in polar solvents and insoluble in non-polar solvents.

(f) Ionic compounds are very stable due to high lattice energy.

(g) Ionic compounds are brittle.

(h) Ionic compounds have high density due to a high mass being concentrated in a small volume (see figure below of NaCl crystal).

(i) Reactions of ionic compounds are very fast in aqueous solutions and fused state. Ionic reactions are fast as the transfer of electrons takes place.

Conductivity of ionic compounds

In the solid state, as the ions are held rigidly and cannot move about, ionic compounds cannot conduct electricity.

In the fused state, as temperature increases, the kinetic energy of the ions increases and the electrostatic forces of attraction are overcome. The rigid network is broken. The ions are able to move and hence conduct electricity.

In aqueous solution, the rigid network of ions is broken as the ionic compound dissociates setting the ions free which are available for conducting electricity.

6.4 COVALENT BOND, 1916, LEWIS

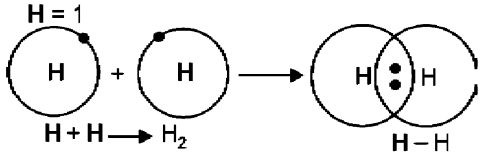
As stated earlier, chemical bonds form due to redistribution of electrons. The first type of redistribution that we saw in ionic bonds is by transfer of electrons. The second type of redistribution is by sharing of electrons.

A covalent bond is a chemical bond formed between two atoms (usually non-metal atoms) by sharing of valence electrons. (Covalent Bonds are also known as Atomic Bond or Electron Pair Bond)

The sharing of electrons takes place in such a way that both the atoms attain inert gas configuration. *The actual bond is formed due to the nuclei of the two atoms attracting the shared pair.*

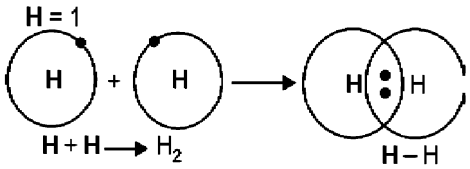
Examples:

(a) Hydrogen molecule



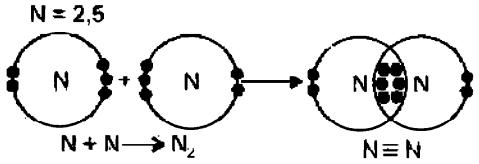
Molecule of Hydrogen – One shared pair

(b) Oxygen molecule



Molecule of Hydrogen – One shared pair

(c) Nitrogen molecule



Molecule of Nitrogen - Three shared pairs

Lewis Dot diagram

To represent the valence shell electrons, dots or crosses around the symbol of the atom are used. This is called Lewis electron dot diagram. The diagram is used to represent the bond formation between atoms more conveniently.

Lewis dot representation of valence shell electrons

Element:	$H^{\bullet}$	$:\dot{C}:$	$:\dot{N}:$	$:\ddot{O}\cdot$	$\cdot\ddot{Cl}:$
Electronic configuration:	(1)	(2,4)	(2,5)	(2,6)	(2,8,7)

Lewis dot representation of covalent molecules:

H_2 molecule :	$H^{\bullet} + {}^{\bullet}H \rightarrow H:H$	or H H	or H_2
Cl_2 molecule :	$:\ddot{Cl}\cdot + \cdot\ddot{Cl}: \rightarrow :\ddot{Cl}:\ddot{Cl}:$	or Cl–Cl	or Cl
HCl molecule :	$H^{\bullet} + \cdot\ddot{Cl}: \rightarrow H:\ddot{Cl}:$	or H–Cl	or HCl
N_2 molecule :	$:\dot{N}: + :\dot{N}: \rightarrow :N::N:$	or N≡N	or N_2
H_2O molecule :	$H^{\bullet} + \cdot\ddot{O}\cdot + {}^{\bullet}H \rightarrow H:\ddot{O}:H$	or H–O	or H_2O
CO_2 molecule :	$:\ddot{O}: + :\dot{C}: + :\ddot{O}: \rightarrow :\ddot{O}::C::\ddot{O}:$	or $O=\overset{H}{C}=O$	or CO_2

Factors favouring covalent bond formation

Two atoms X and Y will form a covalent bond if

(a) both atoms do not give up electrons easily

(b) both atoms attract the shared electron pair equally or almost equally.

(c) the number of valence electrons in the two atoms is 5, 6 or 7 (the only exception being hydrogen atom – that has 1 valence electron only but forms covalent bonds).

Properties of covalent compounds:

Covalent compounds exhibit some common and characteristic properties. They are given below.

(a) Most covalent compounds are gases or liquids. Some are soft solids. (Diamond is an exception. Diamond has only covalent bonds and is a very hard substance).

(b) Most covalent compounds have low melting and boiling points (Diamond again is an exception. Its melting point is high).

(c) Most covalent compounds are poor conductors of electricity. (Graphite has covalent bonds and is an exception as it is a good conductor. Some covalent compounds like HCl that ionise in aqueous solution are good conductors).

(d) Covalent compounds are soluble in non-polar solvents and are insoluble in polar solvents.

(e) Covalent compounds are neither hard nor brittle.

(f) Reactions of covalent compounds are slow.

6.5 CLASSIFICATION OF COVALENT BONDS

Covalent bonds can be classified based on certain factors as follows:

(a) Based on covalency.

(b) Based on combining atoms

(c) Based on electronegativity differences of the atoms.

(a) Based on covalency

Covalency is the number of shared pairs of electrons between atoms in a covalent bond. It is never greater than 3.

Single Covalent Bond molecules have one shared pair of electrons. Their covalency is 1.

Examples: H_2, Cl_2

Double Covalent Bond molecules have two shared pairs of electrons. Their covalency is 2.

Example: O_2

Triple Covalent Bond molecules have three shared pairs of electrons. Their covalency is 3.

Example: N_2

(b) Based on combining atoms

If both atoms in a covalent bond belong to same element then the bond is said to be **Homoatomic** (e.g. H_2, Cl_2) and if they belong to different elements, the bond is said to be **Heteroatomic** (e.g. HCl, NH_3)

(c) Based on electronegativity differences of the atoms

If the participating atoms in a covalent bond belong to different elements then there will be electronegativity difference between the two atoms.

Due to the electronegativity differences of the participating atoms, the more electronegative atom in the covalent bond attracts the shared pair of electrons towards itself setting up a small charge on the atom.

The residual + charge is set up on the other less electronegative atom.

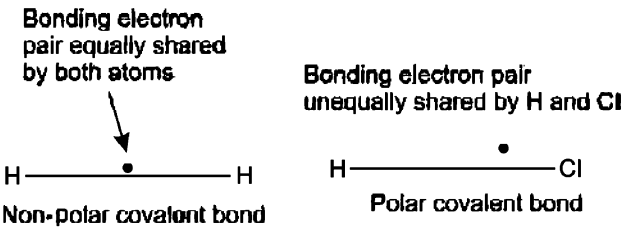
Thus two poles result in the molecule. This is called a *dipole*.

Such covalent bonds with dipoles in them are known as *Polar Covalent Bonds*.

Examples: HCl, H_2O.

If however the electronegativity difference is very small or both atoms have equal electronegativity, then no dipole can be formed and such bonds are known as *Non-polar Covalent Bonds*.

Example: H_2



Polar covalent bond is formed by unequal sharing of electrons

6.6 IONIC COMPOUNDS AND COVALENT COMPOUNDS

Given below is a comparison between ionic compounds and covalent compounds.

Ionic compounds	Covalent compounds
1. Ionic compounds are crystalline, brittle and hard solids at room temperature.	1. Covalent compounds are soft solids or gases or liquids at room temperature. They are neither hard nor brittle.
2. Ionic compounds are generally poor conductors of electricity in the solid state. In molten (fused) state or in aqueous solution the ions conduct electricity.	2. Most covalent compounds are poor conductors of electricity.
3. Ionic compounds have high melting point and boiling point.	3. Most covalent compounds have low melting and boiling points
4. Ionic compounds are soluble in polar solvents and insoluble in non-polar solvents.	4. Covalent compounds are soluble in non-polar solvents and are insoluble in polar solvents.
5. Ionic compounds are very stable.	5. Covalent compounds are unstable.
6. Ionic compounds have high density due to a high mass being concentrated in a small volume.	6. Covalent compounds have low density.
7. Reactions of ionic compounds are very fast in aqueous solutions and fused state. Examples: $NaCl$, $CaCl_2$, MgO, Al_2O_3	7. Reactions of covalent compounds are slow. Examples: Cl_2, H_2, O_2, NH_3, CH_4

6.7 COORDINATE COVALENT BOND (DATIVE BOND)

Coordinate Covalent Bond is a special covalent bond in which both the electrons in the shared pair are donated by one of the atoms known as the *donor*. The other atom is known as *acceptor*.

Examples:

1. **Formation of Hydronium ion**: Water molecule has one oxygen atom and two hydrogen atoms. Oxygen atom has the configuration (2, 6). The six valence electrons are shown using Lewis dot diagram in the figure below. The bond formation is illustrated below.

   (i) The first valence electron of oxygen atom (represented by a dot) is shared with the valence electron of the first hydrogen atom (represented by a cross).

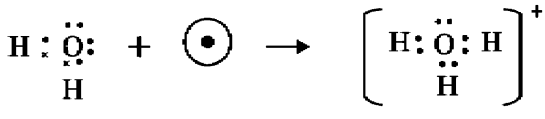
   (ii) The second valence electron of oxygen atom (represented by another dot) is shared with the valence electron of the second hydrogen atom (represented by a cross).

   (iii) The oxygen atom has four more electrons or two pairs shown by the remaining dots in the figure below.

   (iv) One of the 'free' pair of electrons of oxygen atom is donated to H^+ ion which has an empty shell.

   (v) The empty shell of H^+ ion is now 'filled' with two electrons and thus gets duplet configuration and becomes stable.

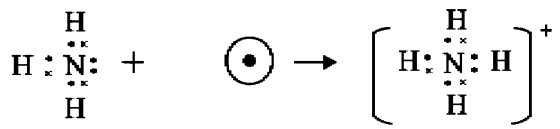
   (vi) In the combined entity, the number of electrons is one less than the number of protons and the entity acquires a positive charge. Hydronium ion thus has one positive charge.



Water molecule H^+ ion Hydronium ion

In hydrogen ion the first and only shell in empty. This shell can accept two electrons from oxygen atom in water molecule.

2. **Formation of Ammonium ion**: Ammonia molecule has one nitrogen atom and three hydrogen atoms. Nitrogen atom has the configuration (2, 5). The five valence electrons are shown using Lewis dot diagram in the figure. The bond formation is illustrated below.
   (i) The first valence electron of nitrogen atom (represented by a dot) is shared with the valence electron of the first hydrogen atom (represented by a cross).
   (ii) The second valence electron of nitrogen atom (represented by another dot) is shared with the valence electron of the second hydrogen atom (represented by a cross).
   (iii) The third valence electron of nitrogen atom (represented by another dot) is shared with the valence electron of the third hydrogen atom (represented by a cross).
   (iv) The nitrogen atom has two more electrons or one pair shown by the remaining dots in the figure.
   (v) The 'free' pair of electrons of nitrogen atom is donated to H^+ ion which has an empty shell.
   (vi) The empty shell of H^+ ion is now 'filled' with two electrons and thus gets duplet configuration and becomes stable.
   (vii) In the combined entity, the number of electrons is one less than the number of protons and the entity acquires a positive charge. Ammonium ion thus has one positive charge.



In hydrogen ion the first and only shell is empty. This shell can accept two electrons from nitrogen atom in ammonia molecule.

Once a co-ordinate covalent bond is formed it cannot be distinguished from a pure covalent bond as the electrons cannot be differentiated. In a covalent bond, there is a redistribution of electron density between the two atoms. The electrons are shared by two nuclei and revolve around both of them in the redistributed space. The valence configuration and process of formation of a co-ordinate covalent bond as illustrated above in the two examples tell us that one atom has donated both electrons. After the formation, it is designated as a strong bond formed by sharing of electrons by the atoms.

A co-ordinate covalent bond is thus a modified form of a covalent bond.

6.8 VAN DER WAAL'S FORCES

In chemical bonds the forces of attraction are within a molecule.

When two molecules come close together, they exert attractive forces that are physical forces.

Intramolecular bonding versus Intermolecular bonds

Intramolecular attractions are the forces of attraction which hold an individual molecule together (for example, the covalent bonds).

Intermolecular attractions are attractions between one molecule and a neighbouring molecule. They are weaker forces than the chemical forces present in ionic or covalent compounds. These forces are very short-lived intermolecular attractive forces that exist between all kinds of atoms, molecules or ions *when they are close together*. They are usually set up due to *induced dipoles.*

These two words are so confusingly similar that it is safer to never use one of them. The term "intramolecular" won't be used again in this chapter.

The origin of van der Waal's forces – Temporary fluctuating dipoles

The electrons revolve around the nucleus of atoms *at very large speeds* creating a cloud like appearance.

Illustration:

When a fan is stationary the three blades can be seen separately. As the fan starts revolving and picks up speed only a blur can be seen. The blades are no longer visible separately. The electron when it revolves at large speeds around the nucleus appears like a blur or cloud. This cloud is called electron density.

Consider hydrogen molecule as shown. The lozenge-shaped diagram represents a small symmetrical molecule – Hydrogen or Bromine molecule. The even shading shows that the electron density of the valence electrons is even and symmetrical.

The electrons being mobile, due to some disturbance at any one instant, might find themselves towards one end of the molecule, making that end δ–.



The other end will be temporarily short of electrons and so becomes δ^+.



An instant later the electrons may well have moved up to the other end, reversing the polarity of the molecule.

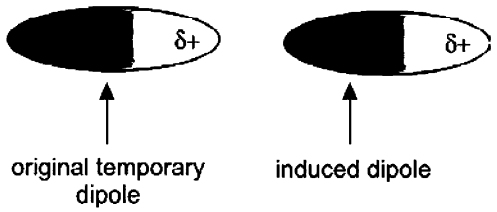
This causes rapidly fluctuating dipoles in all molecules. It even happens in monatomic molecules – molecules of noble gases, like helium, which consist of a single atom.

If both the helium electrons happen to be on one side of the atom at the same time, the nucleus is no longer properly covered by electrons for that instant.



How temporary dipoles give rise to intermolecular attractions

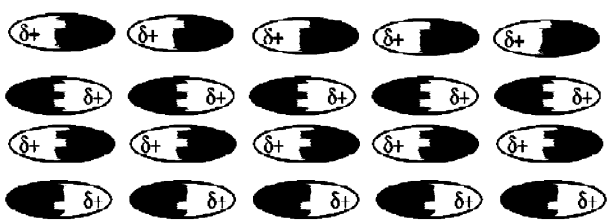
Imagine a molecule which has a temporary polarity as shown above. If another molecule which happens to be entirely non-polar comes close to it, the electrons in the non-polar molecule will tend to be attracted by the slightly positive end of the left hand one.



This sets up an ***induced dipole*** in the approaching molecule, which is orientated in such a way that the positive end of one is attracted to the negative end of the other.

As long as the molecules stay close to each other the polarities will continue to fluctuate in synchronisation so that the attraction is always maintained.

There is no reason why this has to be restricted to just two molecules. As long as the molecules are close together this synchronised movement of the electrons can occur over huge number of molecules.



The diagram shows how a whole lattice of molecules could be held together in a solid using van der Waal's dispersion forces. This arrangement is not a permanent arrangement – an instant later, a quite different arrangement of the distribution of the electrons is possible.

Noble gases and van der Waal's forces:

Noble gases are very stable and tend not to interact, making them difficult to condense into liquid. However, *the larger the atom of the noble gas, the easier it is to condense the gas.* This is because a larger atom has a larger electron cloud, which more readily forms an ellipsoid, making the atom a temporary dipole. The electron cloud at any moment takes the shape of an ellipsoid around the nucleus, leaving a slight negative charge on one side of the major axis and a slight positive charge on the other. The dipole induces the same shift in neighboring atoms and spreads from one atom to the next. Unlike charges attract, and the induced dipoles are held together by van der Waal's force.

The reason that the boiling points increase as you go down the group is that the number of electrons increases, and so also does the radius of the atom. The more the number of electrons and more the distance over which they can move, the bigger the possible temporary dipoles and therefore the bigger the dispersion forces.

Because of the greater temporary dipoles, xenon molecules are *'stickier'* than neon molecules. Neon molecules will break away from each other at much lower temperatures than xenon molecules – hence neon has a lower boiling point.



This is the reason that (all other things being equal) bigger molecules have higher boiling points than small ones. Bigger molecules have more electrons and more distance over which temporary dipoles can develop and so the bigger molecules are *'stickier'*.

Applications of van der Waal's Forces

(a) van der Waal's forces cause inert gases and halogens to condense to liquids when cooled. All molecules experience intermolecular attractions, although in some cases these attractions are very weak. Even in a gas like hydro-

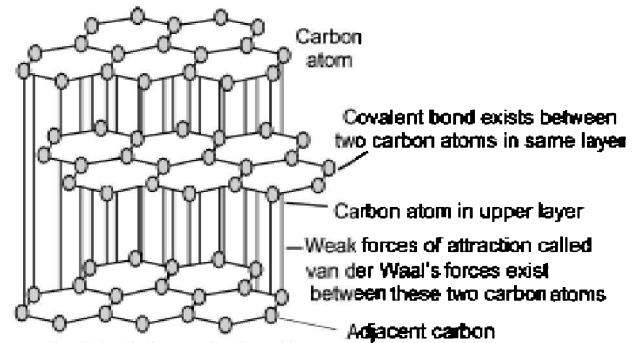
gen, if the molecules are slowed by cooling the gas, the attractions are large enough for the molecules to stick together eventually to form a liquid and then a solid.

In the case of hydrogen the attractions are so weak that the molecules have to be cooled to 21 K (–252°C) before the attractions are strong enough to condense the hydrogen into a liquid. Helium's intermolecular attractions are even weaker – the molecules won't stick together to form a liquid until the temperature drops to 4 K (-269°C).

(b) Gases show deviation from Gas Laws due to van der Waal's Forces. The weak forces of attraction slow down the molecules that are about to collide with the walls of the container. The force with which the molecules collide with the walls of the container is actually less due to these forces of attraction between gas molecules. The *measured pressure* is weaker than the *ideal pressure*. Hence in reality the inverse proportionality between volume and pressure (Boyles' law) and direct proportionality between volume and absolute temperature (Charles' Law) in *gases is affected to some extent*.

Due to van der Waal's forces, gases that strictly obey the gas laws, or show ideal behaviour do not exist.

(c) Some crystals are soft and have low MP as the atoms / molecules are held only by weak van der Waal's Forces.



van der Waal's forces in Graphite

Example: In graphite, carbon atoms in adjacent layers are held by van der Waal's forces (see figure).

6.9 BOND PAIRS AND LONE PAIRS

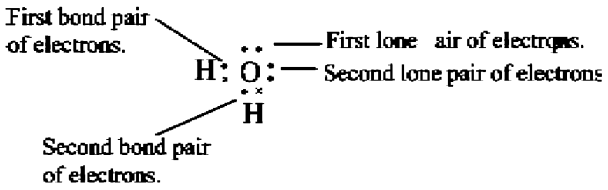
Bond Pairs: Bond pair is the pair of electrons that is shared. This pair belongs to and is attracted by two nuclei. In a bond pair one electron belongs to one atom and the other electron belongs to the other atom in the bond.

For example in water there are two bond pairs (see illustration 1 below).

Lone pairs: Lone pair of electrons is the pair of electrons that belongs to and is attracted by one nucleus.

For example in ammonia there is a lone pair on nitrogen (see illustration 2 below).

Illustration 1



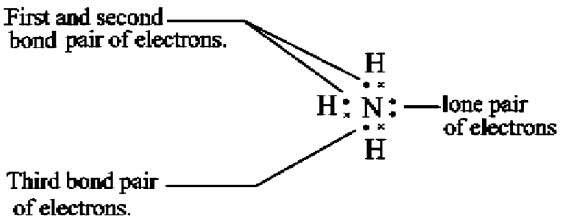
Water molecule

(i) *The first valence electron of oxygen atom (represented by a dot) is shared with the valence electron of the first hydrogen atom (represented by a cross). This represents the first bond pair.*

(ii) *The second valence electron of oxygen atom (represented by another dot) is shared with the valence electron of the second hydrogen atom (represented by a cross). This represents the second bond pair.*

(iii) *The oxygen atom has thus four more electrons or two pairs shown by the remaining dots in the figure above. These represent the two lone pairs of electrons of oxygen atom.*

Illustration 2



Ammonia molecule

(i) *The first valence electron of nitrogen atom (represented by a dot) is shared with the valence electron of the first hydrogen atom (represented by a cross). This represents the first bond pair.*

(ii) *The second valence electron of nitrogen atom (represented by another dot) is shared with the*

valence electron of the second hydrogen atom (represented by a cross). This represents the second bond pair.

(iii) *The third valence electron of nitrogen atom (represented by another dot) is shared with the valence electron of the third hydrogen atom (represented by a cross). This represents the third bond pair.*

(iv) *The nitrogen atom has two more electrons or one pair shown by the remaining dots in the figure below. This represents the lone pair of electrons of nitrogen atom.*

EXERCISE

I. Choose the correct answer

1. Reactive elements have _____electrons in the valence shell.
(a) less than eight (b) more than eight
(c) eight (d) two

2. Single shell atoms try to acquire _____configuration.
(a) octet (b) triplet
(c) duplet (d) none of these

3. The molecule having pyramidal shape is
(a) ammonia (b) water
(c) methane (d) carbon dioxide

4. The molecule having V-shape is
(a) ammonia (b) water
(c) methane (d) carbon dioxide

5. The molecule having linear shape is
(a) ammonia (b) water
(c) methane (d) carbon dioxide

6. The molecule having tetrahedral shape is
(a) ammonia (b) water
(c) methane (d) carbon dioxide

7. Reactions involving ionic compounds are usually
(a) fast (b) slow
(c) can be fast or slow (d) are very slow

8. Reactions involving covalent compounds are usually
(a) fast (b) slow
(c) can be fast or slow (d) are very slow

9. The unit cell of NaCl has
(a) greater number of Na^+ ions than Cl^- ions.
(b) unequal number of Na^+ and Cl^- ions.
(c) equal number of Na^+ and Cl^- ions.
(d) smaller number of Na^+ ions than Cl^- ions.

10. The number of lone pairs in CH_4 molecule is
(a) one (b) two
(c) three (d) zero

11. The number of lone pairs in water molecule is
(a) one (b) two
(c) three (d) none

12. The number of lone pairs in NH_3 molecule is
(a) one (b) two
(c) three (d) none

13. The number of bond pairs in CO_2 molecule is
(a) one (b) two
(c) three (d) four

14. The number of bond pairs in H_2O molecule is
(a) one (b) two
(c) three (d) none

15. The molecule that has equal number of lone pairs and bond pairs is
(a) CO_2 (b) CH_4
(c) NH_3 (d) H_2O

16. The molecule that has more bond pairs than lone pairs is
(a) NH_3 (b) H_2O
(c) both (a) and (b) (d) neither (a) nor (b)

17. The boiling point of elements as we go down a given group increases due to increased
(a) size
(b) electron density
(c) van der Waal's forces
(d) all three options (a, b and c)

18. The correct order of boiling points of inert gases is
(a) Ne < Ne > Ar (b) He > Ne > Ar
(c) He < Ne > Ar (d) He < Ne < Ar

19. Ionic compounds have high density as
(a) a large volume has a small mass
(b) metals combine with non-metals
(c) a large mass is concentrated in a small volume
(d) all three options.

20. Dipoles exist in
(a) HCl molecule (b) water molecule
(c) hydrogen molecule (d) (a) and (b) options

21. As electronegativity difference between combining atoms increases
(i) covalent character increases
(ii) covalent character decreases
(iii) ionic character increases
(iv) ionic character decreases
(a) (i) and (iii) (b) (ii) and (iv)
(c) (ii) and (iii) (d) (i) and (iv)

22. Small charged ends of water molecules attract small but oppositely charged ends present on

other water molecules. These attractions are primarily responsible for

(a) high boiling point of water
(b) water being a liquid
(c) both (a) and (b)
(d) none of these

23. A metal and non-metal are likely to combine to form _____ bond.

(a) covalent (b) ionic
(c) dative (d) polar covalent

24. On combining with other elements chlorine acquires

(a) configuration of argon
(b) configuration of helium
(c) configuration of neon
(d) configuration of krypton

25. Magnesium acquires inert gas configuration in simple compounds formed by it by

(a) losing electrons (b) gaining electrons
(c) sharing electrons (d) It is non-reactive.

26. Water molecule has

(a) two covalent bonds
(b) one covalent and ionic bond
(c) two ionic bonds
(d) no bonds

27. The formula of the compound formed by an element, M in Group 13 (three valence electrons) with an element X in Group 17 (seven valence electrons) will be

(a) MX (b) M_2X_3
(c) MX_3 (d) MX_2

28. Inert gas elements are non-reactive because of

(a) octet configuration
(b) duplet configuration
(c) duplet configuration or octet configuration
(d) duplet configuration and octet configuration

29. Ionic bonds are strong (do not break easily) due to

(a) electrostatic force of attraction
(b) metals combining with non-metals
(c) non-metals combining with non-metals
(d) none of these

30. Covalent bonds are strong due to

(a) non-metals combining with non-metals
(b) metals combining with non-metals
(c) two nuclei attracting the shared pair of electrons
(d) none of these

II. Give one word answers

(i) The bond formed between sodium ion and chloride ion
(ii) The bond between NH_3 and H^+ in ammonium ion
(iii) The number of shared pairs between atoms participating in a bond.
(iv) The state in which a pure ionic compound conducts electricity
(v) Two small but opposite charges present in a covalent molecule.

III. Match the following.

This section contains three questions. Each question contains statements given in two columns, which have to be matched. Statements (i, ii, iii, iv) in Set A have to be matched with Statements (a, b, c, d) in Set B. More than one match or no match between options in the Sets is possible.

1. Match the property of substance in Set A with type of bond in Set B.

Set A	Set B
(i) polarity	(a) van der Waal's forces
(ii) high density	(b) polar covalent bond
(iii) gases or soft solids	(c) covalent bond
(iv) soluble in non-polar solvents	(d) ionic bond

2. Match the type of bond or compound present in Set A with compound or bond in Set B.

Set A	Set B
(i) Ionic	(a) sharing of electrons
(ii) Covalent	(b) transfer of electrons
(iii) CH_4	(c) CO_2
(iv) NH_3	(d) NaCl

3. Match the element in Set A with the configuration in Set B.

Set A	Set B
(i) Carbon	(a) (2)
(ii) Helium	(b) (2, 8, 7)
(iii) Oxygen	(c) (2, 4)
(iv) Chlorine	(d) (2, 6)

IV. State whether the following statements are True or False.

(i) Covalent bonds are formed by sharing electrons.
(ii) In co-ordinate covalent bond both the electrons are equally shared.
(iii) Ammonium chloride molecule has one ionic, three covalent and one dative bond.
(iv) The covalency of oxygen is three.
(v) Ionic compounds dissolve in water as water is a polar solvent.

V. Fill in the blanks

1. The force of attraction that exists between atoms is called _____.
2. The electrostatic force of attraction between oppositely charged ions is known as _____.
3. The bonds between NH_3 and H^+ in NH_4^+ and between H_2O and H^+ in H_3O^+ are _____.
4. The maximum number of covalent bonds formed between two atoms never exceeds _____.
5. A coordinate bond is a _____covalent bond. Both electrons in a co-ordinate bond are provided by _____.
6. Covalency of the bond(s) present in H_2, F_2, Cl_2 is _____and in O_2, it is _____ while in N_2 it is_____
7. Generally, covalent bond is formed between two _____ atoms and ionic bond is formed between _____ and _____atoms.
8. Electrons in _____shell take part in bonding.
9. Ionic compounds have _____melting point and are _____ solids.
10. The number of electron pairs shared between two atoms in a covalent bond is known as _____.
11. An element with Z = 20 is likely to chemically combine with an element whose Z = _____?
    11 / 14 / 16 / 10
12. The total number of electrons that take part in bonding in Nitrogen molecule is _____
13. An example of a compound that has covalent, ionic and dative bonds in it is _____
14. An element Y has a ground state configuration of 2, 8, 8. The type of bond that exists between atoms of Y is _____
15. Outermost orbitals of two elements 'X' and 'Y' have 7 and 3 electrons respectively. The expected formula of their compound is _____.

VI. Each question contains two statements. Statement A is an Assertion Statement and Statement R is the Reason Statement. Each question has four choices out of which only one is correct. Choose the correct one.

Choose A if Statement A is true, Statement R is true; Statement R is correct explanation for statement A

Choose B if Statement A is true, statement R is true; Statement R is not correct explanation for statement A

Choose C if Statement A is true, Statement R is false

Choose D if Statement A is false, statement R is true

1. **Statement –A:** Elements combine chemically to form compounds.

   Statement –R: Compounds are more stable than the combining elements in their free state.
2. **Statement –A:** Atoms with 5, 6, or 7 valence electrons form either ionic or covalent bond.

   Statement –R: Atoms with 1, 2, or 3 valence electrons form ionic bond.
3. **Statement –A:** Ionic bonds are generally formed between metals and non-metals.

   Statement –R: Metals lose electrons, non-metals gain electrons.
4. **Statement –A:** Ionic compounds have empirical formula.

   Statement –R: Electrostatic forces of attraction in ionic bonds are weak.

VII. Answer the following questions.

1. Why do some elements exist in the free state while others exist as compounds?
2. How do atoms acquire duplet or octet configuration?
3. Why are ionic bonds formed between metal and non-metal atoms?
4. Ionic compounds have high density and melting points. Why?
5. Elements A, B, C, D, E have atomic numbers 3, 9, 10, 16, 19 respectively. Which of these elements would you expect to form ionic bonds?
6. Ionic compounds are poor conductors of electricity in the solid state but excellent conductors in the liquid state. Why?
7. Explain the factors favouring covalent bonds.
8. Why are covalent compounds poor conductors of electricity?
9. Tabulate the difference between ionic and covalent compounds.
10. Show NH_3, CH_4 molecule using electron dot diagram.
11. Explain van der Waal's Forces bringing out clearly why they are temporary and surface forces.
12. State one similarity between ionic compound in solid state and covalent compounds.

KEY TO EXERCISE

I.

1. a	2. c	3. a	4. b	5. d	6. c	7. a	8. b
9. c	10. d	11. b	12. a	13. d	14. b	15. d	16. a
17. d	18. d	19. c	20. d	21. c	22. c	23. b	24. a
25. a	26. a	27. c	28. c	29. a	30. c		

II. i. ionic bond ii. dative bond iii. covalency iv. fused state v. dipole

III.

1.	i. b, d	ii. d	iii. a, c	iv. c
2.	i. b, d	ii. a, c	iii. a	iv. a
3.	i. c	ii. a	iii. d	iv. b

IV. i. true ii. false iii. true iv. false v. true

V.

1. Chemical Bond 2. Ionic Bond 3. Dative Bonds 4. three
5. modified, one atom 6. one, two three 7. non-metals atoms, metal and non-metal atoms
8. valence shell 9. high, crystalline solids 10. covalency
11. Z = 16 12. six 13. Ammonium Chloride
14. Van der Waal's' forces 15. X_3Y

VI. 1. A 2. B 3. A 4. C

7 Chemical Kinetics

7.0 INTRODUCTION

Chemical reactions are chemical changes. They differ from physical changes in that they bring about a change in the chemical composition of the reacting substances. This usually takes place under the action of heat.

For a chemical reaction to take place the following conditions must be present.

(i) The reacting substances must come in close contact.

(ii) For some reactions to take place heating may be required.

(iii) Pressure may have to be applied on the reactants for the reaction to take place.

(iv) Some reactions require the presence of light.

(v) Some reactions require the presence of a catalyst.

A chemical reaction starts as soon as the minimum conditions for it to take place exist. The rate at which the chemical change takes place can vary from very slow to instantaneous. For example fermentation reactions like curdling of milk are very slow. A white precipitate of silver chloride forms almost instantaneously when a drop of silver nitrate solution is added to a solution of sodium chloride.

Even if the conditions for the chemical reaction to take place exist the rate at which the chemical reaction takes place depends on several factors. Chemical Kinetics is a branch of chemistry that deals with

(i) a study of reaction rates,

(ii) a study of reaction paths and

(iii) a study of reaction mechanisms.

7.1 REVERSIBLE AND IRREVERSIBLE CHEMICAL REACTIONS

Irreversible Reactions are chemical reactions that cannot be reversed. In other words once the reactants have changed to products, the products cannot form the reactants again under the same conditions. They are represented using single directional arrow: $\rightarrow$

Example: $C + O_2 \rightarrow 2CO_2$

(CO_2 cannot decompose under same conditions to form C and O_2)

Reversible Reactions are chemical reactions that can be reversed under the same conditions in which it has taken place. In other words the products can form the reactants again under the same conditions. They are represented using bi-directional arrow: $\rightleftharpoons$

Examples: $N_2 + 3H_2 \rightleftharpoons 2NH_3$

(NH_3 decomposes to give N_2 and H_2 under the same conditions)

$$2SO_2 + O_2 \rightleftharpoons 2SO_3$$

$$CaCO_3 \rightleftharpoons CaO + CO_2$$

Exothermic reactions are chemical reactions in which there is a release of heat energy.

Examples: $C + O_2 \rightarrow CO_2$ + Heat released

$2Na + Cl_2 \rightarrow 2NaCl$ + Heat released

Endothermic reactions are chemical reactions in which there is an absorption of heat energy.

Examples:

$CaCO_3 \rightarrow CaO + CO_2$ + Heat absorbed

$N_2 + O_2 \rightarrow 2NO$ + Heat absorbed

7.2 RATES OF A CHEMICAL REACTION

The rate of a chemical reaction can be expressed in many ways.

(i) **Rate of a chemical reaction:** The rate of a chemical reaction is given by the change in concentration of one of the reactants or products. It is expressed in moles / litre / second.

For the general reaction : $A + B \rightarrow C + D$

The rate will be given by the rate of change in concentration (change in concentration) of any one of the reactants or products over a time period (time interval).

Expressing concentration

The concentrations of A, B, C and D in the reaction

$$A + B \rightarrow C + D$$

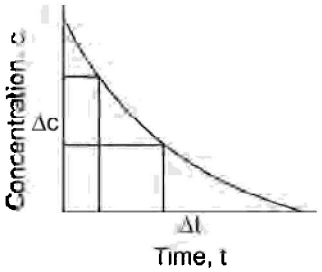
are written in short form as [A], [B], [C] and [D].

The short form notation [A] is read as concentration of A and so on.

Thus, rate $= -\frac{d[A]}{dt} = -\frac{d[B]}{dt} = \frac{d[C]}{dt} = \frac{d[D]}{dt}$

If the concentration of reactants is taken to express rate, then the rate of reaction is a *negative quantity* as the concentration of reactants decreases as the reaction progresses.

If the concentration of products is taken to express rate, then the rate of reaction is a *positive quantity* as the concentration of products increases as the reaction progresses.

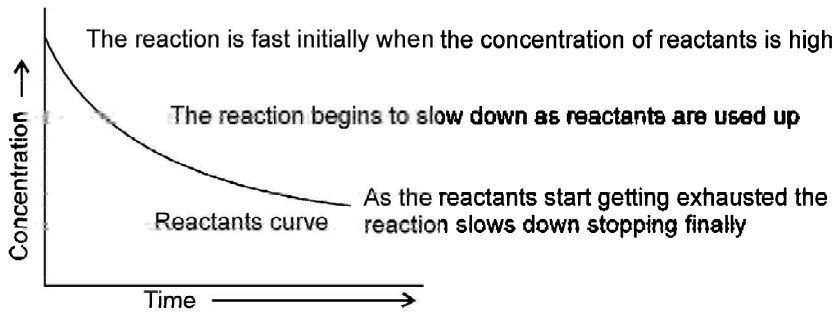


Graphical representation of Rates of chemical reaction

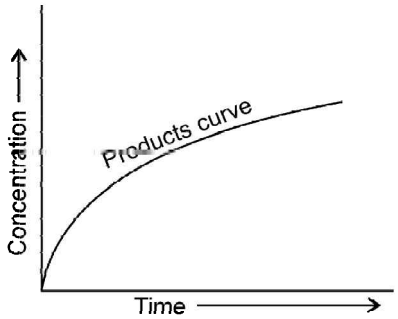
After a chemical reaction has started, if the concentration of either reactants or products are measured at frequent time intervals and the data plotted the graph will look as shown below.

Consider the general reaction : Reactants → Products (R → P)

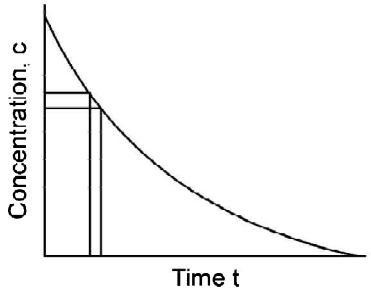
Reactant Graph



Variation in the concentration of reactants with time in a chemical reaction.



Variation in the concentration of reactants with time in a chemical reaction.



Graph to calculate instantaneous rate

(ii) Instantaneous rate is the rate of a chemical reaction at a particular instant. To calculate the *instantaneous rate*, a particular instant is taken after the start of a reaction. The concentration at a *very short time interval* before this instant is taken and the change in concentration of either reactant or product in this interval of time is measured. This is divided by the time interval to arrive at the *instantaneous rate* – shorter the time interval taken, more the accuracy.

Instantaneous rate

$$= \frac{\text{Small decrease in the concentration of reactant}}{\text{Small interval of time}} = \frac{\delta[R]}{\delta t}$$

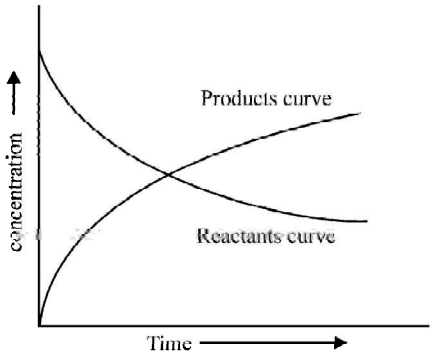
(iii) Average Rate is a more practical way of expressing rate. To calculate the average rate of a chemical reaction the total change in concentration of either the reactant or product is divided by the time taken from start to finish.

Average rate, $r = \frac{\Delta[A]}{\Delta t}$, where ΔA is the total change in concentration of either the reactant or product over the entire reaction period.

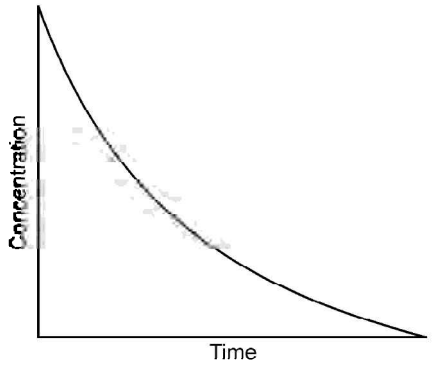
The overall progress of reaction can be shown in a single graph as follows.

The slope of either the reactant curve or the product curve gives the rate of the chemical reaction.

The change in concentration can be perceived either by a change in colour of the reactants / products or pressure changes or pH changes or actual volumetric analysis.



Variation in the concentration of reactants and with time in a chemical reaction.



Graph to calculate Average rate

We can talk about the rate of formation or loss of any species – reactant or product. It is, however, important to specify which species we are talking about. The rate can be positive or negative. A positive rate means that the concentration is increasing with time e.g., concentration of a product; a negative rate means that the concentration is falling with time e.g., concentration of a reactant. The rate may vary with time (and concentration), so it is usual to define the rate over a very small time, *t*.

Thus, rate is the *derivative* of concentration *with respect to time*. This derivative is the slope of the graph of concentration against time, taken at a particular time.

From its definition it is clear that the units of rate are concentration per unit time, for example mol dm^{-3} s^{-1}. It is also common to express concentration not in *moles* per unit volume but in *molecules* per unit volume, so the rate would be expressed in molecules dm^{-3} s^{-1} or molecules cm^{-3} s^{-1}.

7.3 PATH OF A CHEMICAL REACTION

Besides a study of the rate of a chemical reaction, a kinetic study also includes a study of the path. This involves a study of the energy required for the reaction to take place and the net energy changes that take place. A study of the path of a chemical reaction helps in predicting whether a reaction is exothermic or endothermic. One of the essential conditions for a chemical reaction to take place when two or more reactants are involved is that the reactants must come in close contact.

7.4 COLLISION THEORY

A chemical reaction takes place when reacting molecules collide with each other. But, all collisions do not change the reactants to products. Thus, there are collisions that are *effective* and collisions that are *ineffective*.

Effective collisions are those collisions that result in formation of products.

Ineffective collisions are those collisions that do not result in formation of products.

Conditions for effective collisions

For a collision to be effective there are two main conditions to be satisfied.

(i) The reactants must collide in a particular direction or the *orientation of the collision* should be correct and

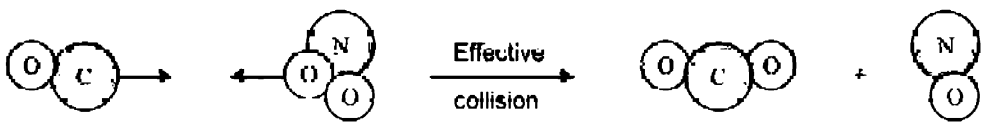
(ii) the reactants colliding must *have sufficient energy*.

When the reactants collide with each other *if the energy of the collision is sufficient and orientation of the collision is proper, then the collision is effective.*

Example : The reaction between CO and NO_2 is as follows :

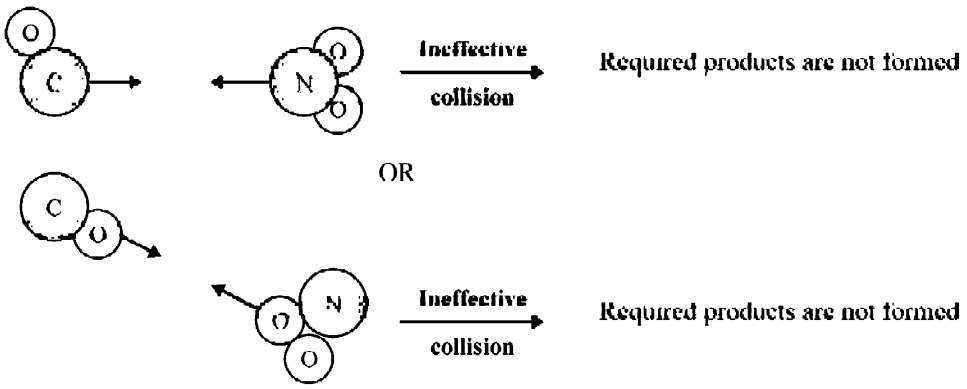
$CO + NO_2 \rightarrow CO_2 + NO$

Illustration of effective collision



CO and NO_2 are properly oriented and hence collide effectively to form CO_2 and NO.

Illustration of ineffective collision



CO and NO_2 are not properly oriented and hence do not make effective collision i.e., required products are not formed.

Activated Complex

After an effective collision the reactants form an intermediate stage called activated complex. This stage is a high-energy temporary stage. Once this stage is reached there is a rapid formation of products.

Conversely, when the reactants collide with each other and *if the energy in the collision is NOT sufficient or orientation of the collision is NOT proper, then the reactants cannot form products as the intermediate stage is not reached.*

The chemical reaction takes place in the following sequence :

Reactants $\xrightarrow[\text{proper orientation of the collision}]{\text{enough energy in the collision}}$ **Activated complex** $\xrightarrow{\text{changes rapidly to}}$ **Products.**

Illustration of formation of Activated Complex

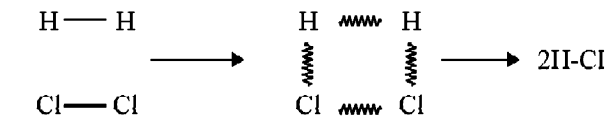
Hydrogen reacts with chlorine to form hydrogen chloride. In this reaction, a molecule of hydrogen collides with a molecule of chlorine to form a hydrogen chloride molecule.

Thus, the path of this chemical reaction is as follows.

1st stage: Existing bonds between hydrogen atoms in hydrogen molecule (H – H) and chlorine atoms in chlorine molecule (Cl – Cl) are broken.

2nd stage: New bond between hydrogen atom and chlorine atom forms. The *activated complex is that state* in which existing bonds are breaking and new bonds are forming.

3rd stage: The *activated complex* changes to the product (H-Cl).



Existing Bonds

Existing Bonds being broken, new ones being formed Activated Complex

7.5 INTERNAL ENERGY, ACTIVATION ENERGY AND THRESHOLD ENERGY

Internal Energy: Every particle possesses some energy called *internal energy*. The *internal energy* of a reactant (denoted for the purpose of this discussion as IE_R) is the sum of all the energies – potential energy, kinetic energy, vibration energy and rotation energy of the reactants. Similarly the products will also have a certain *internal energy* (denoted for the purpose of this discussion as IE_p).

Threshold energy: The reactants must have sufficient energy for an effective collision. The minimum energy that reactants *must have* to form an activated complex which then forms products is called *threshold energy*.

Activation energy: The *additional energy* that a reactant must absorb to get *threshold energy* is called *activation energy*.

Energy considerations in an effective collision

When a chemical reaction takes place the reactants absorb sufficient energy to reach threshold energy values and then form the activated complex, which then changes rapidly to products.

Thus,

(i) **Internal energy** is the energy possessed by the reactants.

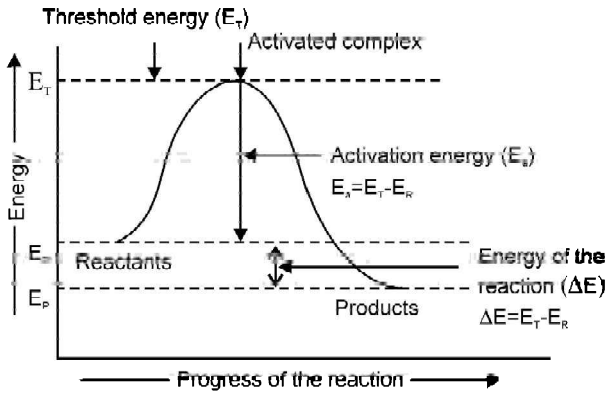
(ii) Activation energy is the energy absorbed by the reactants to reach the *activated phase.*

(iii) Threshold energy is the energy possessed by the *activated complex.*

threshold energy = internal energy + activation energy

$$E_T = IE_R + E_a$$

Illustration



Concept of activated complex

7.6 ENERGY CHANGES IN A CHEMICAL REACTION

A *reacting system* is the environment in which the reaction takes place. All chemical reaction take place with energy changes. Existing bonds have to be broken in the reactants – this requires absorption of energy. New bonds have to be formed in the products – this releases energy.

If E_R is the energy of the reactants and E_P is the energy of the products, the energy change (ΔE). in a chemical reaction is given by

$$\Delta E = E_P - E_R$$

Sign convention

Endothermic reactions

In endothermic reactions, heat is absorbed – the value of ΔE will be positive as $E_P > E_R$ (the energy of products is greater than the reactants). The products must have *absorbed this energy* from the reacting system.

A chemical reaction is said to be *endothermic* if heat is absorbed. In such reactions **ΔE** *is positive.*

Example: $CaCO_3 \rightarrow CaO + CO_2 + \Delta E$

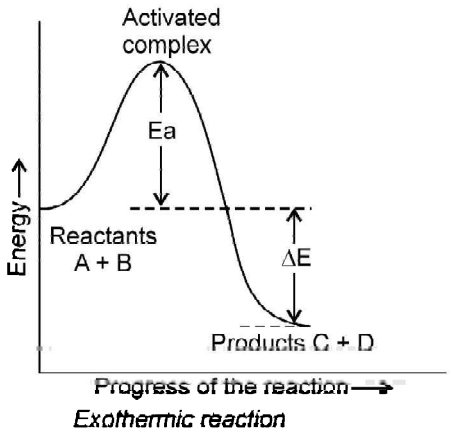
Exothermic reactions

In exothermic reactions, heat is released – the value of ΔE will be negative as $E_P < E_R$ (the energy of products is smaller than the reactants). The reacting system must have *released this energy* into the environment.

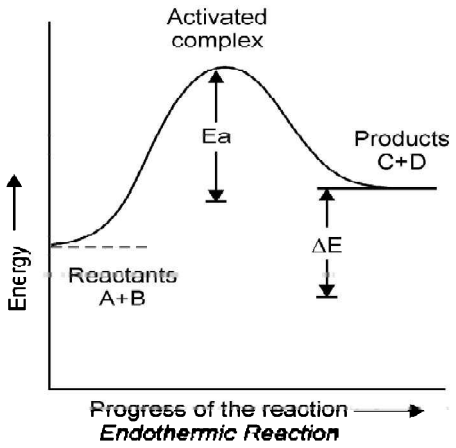
A chemical reaction is said to be *exothermic* if heat is released. In such reactions **ΔE** *is negative.*

Example: $2CO + O_2 \rightarrow 2CO - \Delta E$

Graphical representation of exothermic and endothermic reactions



Exothermic reaction



Endothermic Reaction

7.7 EFFECT OF CATALYST ON REACTION PATHS

A catalyst has the following effect on reaction paths :

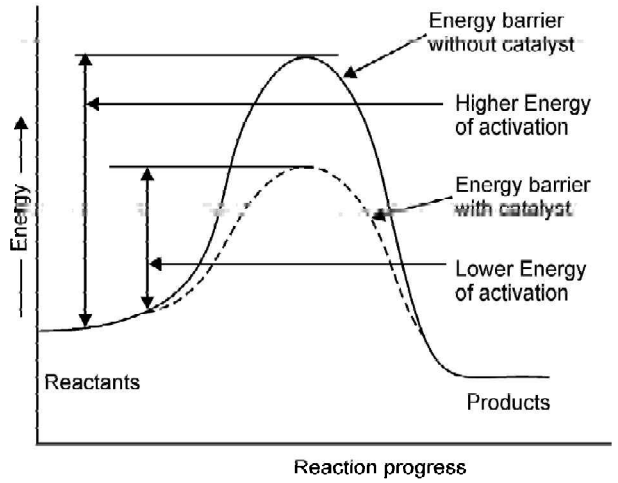
(a) A positive catalyst *decreases the threshold energy value*, so more number of effective collisions takes place. This increases the rate of reaction.

(b) The catalyst *forms an intermediate complex* increasing surface area of reactants so that the number of collisions increases.

(c) A negative catalyst however slows down a chemical reaction by *increasing the threshold energy value* so that less the number of effective collisions takes place. This reduces the rate of reaction.

Illustration

The effect of a positive catalyst is shown graphically for exothermic reactions:



The role of a catalyst in reducing the activation barrier

7.8 FACTORS AFFECTING REACTION RATES

The rate of a chemical reaction is affected by the following factors:

(a) Nature of reactants,

(b) Concentration of reactants,

(c) Surface area of reactants,

(d) Temperature of system,

For every 10°C rise in temperature the reaction rate doubles for most reactions.

(e) Light and (f) Catalysts.

7.9 LAW OF MASS ACTION (RATE LAW)

Consider a reacting system with more than one reactant represented by the stoichiometric equation :

$$aA + bB \rightarrow cC + dD$$

$$r \propto [A]^a[B]^b.$$

Mathematically, $r = k[A]^a[B]^b$,

where k is known as the rate constant or velocity constant.

Value of rate constant

*The value of rate constant will be equal to the rate of the reaction when the concentration of the reactants is unity (*1 mole/*l).*

Let [A] = [B] = 1 mole/*l*

Substituting [A] = [B] = 1 in rate law we get, r = k

Special cases

(i) The concentration for solid reactants is taken as unity as in solids the rate will depend on surface area and not on the concentration of reactants.

(ii) For gases the concentration is expressed in moles/litre.

Example: For the reaction : $S + O_2 \rightarrow SO_2$
the rate $\propto [O_2]$

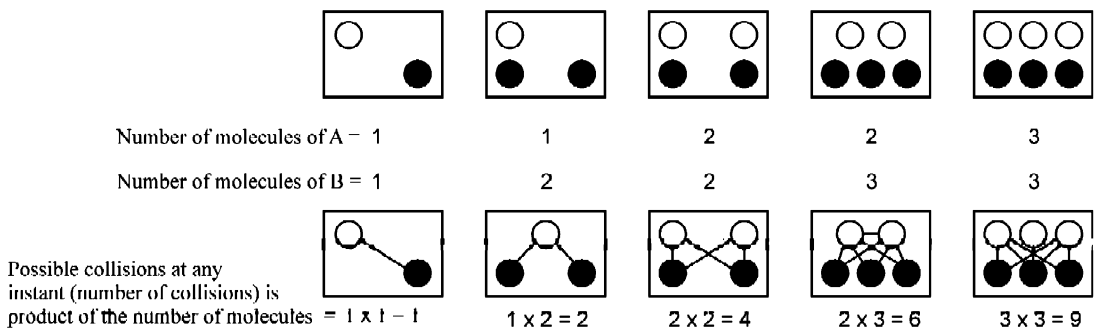
(iii) The *value of rate constant is dependent on temperature*. Its value is equal to rate when concentrations are taken to be unity. Since an increase in temperature will increase the number of collisions due to an increase in kinetic energy, rate constant is temperature dependent but not concentration dependent.

Explanation of rate of reaction in terms of collision theory

The possible collisions are shown in the illustration below.

Illustration

The number of possible collisions that can be effective is illustrated below. In the illustration, reactant A is represented by ● and reactant B is represented by ○.



The chances of collision are numerically equal to the product of the number of molecules of the reacting species at any instant.

From the above illustration it can be seen that the rate of reaction is proportional to the product of the concentration of the reactants. Each concentration in the rate law is raised to the power equal to the coefficient of that reactant in the stoichiometric equation.

The rate of reaction is proportional to the product of the concentration of the active mass and not the sum as the probability of collisions is a product of the actual number of different reacting species available. As concentration increases, the number of collisions increases, and therefore the rate of reaction increases.

7.10 REACTION MECHANISMS

When a chemical reaction takes place in more than one step, rate of reaction will depend on the slowest intermediate step.

To illustrate let us take the example of a toy manufactured in three steps :

Step 1 – This is a fast step so takes very little time.

Step 2 – This step is the slowest step so takes most time.

Step 3 – This step is the fastest step so takes the least time.

The rate of manufacturing the toy will depend on the step 2 as it is the slowest of all steps. Even if Step 1 is over quickly, Step 2 will take time and hold up Step 3. But once Step 2 is over Step 3 takes very little time.

Thus in a multi-stage reaction, the slowest intermediate stage will determine the overall rate of the reaction.

Illustrative examples

Example 1: $HIO_3 + 3H_2SO_3 \rightarrow HI + 3H_2SO_4$

According to the Law of Mass Action, the rate of this chemical reaction should be

$$r = k\,[HIO_3][H_2SO_3]^3$$

i.e., 1 molecule of HIO_3 has to collide with 3 molecules of H_2SO_3 for the reaction to take place. This seems to be unlikely.

The above reaction actually takes place in the following steps:

First step: $HIO_3 + H_2SO_3 \rightarrow HIO_2 + H_2SO_4$ –(1) slow step

Second step: $HIO_2 + H_2SO_3 \rightarrow HIO + H_2SO_4$ – (2) fast step

Third step: $HIO + H_2SO_3 \rightarrow HI + H_2SO_4$ –(3)fast step

So, in the above steps, the first step, which is slow determines the overall rate of the reaction.

In fact, the rate law for this reaction is found to be

$$r \propto [HIO_3]\,[H_2SO_3]$$

Example 2: $4HBr + O_2 \rightarrow 2H_2O + 2Br_2$

This reaction takes place in the following three steps:

First step: $HBr + O_2 \rightarrow HO_2Br$ – (1) slow step

Second step: $HO_2Br + HBr \rightarrow 2HOBr$ – (2) fast step

Third step: $HOBr + HBr \rightarrow H_2O + Br_2$– (3) × 2 fast step

So, in the above three steps, the first step, which is slow determines the overall rate of the reaction.

In fact, the rate law for this reaction is found to be $r \propto [HBr][O_2]$

Thus, the stoichiometric equation does not show how actually the reaction proceeds. To study the mechanism and find out the rate of a chemical reaction the reaction has to be studied kinetically.

SOLVED PROBLEMS

1. For a chemical reaction the concentration of reactants initially at 2.5 moles/*l* is decreased to 0.7moles/*l* after 3hrs. What is the rate of reaction?

Ans: Initial concentration 2.5 moles/*l*

Final concentration 0.7 moles/*l*

Change in concentration = 2.5 – 0.7 = 1.8 moles/*l*

Time taken = 3 hrs = 3 × 60 × 60 seconds = 10800 seconds

Rate = change in concentration in unit time= 1.8/ 10800 moles/*l*/ sec= 0.0002 moles/*l*/ sec

2. The rate constant of a reaction depends on:

(a) temperature (b) mass

(c) weight (d) time

Ans: temperature

3. Rate of a reaction doubles for every rise in temperature by

(a) 10C° (b) 20 C°

(c) 30 C° (d) 40 C°

Ans: 10C°

4. The excess energy required for an effective collision is:

(a) threshold energy (b) activation energy

(c) free energy (d) internal energy

Ans: activation energy

5. The rate of a reaction can be increased by all factors except:
   (a) using catalyst
   (b) increase of temperature
   (c) increase of activation energy
   (d) increase in concentration of reactants.
   Ans: increase of activation energy

6. The reaction $2NO + O_2 \rightarrow 2NO_2$ follows rate law : $r = k[NO]^2[O_2]$. By what factor does the rate change if the volume of the vessel is reduced to one-third of its initial volume?
   Ans: If the volume of the vessel is reduced to one-third, the same numbers of moles occupy one-third the volume. Thus, the concentration of each reactant becomes three times.

   Since $r = k[NO_2]^2[O_2]$

   Tripling the concentration of NO_2 would increase the rate by 9 times and tripling the concentration of O_2 would increase it by three times:

   $r' = k[3NO_2]^2[3O_2]$

   or, $r' = k\ 3^2[NO_2]^2 . 3[O_2]$

   $= 27 . k[NO_2]^2[O_2] = 27\ r$

   Thus rate increases by 27 times.

EXERCISE

I. Choose the correct answer

1. **A chemical reaction was carried out at 300 K and 280 K. The rate constants were found to be k_1 and k_2 respectively. Then**
   (a) $k_2 = 4k_1$ (b) $k_2 = 0.5\ k_1$
   (c) $k_2 = 0.25k_1$ (d) $k_2 = 0.05\ k_1$

2. **The rate of reaction is doubled for 10° C rise in temperature. The increase in the reaction rate as a result of temperature rise from 10° C to 100° C is**
   (a) 112 (b) 512
   (c) 1400 (d) 614

3. **The following mechanism has been proposed for the reaction $2A + B \rightarrow D + E$**

   $A + B \rightarrow C + D$ (slow)

   $A + C \rightarrow E$ (fast)

   The rate law expression for the reaction is
   (a) $r = k[A]^2[B]$ (b) $r = k[A][B]$
   (c) $r = k[A]^2$ (d) $r = [A][C]$

4. **Consider the chemical reaction, $N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$. The rate of this reaction can be expressed in terms of derivative of concentration of $N_2(g)$, $H_2(g)$ or $NH_3(g)$. Identify the correct relationship amongst the following rate expressions.**

   (a) Rate $= \frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$

   (b) Rate $= -\frac{d[N_2]}{dt} = -3\times\frac{d[H_2]}{dt} = 2\times\frac{d[NH_3]}{dt}$

   (c) Rate $= \frac{d[N_2]}{dt} = \frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$.

   (d) Rate $= -\frac{d[N_2]}{dt} = -\frac{d[H_2]}{dt} = \frac{d[NH_3]}{dt}$

5. **On increasing the temperature by 10 K, the rate of reaction becomes double. Which of the following is the most appropriate reasons?**
   (a) With increase of temperature, velocities increase and hence the number of collisions are appreciably increased.
   (b) The activation energy decreases with increase of temperature.
   (c) The bonds between the atoms of the reacting molecules become weak at higher temperature.
   (d) The collisions become more energetic at higher temperatures.

6. **The role of catalyst in a chemical reaction is to change**
   (a) heat of reaction (b) products of reaction
   (c) activation energy (d) equilibrium constant

7. **The ionic reactions are usually very fast because**
   (a) The energy of interaction between charged ions is greater than between neutral molecules
   (b) it does not involve bond breaking
   (c) the number of collisions between ions per unit volume per second are very large.
   (d) these reactions are highly exothermic

8. **In the presence of a catalyst, the heat evolved or absorbed during reaction**
   (a) decreases
   (b) increases
   (c) remains unaffected
   (d) may increase or decrease

9. **A catalyst**
   (a) increase the average kinetic energy of the reaction molecules.
   (b) decreases the value of enthalpy change in the reaction
   (c) reduces the time required for reaching the equilibrium state in the reaction
   (d) decreases the rate of reverse reaction

10. In the reversible reaction $2\ NO_2 \underset{k_2}{\overset{k_1}{\rightleftharpoons}} N_2O_4$, the rate of disappearance of NO_2 is equal to

(a) $\frac{2k_1}{k_2}[NO_2]^2$

(b) $2k_1[NO_2]^2 - 2k_2[N_2O_4]$

(c) $2k_1[NO_2]^2 - k_2[N_2O_4]$

(d) $(2k_1 - k_2)[NO_2]$

11. The rate of gaseous reaction is given by the expression k [A] [B]. If the volume of the reaction vessel is suddenly reduced to ¼ th of the initial volume, the reaction rate relating to original rate will be

(a) 1/10 (b) 1/8

(c) 8 (d) 16

12. Rate constant depends on

(a) temperature (b) time

(c) initial concentration (d) none

13. The reaction $2SO_2\ (g) + O_2\ (g) \rightleftharpoons 2SO_3\ (g)$ is carried out in 1 dm^3 vessel and 2 dm^3 vessel separately .The ratio of the reaction velocities will be

(a) 1 : 8 (b) 1 : 4

(c) 4 : 1 (d_ 8 : 1

II. Give one word answers

1. The change in concentration of the reactants in unit time
2. The type of reaction that takes place with respect to energy change if the internal energy of the products is greater than that of the reactants.
3. The type of reaction that takes place with respect to energy change if the internal energy of the products is smaller than that of the reactants.
4. The minimum energy that an activated complex possesses.
5. The collision that does not result in formation of product.

III. Match the following.

1. Match the property in Set A with the type of reaction in Set B

Set A	Set B
(i) Heat released	(a) reversible reaction
(ii) Heat absorbed	(b) irreversible reaction
(iii) R ← → P	(c) endothermic reaction
(iv) R → P	(d) exothermic reaction

2. Match Set A with Set B.

Set A	Set B
(i) Activation energy	(a) reactants do not change to products
(ii) Threshold energy	(b) energy required for effective collision
(iii) Effective collision	(c) energy to be provided for effective collision
(iv) Ineffective collision	(d) reactants change to products

3. Match the reactions in Set A with the shapes / structure in Set B

Set A	Set B
(i) $\Delta E > 0$	(a) no heat released or absorbed
(ii) $\Delta E < 0$	(b) endothermic reaction
(iii) IE of reactants < IE of products	(c) exothermic reaction
(iv) IE of reactants > IE of products	(d) catalytic reaction

IV. State whether the following statements are True or False

1. An effective collision between reacting molecules has proper orientation and threshold energy.
2. The activated phase in a chemical reaction is a temporary, high energy and unstable state.
3. The rate of a chemical reaction doubles for most reactions for every 100 C° rise in temperature

V. Read the paragraph below and answer the questions that immediately follow each paragraph. Each question has four options out of which only one option is correct. Choose the correct one.

Reversible reactions take place in closed containers only. Such reactions take place in both directions. In the beginning, the forward reaction will be fast, as the concentration of the reactants is high. More collisions take place between the reactants. So the probability of effective collisions is high and hence the rate is high. As the reaction progresses, the rate slows down. As the reactant molecules get converted to products, their concentration decreases. The concentration of the products increases. The backward reaction rate picks up. At a certain stage, the rates of forward reaction and backward reaction will become equal.

1. Closed containers are essential for reversible reaction to take place as

(i) reactants do not escape

(ii) reactants concentration can be kept constant

(iii) products do not escape
(iv) products concentration can be kept constant
(a) i and iii (b) ii and iii
(c) iii and iv (d) i and ii

2. **When the rate of forward and backward reaction are same, the concentrations of reactants and products are**
   (a) not constant
   (b) may or may not be constant
   (c) constant
   (d) can not say
3. **Effective collision rate will be high if**
   (a) concentrations are high
   (b) temperature is high
   (c) catalyst is present
   (d) all three (a, b and c)

VI. Fill in the blanks

1. The rate of a chemical reaction depends on _____, _____ and _____ of the reactants.
2. If the temperature of a reaction that is taking place decreases by 10° C then the rate of the reaction becomes _____.
3. Effective collisions have both _____ energy and _____ orientation of collision.
4. The rate at a given moment during a chemical reaction is called _____ rate.
5. The slope of the reactant curve in a concentration versus time graph for a chemical reaction gives the _____.

VII. Each question contains two statements. Statement A is an Assertion Statement and Statement R is the Reason Statement. Each question has four choices out of which only one is correct. Choose the correct one.

Choose A if Statement A is true, Statement R is true; Statement R is correct explanation for statement A

Choose B if Statement A is true, statement R is true; Statement R is not correct explanation for statement A

Choose C if Statement A is true, Statement R is false

Choose D if Statement A is false, statement R is true

1. **Statement –A:** The rate of a chemical reaction in solid phase is independent of pressure.

   Statement –R: Exerting pressure has no effect on solid reactants.
2. **Statement –A**: In an endothermic reaction heat is absorbed.

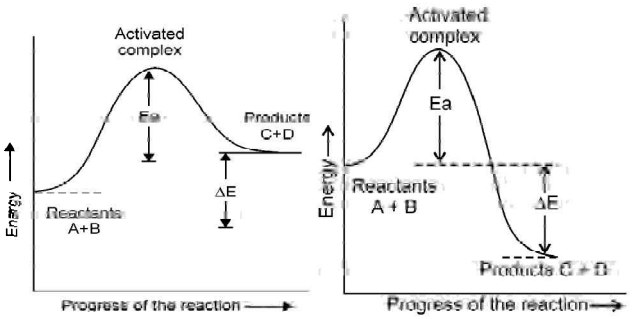
   Statement –R: Internal energy of reactants is smaller than that of products in endothermic reactions.
3. **Statement –A**: Catalysts can slow down chemical reactions.

   Statement –R: Negative catalysts decrease threshold energy values.
4. **Statement –A**: An increase in temperature decreases number of effective collisions.

   Statement –R: As temperature increases, kinetic energy of the reactant molecules increases.

VIII. Answer the following questions.

1. Predict the nature of the thermochemical reaction based on the following graphs.



2. For the reaction A + B → Products, the rate law is given by $r = k[A]^{1/3}[B]^2$. What are the units of the rate constant k?
3. The temperature of a reaction increases from 30° → 80°C. By what factor does the rate of reaction increase?
4. Consider the reaction A + B → C + D. The forward reaction is endothermic with 20kJ / mole of heat energy being absorbed. If the AE for forward reaction is 85kJ / mole, find graphically, the activation energy of backward reaction assuming that the reaction is reversible.
5. The experimental rate law is rate = $k[NO_2^-][H^+]^2[I^-]$ for the reaction

   $2NO_2^- + 4H^+ + 2I^- \rightarrow I_2 + 2NO + 2H_2O$

   How does the rate alter if
   (i) both $[H^+]$ and $[I^-]$ are kept constant but $[NO_2^-]$ is doubled.
   (ii) $[NO_2^-]$ and $[I^-]$ are kept constant but $[H^+]$ is doubled.
   (iii) All concentrations are doubled.

KEY TO EXERCISE

I. 1. c 2. b 3. b 4. b 5. a 6. c 7. c 8. b
9. c 10. c 11. d 12. a 13. a

II. 1. rate of chemical reaction 2. endothermic reaction 3. exothermic reaction
4. threshold energy 5. ineffective collision

III. 1. i. d ii. c iii. a iv. b 2. i. c ii. b iii. d iv. a
3. i. b ii. c iii. b iv. c

IV. 1. true 2. true 3. fasle

V. 1. a 2. c 3. d

VI. 1. nature, concentration and surface area 2. half
3. threshold, proper 4. instatntaneous
5. rate of chemical reaction

VII. 1. A 2. A 3. C 4. D

VIII. 1. Endothermic, exothermic 2. $(l$ / mole$)^{4/3}$sec^{-1} 3. 32 times
5. (i) rate doubles, (ii) 4 times, (iii) 16 times

8 Chemical Equilibrium

8.0 INTRODUCTION

Chemical equilibrium deals with reversible chemical changes that take place in closed containers in which reactants change to products and the products change to reactants under the same conditions. A study of chemical equilibrium involves a study of how changing the concentration of reactants and products, pressure and temperature in a closed container affect the forward reaction or the backward reaction. It studies the best conditions for increasing or decreasing one side of the reversible reaction – forward or backward. It helps the engineer in a chemical manufacturing unit to optimize the production. For example a study of the chemical equilibrium $N_2 + 3H_2 \rightleftharpoons 2NH_3$ tells the engineer the best conditions under which more ammonia can be produced in the least time.

8.1 REVERSIBLE AND IRREVERSIBLE CHANGES

A chemical reaction is a chemical change where a substance changes to a new substance under certain conditions of temperature and pressure, with or without the presence of a catalyst. The change can be *reversible* or *irreversible*.

A chemical change represented by Reactants → Products is considered to be *reversible* if under the same conditions the products change to reactants, i.e., Products → Reactants. Such changes are represented using a double-headed arrow: $\rightleftharpoons$

Example: Decomposition of nitrogen tetraoxide into nitrogen dioxide is a reversible change. This change is represented as $N_2O_4 \rightleftharpoons 2NO_2$

If on the other hand the products cannot be changed to reactants again under the same conditions then the reaction is said to be *irreversible*. Such changes are represented using a single-headed arrow: →

Example: Zinc granules react irreversibly with dilute hydrochloric acid to liberate hydrogen.

$Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$

8.2 FORWARD REACTION AND BACKWARD REACTIONS

In a reversible reaction, the change in which reactants change to products (R → P) is the *forward reaction* and the reaction in which products change back to reactants (P → R) is the *backward reaction.*

8.3 STATE OF CHEMICAL EQUILIBRIUM

Consider a reversible reaction

$$A + B \rightleftharpoons C + D$$

When some quantity of A and B are mixed in a closed container, the forward reaction starts taking place. Initially, the concentration of reactants, A and B is maximum and hence only the forward reaction takes place. *The rate of forward reaction is high.*

As concentration of C and D increase, they start reacting to give A and B back. *The rate of backward reaction slowly starts increasing.*

After sometime as the concentration of the reactants starts decreasing, the rate of forward reaction slowly starts decreasing and as concentration of products starts increasing, the rate of backward reaction starts increasing.

After some time, the rate of forward reaction becomes equal to the rate of backward reaction. At this stage there is no change in either the rate of forward or backward reactions – they remain equal as long as the conditions do not change. At this stage, the concentration of reactants and products become constant and do not change. The concentration of reactants and products do not change because, as the reactant change to become products, simultaneously the products change back to reactants.

In a reversible chemical change, the state of chemical equilibrium is the stage in which the rate of forward reaction is equal to the rate of backward reaction.

Illustration

Consider the same reversible chemical reaction.

	A +	B $\rightleftharpoons$	C +	D
Concentration initially, $t = 0$	max	max	0	0
Concentration at t second	***decreases***	***decreases***	***increases***	***increases***
Concentration at equilibrium	***constant***	***constant***	***constant***	***constant***

8.4 DYNAMIC NATURE OF CHEMICAL EQUILIBRIUM

State of chemical equilibrium is a *dynamic state and* not a static state. Apparently everything seems to be at rest or static as nothing seems to be happening in the reaction container. *But at any given moment both the reactions – forward and backward reactions are taking place - only their rates are same and so no change seems to take place.*

Illustration of dynamic and static equilibriums

Consider two specially designed water tanks as shown below.

In the first water tank the rate of flow of water into the tank is equal to the rate of flow of water out of the tank. Hence, there is no change in the height of water column in the tank. Though fresh water is entering the tank continuously and the old water is leaving the tank continuously there seems to be no change as the level of water in the tank is constant. The water in this tank is said to be in a state of ***dynamic equilibrium*** – an equilibrium in which though there is a change taking place *there is no net change.*

Chemical equilibriums are dynamic equilibriums.

In the second tank however, water is neither entering the tank nor leaving the tank. The height of the water in the two parts of this tank is therefore constant. The water in this tank is said to be in a state of ***static equilibrium.***

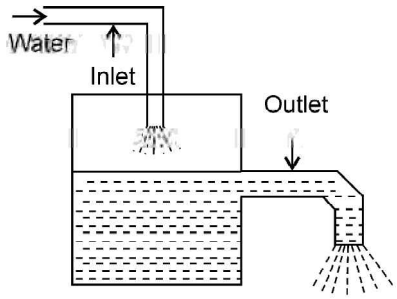
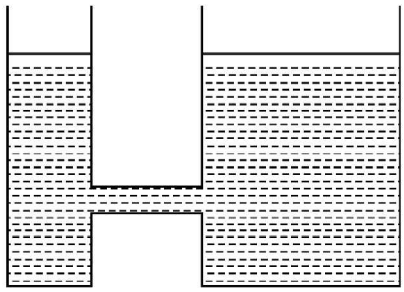


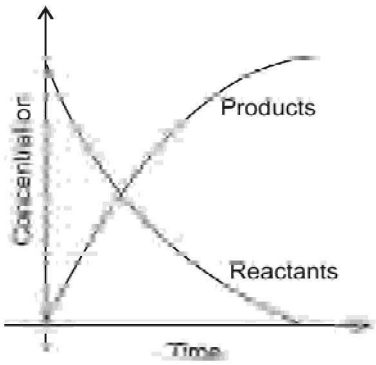
Illustration of dynamic equilibrium


Illustration of Static equilibrium

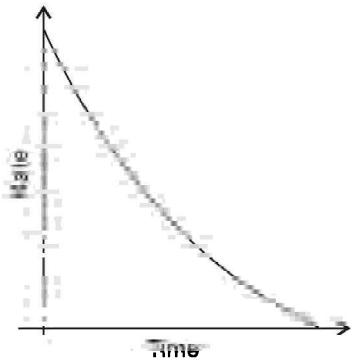
8.5 GRAPHICAL REPRESENTATION OF IRREVERSIBLE REACTIONS

An *irreversible reaction* is represented graphically as follows.

Concentration graph



Rate graph



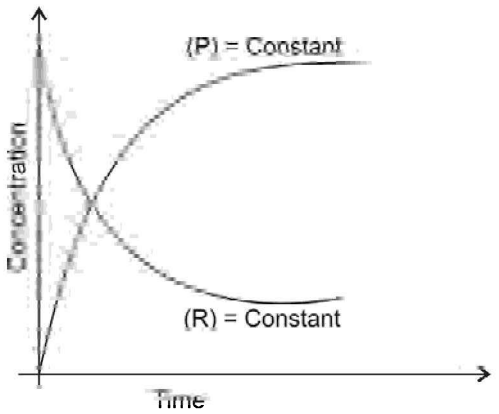
Concentration Graph: In an irreversible reaction, [R] decreases with time and [P] increases with time. After some time [R] = 0 and the reaction stops.

Rate Graph: In an irreversible reaction, only the forward reaction takes place. The rate of the forward reaction is high initially and then decreases to become zero as the reaction stops.

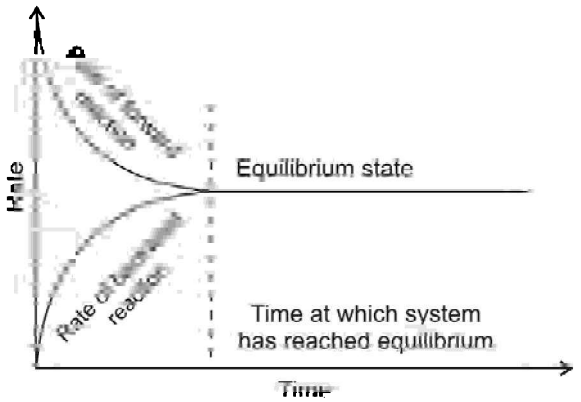
8.6 GRAPHICAL REPRESENTATION OF THE STATE OF CHEMICAL EQUILIBRIUM

A *reversible reaction* is represented graphically as follows.

Concentration graph



Rate graph



Concentration Graph: In a reversible reaction taking place in a closed container, [R] decreases with time and [P] increases with time. In the state of chemical equilibrium [R] and [P] become constant. Both forward and backward reactions take place simultaneously.

Rate Graph: In a reversible reaction taking place in a closed container, the rate of forward reaction is high initially – decreasing with time. The rate of the backward reaction is zero initially and slowly increases. After sometime both the rates become equal.

8.7 CHARACTERISTICS OF CHEMICAL EQUILIBRIUM

For a chemical reaction to reach equilibrium following characteristics are needed.

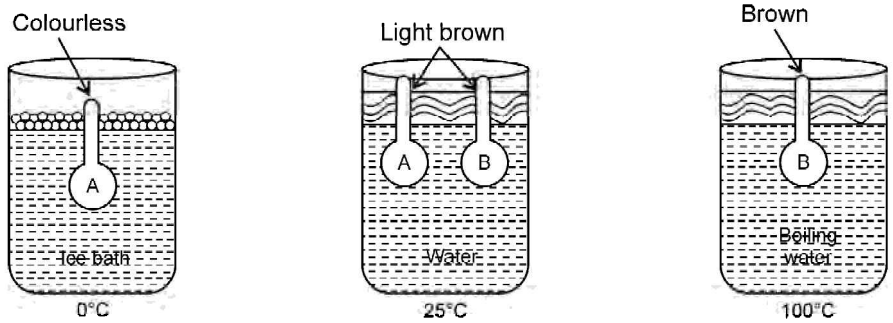
1. For a reversible chemical reaction to reach a state of chemical equilibrium *the reaction must take place in a closed container*, so that reactants and products do not escape.
2. In the state of chemical equilibrium, the *physical conditions like temperature, pressure and concentration remain unchanged.*
3. Chemical equilibrium is *dynamic* in nature. The reaction seems to have stopped. But, at any given moment, both reactions are going on.
4. *The rate of forward reaction = rate of backward reaction.*
5. *As long as the physical conditions are not changed the composition of the equilibrium mixture does not change with time.*
6. *Equilibrium can be attained from both sides*, i.e., starting with reactants or starting with products. We can start with reactants or products to reach the state of chemical equilibrium.

Illustration

The equilibrium $\underset{\text{Reddish brown}}{2NO_2} \rightleftharpoons \underset{\text{Colourless}}{N_2O_4}$ can be attained starting with NO_2 or starting with N_2O_4.

In other words, the equilibrium can also be shown as $N_2O_4 \rightleftharpoons 2NO_2$

Experiment: Take two identical glass bulbs A and B and fill them with NO_2 gas at equal pressures. Immerse the bulb A in an ice-bath at 0°C and bulb B in boiling water bath at 100°C. After some time the gas in bulb A at 0°C becomes colourless while that in bulb B at 100°C becomes reddish brown. Obviously in A the gas in N_2O_4 and in B it is NO_2.



Chemical equilibrium can be attained from either side of a reversible reaction

Now take out both the bulbs and put them in a water-bath at 25°C. After some time the colour of both the gases begins to change and ultimately becomes equally brown, i.e., the colour of the gas in both the bulbs becomes deeper than that at 0°C and lighter than that at 100°C. This is a visual observation by which we have been able to see the state of equilibrium being reached. This experiment shows that chemical equilibrium can be attained from either side of a reversible reaction.

8.8 TYPES OF CHEMICAL EQUILIBRIUM

Chemical equilibrium are of two types depending on the physical states of the reactants and products.

Homogeneous equilibrium is an equilibrium in which all reactants and products are in a single physical state.

Example $H_{2(g)} + I_{2(g)} \rightarrow 2HI_{(g)}$

Heterogeneous equilibrium is an equilibrium in which the reactants and products are not in same physical states.

Example $CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$

8.9 LAW OF CHEMICAL EQUILIBRIUM (LAW OF MASS ACTION)

Consider the chemical equilibrium

$A + B \rightleftharpoons C + D$

Applying Law of Mass Action to forward reaction $r_f \propto [A][B]$

$\therefore r_f = k_f [A][B]$, where k_f *is rate constant of forward reaction*

Similarly, *applying Law of Mass Action to backward* $r_b \propto [C][D]$

$r_b = k_b [C][D]$, where k_b *is rate constant of backward reaction.*

Since at equilibrium,

rate of forward reaction = rate of backward reaction

$$\text{i.e., } r_f = r_b$$
$$k_f[A][B] = k_b[C][D],$$

Rearranging, $K_c = \frac{k_f}{k_b} = \frac{[C][D]}{[A][B]}$ where K_c *is the Equilibrium constant.*

Thus, K_c is a ratio of the rate constants of forward and backward reactions.

Consider the general equation

$aA + bB \rightleftharpoons cC + dD$.

Representing concentrations in short form

The concentration of reactants is expressed by putting the molecular formula in between square brackets.

For example, $[NH_3]$ means concentration of ammonia.

The expression for Equilibrium Constant is given by $K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$

This expression is the Law of Chemical Equilibrium.

8.10 SIGNIFICANCE OF EQUILIBRIUM CONSTANT

1. Knowledge of the equilibrium constant, K_c is of immense use to chemical engineers. A large K_c value indicates forward reaction has proceeded to a large extent before equilibrium state was reached and vice-versa. This is used to predict whether it is practical to prepare a chemical by a given process.
2. If equilibrium constant, K_c and initial concentration of reactants are known, the concentration of all substances in the reaction system at equilibrium can be calculated. This will be illustrated in the problems below.

8.11 LE CHATELIER'S PRINCIPLE

The variables in a chemical equilibrium are *temperature, pressure, concentration, and catalyst.*

According to **Le Chatelier's Principle** *if any of the physical conditions of a chemical equilibrium are changed, the equilibrium shifts in the direction that will nullify the effect of that change.*

Explanation of the principle

The physical properties that can be changed are temperature or pressure or concentration of any component or all of these in the chemical equilibrium.

If the direction of equilibrium shifts backward, *the rate of backward reaction increases.*

If the direction of equilibrium shifts forward, *the rate of forward reaction increases.*

8.12 EFFECTS OF CHANGING PHYSICAL CONDITIONS OF A CHEMICAL EQUILIBRIUM

The change in number of molecules between reactants and products can be denoted by Δn. Some

reactions proceed with an increase in number of molecules while others proceed with a decrease and yet others with no change in the number of molecules.

Examples: $\Delta n = 0$ for the reaction

$N_2 + O_2 \rightleftharpoons 2NO$

$(\Delta n = 2 - 2 = 0)$

$\Delta n < 0$ for the reaction

$N_2 + 3H_2 \rightleftharpoons 2NH_3$

$(\Delta n = 2 - 4 = -2)$

$\Delta n > 0$ for the reaction

$2SO_3 \rightleftharpoons 2SO_2 + O_2$

$(\Delta n = 3 - 2 = 1)$

(a) Effect of changing pressure

Consider a chemical equilibrium where $\Delta n < 0$ (a decrease in number of moles in the forward direction). In the forward reaction there would be a decrease in pressure as the number of moles decreases.

If the pressure in the reacting system is increased, then the equilibrium shifts in the forward direction to offset the increase in pressure.

This happens because in the forward direction there is a decrease in pressure, which nullifies the change in the form of increase in pressure.

Example: Consider the reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$.

In the forward direction, there is a decrease in volume (no. of moles).

1 vol of N_2 reacts with 3 vol of H_2 to give only 2 vol of NH_3

So the effect of increase in pressure is nullified when the *rate of forward reaction increases*.

In other words, *equilibrium shifts in the forward direction*.

Thus, if a chemical engineer wants to increase the production of ammonia he should increase the pressure in the reacting system.

In fact, the pressure used in Born-Haber process for synthesis of ammonia is 200-900atm.

(b) Effect of changing temperature

A reaction that is exothermic in the forward direction in a chemical equilibrium will be endothermic in the backward direction and vice-versa.

For a reaction, which is exothermic in the forward direction an increase in temperature will favour the backward reaction as in this direction there is a decrease in temperature.

Example: Consider the reaction

$CaCO_3 \rightleftharpoons CaO + CO_2 + \Delta H$ (endothermic).

In the forward direction the reaction is endothermic. So the effect of increasing the temperature is nullified when *rate of the forward reaction increases*. In other words, *equilibrium shifts in forward direction*.

(c) Effect of changing concentration

While for solid reactants concentration is taken to be unity, the effect of changing concentration for non-solid reactants or products is as illustrated in the examples.

In the equilibrium $N_2 + 3H_2 \rightleftharpoons 2NH_3 - \Delta H$,

increasing the concentration of N_2 will favour formation of more ammonia. Alternatively, if ammonia (NH_3) is removed from the equilibrium mixture the equilibrium will shift in forward direction.

(d) Effect of catalyst

A catalyst only speeds up a chemical reaction and helps to attain the equilibrium faster. The equilibrium itself is not affected. In the presence of a catalyst the rate of forward reaction increases. The products are formed faster. *This increases the rate of backward reaction.*

Thus the presence of a catalyst, only establishes the equilibrium state faster but has ***no net effect on the equilibrium.***

To illustrate, a reversible chemical reaction $A + B \rightleftharpoons C + D$ reaches chemical equilibrium without catalyst in 30 minutes and with a catalyst in 5 minutes.

8.13 APPLICATION OF LE CHATELIER PRINCIPLE

The effect of changing each of the physical condition of a chemical reaction is tabulated below. How the data given is to be read is illustrated for the first example. Other examples can be read from the table similarly.

(a) If pressure is increased in the chemical equilibrium $N_2 + 3H_2 \rightleftharpoons 2NH_3$, since $\Delta n < 0$, the equilibrium shifts in the forward direction.

Variable	Nature	Example	Change	Equilibrium shift
(a) Pressure	$\Delta n < 0$	$N_2 + 3H_2 \rightleftharpoons 2NH_3$	increase in pressure	forward
(b) Pressure	$\Delta n < 0$	$N_2 + 3H_2 \rightleftharpoons 2NH_3$	decrease in pressure	backward
(c) Pressure	$\Delta n > 0$	$PCl_5 \rightleftharpoons PCl_3 + Cl_2$	increase in pressure	backward
(d) Pressure	$\Delta n > 0$	$PCl_5 \rightleftharpoons PCl_3 + Cl_2$	decrease in pressure	forward
(e) Temperature	Endothermic	$CO+H_2O \rightleftharpoons CO_2+H_2 - \Delta H$	increase in temperature	forward
(f) Temperature	Endothermic	$CO+H_2O \rightleftharpoons CO_2+H_2 - \Delta H$	decrease in temperature	backward
(g) Temperature	Exothermic	$2S + 3O_2 \rightleftharpoons 2SO_3 - \Delta H$	increase in temperature	backward
(h) Temperature	Exothermic	$2S + 3O_2 \rightleftharpoons 2SO_3 - \Delta H$	decrease in temperature	forward
(i) Concentration	Reactant	$2S + 3O_2 \rightleftharpoons 2SO_3$	increase in reactants	forward
(j) Concentration	Reactant	$2S + 3O_2 \rightleftharpoons 2SO_3$	decrease in reactants	backward
(k) Concentration	Product	$2S + 3O_2 \rightleftharpoons 2SO_3$	increase in products	backward
(l) Concentration	Product	$2S + 3O_2 \rightleftharpoons 2SO_3$	decrease in products	forward

8.14 SOME SPECIFIC REACTIONS AND FAVOURABLE CONDITIONS

1. **Born-Haber Process for manufacture of Ammonia**

Nature – exothermic,

Reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3 - \Delta H$

So, the conditions, which favour formation of NH_3 are

(i) low temperature – actual temperature used is 450-500°C

(ii) high pressure – actual pressure used is 200 – 900 atm

(iii) concentration – excess of N_2 or H_2 (1 : 3 mixture of N_2 and H_2 is used)

(iv) catalyst - Molybdenised iron

2. **Contact process for manufacture of H_2SO_4** (The first step of this process is burning of sulphur)

Nature – exothermic

Reaction $S + O_2 \rightleftharpoons SO_2 - \Delta H$

So, the conditions, which favour SO_2 formation, are

(i) low temperature, and

(ii) high pressure

(iii) with excess of O_2.

3. **Birkeland-Eyde Process for manufacture of HNO_3** (The first step of this process is oxidation of nitrogen)

Nature – endothermic,

Reaction $N_2 + O_2 \rightleftharpoons 2NO + \Delta H$.

So, the conditions, which favour NO formation, are

(i) high temperature, (Note: Pressure will not have any effect as $\Delta n = 0$)

(ii) excess of N_2 / O_2.

4. **Bosch process for the manufacture of water gas**

Nature – exothermic,

Reaction $CO + H_2O \rightleftharpoons CO_2 + H_2 - \Delta H$

So, the conditions, which favour water gas formation, are

(i) low temperature,

(ii) excess of steam.

SOLVED QUESTIONS

1. The following equilibrium concentrations were measured at 800K: $[SO_2] = 3 \times 10^{-3}$, $[O_2] = 3.5 \times 10^{-3}$ and $[SO_3] = 5 \times 10^{-2}$ moles/*l*. Find K_c at 800K for the chemical reaction at equilibrium in both directions. The reaction is

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$$

Ans: $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$

At equilibrium $[SO_2] = 3 \times 10^{-3}$,

$[O_2] = 3.5 \times 10^{-3}$ and

$[SO_3] = 5 \times 10^{-2}$ moles/*l* at 880K

$$= K_c = \frac{[SO_3]^2}{[SO_2]^2[O_2]} = \frac{(5\times10^{-2})^2}{(3\times10^{-3})^2(3.5\times10^{-3})}$$

$$= \frac{25\times10^5}{31.5} = 7.94\times10^4$$

For reverse reaction, $K_c = \frac{1}{7.9 \times 10^4} = 1.2 \times 10^{-5}$

2. PCl_5 dissociates into PCl_3 and Cl_2 in a 2*l* flask at about 600°C. At equilibrium, the mixture is found to contain 1 mole PCl_5 and 2 moles each of PCl_3 and Cl_2. Find

(a) Equilibrium molar concentrations, and (b) Equilibrium constant, K_c for the chemical equilibrium:

$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

Ans: $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

At equilibrium $[PCl_5] = 1$ mole in 2*l* flask $= 0.5$ mole/*l*

$[PCl_3] = 2$ mole in 2*l* flask $= 1$ mole/*l*

$[Cl_2] = 2$ mole in 2*l* flask $= 1$ mole/*l*

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{1 \times 1}{0.5} = 2$$

EXERCISE

I Choose the correct answer

1. For the reaction $2HI \rightleftharpoons H_2 + I_2$
(a) $K_p > K_c$ (b) $K_c > K_p$
(c) $K_p = K_c$ (d) none of these

2. If $N_2 + 3H_2 \rightleftharpoons 2NH_3$ has equilibrium constant K and $2N_2 + 6H_2 \rightleftharpoons 4NH_3$ has equilibrium constant K′, then K′ =
(a) K (b) $\sqrt{K}$
(c) $\frac{1}{\sqrt{K}}$ (d) $\frac{1}{K^2}$

3. The equilibrium constant for the reaction $3A + 2B \rightarrow C$ will be
(a) $\frac{[3A][2B]}{[C]}$ (b) $\frac{[C]}{[3A][2B]}$
(c) $\frac{[C]}{[A]^2[B]^2}$ (d) $\frac{[C]}{[A]^3[B]^2}$

4. In the reaction N_2 (g) + O_2 (g) $\rightleftharpoons$ 2NO (g), H = + 180.7 kJ. On increasing the temperature the production of NO
(a) increases (b) decreases
(c) remains same (d) cannot be predicted

5. If K_1 and K_2 are the equilibrium constants of the equations (i) and (ii) respectively given below. What is the relationship between the two constants?
(i) $SO_{2(g)} + \frac{1}{2} O_{2(g)} \rightleftharpoons SO_{3(g)}$; K_1
(ii) $2SO_{3(g)} \rightleftharpoons 2SO_{2(g)} + O_{2(g)}$; K_2
(a) $(K_1)^2 = \frac{1}{K^2}$ (b) $K_2 = (K_1)^2$
(c) $K_1 = \frac{1}{K^2}$ (d) $K_1 = K_2$.

6. What are the most favourable conditions for the reaction $SO_2 + \frac{1}{2} O_2 \rightleftharpoons SO_3$ to occur; H is negative.
(a) low T, high P (b) low T, and low P
(c) high T, and low P (d) high T, high P

7. Which of the following favours the reverse reaction in chemical equilibrium?
(a) Increasing the concentration of the reactant
(b) Removal of at least one of the products at regular intervals
(c) Increasing the concentration of one or more of the products
(d) Increasing the pressure

8. A reversible reaction is one which
(a) proceeds in one direction
(b) proceeds in both direction
(c) proceeds spontaneously
(d) all the statements are wrong

9. The law of mass action was proposed by
(a) Guldberg and Waage
(b) Le Chatelier and Braun
(c) Kossel and Lewis
(d) van't Hoff

10. Theory of active phase indicates that the rate of a chemical reaction is directly proportional to the
(a) equilibrium constant
(b) properties of reactants
(c) volume of apparatus
(d) concentration of reactants

11. The equilibrium concentration of x, y and yx_2 are 4, 2 and 2 respectively for the equilibrium
$2x + y \rightleftharpoons yx_2$. The value of equilibrium constant, K_c is
(a) 0.625 (b) 6.25
(c) 0.0625 (d) 62.5

12. For a reversible reaction if the concentrations of the reactants are doubled at a given temperature, then equilibrium constant will
(a) also be doubled (b) be halved
(c) become one-fourth (d) remain the same

13. In which of the following, the reaction proceeds to almost completion?
(a) K =1 (b) $K = 10^{-2}$
(c) K = 10 (d) $K = 10^3$

14. In the reaction, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $\Delta H = -93.6$ kJ the yield of ammonia does not increase when

(a) pressure is increased
(b) pressure is decreased
(c) temperature is lowered
(d) volume of the reaction vessel is decreased

15. A cylinder provided with a piston has some PCl_5 which is in equilibrium with PCl_3 and Cl_2. The system is compressed with the help of piston. Indicate the correct statement.

(a) Some more PCl_5 will decompose
(b) The system remains unaffected
(c) PCl_3 and Cl_2 will combine to form PCl_5
(d) Explosion occurs

16. In an exothermic reaction, a 10^0C rise in temperature will

(a) decrease the value of equilibrium constant
(b) double the value of equilibrium constant
(c) not produce any change in equilibrium constant
(d) produce some increase in equilibrium constant

17. Consider the reaction $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ in closed container at equilibrium. What would be the effect of addition of $CaCO_3$ on the equilibrium concentration of CO_2?

(a) increases (b) decreases
(c) remains unaffected
(d) data is not sufficient to predict it.

18. According to Le Chatelier's principle, increasing temperature of a solid and liquid in equilibrium will cause the

(a) amount of solid to decrease
(b) amount of liquid to decrease
(c) temperature to rise
(d) temperature to fall

19. In a vessel containing SO_3, SO_2 and O_2 at equilibrium $[2SO_{3(g)} \rightleftharpoons 2SO_{2(g)} + O_{2(g)}]$ some helium gas is introduced so that total pressure increases while temperature and volume remain the same. According to Le Chatelier's principle, the dissociation of SO_3

(a) increases (b) decreases
(c) remain unaltered
(d) changes unpredictably

20. Reaction A + B → C + D + 38 kcal has activation energy 20 kcal. Activation energy for the reaction C + D → A + B is

(a) 20 kcal (b) – 20 kcal
(c) 18 kcal (d) 58 kcal

21. State of chemical equilibrium is

(a) dynamic (b) static
(c) difficult to maintain
(d) may be dynamic or static

22. Equilibrium constant depends on

(a) temperature
(b) the presence of catalyst
(c) the absence of catalyst
(d) concentration

23. A large increase in the rate of reaction on increasing temperature is due to

(a) the increase in number of collisions
(b) increase in number of activated phase
(c) increase in products
(d) lowering of activation energy

II. Give one word answers

1. A chemical reaction in which the rate of forward reaction is equal to the backward reaction.
2. The composition of the reaction mixture of a chemical reaction at equilibrium.
3. The direction in which a chemical reaction at equilibrium shifts if the concentration of products is increased.
4. The direction in which an endothermic chemical reaction at equilibrium shifts if the temperature is lowered.
5. The direction in which a chemical reaction at equilibrium shifts if $\Delta n < 0$, if the pressure is increased.

III. This section contains three questions. Each question contains statements given in two columns, which have to be matched. Statements (i, ii, iii, iv) in Set A have to be matched with Statements (a, b, c, d) in Set B. More than one match or no match between options in the Sets is possible.

1. Match the change in property in Column I of a chemical equilibrium R → P where $\Delta n = 0$ and the reaction is endothermic in forward direction with the change that takes place in rate in Column II.

Set A	Set B
(i) Increasing pressure	(a) forward reaction rate increases
(ii) Decreasing pressure	(b) forward reaction rate decreases
(iii) Increasing temperature	(c) backward reaction rate increases
(iv) Decreasing temperature	(d) backward reaction rate decreases

2. Match the reactions at equilibrium in Column I with the conditions in Column II that will slow down the forward reaction.

Set A	Set B
(i) $N_2 + 3H_2 \rightarrow 2NH_3$	(a) increasing pressure
(ii) $PCl_5 \rightarrow PCl_3 + Cl_2$	(b) decreasing pressure
(iii) $CO + H_2O \rightarrow CO_2 + H_2$ at 100^0C	(c) increasing temperature
(iv) $2S + 3O_2 \rightarrow 2SO_3$	(d) decreasing temperature

IV. State whether the following statements are True or False

1. The state of chemical equilibrium is static rather than dynamic.
2. Increasing the pressure of a chemical equilibrium in which $\Delta n = 0$ will increase the rate of forward reaction.
3. A catalyst only helps in reaching the equilibrium state faster.

V. Each question contains two statements. Statement A is an Assertion Statement and Statement R is the Reason Statement. Each question has four choices out of which only one is correct. Choose the correct one.

Choose A if Statement A is true, Statement R is true; Statement R is correct explanation for statement A

Choose B if Statement A is true, statement R is true; Statement R is not correct explanation for statement A

Choose C if Statement A is true, Statement R is false

Choose D if Statement A is false, statement R is true

1. **Statement –A:** Concentrations in an equilibrium mixture remain constant.

   Statement –R: The rates of forward reaction and backward reaction are equal.
2. **Statement –A**: The velocity constant of reaction is dependent on temperature.

   Statement –R: Velocity constant is a constant, which does not change for a reaction.
3. **Statement –A**: An exothermic reaction at equilibrium will proceed faster in the forward direction if temperature is decreased.

   Statement –R: The change in equilibrium is nullified if rate of forward reaction is increased.
4. **Statement –A**: If $\Delta n = 0$ then the equilibrium is effected by increase or decrease in pressure.

   Statement –R: Increase or decrease in pressure changes the concentration.

KEY TO EXERCISE

I.	1. c	2. a	3. d	4. a	5. c	6. a	7. c	8. b
	9. a	10. d	11. c	12. d	13. d	14. b	15. c	16. a
	17. c	18. a	19. c	20. d	21. a	22. a	23. b	
II.	1. chemical equilibrium		2. constant	3. forward	4. backward	5. forward		
III.	1. i. b, c	ii. a, d	iii. a, d	iv. b, c	2. i. b, d	ii. a, c	iii. d	iv. b, c
IV.	1. false	2. false	3. true					
V.	1. A	2. C	3. A	4. D				

9 Metallurgy

9.0 INTRODUCTION

Elements can be broadly classified into metals, non-metals, metalloids and inert gas elements. Many metals are used in our day-to-day lives.

Some of the metals that are useful to man in the elementary state are iron, aluminium, copper, zinc, gold and silver. Most metals that are useful for man are not found in the elementary state but as compounds in *ores*. The compounds of the metals found in nature are called *minerals*. The naturally occurring minerals from which metals can be conveniently and economically extracted are called *ores*. Metallurgy is the branch of chemistry that deals with extraction of metal from its *ore*.

Almost 80% of the known elements are metals. There are approximately 80 metals that occur naturally in the earth's crust.

9.1 CHARACTERISTICS OF METALS

In general, metals exhibit the following characteristics:

(i) All metals are solids at room temperature excluding mercury which is a liquid.

(ii) Metals are good conductors of heat and electricity.

(iii) Metals are malleable, ductile and sonorous and have lustre.

(iv) Metals occur to the left and centre in the periodic table. Hydrogen is the only non-metal placed to the left in the periodic table.

(v) Metals lose electrons relatively more easily when compared to non-metals to form cations. (Hydrogen is the only element which is not a metal but which also loses electrons).

(vi) The oxides of metals are mostly basic in nature. Some metal oxides are amphoteric in nature. (Example: Sodium oxide – Na_2O, is basic while aluminium oxide – Al_2O_3, is amphoteric).

9.2 CHARACTERISTICS OF NON-METALS

In general, non-metals exhibit the following characteristics:

(i) Non-metals are soft solids or gases. The only liquid non-metal is bromine.

(ii) Non-metals are poor conductors of heat and electricity.

(iii) Non-metals are not malleable, not ductile and are not sonorous. They do not have lustre.

(iv) Non-metals occur to the right in the periodic table. Hydrogen is the only non-metal placed to the left in the periodic table.

(v) Non-metals gain electrons relatively more easily when compared to metals to form anions.

(vi) The oxides of non-metals are mostly acidic in nature. Some non-metal oxides are neutral in nature. (Example: Sulphur dioxide – SO_2 is acidic while N_2O is neutral).

9.3 OCCURRENCE

The metals occur in the earth's crust as minerals or ores. The order of abundance of metals in the earth's crust is

aluminium > iron > calcium > sodium > potassium > magnesium > titanium.

Differentiating between minerals and ores

Minerals are natural materials found in the earth's crust.

Ores are minerals from which metals can be extracted conveniently and economically.

Most metals occur in the combined state as minerals. Out of these minerals some are *ores*.

Ores, from which metals can be extracted conveniently and economically, occur as oxides, carbonates, sulphides, sulphates, halides, nitrates, etc. Some metals like gold, silver, copper and platinum occur free as they do not react easily with water, carbon dioxide, etc.

Examples of ores

(a) Examples of oxide ores

(i) Cuprite – Cu_2O

(ii) Haematite – Fe_2O_3

(iii) Corundum – Al_2O_3

(iv) Cassiterite – SnO_2

(v) Magnetite – Fe_3O_4

(vi) Zincite – ZnO

(vii) Lead ochre – PbO (Litharge)

(b) Examples of carbonates ores

(i) Limestone – $CaCO_3$

(ii) Siderite – $FeCO_3$

(iii) Magnesite – $MgCO_3$

(iv) Dolomite – $CaCO_3.MgCO_3$

(v) Malachite – $CuCO_3.Cu(OH)_2$

(vi) Calamine – $ZnCO_3$

(vii) Cerussite – $PbCO_3$

(c) Examples of sulphides ores

(i) Zinc blende – ZnS

(ii) Cinnabar – HgS

(iii) Galena – PbS

(iv) Copper pyrites – $CuFeS_2$

(v) Iron pyrites – FeS_2

(d) Examples of sulphates ores

(i) Gypsum – $CaSO_4.2H_2O$

(ii) Epsom salt – $MgSO_4.7H_2O$

(iii) Anglesite – $PbSO_4$

(iv) Barytes – $BaSO_4$

(v) Alunite – $K_2SO_4.Al_2(SO_4)_3.4Al(OH)_3$

(e) Examples of halides ores

(i) Rock salt – NaCl

(ii) Fluorspar – CaF_2

(iii) Carnalite – $MgCl_2.KCl$

(iv) Horn silver – AgCl

(v) Cryolite – Na_3AlF_6

(f) Examples of nitrates ores

(i) Chile salt petre – $NaNO_3$

(ii) Bengal nitre or Petre – KNO_3

(g) Examples of Hydrated oxides ores

(i) Limonite – $Fe_2O_3.H_2O$

(ii) Bauxite – $Al_2O_3.2H_2O$

9.4 METALLURGY

Some important terms used in *metallurgy* are given below.

(i) **Minerals** are the compounds of metals found naturally in the earth's crust.

(ii) **Ores** are minerals from which metals can be conveniently and economically extracted.

Ranking	Atomic Number	Name and Symbol of element		% in earth's crust by weight
1	8	Oxygen	O	46.71
2	14	Silicon	Si	27.69
3	13	Aluminum	Al	8.07
4	26	Iron	Fe	5.05
5	20	Calcium	Ca	3.65
6	11	Sodium	Na	2.75
7	19	Potassium	K	2.58
8	12	Magnesium	Mg	2.08
9	22	Titanium	Ti	0.62
10	1	Hydrogen	H	0.14
11	15	Phosphorus	P	0.13
12	6	Carbon	C	0.094
13	25	Manganese	Mn	0.09
14	16	Sulphur	S	0.052
15	56	Barium	Ba	0.05
16	17	Chlorine	Cl	0.045
17	24	Chromium	Cr	0.035
18	9	Fluorine	F	0.029
19	40	Zirconium	Zr	0.025
20	28	Nickel	Ni	0.019

(iii) **Gangue or matrix** is the unwanted impurity in the ore such as mud, stones, sand, etc.

(iv) **Metallurgy** is a branch of chemistry which deals with the extraction of pure metal from its ore.

(v) **Leaching** is a widely used extraction method that converts the ore into aqueous solution.

(vi) **Smelting** is the reduction of the ore in a blast furnace.

(vii) **Matte** is a term used for the molten metal sulphides formed during smelting of copper, nickel and iron ores.

(viii) **Calcination** is the roasting of ore in limited supply of air to remove volatile impurities and make the ore porous to increase its surface area.

(ix) **Roasting** is the heating of ore to convert impurities into oxides and also to convert the ore to its oxide form.

9.5 STEPS IN METALLURGY

The process for extracting metal from the ore can be divided into four steps:

Step 1: *Dressing and concentration of ore*

Step 2: *Conversion of ore into oxide (or a suitable compound)*

Step 3: *Extraction of metal*

Step 4: *Purification or refining of metal.*

9.6 DRESSING AND CONCENTRATION OF ORE

The ores contain a large amount of gangue / matrix. The matrix has to be removed before the metal is extracted. This is done in two steps.

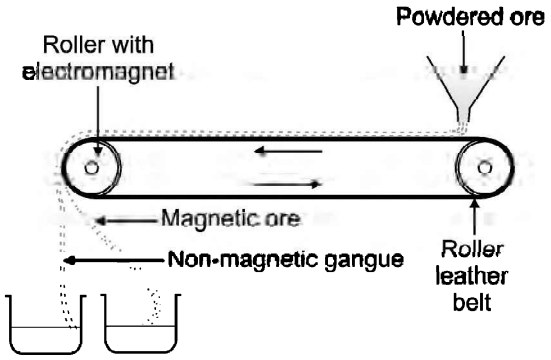
(a) Dressing: The ore is *dressed* using one or more of the following methods:

(i) handpicking, grinding and crushing, and

(ii) pulverising (grinding of ore to fine powder to increase surface area of the ore).

(b) Concentration: The ore is next *concentrated* to increase the percentage of metal in the ore using one of the methods given below.

(i) Electromagnetic process is used primarily for iron ores. The pulverized ore is put on a moving conveyor belt. One of the rollers of the conveyor belt is magnetic. As the pulverized ore moves over the magnetic roller, the iron ore particles get attracted till the last point on the roller and fall into a separate heap. The impurities that are non-magnetic fall vertically down in a separate heap (see schematic diagram and photo below).

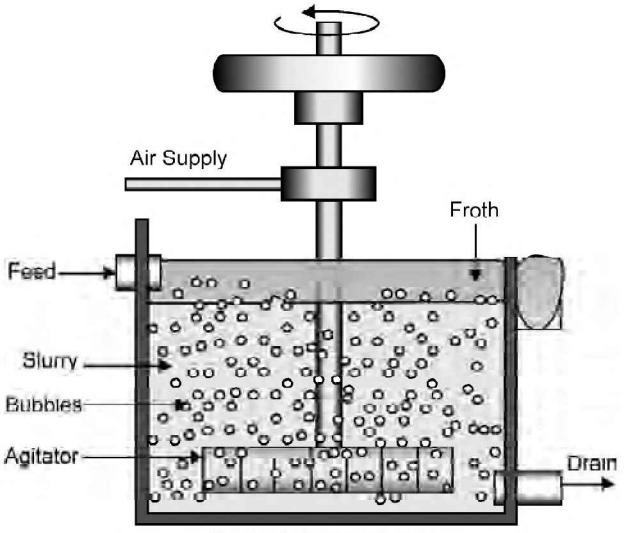


Schematic diagram showing electromagnetic separation



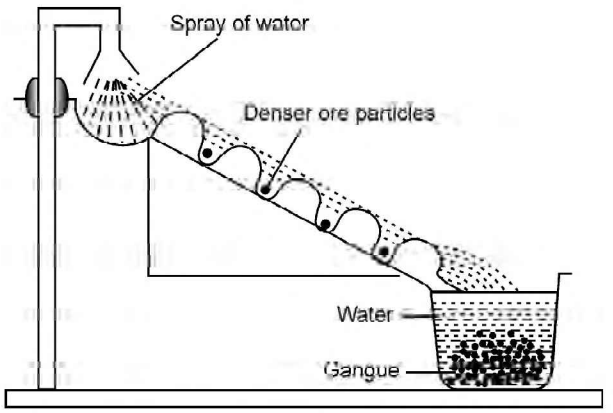
Actual photo of concentration of iron ore by electromagnetic separation

(ii) Froth floatation process [used primarily for sulphide ores such as galena (PbS), zinc blende, (ZnS) and copper pyrites ($CuFeS_2$)]. In the Froth Floatation Process the ore is mixed with water and pine oil and churned with a strong jet of air. The pine oil wets the sulphide ore particles while water wets the remaining gangue particles. Due to churning a froth of oil with ore concentrate is formed. The froth floats at the top and is removed and the sulphide ore recovered. The gangue is left behind at the bottom (See diagram below).

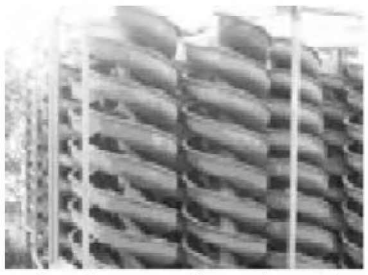


Froth Floatation Process

(iii) Gravity process (except for sulphide ores) in which the pulverized ore is poured over large tables with slanting tops and treated with a jet of water when the lighter matrix is washed away leaving the heavier ore behind (see figures).



Gravity separation using water and vibrating table.



Gravity separation using water (actual photo)

9.7 CONVERSION OF ORE INTO OXIDE

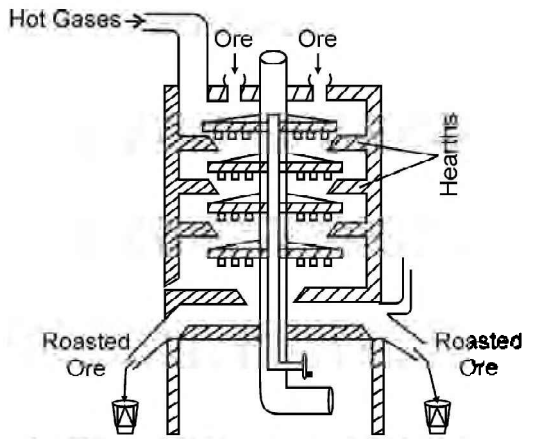
The ore is next treated for removal of volatile impurities and converted into its oxide for extraction of metal easily. Two methods are outlined below.

(i) **Calcination:** The ore is heated in a limited supply of air (taking care that the ore does not melt) to remove volatile constituents including moisture. This also makes the ore porous.

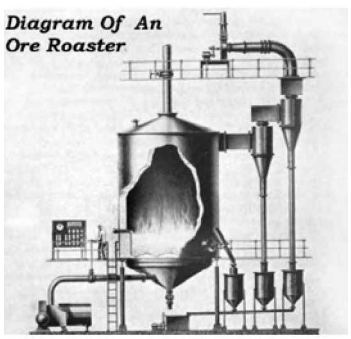
This method is used for carbonates ores and hydrated metallic oxides.

$$ZnCO_3 \rightarrow ZnO + CO_2$$

(ii) **Roasting:** The ore is strongly heated to a high temperature in excess of air to remove volatile impurities. This converts it into oxides. This method is used for sulphide ores.



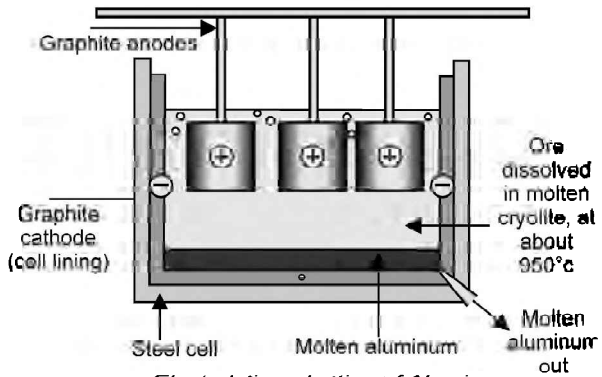
Multiple layered hearth furnace used for roasting



9.8 REDUCTION OF ORE TO PURE METAL (EXTRACTION OF METAL)

The exact method used depends on the position of the metal in the Metal Activity Series.

(i) **Electrolytic process:** This is used for extraction of more electropositive metals like sodium, potassium, calcium, magnesium and aluminium. The fused oxide or salt is used as the electrolyte. On electrolysis, the metal is deposited at the cathode (see figure below).



Electrolytic reduction of Aluminum

(ii) **Reduction with carbon:** This method is used for the metals lower in the electrochemical series like zinc, iron, lead, tin and copper. The ore is mixed with coke (reducing agent), limestone (which acts as *flux*). This mixture is then fired using a hot blast of air. The carbon reduces the metallic oxide to metal. The limestone decomposes to calcium oxide which reacts with sand present in the ore to form calcium silicate – *slag*. The *slag* floats on top of the liquid metal. Both are tapped separately.

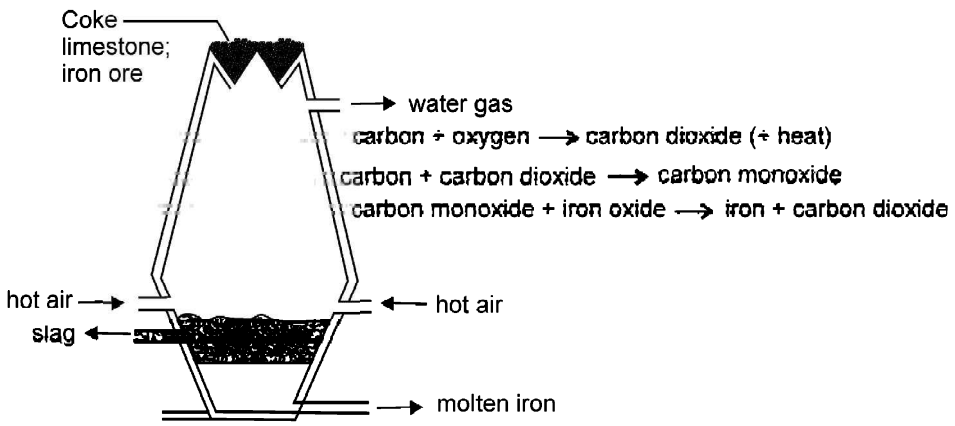
The reactions: $CuO + C \rightarrow Cu + CO$;

$$CaCO_3 \rightarrow CaO + CO_2;$$

$$CaO + SiO_2 \rightarrow CaSiO_3 \text{ (slag)}.$$

This method is commonly used for metallurgy of copper and lead.

(iii) **Reduction using CO gas:** The reducing agent used is carbon monoxide instead of coke. This method is used for the extraction of iron and zinc.



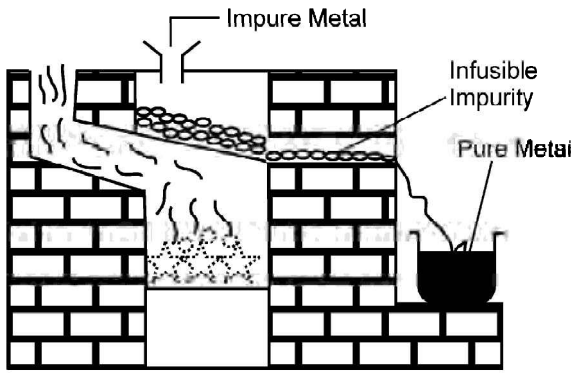
The blast furnace used to extract from iron ore

$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2;$$
$$ZnO + CO \rightarrow Zn + CO_2$$

(iv) Reduction using Al powder: Metals like chromium and manganese cannot be extracted from their oxides by heating them with reducing agents like coke or carbon monoxide as they have strong affinity for oxygen. They can be extracted using electrolytic method or by using aluminium powder as reducing agent.

$$Cr_2O_3 + Al \rightarrow Al_2O_3 + 2Cr$$

9.9 PURIFICATION AND REFINING OF METALS



Liquation Process

Metals extracted by the above methods contain small quantities of impurities. The impurities can be other metals or their oxides and non-metals and gases. They are further refined as follows:

Distillation: The impure metal is strongly heated in absence of air when the metal vapourizes leaving the impurities behind. The vapours are condensed to get pure metal. (This method is used for metals with low melting point like zinc and mercury).

Liquation: The solid impure metal is heated in a hearth with a sloping floor in absence of air. The pure metal liquefies and flows down the sloping floor leaving solid impurities with higher melting point behind. (Used for metals with low melting points like lead and tin) (see figure).

Poling: In this method, green wooden poles are used to stir molten impure copper. The heat of the molten copper makes the green wooden poles emit 'wood' gases which reduce impurities like Cu_2O to copper. This method is used for refining copper.

Oxidation: Oxygen is blown through molten iron the when impurities like carbon, arsenic, sulphur, etc., are oxidized to form their gaseous products and are blown out. This method is used for metals like iron.

Electrolytic Refining: The impure metal is made the anode and suspended in an electrolyte containing a *soluble salt of the same metal*. The cathode is made of a thin strip of the pure metal. On electrolysis the pure metal is deposited on the cathode. The impurities pass into solution or are deposited near the anode as *anode mud*. (This method of refining is used for copper, tin, lead and aluminium).

9.10 METALLURGY OF ALUMINIUM (HALLS PROCESS)

The metallurgy of aluminium by *Hall's process* is outlined below.

Ore: Bauxite ($Al_2O_3.2H_2O$).

Method: The ore is converted to alumina (pure Al_2O_3) and electrolytically reduced.

After dressing of ore and concentration by gravity process, the ore is pulverized and heated to remove volatile impurities.

Step 1: Conversion of ore to alumina by Baeyer's process

The powdered ore is heated with a concentrated solution of sodium hydroxide (NaOH) under pressure

for 2 – 8 hours. The impurities mainly, ferric oxide (Fe_2O_3) and silicon dioxide (SiO_2), *being basic oxides*, do not react with NaOH and settle as a precipitate called *red mud* and is removed by filtration.

Step 2: Aluminium oxide (Al_2O_3) present in the ore *being an amphoteric oxide* reacts and forms a soluble salt – sodium aluminate.

$$Al_2O_3.2H_2O + 2NaOH \xrightarrow{140^0-150^0C,\ 2-8\ hours} 2NaAlO_2 + 3H_2O$$

The sodium aluminate is diluted with water and mildly heated to a temperature of 50 – 60°C when aluminium hydroxide – $Al(OH)_3$ precipitates out.

$$NaAlO_2 + 2H_2O \xrightarrow{50^0-60^0C} NaOH + Al(OH)_3\downarrow$$

The aluminium hydroxide [$Al(OH)_3$] is next filtered. The NaOH obtained as filtrate is reused in the above step. The precipitate is washed, dried and heated to 1100°C to get *pure alumina*.

$$2Al(OH)_3 \xrightarrow{1100^0C} Al_2O_3 + 3H_2O$$

(Pure alumina)

Step 3: Reduction of alumina by electrolysis

As aluminium has a high affinity for oxygen, alumina is a very stable compound. Alumina cannot be reduced to pure aluminium *conveniently* by using reducing agents such as carbon, carbon monoxide or hydrogen. Hence electrolytic reduction is used.

Adding flux

Alumina is mixed with flux – a mixture of chemicals. The flux used is cryolite and fluorspar.

The ratio of alumina to cryolite to fluorspar is 1:3:1.

Alumina (Al_2O_3)	–	20%
Cryolite (Na_3AlF_6)	–	60%
Fluorspar (CaF_2)	–	20%

Reason for adding flux: The flux is added to the ore to

(i) increase its conductivity by providing more ions

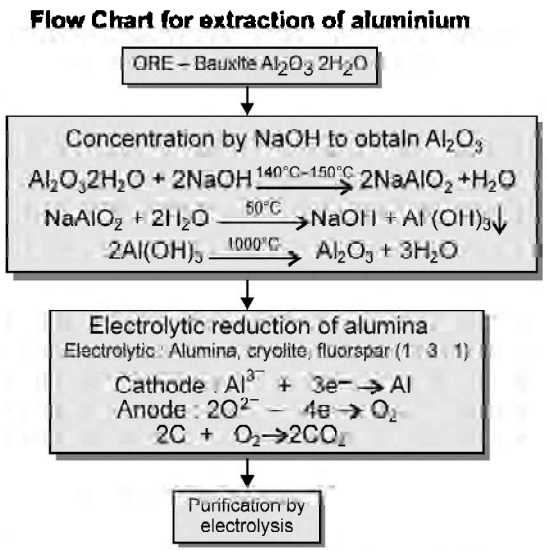
(ii) reduce the melting point of pure alumina to 950°C from 2050°C so that fuel can be saved.

(iii) to facilitate dissolution of alumina as the flux melts before the alumina.

Electrolytic reduction

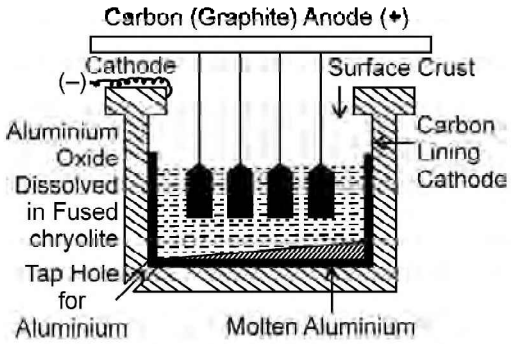
The mixture is heated to 900°C when it melts. The mixture is now put in an iron pot and reduced by electrolysis. The positive ions in the mixture are Na^+, Ca^{2+} and Al^{3+}. The positive ions move towards the cathode. At the cathode aluminium is preferentially discharged.

$$Al^{3+} + 3\ e^- \rightarrow Al$$



Reason for preferential discharge of aluminium ions at the cathode

Metals lower in the electrochemical series are preferentially discharged at the cathode during electrolysis, as these metal ions will have ***higher reduction potential – tendency to gain electrons.*** Due to the position of alumnium in the electrochemical series, aluminium is discharged preferentially at the cathode.

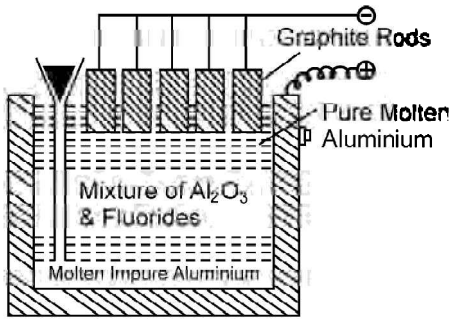


Extraction of aluminium by electrolytic reduction of fused aluminia

As melting point of aluminium is 660°C, the discharged aluminium is in the molten state. Aluminium being heavier than the electrolyte mixture sinks to the bottom. As more aluminium gets discharged at the cathode, the concentration of alumina decreases during electrolysis. This increases the conductivity of the electrolyte. To indicate this, a bulb is connected. When the bulb glows more brightly more alumina is added.

Step 4: **Refining** is done by further electrolysis.

The process uses an electrolytic cell which contains three layers of molten substance of differing densities. Molten impure aluminium forms the bottom layer and serves as anode, while pure aluminium serves as cathode in the top layer. The middle layer consists of a mixture of fluorides of sodium, barium and aluminium. On electrolysis, aluminium from the middle layer passes to the top layer. An equivalent amount of aluminium shifts from the lower layer to the middle layer. Pure aluminium floats on the top and is tapped regularly.



Electrorefining of aluminium

Properties of aluminium

Physical properties

(a) Aluminium is a silvery white metal with good lustre. Over time it appears dull due to formation of an oxide layer.

(b) Aluminium is a strong metal with high tensile strength.

(c) Aluminium is ductile and malleable.

(d) Aluminium is a good conductor of heat and electricity.

(e) Aluminium melts at 660°C and boils at 2050°C.

Chemical properties

(a) Aluminium reacts with oxygen in the air forming a layer of aluminium oxide.

$$2Al + 3O_2 \rightarrow Al_2O_3$$

On heating to 800°C aluminium burns with a bright light forming both aluminium oxide and aluminium nitride.

$$4Al + 3O_2 \rightarrow 2Al_2O_3$$

$$2Al + N_2 \rightarrow 2AlN$$

(b) Pure aluminium reacts with steam liberating hydrogen. Ordinary aluminium does not react as it is rendered passive due to an oxide layer.

$$2Al + 3H_2O \rightarrow Al_2O_3 + 3H_2$$

(c) Aluminium reacts with strong bases and acids showing its amphoteric nature.

$$2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2\uparrow$$

$$2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2\uparrow$$

The reaction is slow in the beginning due to the oxide layer present on the surface. After the layer gets dissolved the main reaction proceeds faster. With dilute and concentrated sulphuric acid aluminium liberates hydrogen and sulphur dioxide respectively.

$$2Al + 3H_2SO_4 \xrightarrow{\text{dilute}} Al_2(SO_4)_3 + 3H_2\uparrow$$

$$2Al + 6H_2SO_4 \xrightarrow{\text{concentrated}} Al_2(SO_4)_3 + 6H_2O + 3SO_2\uparrow$$

Dilute and concentrated nitric acid makes aluminium passive due to formation of the oxide layer on the surface.

(d) Aluminium reacts with non-metals like nitrogen, chlorine, sulphur, and carbon as follows.

$$2Al + N_2 \xrightarrow{\Delta} 2AlN$$

$$2Al + 3Cl_2 \xrightarrow{\Delta} 2AlCl_3$$

$$2Al + 3S \xrightarrow{\Delta} Al_2S_3$$

$$4Al + 3C \xrightarrow{\Delta} Al_4C_3$$

(e) Reducing property: Aluminium is a good reducing agent. Aluminium reduces ferric oxide (Fe_2O_3) when ignited to iron in an extremely exothermic reaction. The reduced iron will be in the molten form.

$$Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3 + \text{heat released}$$

Uses of aluminium

Aluminium is used in the making of alloys as it is light, strong and corrosion resistant.

As aluminium is light, malleable, ductile with high tensile strength it is used in making cooking utensils, window and door frames, thin foils for wrapping food articles, etc.

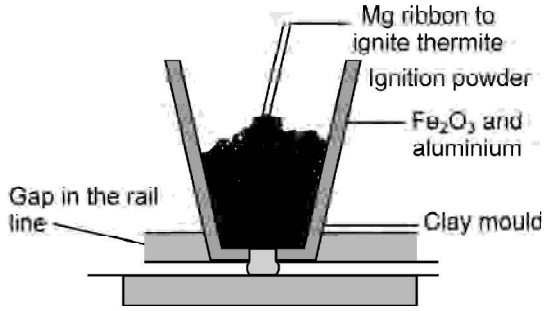
Aluminium is used in making ammonal an explosive (aluminium powder with ammonium nitrate).

Aluminium is used *Thermite welding*: Thermite is a mixture of three parts ferric oxide and one part aluminium powder.

When this is heated the more reactive aluminium displaces iron.

$$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe + \text{Heat Released}$$

The heat released is so high that it can melt the iron. This is used to weld railway lines.



Welding by Thermite process



Welding of rails using alumino thermite process in progress

9.11 METALLURGY OF IRON

The metallurgy of iron is outlined below.

Ores: Haematite (F_2O_3), Limonite ($F_2O_3.H_2O$), Magnetite (Fe_3O_4), Siderite ($FeCO_3$), iron pyrites (FeS_2) and copper pyrites ($CuFeS_2$).

Method: The ore is converted to oxide and reduced with CO.

The dressed ore is concentrated by electromagnetic process and then by gravity processes.

The ore is next roasted in excess air. Roasting has the following benefits:

(i) removes volatile impurities,

(ii) oxidises ferric oxide to ferrous oxide,

(iii) converts carbonate ores into oxide and

(iv) the ore mass becomes porous.

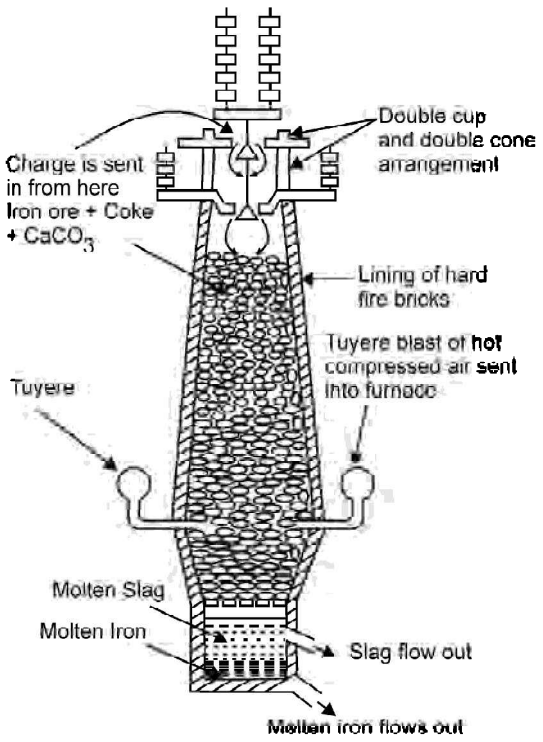
Next, the ore is smelted by reduction in a blast furnace.

The ore is mixed with coke and limestone ($CaCO_3$) in the ratio 3 : 2 : 1. This mixture is called ***charge***.

Carbon monoxide acts as reducing agent, while limestone acts as flux.

Blast furnace

Blast furnaces differ in construction. The one used in the production of iron consists of a chimney like structure (usually 24-30 m high) made of iron or steel and lined with firebrick. It is narrow at the top, increasing in diameter downward, but narrowing again at the bottom. The blast furnace is a specially constructed huge tower with ceramic bricks lining the inner walls to prevent corrosion. The charge is dropped from the *top of the furnace*, while hot, compressed air is sent from the tuyeres (pronounced tweeyer) *at the bottom*.



Blast furnace for the manufacture of cast iron

Reactions

The reactions that take place in the blast furnace are as follows:

Lower part of blast furnace:

$C + O_2 \rightarrow CO_2 + \Delta$ (heat released)

The hot carbon dioxide rises up the tower.

Middle part of blast furnace:

$CO_2 + C \rightarrow 2CO - \Delta$ (heat absorbed)

$CaCO_3 \rightarrow CaO + CO_2$

$CaO + SiO_2 \rightarrow CaSiO_3$ (slag)

The slag trickles down the blast furnace.

Upper part of blast furnace:

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

Melting point of iron is 1200°C. The iron melts

as it reaches the lower part of the furnace along with slag. In the lower parts of the furnace the slag also melts. The slag being lighter floats on the molten iron. Both the slag and the molten iron are tapped separately.

Types of iron

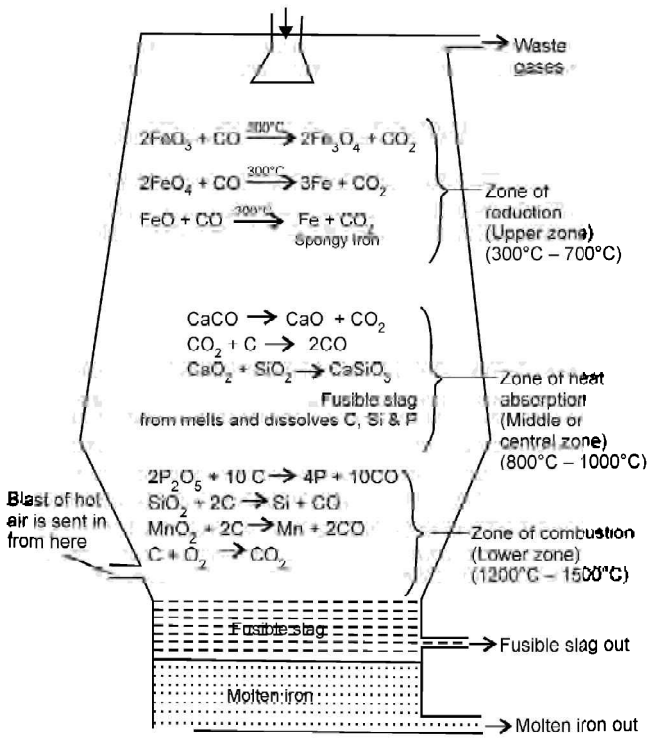
Cast iron is the most impure form of iron containing carbon, sulphur, phosphorous and manganese as impurities. It is hard and brittle.

Pig iron: When molten cast iron is suddenly cooled it changes to *Pig Iron*.

Wrought iron is the *purest form of iron* and contains least amount of carbon and other impurities.

Steel is a mixture of iron with carbon and other elements. (See table below)

Bessemer Converter: Pig iron is converted to steel in a special furnace called *Bessemer Converter* (see diagram). A blast of hot air is forced into molten pig iron. This oxidises all the impurities. The air supply is cut off. Calculated amount of *Spiegel* (an alloy of carbon and manganese) is added to form ***steel***.



Reaction taking place in the blast furnace

Alloys of iron

Alloys of iron	Metals present	% composition
(a) Spiegeleisen	Fe, Mn, C	74, 20, 6
(b) Stainless steel	Fe, Cr, Ni	82, 12, 1
(c) Manganese steel	Fe, Mn, C	84, 15, 1
(d) Tungsten steel	Fe with Tungsten (W)	10% W
(e) Chrome steel	Fe with Cr	4% Cr
(f) Vanadium steel	Fe with V	0.5%V

Hardening and Tempering of Steel

When steel is heated to a temperature of 750°C to 800°C and then suddenly cooled by plunging into cold water or cold oil, the cooled steel is brittle and cannot withstand sharp jerks. This process is called ***hardening***.

If the hardened steel is heated to a controlled temperature between 220°C to 300°C and then allowed to cool very slowly over four hours its hardness remains but the steel becomes more ***tensile***, i.e., it becomes capable of bearing heavy loads or strain without breaking. This process is called ***Tempering*** or ***Annealing***. During tempering the atoms of the metals arrange in such a way that they can take strain.

The various applications of end product and the temperature to which they are heated during tempering and annealing are listed below.

Temperature	Use
300°C	In the manufacture of springs
290°C	In the manufacture of screwdrivers
270°C	In the manufacture of drill knives
245°C	In the manufacture of scissors
220°C	In the manufacture of blades

Physical properties of iron

(a) With dry air iron has no reaction, but moist air turns iron into rust over time.

$$4Fe + 3O_2 + 2xH_2O \rightarrow 2Fe_2O_3.xH_2O$$

With hot air, iron burns giving green sparks forming ferroso-ferric oxide.

$$3Fe + 2O_2 \rightarrow Fe_3O_4\ (FeO.Fe_2O_3)$$

(Magnetic oxide of iron)

(b) With steam, iron forms magnetic oxide of iron

$$3Fe + 4H_2O \rightleftharpoons Fe_3O_4 + 4H_2$$

(c) With dilute acids

Iron forms respective salts with dilute acids with the liberation of hydrogen.

$$Fe + 2HCl \rightarrow FeCl_2 + H_2\uparrow$$

$$Fe + \underset{\text{dilute}}{H_2SO_4} \rightarrow FeSO_4 + H_2\uparrow$$

With cold dilute nitric acid, ammonium nitrate is also formed.

$$4Fe + \underset{\text{dilute}}{10HNO_3} \rightarrow 4Fe(NO_3)_2 + NH_4NO_3 + 3H_2O$$

(d) With concentrated acids

Iron reacts with concentrated sulphuric acid to liberate sulphur dioxide.

$$2Fe + 6H_2SO_4 \rightarrow Fe_2(SO_4)_3 + 3SO_2 + 6H_2O$$

Concentrated nitric acid makes iron *passive* due to the fact that the conc. acid is a powerful oxidising agent and it forms a protective layer of ferric oxide over the surface of iron.

(e) With non-metals iron reacts as follows:

$$2Fe + 3Cl_2 \rightarrow 2FeCl_3$$

$$Fe + S \rightarrow FeS$$

(f) Iron displaces other less reactive metals from aqueous solution of their salts.

$$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$$

Uses of iron

Iron is used in the manufacture of drain pipes, manhole covers, railings, shutters, gates, machine parts, etc. Steel, an alloy of iron, is used in buildings, machines, automobiles and in other devices.

9.12 METALLURGY OF ZINC

The metallurgy of zinc is outlined below.

Ores: Zincite (ZnO), Zinc Blende (ZnS) and Calamine ($ZnCO_3$).

Method: The dressed ore is concentrated by froth flotation if it is zinc blende else by gravity process. Next the ore is roasted in excess air when it gets converted to oxide. The oxide is reduced by coke.

$$ZnO + C \rightleftharpoons Zn + CO.$$

When carbon burns in air, carbon dioxide is formed which tends to convert zinc back to its oxide. To avoid this zinc vapours are absorbed in lead. The two are later separated by distillation. The CO formed as a by-product is used as fuel for heating. The zinc so obtained is impure and is called ***spelter.*** The impure *spelter* is purified by controlled distillation or electrolysis.

Physical properties

(a) Zinc is a bluish white metal, hard and brittle that becomes malleable and ductile at 100°C.

(b) Zinc has good tensile strength.

(c) Zinc melts at 419°C.

(d) Zinc is a good conductor of heat and electricity.

Chemical properties

(a) Zinc does not react with dry air but when heated to above 500°C in moist air it burns with a bluish flame forming zinc oxide.

$$Zn + O_2 \xrightarrow{500°C} 2ZnO$$

(b) Zinc has no reaction with cold water. With steam it liberates hydrogen.

$$Zn + H_2O \xrightarrow{\text{steam}} ZnO + H_2\uparrow$$

(c) With dilute acids zinc forms respective salts with the liberation of hydrogen.

$$Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$$

$$Zn + \underset{\text{Dilute}}{H_2SO_4} \rightarrow ZnSO_4 + H_2\uparrow$$

But with dilute nitric acid zinc forms nitric oxide.

$$3Zn + \underset{\text{Dilute}}{8HNO_3} \rightarrow 3Zn(NO_3)_2 + 4H_2O + 2NO\uparrow$$

(d) With concentrated nitric and sulphuric acid, nitrogen dioxide and sulphur dioxide respectively are formed.

$$Zn + \underset{\text{Concentrated}}{2H_2SO_4} \rightarrow ZnSO_4 + 2H_2O + SO_2\uparrow$$

$$Zn + \underset{\text{Concentrated}}{4HNO_3} \rightarrow Zn(NO_3)_2 + 2H_2O + 2NO_2\uparrow$$

(e) With sodium hydroxide zinc exhibits amphoteric nature.

$$Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2\uparrow$$

(f) With non-metals zinc reacts as follows.

$$Zn + Cl_2 \xrightarrow{\Delta} ZnCl_2$$

$$Zn + S \xrightarrow{\Delta} ZnS$$

(g) Zinc displaces other less reactive metals from an aqueous solution of their salts.

$$Zn + FeSO_4 \rightarrow ZnSO_4 + Fe$$

$$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$$

$$Zn + Pb(NO_3)_2 \rightarrow Zn(NO_3)_2 + Pb$$

(But, $CaCl_2 + Zn \rightarrow$ No reaction as calcium is more reactive than zinc).

Uses of zinc

(a) **Galvanisation**: The process of coating iron sheets with zinc metal to prevent rusting is called *galvanization*. When zinc is coated over iron, it acts as a sacrificial metal getting preferentially oxidized than iron. This prevents the oxidation of iron and formation of rust.

(b) Zinc is used to make alloys like brass, bronze and German silver.

(c) Zinc is used as an electrode in dry cells. The outer coating of dry cells is made of zinc.

(d) Zinc used in special printing.

(e) Zinc is used as reducing agent

(f) Zinc is used in drugs and dyes.

(g) Zinc is used in cosmetics and face creams.

9.13 ALLOYS

Alloy	Metals present	Percentage composition	Uses
Duralumin (Dural)	Al, Cu, Mg, Mn	95, 4, 0.5, 0.5	Aircraft frames, windows
Magnalium	Al, Mg,	95, 5	Cheap balances
Y-Metal	Al, Cu, Ni, Mg	93, 4, 2, 1	Currency coins
Alnico	Al, Fe, Ni, Co	20, 50,20,10	Strong magnets
Aluminium Bronze	Al, Cu	10, 90	Decorative items
Bronze	Cu, Zn, Sn	80, 10, 10	Medals, statues, utensils
Brass	Cu, Zn	70, 30	Decorative items, handles, screws, utensils
German Silver	Cu, Zn, Ni	40, 30, 30	Decorative articles, electric heaters, rheostats
Bell metal	Cu, Sn	80, 20	Bells, gongs
Solder	Sn, Pb	50, 50	Welding, fuse

9.14 CORROSION – A SPECIAL DISCUSSION

Metal surfaces are liable to react with oxygen, carbon dioxide, moisture and other pollutants in the atmosphere. This is the main reason why metals corrode. While most metals suffer corrosion, some metals are more corrosive.

(a) **Corrosion of Iron**: Iron corrodes over time when exposed to moist air forming a powdery coating of brown rust. Over time the entire piece of iron rusts. Rusting is the corrosion of iron.

Rusting takes place as follows:

$$4Fe + 3O_2 \rightarrow 2Fe_2O_3$$

$$Fe_2O_3 + xH_2O \rightarrow Fe_2O_3.xH_2O$$

(rust)

Conditions for rusting: Presence of air or oxygen and moisture are essential. Presence of only oxygen without moisture or vice-versa cannot form rust.

Prevention of rusting

(i) **Coating** the iron surface with paint or oil or grease prevents moist oxygen from coming in contact with the metal and thus effectively prevents rusting of iron.

(ii) **Galvanisation**: Iron is blasted with fine sand to make the surface rough and dipped in molten zinc and then cooled. A thin layer of zinc forms on the iron surface. Since zinc is more reactive than iron, it acts as a sacrificial metal and is preferentially oxidised thus preventing oxidation of iron.

(iii) **Electroplating** with tin, nickel or chromium also prevents rusting.

(iv) **Alloying** (mixing iron in its molten state with other metals) prevents rusting. Stainless steel is an alloy of iron with chromium or nickel.

(b) **Corrosion of aluminium**: Aluminium sheets appear dull in colour due to the formation of a thin oxide layer. This oxide layer actually prevents aluminium from further corrosion.

(c) **Corrosion of copper**: Copper over a period of time reacts slowly with carbon dioxide and

moisture present in the atmosphere to form a greenish coat of basic copper carbonate [$Cu(OH)_2.CuCO_3$] corroding it. It is a very slow process.

(d) **Corrosion of silver**: Tarnishing of silver happens over time due to the formation of a thin *sulphide* layer (not oxide as silver is relatively non-reactive). Air contains very minute amounts of H_2S or SO_2 as pollutants. This reacts with silver articles forming sulphide (Ag_2S). So, silver blackens over time. Simple heating and polishing removes this layer.

Gold normally is resistant to corrosion as it is very non-reactive.

Titanium is another metal which is resistant to corrosion. Titanium is used extensively in hip replacement and knee replacement surgeries as it is resistant to corrosion. *Titanium is a future metal.*

EXERCISE

I. Choose the correct answer

1. Which is not the property of a metal?
(a) Malleable (b) Ductile
(c) Form acidic oxides (d) Conductor

2. Which is a property of a non-metal?
(a) Form cations (b) Have tensile strength
(c) High density (d) Form acidic oxides

3. Calcination of an ore results in
(i) removal of moisture
(ii) porosity of ore
(iii) removal of volatile impurities
(iv) convertion of carbonate ore to oxide
(v) removal of water of crystallisation from hydrated ores
(a) (i) and (iv) (b) (iii) and (iv)
(c) (iii) (d) all five options

4. Roasting of an ore results in
(a) conversion of ore to oxide
(b) melts it
(c) makes it conducting
(d) makes it soluble

5. Froth flotation is a method used for
(a) carbonate ores (b) hydrated ores
(c) sulphide ores (d) oxide ores

6. Dressing and concentration of an ore results in
(a) increase of metal ore concentration
(b) decrease in metal ore concentration
(c) conversion of ore to suitable form
(d) cleans the ore

7. Extraction of metal is done by
(i) electrolytic process ii. using carbon
(iii) using CO iv. using Al powder
(a) (i) and (iv) (b) (i) & (ii)
(c) (i) & (iii) (d) all four options

8. Highly electropositive metals cannot be reduced by
(i) electrolytic process (ii) using carbon
(iii) using CO (iv) using Al powder
(a) all except i (b) all except ii
(c) all except iii (d) all except iv

9. A flux has all or some of these benefits except
(a) lowering melting point
(b) improving conductivity
(c) reacting chemically with impurities like sand
(d) refining the metal

10. The purest form of iron is
(a) cast iron (b) pig iron
(c) wrought iron (d) Spiegeleisen

II. Read the paragraph below and answer the questions that immediately follow each paragraph. Each question has four options out of which only one option is correct. Choose the correct one.

Paragraph A

In the extraction of metals, the less electropositive metals are reduced by chemical methods while the more electropositive metals are reduced by electrolysis. In general ores are converted to their oxides and then reduced by chemical reducing agents. The more electropositive metals are reduced with difficulty by chemical methods. For such metals extraction by electrolytic reduction is preferred. During electrolysis of these metals compounds, the molecules dissociate into ions. The metal ions migrate to the cathode where they get reduced by accepting electrons. However for the less electropositive metals, electrolytic reduction is relatively more expensive. The chemical method using reducing agents like carbon or carbon monoxide is preferred.

1. Metals that are easily reduced from their oxides are _____ reactive.
(a) more (b) less
(c) as (d) not

2. In electrolysis, reduction takes place at _____ (electrode).
(a) cathode (b) anode
(c) anti-cathode (d) bottom

3. In general the electrolyte used in the electrolytic extraction of metals is in the fused state rather than an aqueous solution. This is because
(a) the salt is insoluble
(b) to keep metal dry
(c) water is poor conductor
(d) to eliminate H^+ ions

Paragraph B

'Alumina (aluminium oxide) has a very high melting point of over 2000°C so that it cannot be readily liquefied. However, conversion of alumina to aluminium and oxygen, by electrolysis, can occur when it is dissolved in some other substance.'

4. **Which solution is used to react with bauxite as a first step in obtaining pure aluminium oxide?**
   (a) Solution of NaOH (b) Solution of KOH
   (c) Salt solution (d) Sugar solution
5. **The aluminium oxide for the electrolytic extraction of aluminium is obtained by heating aluminium hydroxide. The unbalanced equation for the reaction is**
   (a) $Al(OH)_3 \rightarrow Al_2O_3.2H_2O + H_2O$
   (b) $Al_2O_3.2H_2O \rightarrow Al(OH)_3 + H_2O$
   (c) $Al(OH)_3 \rightarrow Al_2O_3 + H_2O$
   (d) $Al + H_2O \rightarrow Al(OH)_3$
6. **The element which serves both as the anode and cathode in the extraction of aluminium is**
   (a) platinum (b) iron
   (c) aluminium (d) carbon
7. **The equation for the reaction that occurs at the cathode during the extraction of aluminium by electrolysis is**
   (a) $2Al - 6e^- \rightarrow 2Al^{3+}$ (b) $2Al^{3+} + 6e^- \rightarrow 2Al$
   (c) $Na^+ + e^- \rightarrow Na$ (d) $Na - e^- \rightarrow Na^+$
8. **The equation for the reaction which occurs at the anode when aluminium is purified by electrolysis is**
   (a) $O - 2e^- \rightarrow O^{2-}$ (b) $2F^- + 2e^- \rightarrow F_2$
   (c) $O^{2-} - 2e^- \rightarrow O$ (d) $6O^{2-} - 6e^- \rightarrow 3O_2$

III. Match the following.

1. **Match the process given in Set A with one of the four major steps of metallurgy in Set B.**

Set A	Set B
(i) Liquation	(a) Refining
(ii) Magnetic separation	(b) Conversion to metal by reduction of ore
(iii) Reduction by CO or C	(c) Concentration
(iv) Roasting	(d) Conversion of ore to oxide

2. **Match the ores given in Set A with the metals in Set B.**

Set A	Set B
(i) Carnalite	(a) Zinc
(ii) Bauxite	(b) Iron
(iii) Calamine	(c) Magnesium
(iv) Haematite	(d) Aluminium

3. **Match the metal in Set A with method of reduction in Set B.**

Set A	Set B
(i) Aluminium	(a) Reduced using coke
(ii) Zinc	(b) Electrolytic reduction
(iii) Iron	(c) Reduced using CO
(iv) Chromium	(d) Reduced using Al powder

IV. State whether the following statements are True or False

1. Metals high up in the metal activity series are extracted by electrolysis.
2. Metals that have a very strong affinity for oxygen are reduced by using coke as reducing agent.
3. To refine metals after extraction liquation is used if the melting points are high.
4. Bauxite is converted to alumina by using the amphoteric nature of aluminium oxide.
5. Pig iron is converted to steel in a Bessemer converter.

V. Find the odd one out

1. calcination, magnetic separation, froth floatation, pulverization, gravity separation
2. electrolysis, carbon monoxide, coke, aluminium powder
3. duralumin, magnalium, Y-metal, alnico, brass

VI. Fill in the blanks

1. In roasting the pulverized ore is heated strongly in the _____ of air.
2. In calcination the pulverized ore is heated strongly in the _____ of air.
3. The flux used in the metallurgy of aluminium is a mixture of _____, _____and _____.
4. The flux used in the metallurgy of iron is _____.
5. Roasting converts carbonate and sulphide ores to the respective _____.

VII. Each question contains two statements. Statement A is an Assertion Statement and Statement R is the Reason Statement. Each question has four choices out of which only one is correct. Choose the correct one.

Choose A if Statement A is true, Statement R is true; Statement R is correct explanation for statement A

Choose B if Statement A is true, statement R is true; Statement R is not correct explanation for statement A

Choose C if Statement A is true, Statement R is false

Choose D if Statement A is false, statement R is true

1. **Statement –A**: Alkali earth metals are more basic than corresponding alkali metals.

   Statement –R: Alkali earth metals ions have high hydration energy.

2. **Statement –A**: Addition of fluorspar and cryolite to alumina in metallurgy of aluminium improves conductivity.

   Statement –R: Fluorspar and cryolite act as flux reducing the melting point of alumina.

3. **Statement –A**: Calcination and roasting of ores are important steps in metallurgy.

   Statement –R: Calcination and roasting reduce the ore to metal.

4. **Statement –A**: Alkali earth metals are extracted by electrolytic reduction.

   Statement –R: Alkali earth metals are easily reduced.

VIII. Answer the following questions.

1. Differentiate between a mineral and an ore.
2. Enlist the methods of dressing and concentration of an ore. Mention the reasons for dressing and concentration..
3. Name three ores each, that have the metal as carbonate, sulphate and oxide.
4. Differentiate between roasting and calcination. How are they beneficial in metallurgy?
5. Given below are terms used in metallurgy. Identify the step in which each of these terms occur.

   Polling, electrolytic reduction, calcination, gravity separation, distillation, liquation, handpicking, flux, gangue, matrix.
6. Write the equations involved in the concentration of bauxite to pure alumina by both methods.
7. What is the flux used in the metallurgy of aluminium? What is its role?
8. Write a brief outline of metallurgy of zinc and iron with reactions. What is the reducing agent in each?
9. Certain metals are extracted by electrolytic methods while others are extracted by chemical methods. When is a particular method used?
10. Differentiate between the following forms of iron: *cast iron, pig iron, wrought iron and steel.*
11. (i) Name three alloys of tin (Sn).
    (ii) Differentiate between brass and bronze.
12. Why is the alloy alnico used for making strong magnets? (Hint: see its composition)
13. Tempering of steel increases its tensile strength. This is done so that in its end use it can bear heavy strain. How is tempering done?
14. Gold and titanium do not undergo corrosion. Why?
15. Can Aluminium be galvanised with zinc to prevent its corrosion and why?
16. Sulphide ores are roasted while carbonate ores are calcined before reduction. State why?
17. Outline the method of preparing steel in a Bessemer converter.

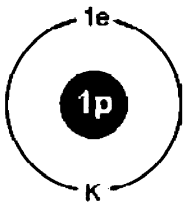
KEY TO EXERCISE

I.	1. c	2. d	3. d	4. a	5. c	6. a	7. d	8. a
	9. d	10. c						
II.	1. b	2. a	3. d	4. a	5. c	6. d	7. b	8. d
III.	1. i. a ii. c iii. b iv. d		2. i. c ii. d iii. a iv. b		3. i. b ii. a iii. a iv. d			
IV.	1. true	2. false	3. false	4. true	5. true			
V.	1. calcinations		2. electrolysis		3. brass			
VI.	1. presence	2. absence	3. cryolite, fluorspar, alumina					
	4. limestone	5. oxides						
VII.	1. D	2. B	3. C	4. C				

10 Study of Group IA Element—Hydrogen

10.0 INTRODUCTION TO HYDROGEN

Hydrogen was discovered by Henry Cavendish in 1766 and named by Lavoisier as *Hydrogen*. Hydrogen is the lightest of all elements and is the first element in the Periodic Table. Hydrogen is placed in the first period and Group IA of the Periodic Table.



Hydrogen atom

10.1 OCCURRENCE

Hydrogen is the *most abundant element in the universe* but in the earth's crust it is the ninth most abundant element. Around 0.9% of earth's weight is hydrogen.

Free state: Hydrogen being very reactive does not exist free in nature except in minute traces in coalmines as natural gas.

Combined state: Examples of compounds containing hydrogen are water (one-ninth the part by weight), methane, all organic compounds and petroleum products, etc. The main composition of the sun is hydrogen gas. The sun produces light and heat energy that is responsible for life on earth by nuclear fusion of hydrogen to give helium.

10.2 LABORATORY PREPARATION OF HYDROGEN

Hydrogen can be prepared in the laboratory by using any of the following three reactions.

(i) By reacting potassium (K), sodium (Na), calcium(Ca) with cold or warm water.

(ii) By displacement of hydrogen from certain acids by certain metals

(iii) By reacting certain metals with concentrated alkalis.

The preferred method for the laboratory preparation of hydrogen is by reacting zinc granules with dilute HCl or dilute H_2SO_4. The reaction that takes place is as given below.

$$Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$$

HYDROGEN

Symbol – H

Molecular formula – H_2

Atomicity – 2

Atomic Number – 1

Mass Number – 1

Atomic Weight – 1

Molecular Weight – 2

Hydrogen belongs to Group IA and 1st period of the Periodic Table.

Electronic Configuration: (1)

Valence electrons = 1

Valency = 1+

Physical state – a gas at room temperature and pressure

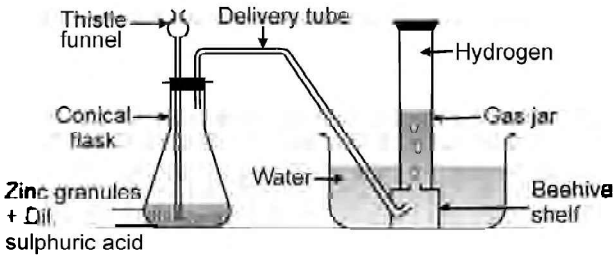
Non-metal

Vapour density – 1

Method of collection of hydrogen

Hydrogen is almost insoluble and lighter than air. Hence the preferred method is by downward displacement of water or over mercury. (See box).

It is *not* collected by downward displacement of air as it forms an explosive mixture with air.



Laboratory preparation of hydrogen

Important points to note in the laboratory preparation of hydrogen

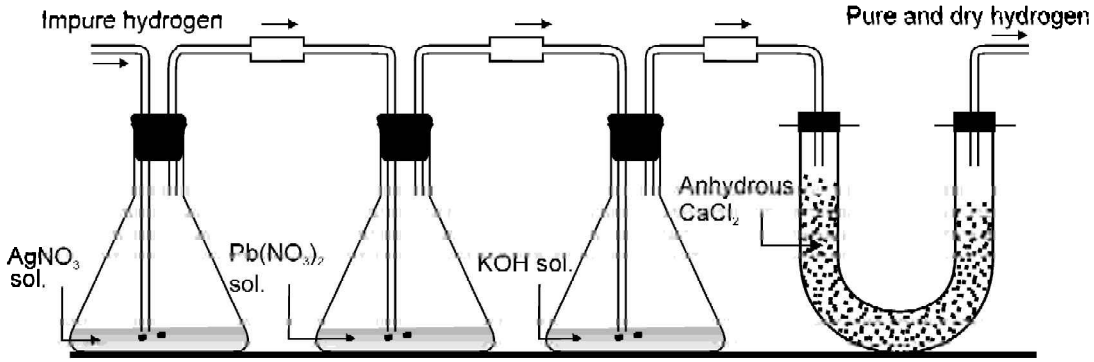
– Zinc is used in granulated form to increase the surface area. The granules are prepared by pouring molten zinc into cold water.

- The granules contain copper as impurity. The copper present in granulated zinc acts as a catalyst.
- Zinc is preferred over other metals (like sodium, potassium) as it is less expensive and the evolution of hydrogen is controlled.
- The hydrogen that is produced using granulated zinc contains impurities such as arsine, phosphine, hydrogen sulphide, sulphur dioxide, carbon dioxide, nitrogen dioxide and water vapour. These impurities can be removed by passing the gas through different solutions as follows.

Methods of Collection of Gas

In the laboratory preparation, a gas can be collected by the following methods.

1. Downward displacement of water (as shown in the figure for hydrogen gas). This method is preferred if the gas is insoluble in water and lighter than air.
2. Downward displacement of mercury (instead of water, mercury is used). This method is preferred if dry gas is required.
3. Upward displacement of air. This method is preferred if the gas is soluble in water and heavier than air. (Example: HCl gas)



Purification of hydrogen gas the in laboratory

(i) The impure hydrogen is passed through silver nitrate ($AgNO_3$) solution to remove arsine and phosphine.

(ii) Next the impure hydrogen is passed through lead nitrate [$Pb(NO_3)_2$] solution to remove hydrogen sulphide gas.

(iii) Next the hydrogen gas is passed through potassium hydroxide (KOH) solution to remove carbon dioxide, sulphur dioxide and nitrogen dioxide present as impurities.

(iv) The hydrogen gas is then passed over anhydrous calcium chloride ($CaCl_2$) to remove water vapour to make hydrogen gas dry.

- Care should be exercised while collecting hydrogen, as it is a combustible gas. The apparatus should be airtight.

10.3 INDUSTRIAL MANUFACTURE OF HYDROGEN

Hydrogen is prepared on a large scale either by Bosch Process or from natural gas.

(a) Bosch process

This process involves three steps.

(i) Superheated steam at 170°C is passed over white-hot coke to form water gas.

$$\underset{\text{(White hot coke)}}{C} + \underset{\text{(Super heated steam)}}{H_2O} \xrightarrow{170^0C} \underset{\text{(water gas)}}{CO + H_2}$$

(ii) *Water Gas shift reaction*: Since the desired component is hydrogen, the next step involves separation of CO by oxidizing it first to carbon dioxide. The water gas is mixed with an equal volume of steam when CO is oxidized to CO_2. (See box in chemical properties for oxidation and reduction reactions)

$$[CO + H_2] + H_2O \rightarrow CO_2 + 2H_2$$

This reaction is also known as *water gas shift reaction*.

(iii) The CO_2 thus produced is removed by passing the mixture of gases ($CO_2 + 2H_2$) under a pressure of 30 atm through water when CO_2 dissolves. Hydrogen that bubbles out is collected in cylinders.

(b) From natural gas by steam reforming process

Natural gas is obtained as a by-product of petroleum refining. Natural gas consists mainly of methane (CH_4) from which hydrogen is separated as follows.

Natural gas is mixed with excess steam at 800°C and 30atm pressure and passed over powdered nickel that acts as a catalyst. Hydrogen is formed in two reactions.

$$CH_4 + H_2O \xrightarrow{Ni-800^0C} CO + 3H_2$$

$$CO + H_2O \text{ (excess)} \rightarrow CO_2 + H_2$$

The mixture of these two gases (CO_2 and H_2) is compressed to a pressure of 30 atm and passed through water when CO_2 reacts with water forming H_2CO_3 acid and hydrogen passes out unreacted.

$$H_2O + CO_2 \xrightarrow{30atm} H_2CO_3$$

(c) **Electrolysis :** Hydrogen can also be produced by electrolysis of acidified water or brine (saturated NaCl) solution. This method is used where electricity is cheap.

10.4 PHYSICAL PROPERTIES OF HYDROGEN

(a) Hydrogen is a colourless and odourless gas.

(b) Hydrogen is an insoluble gas (only 2ml dissolves in 100ml of water).

(c) Hydrogen is a very light gas (20 times lighter than air).

(d) Though hydrogen is non-poisonous gas it causes death due to asphyxiation – deprival of oxygen.

(e) Specific properties: MP = – 259°C,
BP = – 252°C.
Density = 0.0899 g/cc

(f) Hydrogen is difficult to liquefy. It liquefies at 20 atm when cooled by liquid air.

(g) Hydrogen exists in three isotopic forms, viz., deuterium and tritium (see box).

The three isotopes have the same number of electrons and protons but different number of neutrons as tabulated below.

Isotopes

When atoms of the same element have different number of neutrons they will have the *same atomic number but different mass number*. The atoms are said to be *Isotopes* of the element and the element is said to exhibit *Isotopy*.

Example: Chlorine gas exists in two isotopic forms: $_{17}Cl^{37}$, $_{17}Cl^{35}$.

Isotope	Symbol	Electrons	Protons	Neutrons	Remarks
Protium	$_1H^1$	1	1	0	Most abundant form of hydrogen
Deuterium	$_1H^2$	1	1	1	Used as oxide (*heavy water* - D_2O) in nuclear reactors.
Tritium	$_1H^3$	1	1	2	Radioactive isotope

(h) ***Adsorption*** of hydrogen: Metals like gold, palladium and platinum have the capacity to absorb gases on their surfaces. Surface absorption is called *adsorption*. Hydrogen is easily adsorbed on the surface of palladium. At room temperature almost 900 volumes of hydrogen is adsorbed by powdered palladium. Palladium enriched by surface adsorption of hydrogen is used for *Hydrogenation of vegetable oils*. (See Chemical Properties below)

10.5 CHEMICAL PROPERTIES OF HYDROGEN

Hydrogen exhibits the following chemical properties.

(a) Hydrogen is a *combustible gas* but does not support combustion.

(b) Hydrogen *burns with a pale blue flame* making a pop sound due to spontaneous combustion.

(c) Hydrogen is a *neutral gas* (it has no action on litmus). (See box)

(d) **Reaction of hydrogen with metals to form hydrides.**

$$Ca + H_2 \rightarrow CaH_2$$

The hydrides react with water liberating H_2 again.

$$CaH_2 + H_2O \rightarrow Ca(OH)_2 + H_2\uparrow$$

In metal hydrides, hydrogen exists as a nega-

Identifying substances as acidic, basic or neutral by the action of litmus

Depending on the action of litmus, substances are classified into three types – acidic, basic and neutral. In general acidic substances are sour to taste while basic substances are bitter. Neutral substances are neither sour nor bitter. Substances can be identified as acidic, basic or neutral by their action on moist litmus paper as follows.

Moist *blue* litmus paper turns *red* in the presence of *acidic substances* but remains unchanged in the presence of basic substances. Moist *red* litmus paper turns *blue* in the presence of *basic substances* but remains unchanged in the presence of acidic substances.

Blue litmus is turned by **A**cids to **R**ed (**B A R**).

tive ion whereas in compounds with non-metals it exists as a positive ion.

(e) **Reaction of hydrogen with non-metals**

(i) With oxygen, hydrogen forms water.

$$2H_2 + O_2 \xrightarrow{burning} 2H_2O$$

(ii) With chlorine, hydrogen forms HCl (in diffused light).

$$H_2 + Cl_2 \xrightarrow{diffused\ light} 2HCl$$

Hydrogen does not react with chlorine in darkness. In bright light the reaction is explosive.

(iii) With fluorine, hydrogen reacts even in darkness.

$$H_2 + F_2 \rightarrow 2HF$$

(iv) With nitrogen, hydrogen forms ammonia.

$$N_2 + 3H_2 \xrightarrow[\text{in the absence of air}]{1000°C} 2NH_3$$

(v) With boiling sulphur, hydrogen forms H_2S.

$$H_2 + S \rightarrow H_2S$$

(vi) With carbon it forms methane at 1100° and C_2H_2 at 2727°C.

$$C + 2H_2 \xrightarrow{1100°C} CH_4$$

$$2C + H_2 \xrightarrow{2727°C} C_2H_2$$

(f) **Hydrogenation of vegetable oils**: Hydrogen reacts with vegetable oils to form vanaspathi.

$$\text{Vegetable oils} + H_2 \xrightarrow{Ni,\ pressure,\ heat} \text{vanaspathi}$$

Combustibility versus Supporter of Combustion

A gas that burns is **a combustible gas**. **Example:** Hydrogen gas burns. Hydrogen is combustible.

A gas that allows burning is **a supporter of combustion**.

Example : Oxygen makes a glowing splinter burn more brightly. Oxygen is a supporter of combustion.

Oxidation versus Reduction

A substance is said to be oxidized, if oxygen is added or hydrogen is removed from it. Substances that are oxidized are *reducing agents*.

E.g.: $2CO + O_2 \rightarrow 2CO_2$

CO is oxidized, and is a reducing agent

$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$

CH_4 is oxidized (and is a reducing agent).

A substance is said to be *reduced*, if oxygen is removed or hydrogen is added to it. Substances that are reduced are *oxidising agents*.

E.g.: $CO_2 + C \rightarrow 2CO$

CO_2 is reduced (and is an oxidizing agent)

$Ca + H_2 \rightarrow CaH_2$

Ca is reduced (and is an oxidizing agent)

Foods cooked using vanaspathi have better taste and can be stored for a longer time.

(g) **Reducing properties of Hydrogen**: When hydrogen gas is passed over heated oxides of less active metals, it reduces them to free metals. *Hydrogen acts as a reducing agent.* (See box)

Hydrogen removes oxygen from metallic oxides and reduces them to free metals.

$$CuO + H_2 \rightarrow Cu + H_2O$$

10.6 TEST FOR HYDROGEN

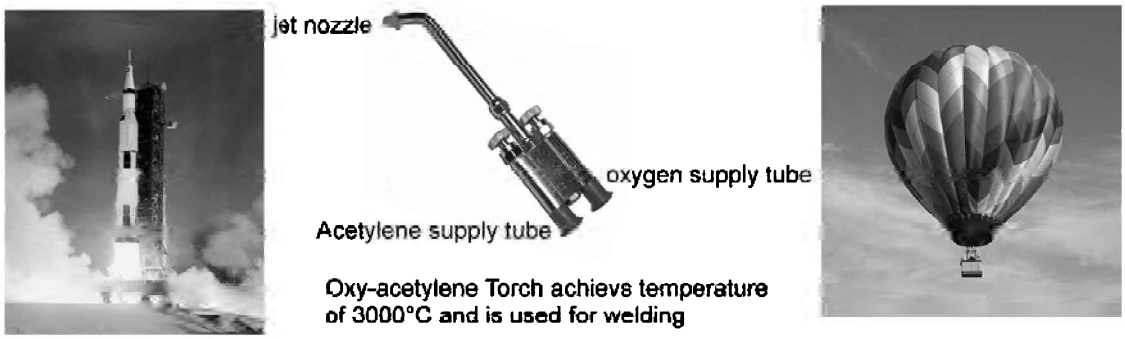
Hydrogen burns with a pop sound in air and a pale blue flame. The only product formed is water seen as small drops on the walls of the container. This unique property of hydrogen is used as a test for detecting its presence.

10.7 USES OF HYDROGEN

Hydrogen has the following uses.

(a) Hydrogen is used as a rocket fuel. *Hydrogen has the highest calorific value amongst all fuels and is extremely light*. Hence, use of

hydrogen, as a fuel in spaceships is very suitable. Its use as a common fuel is not widespread as the storage and transport of hydrogen is very risky. Hydrogen is an extremely dangerous fuel that can explode quite easily. Extreme care has to be taken in its transport. In all probability, hydrogen will be used as fuel in automobiles in the future.



Hydrogen finds varied uses-as a rocket fuel, in oxy-acetylene torches and in weather balloons

(b) Hydrogen is used in welding as it burns in air producing a temperature of 2400°C.

(c) Hydrogen is used in hydrogenation of vegetable oils to convert them to vanaspathi. Vanaspathi gives a better taste and flavour to food cooked in it.

(d) Hydrogen is used in weather balloons that find usage in weather forecasting.

(e) Hydrogen is used to extract less active metals from their oxides. More active metals form more stable oxides from which extraction of free metal using hydrogen is more difficult.

(f) Hydrogen is used for making artificial petrol.

(g) Hydrogen is used in the manufacture of other chemicals like ammonia.

EXERCISE

I. Choose the correct answer

1. Which is a natural isotope of hydrogen?
(a) $_1H^1$ (b) $_1H^4$
(c) $_1H^2$ (d) $_1H^3$

2. Dry hydrogen is collected by upward delivery (downward displacement) as it is
(a) soluble and lighter than air.
(b) insoluble and heavier than air.
(c) soluble and heavier than air.
(d) insoluble and lighter than air.

3. Which process is not used for industrial manufacture of hydrogen?
(i) Electrolysis of water (ii) Bosch process
(iii) Haber's Process (iv) Contact Process
(a) i and ii (b) i and iii
(c) iii and iv (d) all four options

4. Which is a physical property of hydrogen?
(a) Gas (b) A = 1
(c) Z = 1 (d) all the three

5. Hydrogen is a good
(a) oxidant. (b) reductant.
(c) drying agent. (d) catalyst.

6. Hydrogenation of vegetables oils converts
(a) vegetable oils to vanaspathi
(b) vanaspathi to vegetable oils
(c) vegetable oils to ghee
(d) ghee to vegetable oils

7. The chemical name of water is
(a) monoxide of hydrogen.
(b) dioxide of hydrogen.
(c) hydrogen peroxide.
(d) water vapour.

8. When hydrogen combines exclusively with metals the charge it has is
(a) positive. (b) negative.
(c) neutral charge.
(d) can be positive or negative.

9. As hydrogen is explosive the precaution that has to be taken in its laboratory preparation is that
(a) the apparatus should be airtight.
(b) heating appliances should be turned off.
(c) no precaution is necessary.
(d) both (a) and (b)

10. Hydride ion is represented by the formula ____ whereas hydrogen ion is represented by the formula ____

(a) H^-, H^+ (b) H^+, H^+

(c) H^-, OH^- (d) OH^-, H^+

11. The factor that prevents the large-scale use of hydrogen as fuel is

(i) when hydrogen burns large amount of heat energy is released.

(ii) hydrogen burns uncontrollably.

(iii) hydrogen does not catch fire easily.

(iv) hydrogen is very explosive so transportation is difficult.

(a) (i) and (ii) (b) (ii) and (iii)

(c) (iii) and (iv) (d) (i), (ii), (iii) and (iv)

12. Carbon dioxide, sulphur dioxide and nitrogen dioxide present as impurities in hydrogen gas in its laboratory preparation are removed by

(a) passing through silver nitrate ($AgNO_3$) solution.

(b) passing through lead nitrate [$Pb(NO_3)_2$] solution.

(c) passing through potassium hydroxide (KOH) solution.

(d) passing over anhydrous calcium chloride ($CaCl_2$).

13. Hydrogen gas in its laboratory preparation is dried by

(a) passing through silver nitrate ($AgNO_3$) solution.

(b) passing through lead nitrate [$Pb(NO_3)_2$] solution.

(b) passing through potassium hydroxide (KOH) solution.

(d) passing over anhydrous calcium chloride ($CaCl_2$).

14. Hydrogen sulphide gas present as impurity in hydrogen gas in its laboratory preparation is removed by

(a) passing through silver nitrate ($AgNO_3$) solution.

(b) passing through lead nitrate [$Pb(NO_3)_2$] solution.

(c) passing through potassium hydroxide (KOH) solution.

(d) passing over anhydrous calcium chloride ($CaCl_2$).

15. Arsine and phosphine present as impurities in hydrogen gas in its laboratory preparation are removed by

(a) passing through silver nitrate ($AgNO_3$) solution.

(b) passing through lead nitrate [$Pb(NO_3)_2$] solution.

(c) passing through potassium hydroxide (KOH) solution.

(d) passing over anhydrous calcium chloride ($CaCl_2$).

II. Match the following.

1. Match the example in Set A with the type of oxide in Set B

Set A	Set B
(i) CaH_2	(a) Oxide
(ii) H_2O	(b) Halide
(iii) H_2O_2	(c) Hydride
(iv) HF	(d) Super oxide

2. Match the reactants in Set A with the products in Set B.

Set A	Set B
(i) $[CO + H_2] + H_2O$(steam)	(a) C_2H_2
(ii) $ZnO + H_2$	(b) CH_4
(iii) $C + 2H_2 \xrightarrow{1100°C}$	(c) $CO_2 + 2H_2$
(iv) $2C + H_2 \xrightarrow{2727°C}$	(d) $Zn + H_2O$

3. Match the reactions of hydrogen with non-metals given in Set A with a characteristic property of the product formed in Set B.

Set A	Set B
(i) Carbon	(a) rotten egg smell
(ii) Sulphur	(b) fuel
(iii) Chlorine	(c) gives dense white fumes with HCl
(iv) Nitrogen	(d) gives dense white fumes with ammonia

4. Match the reactions given in Set A with the equations in Set B

Set A	Set B
(i) Carbon and steam	(a) $MgH_2 + 2H_2O \rightarrow Mg(OH)_2 + 2H_2$
(ii) Hydrogen and oxygen	(b) $PbO + H_2 \rightarrow Pb + H_2O$
(iii) Reducing property of hydrogen	(c) $2H_2 + O_2 \rightarrow 2H_2O$
(iv) Magnesium hydride with water	(d) $C + H_2O \xrightarrow{170°C} CO + H_2$

5. Match the substances in Set A with its property in Set B.

Set A	Set B
(i) Sulphuric Acid	(a) reducing agent
(ii) Sodium Hydroxide	(b) has no action on red litmus
(iii) Hydrogen	(c) oxidising agent
(iv) Chlorine	(d) has no action on blue litmus

III. State whether the following statements are True or False

1. Hydrogen is the most abundant element in the universe but only available to the extent of 0.9% by weight on the earth.
2. Hydrogen can form both cation and anion by loosing and gaining an electron respectively.
3. Hydrogen is the best fuel in terms of calorific value.
4. Hydrogen is the only element that has no neutrons.
5. In the laboratory preparation hydrogen cannot be collected by downward displacement of air as it forms an explosive mixture with air.

IV. Fill in the blanks

1. Sodium reacts with cold water forming _____ and liberating _____.
2. When steam is passed over red hot coke, a mixture of _____ and _____ is formed.
3. As hydrogen is an extremely _____ gas it is used in weather balloons.
4. One of the industrial uses of hydrogen is in the industrial preparation of _____by combining with nitrogen gas.
5. When hydrogen combines with metals it forms _____.

V. Each question contains two statements. Statement A is an Assertion Statement and Statement R is the Reason Statement. Each question has four choices out of which only one is correct. Choose the correct one.

Choose A if Statement A is true, Statement R is true; Statement R is correct explanation for statement A

Choose B if Statement A is true, statement R is true; Statement R is not correct explanation for statement A

Choose C if Statement A is true, Statement R is false

Choose D if Statement A is false, statement R is true

1. **Statement –A**: Hydrocarbon fuels are non-polluting.

   Statement –R: Hydrocarbons on complete burning form carbon dioxide and water as the only products.
2. **Statement –A**: Hydrogen at present cannot be used as a fuel on a large scale.

   Statement – R: Storage of hydrogen is dangerous as it is a light gas that is very explosive.
3. **Statement –A**: Hydrogen if required dry, is collected by downward displacement of air.

   Statement – R: Hydrogen is a light gas.
4. **Statement – A**: One kilogram of methane gas releases less heat energy than hydrogen gas.

   Statement – R: Hydrogen has the highest calorific value amongst all fuels releasing enormous amounts of heat energy on burning.
5. **Statement – A**: Hydrogen is a non-polluting fuel.

   Statement – R: The only by product of using hydrogen as a fuel, is water.

VI. This section contains a paragraph. Based on the paragraph linked comprehension type multiple-choice questions have to be answered. Each question has 4 choices, out of which only one is correct. Choose the correct one.

A fuel on burning gives heat and light energy. Human beings for their day-to-day requirements can use this energy. A good fuel has the following characteristics.

(i) the fuel should be easily available,
(ii) the fuel should be easily transported,
(iii) the burning of the fuel should be easy to control,
(iv) the fuel should have a low ignition temperature, and
(v) the fuel should have a high calorific value.

Ignition temperature of a fuel is the temperature at which it catches fire. Calorific value of a fuel is the amount of heat energy released when one kg of the fuel is burnt. Hydrogen has the highest calorific value amongst all fuels. It also has a low ignition temperature. However, hydrogen cannot be used on a large scale at present as storage and transport are difficult and hydrogen is very expensive to manufacture.

1. Three fuels that can be used easily are hydrogen, methane (marsh gas) and butane (LPG). The formulae of these fuels are H_2 (hydrogen), CH_4 (marsh gas) and C_4H_{10} (LPG). The fuel out of the given fuels that will have the highest calorific value is
   (a) methane (b) hydrogen
   (c) butane
   (d) all three fuels have same calorific values
2. The increasing order of calorific values of the fuels CH_4, H_2, C_4H_{10} is
   (a) $CH_4 < H_2 < C_4H_{10}$
   (b) $H_2 < CH_4 < C_4H_{10}$
   (c) $C_4H_{10} < CH_4 < H_2$
   (d) $H_2 = CH_4 = C_4H_{10}$
3. The factors which do not favour use of hydrogen as a fuel
   (i) causes pollution
   (ii) cost of manufacture is high
   (iii) storage is difficult
   (iv) transport is risky as it is highly inflammable
   (a) (i) only (b) (ii), (iii) and (iv)
   (c) (i) and (iii)
   (d) all four options (i), (ii), (iii) and (iv)

4. Which of these fuels has a higher calorific value?
   a. Water Gas
   b. Producer Gas
   c. both (a) and (b) have same calorific value
   d. both are not good fuels.
5. Nuclear energy cannot be used as a common fuel because
   (i) nuclear fuel is not easily transported
   (ii) nuclear fuel is very expensive
   (iii) nuclear fuel is difficult to use
   (iv) nuclear reactions require high temperatures to start
   (v) nuclear fuel is not easily available
   (vi) nuclear reactions are not easily controlled
   (vii) disposal of nuclear waste is very difficult and expensive
   (a) (i), (ii) and (iii) (b) (iii), (iv) and (v)
   (c) (v), (vi) and (vii)
   (d) all the given options (i), (ii), (iii), (iv), (v), (vi) and (vii)

VII. Answer the following questions.

1. State the position of hydrogen in the Modern Periodic Table.
2. What is the abundance of hydrogen in the universe and on the earth?
3. What are the isotopic forms of hydrogen and how do they differ from each other?
4. Give the reaction for the laboratory preparation of hydrogen.
5. How and why is hydrogen collected in its laboratory preparation?
6. How are zinc granules prepared? Why is this form of zinc used in the laboratory preparation of hydrogen?
7. Give the reactions involved in the two methods by which hydrogen can be prepared in the industry.
8. What is Water Gas shift reaction?
9. Give reasons why hydrogen is a good fuel.
10. Why is hydrogen not at present used as a fuel for common use?
11. Give a test for hydrogen.
12. Why are oils hydrogenated? How is hydrogenation done?
14. Hydrogen can form positive ions as well as negative ions. Elaborate this statement.
15. Which property makes hydrogen suitable for weather forecasting?
16. The higher the hydrogen content, in a hydrocarbon, more is its calorific value. Arrange the following fuels in increasing order of calorific value:
    CH_4, C_4H_{10}, H_2, C_2H_2, C.
17. Give one reaction where hydrogen is used as a reducing agent. Is hydrogen oxidized or reduced in this reaction? Explain.
18. Hydrogen can be used for the preparation of other chemicals like ammonia. Give the reaction.

Equation practice

Complete and balance the following reactions. Give the reaction conditions where asked.

1. $H_2 + O_2 \rightarrow$
2. $H_2 + Cl_2 \xrightarrow{?}$
3. $H_2 + F_2 \rightarrow$
4. $N_2 + H_2 \xrightarrow{?}$
5. $H_2 + S \rightarrow$
6. $C + H_2 \xrightarrow{?}$
7. $C + H_2 \xrightarrow{?}$
8. $Ca + H_2 \rightarrow$
9. $CaH_2 + H_2O \rightarrow$
10. $CuO + H_2 \rightarrow$

KEY TO EXERCISE

I. 1. a 2. d 3. c 4. d 5. b 6. a 7. a 8. b
9. d 10. a 11. d 12. c 13. d 14. b 15. a

II. 1. i. c ii. a iii. d iv. b 2. i. c ii. d iii. b iv. a
3. i. b ii. a iii. d iv. c 4. i. d ii. c iii. b iv. a
5. i. b ii. d iii. a iv. c

III. 1. true 2. true 3. true 4. true 5. true

IV. 1. sodium hydroxide, hydrogen 2. carbon monoxide, hydrogen
3. light 4. ammonia 5. metal hydrides

V. 1. D 2. A 3. D 4. A 5. A

VI. 1. b 2. b 3. b 4. a 5. d

11 Study of Group IVA Element—Carbon and its Oxides



Antoine Lavoisier in his youth



Carl Wihlem Scheele

11.0 INTRODUCTION TO CARBON

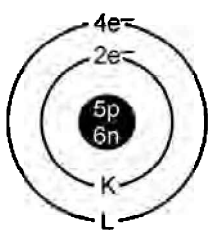
Carbon is one of the three most important elements on earth with hydrogen and oxygen for sustaining life.

Although it forms a large variety of compounds, most forms of carbon are comparatively unreactive under normal conditions. There are over ten lakh carbon compounds known. As there are so many compounds with carbon, the study of carbon compounds has been classified as a separate branch of chemistry called *Organic Chemistry*.

The name carbon comes from Latin *carbo* for coal and charcoal. Carbon was known in the form of soot and charcoal to the earliest human civilisation. Diamond (a form of carbon) was known as early as 2500 BC in China.

In 1772, Lavoisier (see photo) showed that diamonds are a form of carbon. Scheele (see photo) showed that graphite, which had been thought of as a form of lead, was actually a type of carbon. A new allotrope of carbon, fullerene, was discovered in 1985.

Carbon is the fourth most abundant element in the universe by mass after hydrogen, helium and oxygen. It is present in all known life forms. In the human body, carbon is the second most abundant element by mass (about 18.5%) after oxygen. It is the 12th most abundant element on earth. Only 0.09% of earth's crust is carbon.



Structure of carbon atom

Carbon the element

Symbol – C,
Z = 6, A = 12
Physical state: solid, non-metal.
Electronic configuration – 2, 4
Valency – 4
Carbon belongs to Group IV A and 2nd period.
Other elements of the group are Silicon, Germanium, Tin and Lead.

11.1 OCCURRENCE

Free State: In the pure form, carbon occurs as *diamond* and *graphite*. In the impure form, carbon occurs as wood and coal.

Combined state – Carbon occurs in the combined state in the following compounds.

(a) Gaseous compounds: Carbon dioxide (0.03% of air is CO_2), carbon monoxide (in volcanic gases), hydrocarbons, natural gas, marsh gas.

(b) Solid compounds: Carbonates – $CaCO_3$ (limestone, chalk), $MgCO_3$ (magnesite), $CaCO_3.MgCO_3$ (dolomite). Carbon also occurs in the combined state as fats, carbohydrates and proteins.

Charcoal and coke

Charcoal and coke are not natural forms of carbon.

Wood when heated in absence of air forms *charcoal*.

Coal when heated in absence of air forms *coke*.

11.2 ALLOTROPES OF CARBON

Allotropy is the phenomenon in which an element exists in two or more physical forms with same chemical properties but different physical properties. Each form is known as an *allotrope*.

The allotropes of carbon are classified as follows:

Allotropes of Carbon

Crystalline Allotropes	Amorphous Allotropes	
a. Diamond	a. Coal →	i. Coke
b. Graphite		ii. Gas carbon
c. Fullerenes	b. Lampblack or soot	
	c. Charcoal →	i. Wood Charcoal
		ii. Bone Charcoal
		iii. Sugar Charcoal

11.3 CRYSTALLINE ALLOTROPES

Diamond

Diamond is the purest form of carbon. Diamonds are often found near volcanoes. Diamonds are formed deep in the earth's crust where high-pressure conditions are present.

Study of Diamond

Physical Properties

1. Diamond is the hardest naturally occurring substance.
2. Pure diamond is colourless, transparent and brittle.
3. Density of diamond is high at 3.5, due to close packing of carbon atoms.
4. Diamond is a poor conductor of electricity and heat as all valencies are satisfied with covalent bonding. There are no free electrons to conduct electricity.
5. The refractive index, of diamond is high at 2.5. Due to high refractive index, the light ray entering a diamond crystal suffers total internal reflection, due to which it sparkles.
6. Diamond is insoluble in all solvents
7. Diamond is stable when heated in air; it does not melt.
8. If heated in vacuum diamond changes to graphite.

Chemical Properties

1. Diamond is an inert substance as all valencies are satisfied.
2. At 900°C, diamond catches fire. No ash is formed indicating its purity: $C + O_2 \rightarrow CO_2$
3. With fluorine at 750°C, diamond forms CF_4. $C + 2F_2 \rightarrow CF_4$
4. Diamond does not react with acids or bases.
5. Diamond reacts with sodium carbonate at very high temperatures forming CO.

Uses – Diamond has the following uses

1. In making jewellery
2. To cut glass
3. In making drill tips
4. In watches as decorative pieces
5. In polishing stones used in jewellery
6. In producing laser beams

Gems

Presence of impurities in diamonds gives colour to diamonds. Coloured diamonds are called *gems*. Example: *Emerald* is a gem with chromic oxide (Cr_2O_3) as impurity.

Structure of diamond

In diamond, each carbon atom is chemically bonded to four other carbon atoms in a tetrahedral shape. The shape is rigid and hard. The distance between two carbon atoms is 1.54Å (1Å = 1×10^{-8} cm) and C – C – C bond angle is 109°28′ (see figure below).





Preparation of artificial diamonds

Natural diamonds are scarce. There is a huge demand for artificial diamonds. Artificial diamonds are prepared by heating sugar charcoal to a very high temperature (3000 – 3500°C) and compressing it to

a very high pressure. If boron is added to diamonds it becomes a *semi-conductor*. Semi-conductors are used extensively in electronic appliances.

Graphite

Graphite is greyish black solid. Graphite is also called *Black Lead* or *Plumbago*. Graphite is found in Sri Lanka and USA.

Study of Graphite

Physical Properties

1. Graphite is a dark grey, very soft solid; it has metallic lustre.
2. Density of graphite is 2.25gm/cc.
3. Graphite marks paper grey.
4. Graphite is good conductor of electricity due to its free 4th electron.
5. Graphite is insoluble in solvents.

Chemical Properties

1. Graphite is an inactive element though slightly more reactive than diamond as it has one unsatisfied valency.
2. Graphite does not burn in air.
3. In pure oxygen graphite catches fire at 700°C. Some ash is formed, as is not very pure form of carbon. $C + O_2 \rightarrow CO_2 +$ Heat released.
4. Graphite has no action with acids or alkalis.
5. Graphite reacts only with strong oxidizing agents such as boiling nitric acid, forming CO_2.
6. Graphite reacts with sodium carbonate at very high temperatures (fusion temperature) forming CO.

Uses

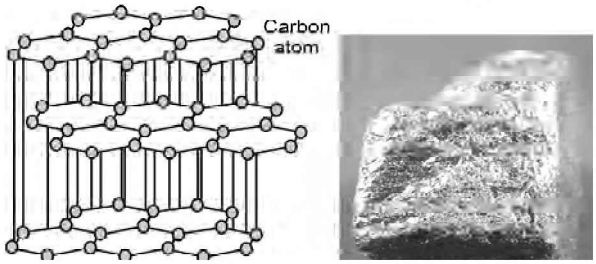
Graphite has the following uses.

1. Graphite is used to make lead pencils. Graphite is mixed with powdered sand and made into 'lead' pencils – more the clay harder the pencil lead.
2. Graphite is used as an electrode in electrolysis.
3. Graphite is used as a dry lubricant.
4. Graphite is used for making refractory crucibles.
5. Graphite is used in nuclear reactors as a *moderator* that reduces the speed of neutrons.

Structure of graphite

Every carbon atom is chemically bonded to three other carbon atoms in a hexagonal ring in one plane. Each carbon atom is also bonded with *one carbon atom in the adjacent plane below* and *one more carbon atom in the adjacent plane above by weak van der Waals' forces*. The bonds between carbon atoms in the same plane are strong bonds while those between carbon atoms in the adjacent planes are weak. The parallel rings can thus slip over each other. This makes graphite a good lubricant.

In the same plane, bond distance between adjacent carbon atoms is 1.42 Å and bond angle is 120°, while the bond length between carbon atoms in adjacent planes is 3.35 Å (see figure below).



Preparation of artificial graphite by Acheson process:

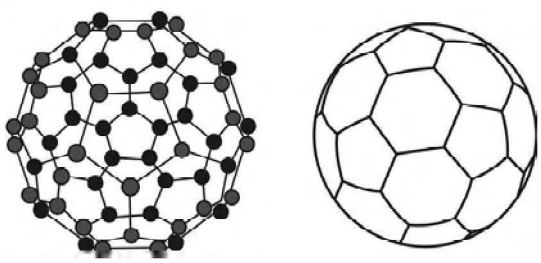
A mixture of coke and sand, are heated in presence of iron (III) oxide as catalyst. An electric current is passed through the mixture for 20-30 hours to create a temperature of 3000 °C. The coke and sand react to form silicon carbide which then decomposes to its elements. Silicon vaporizes leaving *graphite* behind.

$$SiO_2 + 3C \rightarrow SiC + 2CO$$

$$SiC \rightarrow Si\uparrow + \underset{\text{graphite}}{C}$$

Fullerenes (Crystalline Allotrope)

Buckminster Fullerene, C_{60} is a rare allotrope of C with 12 pentagonal and 20 hexagonal rings arranged in the shape of a football. There are a total of 60 carbon atoms arranged in 32 rings. The centre is hollow. Bond length is 1.4 Å. This allotrope has the potential for use in AIDS treatment.



Fullerenes are obtained by condensing vaporized graphite.

Fullerenes are allotropes of carbon found in meteorite craters and some rocks in Russia. Fullerenes with 70, 90 and 120 carbon atoms have been discovered. Some of these fullerenes contain atoms of other elements and are *superconductors*.

11.4 AMORPHOUS ALLOTROPES

(a) Coal

Coal is a natural allotrope while coke, lamp black (soot) and gas carbon are prepared artificially from carbonaceous substances.

Occurrence

Coal forms naturally over millions of years as plants and animals die and the fossil remains get buried over time deep under the earth. Plant tissue is made up of cellulose molecules – $(C_6H_{10}O_5)_n$. Bacterial decomposition of plant tissues results in removal of hydrogen and oxygen leaving the residue rich in carbon.

Coal is an impure form of carbon and is classified based on the percentage of carbon as follows.

Peat – 50 to 60% of carbon
Lignite – 60 to 70% of carbon
Bituminous – 70 to 80% of carbon
Anthracite – 80 to 90% of carbon.

Thus *peat* is the most inferior form while *anthracite* is the most superior form of coal. Coal contains impurities such as nitrogen, sulphur, phosphorous, moisture, etc.

Uses

Coal is used primarily as a fuel. Destructive distillation (heating in absence of air) of coal yields many useful products. The coal loses its volatile constituents leaving coke behind.

$$\text{Coal} \xrightarrow[700^0C-1300^0C]{\textit{destructive distillation}} \text{Coke}$$

(i) *Coal gas* containing $H_2 + CH_4 + CO$ (used as fuel),
(ii) *Ammoniacal liquor*, a concentrated solution of ammonia, ammonium compounds, and sulphur compounds (this is modified and used as a fertilizer).
(iii) C*oal tar* containing benzene, toluene phenol, etc, organic chemicals.
(iv) Coke containing above 90% C (fuel).

(b) Coke

Preparation

Coke is prepared by destructive distillation of bituminous coal at 1300°C when the volatile impurities get vaporized leaving a black residue. This black residue has around 98% carbon and is called *coke*.

Reactions in a fresh coke furnace

When coke is lit to form a fire, the reactions take place in three stages.

(i) **At the bottom** of the stack of coke, *where there is plenty of air due to fanning of the fire*, carbon dioxide forms.

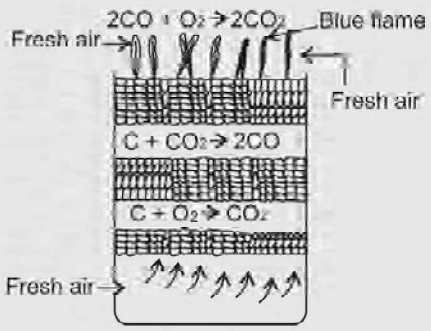
$$C + O_2 \rightarrow CO_2$$

(ii) **In the middle** of the stack of coke, where the supply of air reduces but the temperatures are quite high carbon dioxide is reduced by the hot coke to carbon monoxide.

$$CO_2 + C \rightarrow 2CO$$

(iii) **At the top**: Carbon monoxide rises to the top of the stack of coke where the air supply is unlimited. It burns with a blue flame.

$$2CO + O_2 \rightarrow 2CO_2$$



Gas carbon

When coal is destructively distilled the impurities (mainly hydrocarbon compounds) decompose to give *gas carbon* as a hard grey deposit.

Gas carbon is *not a gas* but a grey solid that is a good conductor of electricity. Gas carbon is used to make electrodes and in arc lamps.

Properties

Coke is porous and black. Coke burns with almost no smoke as coke has very little impurity when compared to coal. Coke is a very strong reducing agent. Coke is a non-conductor of heat.

Uses

Coke is used as a fuel. It is used to make graphite crucibles and calcium carbide. It is also used as a reducing agent in metallurgy. Coke is used for the manufacture of gaseous fuels like producer gas and water gas.

(c) Lampblack

Preparation

Lampblack is obtained by incomplete combustion of carbonaceous fuels like kerosene. A black deposit of *lampblack* can be seen on the inside of the glass holder of a kerosene wick lamp. When a steel plate is introduced into a candle flame it becomes coated with a layer of *lampblack*.

Properties

Lampblack is black and light in weight. It burns with little or no smoke, as it is a pure form of carbon. Lampblack is a non-conductor of electricity.

Uses

Lampblack is used in rubber tyres, black shoe polish; in ink used in printers and typewriter ribbons.

(d) Charcoal

Charcoal is formed when carbon containing compounds are heated in a limited supply of air.

(i) Wood charcoal

Wood charcoal is formed by destructive distillation of dry wood. The products are *wood charcoal*, wood gas, wood tar and pyroligenous acid.

Physical properties of wood charcoal

Wood charcoal is black, porous, soft and brittle. It *adsorbs* gases (see box). Wood charcoal decolorises and deodorises and is a bad conductor of heat and electricity. Charcoal floats on water due to adsorbed gases; if the water is boiled the adsorbed gases escape and the charcoal sinks to the bottom.

Chemical properties of wood charcoal

(1) Wood Charcoal burns in air or oxygen at about 400°C to form carbon dioxide (if excess air is present) or carbon monoxide (if limited air is present).

Excess air: $C + O_2 \xrightarrow{\Delta} CO_2$;

Limited air: $2C + O_2 \xrightarrow{\Delta} 2CO$

(2) When heated with sulphur, charcoal forms carbon disulphide:

$$C + 2S \xrightarrow{\Delta} CS_2$$

(3) When heated with hydrogen, charcoal forms methane:

$$C + 2H_2 \xrightarrow{\Delta} CH_4$$

(4) Wood charcoal is a good reducing agent.

– Charcoal reduces metal oxides to metal:

Adsorption

Adsorption is absorption of a gas on the surface of another solid called the adsorbing medium. It takes place only at the surface of the medium. After adsorption, simple heating of the medium, will release the *adsorbed* gas.

Activated charcoal

Wood charcoal is *heated in steam* to 900°C when it becomes very porous. *Its adsorbing capacity increases manifold.*

Activated charcoal can also be prepared by destructive distillation of coconut shells.

$$ZnO + C \xrightarrow{\Delta} Zn + CO$$

– Charcoal reduces steam to water gas:

$$H_2O + C \xrightarrow{\Delta} [CO + H_2]$$

(Water gas)

– Charcoal reduces CO_2 to CO:

$$CO_2 + C \xrightarrow{\Delta} 2CO$$

– Charcoal reduces oxides of calcium and silicon to respective carbides:

$$CaO + 3C \xrightarrow{\Delta} CaC_2 + CO$$

$$SiO_2 + 3C \xrightarrow{\Delta} SiC + 2CO$$

– Charcoal reduces concentrated sulphuric and nitric acids to respective oxides:

$$C + 2H_2SO_4 \rightarrow CO_2 + 2H_2O + 2SO_2$$

$$C + 4HNO_3 \rightarrow CO_2 + 2H_2O + 4NO_2$$

Uses of wood charcoal

Wood charcoal is used as a fuel. It has a low ignition point and is often used to start a fire in crackers.

Wood charcoal is used as an adsorbent in gas masks used in mining, defence and industry.

Wood charcoal is used as a deodoriser for petroleum products.

Wood charcoal is used as a decolorising agent in sugar solutions, organic preparations, alcohol, etc.

Wood charcoal is used as a constituent in gunpowder and crackers.

(ii) Sugar Charcoal

Sugar charcoal is the purest form of amorphous carbon. It is prepared by the action of concentrated sulphuric acid on cane sugar or by heating cane

sugar in a limited supply of air.

$C_{12}H_{22}O_{11} \rightarrow 12C + 11H_2O$

The residue is heated in chlorine to remove any remaining hydrogen leaving behind carbon.

Use: Sugar charcoal is used in the manufacture of artificial diamonds.

(iii) Bone Charcoal

Bone charcoal is prepared by the destructive distillation of bones. The residue has 10% carbon and 90% impurities like calcium phosphate and carbonate. The impurities are removed by dissolving in dilute hydrochloric acid. A black powder called *ivory black* is left behind. Bone charcoal is porous and adsorbs colouring matter.

Use: Bone charcoal is used in purification of organic compounds, decolourising sugar and in paints.

11.5 FUELS

Carbon Fuels

Fuel is a source of energy for human activity. Combustion of fuels releases energy mainly as heat (a little light is also liberated).

Most fuels contain carbon and hydrogen. They are known as *hydrocarbons*.

Characteristics of a good fuel

A good fuel should have the following characteristics.

(i) The fuel should be *easily available.*
(ii) The fuel should be *economical*.
(iii) The fuel should have *high calorific value* (see box).
(iv) The fuel should *not pollute.*
(v) The fuel should have a *low ignition temperature*.
(vi) The burning of fuel should be *controllable*.

Incomplete combustion of carbon fuels is dangerous as this produces carbon monoxide gas.

Calorific value is the amount of heat released when 1 kg of the fuel is burnt completely. It is measured in kJ/kg for solid and gaseous fuels (see box).

Calorific values of some fuels

Hydrogen	– 150 kJ/g
Coal	– 30 kJ/g
Wood	– 17 kJ/g
Butane (LPG)	– 50 kJ/g
Methane	– 55 kJ/g

The calorific value of hydrogen gas is the *highest* amongst all fuels – 150 kJ/ g. More the hydrogen content in a fuel more is its calorific value.

Ignition temperature is the temperature at which an air-fuel mixture catches fire and continues to burn on its own (see box).

Fuels are of many types. Some types of fuels are given below.

Solid fuels

Wood is mainly cellulose. Wood when burnt releases large amounts of energy.

Coal is a complex mixture of carbon, hydrogen, oxygen, nitrogen, sulphur, etc. It is formed by the action of bacteria on wood. The bacteria remove hydrogen and oxygen. Coal is formed over millions of years.

Ignition temperatures

Fuel	Ignition Temperature (°C)
Bituminous coal	300
Butane	420
Carbon	700
Carbon monoxide	609
Coke	700
Hydrogen	500
Gasoline	280
Kerosene	210
Methane (Natural Gas)	580
Peat	227
Wood	300

Liquid fuels

Crude petroleum is a mixture of hydrocarbons – compounds that contain only hydrogen and carbon. Hydrocarbons occur as underground deposits and are one of the main sources of fuel for the modern civilization.

Gaseous fuels

Coal gas formed by destructive distillation of coal at 1000°C, contains other gases like nitrogen, carbon, hydrogen and carbon dioxide.

Detecting leakages from LPG cylinders

LPG is mixed with ethyl mercaptan (C_2H_5SH), which has a foul smell for easy detection of leakages.

Hydrogen sulphide has a foul smell of rotten eggs but is not used to detect leakages from LPG cylinders as it gets oxidized to sulphur dioxide – a poisonous gas.

LPG is used as kitchen fuel as it liquefies easily under pressure at room temperature. LPG is mainly butane – C_4H_{10} (see box).

Producer gas is formed by heating carbon in insufficient air.

$$C + \text{Air (insufficient)} \rightarrow \underset{\text{Producer Gas}}{[CO + N_2]}$$

Calorific value of Producer Gas is low as around 50% of the fuel is made up of nitrogen, which is not a fuel.

Water gas is formed when superheated steam is passed over white-hot coke in Bosch process.

$$C + H_2O \rightarrow \underset{\text{water gas}}{[CO + H_2]}$$

Biogas is formed due to anaerobic fermentation of cattle dung in presence of water. It contains CH_4, CO_2, H_2 and H_2S.

11.6 COMBUSTION

Combustion or burning is an exothermic chemical reaction between a fuel and an oxidising agent with the release of of heat and light energy. The light energy is seen as a glow or flame.

The heat generated by direct combustion is necessary to maintain the high temperature necessary for burning of the fuel. In a complete combustion reaction, a compound reacts with an oxidizing agent, such as oxygen. Along with heat and light energy, the products formed are compounds of each element in the fuel with the oxidizing agent.

For example in the burning of methane the products are carbon dioxide and water.

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{heat energy} + \underset{\text{light energy}}{}$$

A simpler example can be seen in the combustion of hydrogen and oxygen, the product is water.

$$2H_2 + O_2 \rightarrow 2H_2O(g) + \text{heat energy} + \underset{\text{energy}}{\text{light}}$$

In most combustion reactions, oxygen is the oxidising agent and is obtained from air. The exhaust gases of burning of a fuel in air (called *flue gas*) will therefore contain nitrogen.

Types of combustion

Rapid combustion

Rapid combustion is a form of combustion in which large amounts of heat and light energy are released that often results in a *fire*. An example is an internal combustion engine. A large volume of gas is liberated in combustion besides the production of heat and light. The sudden evolution of large quantities of gas creates excessive pressure that produces a loud noise. Such a combustion is known as an *explosion*.

Slow combustion

Slow combustion is a form of combustion which takes place at low temperatures. Cellular respiration, metabolism are two examples of slow combustion.

Causes of pollution

Pollution is caused mainly by burning of carbonaceous fuels.

In reality combustion is not a complete or perfect process.

Consider two common types of combustions:

(i) Combustion of coal

(ii) Combustion of carbon compounds (for example combustion of hydrocarbons, wood etc.)

In the exhaust gases or flue gases formed both unburned carbon (as soot) and carbon compounds (carbon monoxide and others) will be present. When the combustion takes place in air, some nitrogen can be oxidized to various oxides of nitrogen. Thus, combustion leads to release of soot, carbon monoxide and oxides of nitrogen into the atmsophere polluting it.

Complete combustion

In complete combustion, the reactant burns in oxygen, producing a limited number of products.

For example when a hydrocarbon burns *completely* in oxygen, the reaction will only yield carbon dioxide and water along with heat and light energy.

Incomplete combustion

Incomplete combustion occurs when enough oxygen is not present to allow the fuel (usually a hydrocarbon) to react completely with oxygen. It produces carbon monoxide.

Incomplete burning of hydrocarbons gives a yellow flame due to unburnt carbon particles. Complete combustion of hydrocarbons gives a blue flame.

Incomplete combustion is harmful

Carbon monoxide and soot are the most common byproducts of incomplete combustion. Both cause pollution. Incomplete combustion also leads to less heat energy being evolved than when the fuel burns to undergo complete combustion.

11.7 OXIDES OF CARBON

Carbon forms two oxides: carbon monoxide and carbon dioxide. A lesser-known unstable oxide of carbon is carbon suboxide – C_2O_3. Carbon monoxide though poisonous is an excellent fuel as it burns releasing large amounts of heat with no smoke.

11.8 CARBON MONOXIDE

Preparation of carbon monoxide

The following reactions are used to prepare carbon monoxide.

(a) Reducing oxides of less active metals with charcoal or coke:

$$Fe_2O_3 + 3C \xrightarrow{\Delta} 2Fe + 3CO$$

$$ZnO + C \xrightarrow{\Delta} Zn + CO$$

(b) Passing carbon dioxide through red hot coke:

$$CO_2 + C \xrightarrow{\text{red hot}} CO - \Delta$$

(c) Passing steam through red hot coke :

$$H_2O + C \rightarrow CO + H_2$$

(Super heated steam + white hot coke)

Carbon monoxide

Molecular formula – CO
Molecular weight: 28
A gas at room temperature
Occurrence: Near volcanoes, vehicle fumes, marshy places.
Carbon monoxide is a product of incomplete combustion of carbon fuels.

Poisonous nature of Carbon monoxide

Just 1% of carbon monoxide in air is fatal. Carbon monoxide is 300 times more soluble in blood than oxygen. On inhalation, it forms carboxy-hemoglobin, depriving the tissues of oxygen causing death.

(d) Dehydrating oxalic acid or formic acid by using concentrated sulphuric acid.

$$HCOOH + H_2SO_4 \rightarrow CO + H_2SO_4.H_2O$$

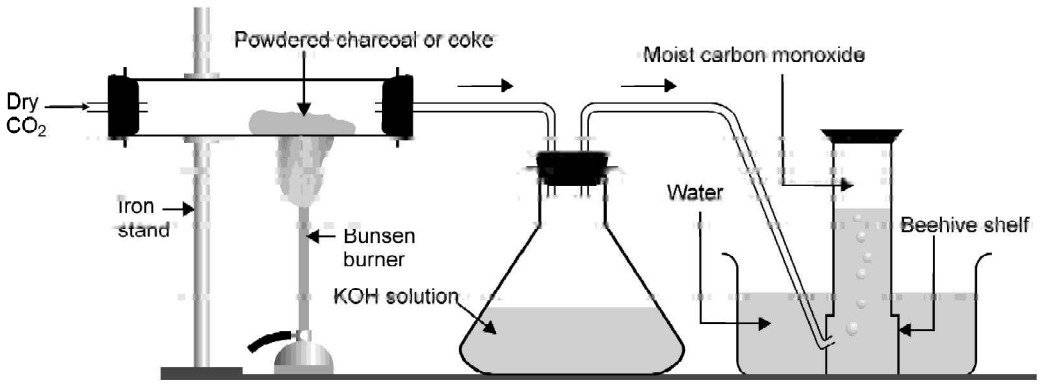
$$(COOH)_2 + H_2SO_4 \rightarrow CO + CO_2 + H_2SO_4.H_2O$$

Laboratory preparation of carbon monoxide

Method 1: From carbon dioxide gas

Dry carbon dioxide gas is passed over red hot charcoal when the gas gets reduced to carbon monoxide. The gaseous product is passed through caustic potash to remove excess carbon dioxide (see diagram). The carbon monoxide is collected by downward displacement of water. The reaction is

$$CO_2 + C \text{ (heat)} \rightarrow 2CO$$



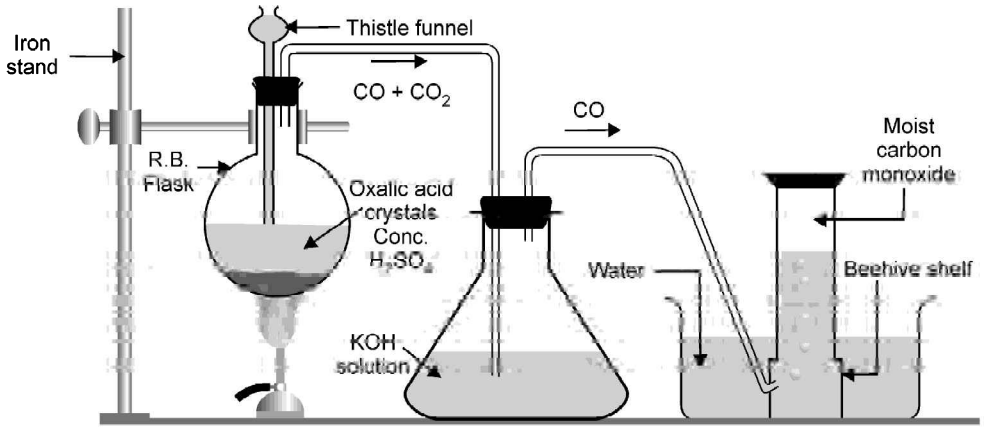
Laboratory Preparation of Carbon Monoxide from Carbon dioxide gas

Method 2: From oxalic acid

Oxalic acid is dehydrated by heating with concentrated sulphuric acid to form carbon monoxide and carbon dioxide gases. The gases are bubbled through caustic potash (see diagram) to remove carbon dioxide and traces of sulphur dioxide that may have formed. The carbon monoxide is collected by downward displacement of water. The reaction is

$$(COOH)_2 + H_2SO_4 \xrightarrow{\Delta} CO + CO_2 + H_2SO_4.H_2O$$

$$KOH + CO_2 \rightarrow K_2CO_3 + H_2O$$



Preparation of Carbon monoxide by the dehydration of oxide acid crysials

Physical properties

The physical properties of carbon monoxide are listed below.

1. Carbon monoxide is colourless, odourless and tasteless.
2. Carbon monoxide is as light as air.
3. Carbon monoxide is insoluble in water.
4. Carbon monoxide has a boiling point of –192°C.
5. Carbon monoxide is very poisonous.

Chemical properties

The chemical properties of carbon monoxide are listed below.

1. Carbon monoxide is a combustible gas, burning with a blue flame
2. Carbon monoxide is a neutral gas.
3. Carbon monoxide *reacts with non-metals* as shown.

   (a) $2CO + O_2 \rightarrow 2CO_2$

   (b) $CO + 2H_2 \xrightarrow{Zn+Cu,450°C} CH_3OH$ (Methanol)

   (c) $CO + Cl_2 \xrightarrow{sunlight} CO.Cl_2$ (Phosgene or carbonyl chloride)
4. Carbon monoxide *reacts with metals* forming *carbonyls* (see box).

   (a) $Fe + 5\ CO \rightarrow Fe(CO)_5$ (Iron carbonyl)

   (b) $Ni + 4CO \rightarrow Ni(CO)_4$ (Nickle carbonyl)
5. Carbon monoxide exhibits reducing property. It reduces metal oxides to the metals as shown.

   (a) $Fe_2O_3 + 3CO \xrightarrow{\Delta} 2Fe + 3CO_2$

   (b) $ZnO + CO \xrightarrow{\Delta} Zn + CO_2$

   (c) $CuO + CO \xrightarrow{\Delta} Cu + CO_2$

   Carbon monoxide reduces superheated steam to hydrogen getting oxidized to carbon dioxide. *This reaction is called Water gas Shift Reaction and is used to manufacture hydrogen on a large scale. The carbon dioxide formed is easily separated from hydrogen by passing the two gases under high pressure through water when carbon dioxide dissolves. The hydrogen is dried and collected.*

   $CO + H_2O \xrightarrow{Fe_2O_3+Cr_2O_3,450°C} CO_2 + H_2\uparrow$
6. When compressed to a high pressure and passed through water, carbon monoxide forms formic acid.

   $CO + H_2O \xrightarrow{high\ pressure} HCOOH$
7. Carbon monoxide reacts with ammonical cuprous chloride to form an unstable product that easily decomposes.

   $CuCl + CO + 2H_2O \rightarrow CuCl.CO.2H_2O$

   $CuCl.CO.2H_2O \xrightarrow{\Delta} CuCl + CO + 2H_2O$

Carbonyls are unstable

Metal carbonyls are poisonous, volatile unstable liquids that decompose very easily to metals. *The carbonyls are used in the refining of certain metals in their metallurgy.*

Method of refining: The impure metal is reacted with carbon monoxide to form the carbonyl of the metal. The carbonyl being a liquid is easily separated. The unstable liquid is heated when it decomposes forming carbon monoxide *leaving pure metal behind.*

This property is used to separate carbon monoxide from other gases that *do not react with ammonical cuprous chloride*, like nitrogen, oxygen, hydrogen, etc.

8. Carbon monoxide reacts with soda lime to form sodium formate.

$$CO + NaOH \xrightarrow{\text{lime - CaO}} \underset{\text{(Sodium formate)}}{HCOONa}$$

Uses

1. Carbon monoxide is used as a fuel. It is a constituent of coal gas, water gas, producer gas. All of these fuels are gaseous fuels with high calorific value.
2. Carbon monoxide is used to refine metals like nickel by converting them to liquid carbonyls.
3. Carbon monoxide is used to extract metals by its reducing action.
4. Carbon monoxide is used to prepare *phosgene* that is used in the dyeing industry and as a war gas.
5. Carbon monoxide is used to manufacture chemicals like sodium formate, methyl alcohol, synthetic petrol, etc.

11.9 CARBON DIOXIDE

Preparation of carbon dioxide

(a) Burning charcoal or hydrocarbons:

$$C + O_2 \rightarrow CO_2$$
$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

(b) Heating of bicarbonates:

$$2NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$$
$$Ca(HCO_3)_2 \rightarrow CaCO_3 + CO_2 + H_2O$$

(c) Heating of carbonates:

$$CaCO_3 \rightarrow CaO + CO_2$$
$$ZnCO_3 \rightarrow ZnO + CO_2$$

(d) Treating carbonates with dilute acids:

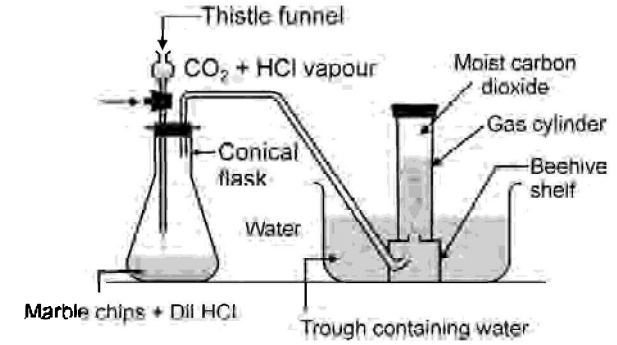
$$Na_2CO_3 + 2HCl \rightarrow 2NaCl + CO_2 + H_2O$$
$$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$$

Laboratory preparation of carbon dioxide

Method 1: Carbon dioxide is prepared by the action of dilute hydrochloric acid on calcium carbonate. The gas is collected by downward displacement of water (see diagram). The reaction is

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2\uparrow + H_2O$$

Dilute sulphuric acid cannot be used as it forms a precipitate of calcium sulphate ($CaSO_4$) which forms a coating on the marble and prevents further action.



Laboratory preparation of CO_2 gas by the action of dil. H_2SO_4 on marble.

Method 2: Carbon dioxide can also be prepared by heating sodium bicarbonate when it decomposes to sodium carbonate and carbon dioxide. The reaction is

$$2NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$$

Antidote to CO poisoning

As carbon monoxide is an extremely poisonous gas, immediate resurrection is needed to anyone who has inhaled the gas. Artificial respiration is given using *carbogen*. (*Carbogen* is a mixture of 95% oxygen and 5% carbon dioxide).

To avoid carbon monoxide poisoning in factories, the workers wear special *masks*. The masks contain special air-filters made of *hopcalite* – a mixture of 50% MnO_2, 30% CuO, 15% Co_2O_3 and 5% Ag_2O which oxidizes CO to CO_2.

Test for carbon monoxide

(a) CO has no effect on limewater.

(b) CO burns with a blue flame.

(c) A filter paper dipped in palladium chloride or platinum chloride solution on exposure to carbon monoxide turns from white to pink with low levels and greenish black with high levels of carbon monoxide.

Carbon dioxide

Molecular formula – CO_2; Mol Wt: 44; gas at room temperature

Occurrence: 0.03% of the atmosphere is carbon dioxide. Carbon dioxide occurs as dissolved gas in water, as carbonates and bicarbonates in the earth's crust.

The gas is collected by upward displacement of air, as carbon dioxide is heavier than air.

Physical properties of carbon dioxide

The physical properties are listed below.

1. Carbon dioxide is an acidic oxide.
2. Carbon dioxide is a colourless, odourless gas with a sour taste.
3. Carbon dioxide is fairly soluble in water (1vol. dissolves in 1vol of water under pressure).
4. Carbon dioxide is heavier than air.
5. Carbon dioxide solidifies at room temperature (25°C) at 70 atm pressure to form *solid carbon dioxide* known as *dry ice*. *Dry ice* on heating sublimes to gaseous state.
6. Carbon dioxide is a non-poisonous gas but causes death due to asphyxiation (deprival of oxygen).

Chemical properties of carbon dioxide

The chemical properties are listed below.

1. Carbon dioxide is not combustible and does not support combustion.
2. Carbon dioxide turns moist blue litmus red showing that it is an acidic gas.
2. When a piece of burning magnesium is inserted in a jar of carbon dioxide, it continues to burn forming magnesium oxide and black particles of carbon.

$$CO_2 + 2Mg \rightarrow 2MgO + C$$

3. Carbon dioxide reacts with water to form a weak acid.

$$CO_2 + H_2O \rightarrow H_2CO_3$$

4. Carbon dioxide reacts with alkalis to form carbonates and in excess to form bicarbonates.

$$2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$$
$$Na_2CO_3 + H_2O + CO_2 \rightarrow 2NaHCO_3$$

When carbon dioxide is bubbled through limewater, the limewater becomes milky. The milky solution becomes clear when excess carbon dioxide is bubbled through the limewater.

The milkiness forms as an *insoluble carbonate* is formed: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3\downarrow + H_2O$

The milkiness disappears as a *soluble bicarbonate* is formed:

$$CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$$

This reaction is often used as a test for the presence of carbon dioxide.

5. Carbon dioxide reacts with non-metals:
$CO_2 + C \rightarrow 2CO$
6. Carbon dioxide reacts with metals, oxidizing them to oxides and getting reduced to carbon.

$$4Na + CO_2 \rightarrow 2Na_2O + C$$
$$2Ca + CO_2 \rightarrow 2CaO + C$$
$$2Mg + CO_2 \rightarrow 2MgO + C$$

7. Carbon dioxide reacts with metal oxides forming carbonates.

$$Na_2O + CO_2 \rightarrow Na_2CO_3$$
$$CaO + CO_2 \rightarrow CaCO_3$$
$$4Al + 3CO_2 \rightarrow 2Al_2O_3 + 3C$$

Removing excess carbon dioxide from chemical reactions

Though limewater turns milky with carbon dioxide, limewater is not used to remove excess carbon dioxide. Limewater contains only 0.19% of calcium hydroxide which is available for reaction with carbon dioxide. Instead caustic potash is used as it contains a higher percentage of potassium hydroxide that is available for reaction with carbon dioxide

Test for carbon dioxide

There are three tests that are to be done to conclusively establish the presence of carbon dioxide.

(a) The gas should turn limewater milky (this test is not conclusive as SO_2 and SO_3 also turn limewater milky).

(b) The gas should put off a burning candle (this test is also not a conclusive test).

(c) The gas should react with burning magnesium leaving a black deposit of carbon. The other two gases, SO_2 and SO_3, will leave yellow sulphur as deposit.

$$2Mg + CO_2 \rightarrow 2MgO + C$$

Gases that turn limewater milky

Like carbon dioxide, sulphur dioxide and sulphur trioxide also turn limewater milky, forming insoluble $CaSO_4$ and $CaSO_3$ respectively. The milkiness disappears on excess gas being bubbled as in the case of carbon dioxide due to the formation of soluble $Ca(HSO_4)_2$ and $Ca(HSO_3)_2$ respectively.

Uses of carbon dioxide

1. Salts like 'ENO' are used for reducing acidity and gas formation in the stomach. 'ENO' is a mixture of dry citric acid and tartaric acid with sodium bicarbonate. When water is added to this mixture, carbon dioxide is released to relieve gas pressure in the stomach.
2. Carbon dioxide under high pressure is bubbled through soft drinks to give them the tingling sensation and fizz.
3. Dry ice or solid carbon dioxide is used as a refrigerant. Dry ice has many advantages over ordinary ice as a refrigerant. Dry ice lasts longer, cools to below –40°C, is non poisonous and sublimes slowly leaving no residue.
4. Carbon dioxide in the form of baking powder is used in the baking industry for making cakes and bread. When heated the baking powder releases carbon dioxide which helps the cakes and breads to become soft.
5. Carbon dioxide is used to manufacture urea. Carbon dioxide and ammonia are compressed to 150 atm pressure and heated to 150°C to form urea.

$$2NH_3 + CO_2 \xrightarrow{150°C,\ 150atm} NH_2.CO.NH_2 + 2H_2O$$

6. Carbon dioxide is used to manufacture *carbogen*, *washing soda*, *baking soda* (as a leavening agent) and *white lead* $\{Pb(OH)_2.PbCO_3\}$
7. Carbon dioxide is heavier than air and incombustible. This property of carbon dioxide makes it a good fire extinguisher.
8. Carbon dioxide is used by plants during photosynthesis.

Comparison of general properties between carbon monoxide and carbon dioxide

	Carbon monoxide	Carbon dioxide
(a) Density	As dense as air	Heavier than air.
(b) Solubility	Insoluble	Partially soluble. 1 volume CO_2 dissolves in 1 volume of water.
(c) Liquefaction	Very difficult to liquefy	Can solidify at room temperature at a pressure of 70 atm.
(d) Poisonous nature	Poisonous	Non-poisonous.
(e) Combustibility	Burns with blue flame	Non-combustible.
(f) Nature	Neutral oxide	Acidic oxide.
(g) Action on lime water	No action	Turns milky. If excess CO_2 is passed the milkiness disappears.
(h) Action on red hot coke	No action	Oxidizes coke to CO: $C + CO_2 \rightarrow 2CO$
(i) Oxidizing action	None	Forms oxide: $2Mg + CO_2 \rightarrow 2MgO + C$
(j) Reducing action	Reduces metallic oxides to metals: $Fe_2O_3 + 2CO \rightarrow 2Fe + 2CO_2$	No reducing action.

11.10 FIRES AND FIRE EXTINGUISHERS

As seen above, in rapid combustion large amounts of heat and light energy are released that often results in a *fire*. A fire that can be controlled is desirable but fires that are uncontrollable lead to destruction of property.

A *fire extinguisher* is used to control a fire by putting it out.

Carbon dioxide gas is often used in fire extinguishers to put out fires. As carbon dioxide is heavier than air it sinks to the bottom forming a blanket cutting off the air supply.

Fires can be caused due to oil spillage or due to electrical short circuit or it may be just an ordinary fire. Fire extinguishers are of three basic types. The fire extinguisher that is used depends on the type of fire.

Liquid CO_2 fire extinguisher

Used for all types of fires.

Liquid CO_2 *fire extinguishers are used to put off all types of fires including oil and electric fires.* Liquid carbon dioxide is stored in cylinders under

pressure. The cylinders are fitted with a valve which when opened lets out a spray of flakes of dry ice (solid carbon dioxide) $CO_{2(liquid)} \rightarrow CO_{2(gas)}$

Foam type fire extinguishers

Used for oil fires but not electric fires.

Foam type fire extinguishers are used to put off oil fires. They cannot be used to put off electric fires. The basic reaction used to produce carbon dioxide is

$$6NaHCO_3 + Al_2(SO_4)_3 \rightarrow 3Na_2SO_4 + 2Al(OH)_3 + 6CO_2\uparrow$$

A diagram showing how a *foam type fire extinguisher* looks is given below. A foam of carbon dioxide ejects out of the nozzle, when the seal is broken. The foam covers the burning oil fire, cutting off the air supply.

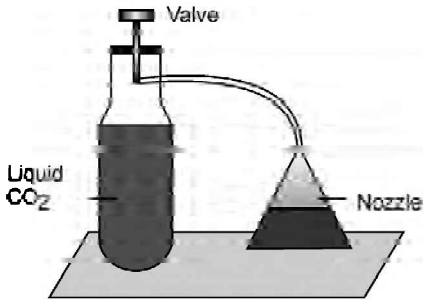
Soda-acid fire extinguishers

Used for ordinary fires but not oil or electric fires.

Soda-acid fire extinguishers are used to put off ordinary fires. They cannot be used to put out oil fires or electrical fires. The basic reaction used to produce carbon dioxide is

$$6NaHCO_3 + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O + 2CO_2\uparrow$$

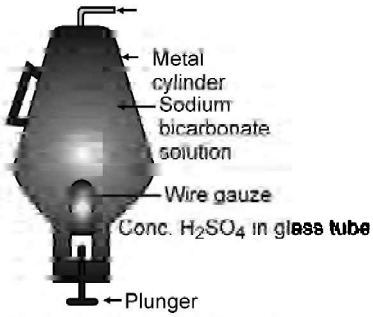
A diagram showing how a *soda-acid fire extinguisher* looks is given below. A solution of conc. H_2SO_4 mixed with $NaHCO_3$ ejects out of the nozzle when the seal is broken. The solution sinks to the bottom of the fire as oil is less dense than the solution. So soda-acid type fire extinguisher cannot be used for oil fires. They cannot be used for electrical fires also as they contain electrolytes which can conduct electricity and cause shock.



Liquid carbon dioxide fire extinguisher



Foam acid type fire extinguisher



Soda acid type fire extinguisher

EXERCISE

I. Choose the correct answer

1. _____ is a rare allotrope of carbon.
(a) graphite (b) diamond
(c) Buckminister Fullerene (d) coal

2. The fuel with the highest calorific value amongst the given fuels is
(a) biogas (b) producer gas
(c) water gas (d) hydrogen

3. Carbon monoxide differs from carbon dioxide in which property?
(a) Reducing property (b) Acidic nature
(c) Combustibility (d) All three

4. Formic acid when heated with conc. sulphuric acid forms
(a) carbon monoxide (b) carbon dioxide
(c) carbonates (d) bicarbonates

5. The important natural sources of carbon compounds is / are
(a) coal (b) petroleum
(c) natural gas (d) all three (a, b, c)

6. Destructive distillation of coal gives
(a) coal gas (b) coal tar
(c) ammonical liquor (d) all three (a, b, c)

7. The artificial form of carbon from the given options is
(i) charcoal (ii) coke
(iii) lampblack (iv) graphite
(a) i, ii and iii (b) ii and iii
(c) i, ii, iii and iv (d) iv only

8. All valencies are not satisfied in
(a) diamond b. graphite
(c) Buckminster fullerene
(d) all the three (a, b, c)

9. Diamonds are coloured due to
(a) presence of carbon
(b) presence of impurities
(c) total internal reflection
(d) all the three (a, b, c)

10. The inferior form of coal is _____ coal.
(a) peat b. bituminous
(c) anthracite d. lignite

11. The form of coal that produces least amount of smoke on burning is

(a) peat coal (b) bituminous coal
(c) anthracite coal (d) lignite coal

12. The option that correctly describes the type of reaction that takes place at the bottom of a fresh coke fire is

(a) oxidation, exothermic
(b) reduction, endothermic
(c) oxidation, endothermic
(d) reduction, exothermic

13. The option that correctly describes the type of reaction that takes place in the middle a fresh coke fire is

(a) oxidation, exothermic
(b) reduction, endothermic
(c) oxidation, endothermic
(d) reduction, exothermic

14. The option that correctly describes the type of reaction that takes place at the top of a fresh coke fire is

(a) oxidation, exothermic
(b) reduction, endothermic
(c) oxidation, endothermic
(d) reduction, exothermic

15. The blue flame at the top of a fresh coke fire is due to

(a) complete burning of carbon.
(b) more air supply
(c) burning of carbon monoxide
(d) reduction of carbon dioxide.

16. The fuel with highest calorific value is

(a) C_2H_6 (b) C_2H_4
(c) C_2H_2 (d) C_2H_5OH

17. The property which makes wood charcoal a good deodoriser is

(a) reducing property.
(b) burns exothermically.
(c) adsorption
(d) combines with other elements.

18. The allotrope of carbon that is used in shoe polish is

(a) graphite. (b) charcoal.
(c) coke (d) lampblack.

19. The allotrope of carbon that is used as an electrode is

(a) graphite. (b) charcoal.
(c) gas carbon. (d) both (a) and (c)

20. The allotrope of carbon that is used as a reducing agent in metallurgy is

(a) graphite. b. charcoal.
(c) coke d. lampblack.

21. The allotrope of carbon that is used for making artificial diamonds is

(a) graphite. (b) sugar charcoal.
(c) bone charcoal. (d) wood charcoal.

22. The allotrope of carbon that is used in nuclear reactors is

(a) graphite. (b) charcoal.
(c) coke (d) lampblack.

23. The allotrope of carbon that is used in producing gaseous fuels like producer gas, water gas is

(a) graphite. (b) charcoal.
(c) coke. (d) lampblack.

24. Water gas is a mixture of

(a) CO and H_2. (b) CO and N_2.
(c) CO, CH_4 and H_2. (d) CO, CH_4 and N_2.

25. Producer gas is a mixture of

(a) CO and H_2. (b) CO and N_2.
(c) CO, CH_4 and H_2. (d) CO, CH_4 and N_2.

26. Between Water gas, biogas and producer gas, the fuel with the highest calorific value is

(a) Water gas (b) Producer gas
(c) Biogas
(d) all three have the same calorific values.

27. The chemical added to butane to detect leakages, while filling LPG cylinders for domestic use is

(a) sulphur dioxide. (b) hydrogen sulphide.
(c) ethyl mercaptan. (d) perfumes.

28. The combustion which is the least polluting is

(a) rapid combustion.
(b) slow combustion.
(c) complete combustion.
(d) incomplete combustion.

29. Carbon monoxide is the acid anhydride of

(a) oxalic acid (b) formic acid
(c) carbonic acid (d) both (a) and (b)

30. The acid anhydride of carbonic acid is

(a) carbon monoxide (b) carbon dioxide
(c) both (a) and (b) (d) none

31. The gas that is neutral to dry litmus is

(a) dry carbon monoxide
(b) dry carbon dioxide
(c) both (a) and (b)
(d) neither (a) nor (b)

32. Carbogen is a mixture of

(a) 95% CO_2 and 5% O_2
(b) 95% O_2 and 5% CO_2
(c) 97% CO_2 and 3% O_2
(d) 97% O_2 and 3% CO_2

33. Hopcalite is a mixture of _____ used in gas masks to prevent inhalation of poisonous gases in factories.

(a) bone charcoal and wood charcoal

(b) 50% MnO_2, 30% CuO, 15% Co_2O_3 and 5% Ag_2O

(c) ferric oxide and coke

(d) 50% MnO_2, 30% CuO and 15% Co_2O_3.

34. Carbonyls are used for metal refining as they

(a) are liquids that can be easily separated

(b) are stable compounds

(c) are unstable liquid compounds that decompose easily on heating

(d) both (a) and (c)

II. State whether the given statements about fire extinguishers are true or false.

1. A liquid fire extinguisher is used for electrical fires only.
2. A foam type fire extinguisher is used for oil fires but not electrical fires.
3. A soda-acid fire extinguisher is used for all fires including electrical and oil fires.

III. Match the following.

1. Match the mixture in Set A with the composition in Set B.

Set A	Set B
(i) Health salts	(a) carbon monoxide with nitrogen
(ii) Carbogen	(b) carbon monoxide with hydrogen
(iii) Producer gas	(c) citric acid with sodium bicarbonate
(iv) Water gas	(d) oxygen with carbon dioxide

2. Match the reaction in Set A with the product obtained in Set B.

Set A	Set B
(i) Table sugar is dehydrated	(a) Carbon monoxide
(ii) Complete combustion of hydrocarbon carbon	(b) Carbon monoxide and dioxide
(iii) Gaseous product of thermal decomposition of calcium bicarbonate is sent over heated coke.	(c) Carbon dioxide and water
(iv) Dehydration of oxalic acid	(d) Carbon

3. Match the reactant from Set B that is required to be added to the reactant in Set A to form the stated product. (reaction conditions are not required.

Set A	Set B
(i) Methyl alcohol from carbon monoxide	(a) calcium hydroxide
(ii) Carbon from carbon dioxide	(b) caustic soda
(iii) Calcium carbonate from carbon dioxide	(c) burning magnesium
(iv) Sodium formate from carbon monoxide	(d) hydrogen

IV. State whether the following statements are True or False

1. Gas carbon is a crystalline allotrope of carbon.
2. Graphite is a bad conductor of electricity.
3. Anthracite is the most inferior form of coal.
4. The allotrope of carbon used for printers ink is lampblack.
5. Activated charcoal is an adsorbent.
6. Diamond is the purest form of carbon.
7. Gems are pure diamonds.
8. Due to its structure graphite can be used as a lubricant.
9. Incomplete combustion of hydrocarbon fuels leads to pollution.
10. A filter paper dipped in a solution of palladium chloride turns the filter paper pink.

V. *In this lesson you have learnt that carbon dioxide is an excellent fire extinguisher. Read the paragraphs below and answer the questions that immediately follow each paragraph. Each question has four options out of which only one option is correct. Choose the correct one.*

Fire extinguishers are divided into four categories, based on different types of fires. Class A extinguishers are for ordinary combustible materials such as paper, wood, cardboard, and most plastics. Class B fires involve flammable or combustible liquids such as gasoline, kerosene, grease and oil. Class C fires involve electrical fires. Class D fire extinguishers are commonly found in a chemical laboratory. They are for fires that involve combustible metals, such as magnesium, titanium, potassium and sodium.

1. Class A fire extinguishers can be used for
   (a) oil fires. (b) electrical fires.
   (c) chemical fires. (d) ordinary fires.
2. Class B fire extinguishers can be used for
   (a) oil fires. (b) electrical fires.
   (c) chemical fires. (d) forest fires.

3. Class C fire extinguishers can be used for
   (a) oil fires. (b) electrical fires.
   (c) chemical fires. (d) forest fires.
4. Class D fire extinguishers can be used for
   (a) oil fires. (b) electrical fires.
   (c) chemical fires. (d) forest fires.

VI. Fill in the blanks

1. When diamond and graphite catch fire, ash is left behind by _____ but not _____.
2. Diamond is _____ whereas graphite is _____ in shape.
3. Destructive distillation of coal gives _____, _____, _____ and _____.
4. Coke is prepared by destructive distillation of _____ coal at 1300°C.
5. Producer gas is a mixture of _____ and _____.
6. Between producer gas and water gas, the fuel having higher calorific value is _____.
7. A burning piece of magnesium leaves a _____ residue with carbon dioxide.
8. Carbon dioxide reacts with ammonia at 150°C and 150 atm to form _____.
9. When carbon dioxide is bubbled in excess through lime water the product formed is a solution of _____.
10. Health salts used for relieving gas pressure are made of _____, _____ and _____

VII. Each question contains two statements. Statement A is an Assertion Statement and Statement R is the Reason Statement. Each question has four choices out of which only one is correct. Choose the correct one.

Choose A if Statement A is true, Statement R is true; Statement R is correct explanation for statement A

Choose B if Statement A is true, statement R is true; Statement R is not correct explanation for statement A

Choose C if Statement A is true, Statement R is false

Choose D if Statement A is false, statement R is true

1. **Statement –A**: Soda-acid fire extinguishers cannot be used in oil fires.

   Statement –R: Soda-acid fire extinguishers form a solution that sinks to the bottom of the fire.
2. **Statement –A**: Soda-acid fire extinguishers are used in electrical fires.

   Statement –R: Soda-acid fire extinguishers form a solution that has electrolytes which can cause electric shock.
3. **Statement –A**: Foam type fire extinguishers cannot be used to put out electrical fires.

   Statement –R: Foam type extinguishers do not cut the air supply to the fire.
4. **Statement –A**: Turning of limewater milky when a gas is bubbled into it is a sure test of the presence of CO_2.

   Statement –R: Lime water reacts with CO_2 forming a precipitate of calcium carbonate.
5. **Statement –A**: Dilute sulphuric acid can be used instead of hydrochloric acid in the laboratory preparation of carbon dioxide gas from marble.

   Statement –R: A precipitate of calcium sulphate forms in the above reaction that prevents the reaction from completion.

VIII. Answer the following questions.

1. Give the reactions in a coke furnace in a free supply of air.
2. Why is carbon monoxide not prepared in the laboratory by burning coke?
3. How can we treat carbon monoxide poisoning victims?
4. Industrial workers working near coal furnaces wear gas masks to prevent carbon monoxide poisoning. What is the chemical used in these masks?
5. Using limewater only how would you prove a given gas is carbon monoxide?
6. How would you distinguish between carbon dioxide and sulphur dioxide?
7. Carbon dioxide is used as a fire extinguisher. Name the different types of fire extinguishers and for which type of fires they are applicable.
8. What type of fire extinguisher is used for electrical fires?
9. How can you separate carbon monoxide from carbon dioxide using a neutral liquid?
10. Compare diamond and graphite with respect to structure, physical properties and chemical properties.
11. List the uses of diamond and graphite.
12. State how carbon compares with other elements of its Group in atomic size, allotropy, catenation, electronegativity, metallic character, non-metallic character, reactivity, and melting and boiling points.
13. Why is the allotrope $C_{Graphite}$ more reactive than $C_{Diamond}$?
14. Diamond sparkles. Give two reasons for it.
15. Why is graphite used as a lubricant?
16. Why is graphite used as an electrode?

17. What is destructive distillation? What products are obtained on destructive distillation of wood?
18. What is activated charcoal? State how it is prepared and its use?
19. Give two reactions showing that carbon can act as a reducing agent.
20. Give the preparation and one application of lampblack, bone charcoal and sugar charcoal.
21. Arrange the following fuels in increasing order of calorific value: C_2H_6, H_2, C_3H_8, C_2H_5OH, CH_4
22. Which gas has the higher calorific value: Biogas or natural gas?
23. Which has higher calorific value: C_4H_{10} or H_2?
24. Why is H_2 not used as fuel instead of C_4H_{10} as a domestic fuel?
25. A family consumes 14.5kg of LPG in 29 days. Calculate the average energy consumed per day if calorific value of LPG is 55kJ / g.
26. Why does a burning matchstick light a splinter of wood but not a log of wood?
27. Which is a better fuel?

Fuel	Ignition temperature	Calorific value
Gas A	20°C	75.7kj/g
Gas B	75°C	50kj/g

KEY TO EXERCISE

I. 1. c 2. d 3. d 4. a 5. d 6. d 7. a 8. b
9. b 10. a 11. c 12. a 13. b 14. a 15. c 16. a
17. c 18. d 19. a 20. c 21. b 22. a 23. c 24. a
25. b 26. a 27. c 28. c 29. d 30. b 31. c 32. b
33. b 34. c

II. 1. false 2. true 3. false

III. 1. i. c ii. d iii. a iv. b 2. i. d ii. c iii. a iv. b
3. i. d ii. c iii. a iv. b

IV. 1. false 2. false 3. false 4. true 5. true 6. true 7. false
8. true 9. true 10. true

V. 1. d 2. a 3. b 4. c

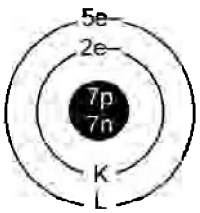
VI. 1. graphite, diamond 2. tetrahedral, hexagonal 3. coal gas, ammoniacal liquor, coal tar, coke
4. bituminous 5. carbon monoxide, nitrogen
6. water gas 7. black 8. urea 9.calcium bicarbonate
10. dry citric acid, tartaric acid, sodium bicarbonate

VII. 1. A 2. D 3. C 4. D 5. D

12 Study of Group VA Element-Nitrogen and its compounds

12.0 INTRODUCTION TO NITROGEN

Daniel Rutherford discovered nitrogen in1772. He called it *noxious air or fixed air*, but it was Lavoisier who named it Azote (meaning inert or lifeless). The terms azote or azide ion are still used for nitrogen and its ion. Chaptal renamed the gas as nitrogen in1823.



Nitrogen atom : A = 14; Z = 7

12.1 OCCURRENCE OF NITROGEN

Free state – It occurs in air to the extent of 78%.

Combined state – It occurs in all the three physical states.

Solid state – as nitre (KNO_3), chile salt petre ($NaNO_3$) and proteins.

Gaseous state – as ammonia in air where vegetables and animals decay, and as vehicle exhaust gas in the form of oxides of nitrogen.

Liquid state – as acid rain.

12.2 GENERAL METHODS OF PREPARATION OF NITROGEN

Nitrogen is prepared in two ways

(i) From chemical compounds

(ii) From air for large scale use

(i) From chemical compounds

(a) By heating Ammonium dichromate

$$(NH_4)_2Cr_2O_7 \rightarrow N_2 + Cr_2O_3 + 4H_2O$$

(b) By treating conc. Ammonium hydroxide (NH_4OH) with bleaching powder

$$2NH_4OH + 3CaOCl_2 \rightarrow N_2 + 3CaCl_2 + 5H_2O$$

(c) By treating excess Ammonia (NH_3) with Chlorine gas

NITROGEN

Symbol – N

Molecular formula – N_2

Atomicity – 2

Atomic Number – 7,

Mass Number – 14,

Atomic Weight – 14

Molecular Weight – 28

Nitrogen belongs to Group VA and 2nd period of the Periodic Table.

Electronic Configuration: (2, 5)

Valence electrons = 5

Valency – 3

Physical state – a gas at room temperature and pressure

Non-metal

Vapour density – 14

Non-poisonous

Inactive gas

$$8NH_3 + 3Cl_2 \rightarrow N_2 + 6NH_4Cl$$

(d) By heating Ammonium nitrite – this method is used in an indirect way in the laboratory preparation. (See below).

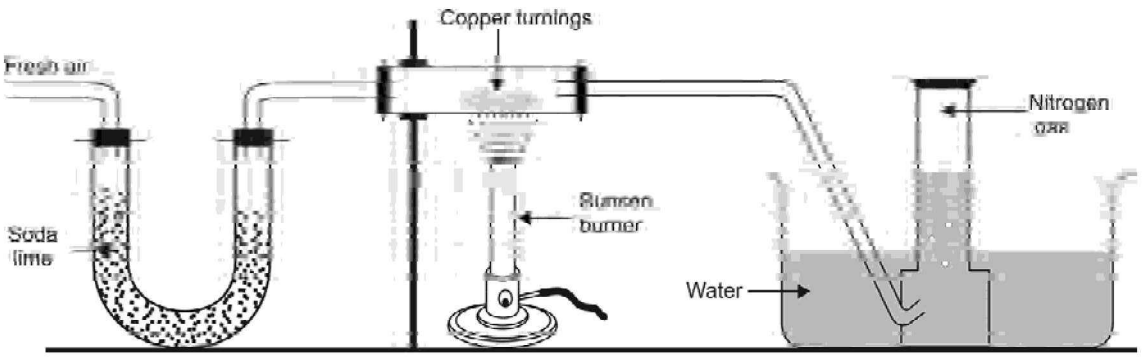
$$NH_4NO_2 \rightarrow N_2 + 2H_2O$$

(e) By oxidation of Ammonia : Ammonia can be oxidized by any of the following methods.

The reaction is

$$2NH_3 + 3CuO \xrightarrow{\Delta} N_2 + Cu + 3H_2O$$

Nitrogen can also be prepared by passing air through caustic potash to remove CO_2 and then passing the residual air over heated copper turnings to remove oxygen. Nitrogen in air comes out unreacted and is collected by downward displacement of water. (This method is sometimes preferred in the laboratory preparation. (See diagram).



By simple burning of Ammonia: $4NH_3 + 3O_2 \xrightarrow{\text{simple burning}} 2N_2 + 6H_2O$

(ii) **From air**: By fractional distillation of liquid air. See box for a summary of fractional distillation of liquid air.

FLOWCHART SHOWING THE FRACTIONAL DISTILLATION OF LIQUID AIR

Air is filtered for removing dust particles

↓

The air is passed through dilute caustic soda when gases like CO_2, SO_2 are removed from air.

↓

Next the air is cooled to below 0°C to remove moisture
(water vapour present in the air solidifies to ice)

↓

The pure air is compressed to a pressure of 200 atm due to which its temperature increases. The hot compressed air is passed into a cooling tank.

↓

Next the cooled air is *repeatedly* compressed and expanded into an evacuated chamber through a fine nozzle. Every cycle of compression and expansion results in a drop in the temperature of air *till it liquefies*. This is called *Joule Thomson Effect*.

↓

The liquid air is at a temperature of –200°C. The liquid air consists mainly of nitrogen and oxygen and traces of inert gases. The boiling point of liquid nitrogen is –195.8°C while that of liquid oxygen is –183°C. The difference in boiling points is used to separate the two liquids by fractional distillation. The liquid air is very slowly warmed to –195°C, when nitrogen starts boiling off.

↓

Nitrogen is collected by compressing into steel cylinders.

12.3 LABORATORY PREPARATION OF NITROGEN

The basic principle of the laboratory preparation is that ammonium nitrite when heated decomposes to nitrogen and water (See box).

Ammonium nitrite is unstable, difficult to store and explosive. Hence direct heating is not done. Ammonium nitrite is prepared *in situ*. Sodium nitrite ($NaNO_2$) and ammonium chloride (NH_4Cl) are dissolved in water and then heated gently in a round bottom flask.

The following reactions take place.

$$NaNO_2 + NH_4Cl \rightarrow NH_4NO_2 + NaCl$$
$$NH_4NO_2 \rightarrow N_2\uparrow + 2H_2O$$

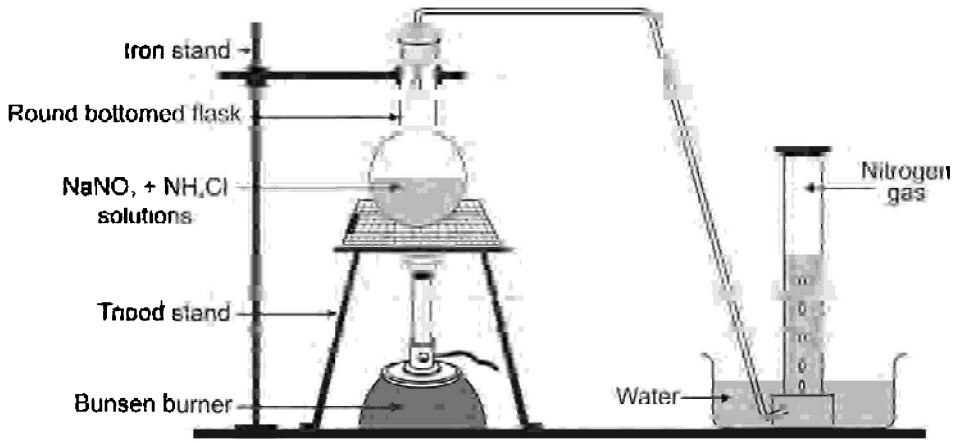
The apparatus is shown below.

Heating ammonium nitrate and nitrite

Ammonium nitrite (NH_4NO_2) when heated decomposes to give nitrogen and water, whereas ammonium nitrate (NH_4NO_3) decomposes on heating to give nitrous oxide and water.

$$NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$$

$$NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$$



Laboratory preparation of nitrogen

Method of collection: As nitrogen is very slightly soluble in water and as heavy as air, it is collected by downward displacement of water.

Some precautions –

(i) The reactants are not used in dry state as ammonium chloride (NH_4Cl) sublimes.

(ii) Ammonium nitrite is not heated directly as it is unstable and difficult to store and is an explosive.

(iii) The reaction is exothermic and may become violent. Heating should be stopped once evolution of nitrogen starts.

12.4 PHYSICAL PROPERTIES OF NITROGEN

The physical properties of nitrogen are listed below.

(i) Nitrogen is a colourless, odourless and tasteless gas.

(ii) Nitrogen is an insoluble gas lighter than air.

(iii) Nitrogen liquefies at high pressure and low temperature.

(iv) The melting point of nitrogen is –210°C and boiling point is –195.8°C.

(v) Nitrogen is a neutral, non-poisonous gas but causes death due to asphyxiation.

12.5 CHEMICAL PROPERTIES OF NITROGEN

The chemical properties of nitrogen are listed below.

(a) Nitrogen is not combustible and does not support combustion.

(b) Nitrogen is neutral to litmus

(c) Nitrogen is a non-reactive element (See box).

(d) Action of nitrogen with non-metals :

(i) Haber's process – nitrogen reacts with hydrogen in the presence of molybdenised iron as catalyst at 450 – 500°C and 200-900 atm pressure to form ammonia.

$$N_2 + 3H_2 \xrightarrow{450\text{-}500°C,\ 200\text{-}900atm,\ Fe\text{-}Mo} 2NH_3$$

(ii) Birkeland Eyde's Process – Nitrogen reacts with oxygen at 3000°C to form nitric oxide.

$$N_2 + O_2 \xrightarrow{3000°C \text{ or lightning discharge}} 2NO$$

(e) Nitrogen reacts with metals to form respective *nitrides*.

$$6Na + N_2 \xrightarrow{\Delta} 2Na_3N$$

$$3Mg + N_2 \xrightarrow{\Delta} Mg_3N_2$$

$$2Al + N_2 \xrightarrow{\Delta} 2AlN$$

$$3Ca + N_2 \xrightarrow{\Delta} Ca_3N_2$$

(f) Nitrogen reacts with calcium carbide to form *Nitrolim*, a fertilizer.

$$CaC_2 + N_2 \xrightarrow{>1000°C} [CaNCN + C]$$

(Nitrolim)

Nitrogen is non-reactive

The atoms of nitrogen have a high affinity for each other in the molecule of nitrogen (N_2). The bonds between the two atoms are very strong and cannot be broken easily. To break the bond between nitrogen atoms in the molecule of nitrogen, high heat energy has to be supplied. Only then will the nitrogen atoms separate and combine with other atoms.

This is the reason for the non-reactive nature of nitrogen.

12.6 TEST FOR NITROGEN

Nitrogen is basically a non-reactive element besides being colourless, tasteless, odourless and a neutral element. Due to this identification of nitrogen is difficult. *Any test to detect the presence of nitrogen has to be an indirect test.*

(i) Presence of nitrogen can be detected by inserting a burning piece of magnesium into a jar of the gas. If the residual ash is moistened with a few drops of water it will give the characteristic smell of ammonia.

(ii) The presence of ammonia can be confirmed by formation of dense white fumes of ammonium chloride when a glass rod inserted in conc. HCl is brought near it.

(iii) Ammonia contains nitrogen. This nitrogen must have come from the residue. The residue forms when magnesium burns in the gas. Hence the gas is nitrogen. Thus presence of nitrogen is confirmed indirectly. The reactions that nitrogen undergoes here are

$$3Mg + N_2 \xrightarrow{\Delta} Mg_3N_2$$

$$Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3\uparrow$$

12.7 USES

(a) Nitrogen is used in the manufacture of ammonia and fertilizers (see box) like *nitrolim*.

(b) Nitrogen is used to provide an inert atmosphere in electric bulbs. Inert atmosphere in an electric bulb prevents oxidation of tungsten filament (Nowadays, inert gases like argon are being used in electric bulbs instead of nitrogen as argon has better heat conductivity).

(c) Nitrogen gas is used in tinned foods to provide inert atmosphere so that food does not get spoilt. It acts as a preservative by preventing the growth of aerobic bacteria (bacteria that thrive in the presence of oxygen).

(d) Nitrogen is used in high temperature mercury thermometers – the space above mercury in such thermometers is filled with nitrogen to prevent the vaporization of mercury at high temperatures.

(e) In the atmosphere (air), nitrogen plays an important role to control the rate of combustion. (Just imagine what would happen if air contained only oxygen and someone lights a matchstick!)

Fertilisers

A good fertilizer should be soluble, have essential elements which can be absorbed by plants, should not change the acidity or basicity of soil (have good buffering action) and should be stable.

Types of fertilizer: There are two types of fertilizers: *natural* and *artificial fertilizers*.

Natural fertilizer: animal waste, sewage, bones, blood, fish scrap, plant waste, kitchen waste, etc. These wastes decompose in absence of air when they change to natural fertilizer.

Artificial fertiliser: They are of three types.

a. N-Type: The main ingredient or element is Nitrogen.
Examples:
Urea – H_2NCONH_2
Ammonium nitrate – NH_4NO_3
Ammonium sulphate – $(NH_4)_2SO_4$
Nitrolim – mixture of calcium cyanamide and carbon [CaNCN + C]

b. P-Type: The main ingredient or element is Phosphorous.
Example: Super phosphate of lime – a mixture of $[Ca(H_2PO_4)_2 + 2CaSO_4]$

c. K-Type: The main ingredient or element is Potassium – also called potash fertilizers.
Examples: KNO_3, KCl

(f) Nitrogen is an essential element in proteins.

12.8 COMPOUNDS OF NITROGEN

Nitrogen forms many important compounds. Some of them are listed below.

(i) Oxides of nitrogen

(ii) Metal nitrates and nitrites

(iii) Ammonia and its compounds

(iv) Nitric acid

(i) Oxides of Nitrogen

Nitrogen forms many oxides with oxygen. Some of these oxides are *acidic while other oxides are neutral*.

When acidic oxides react with water they form acids.

The neutral oxides do not react with acids or bases.

The oxides of nitrogen, their formulae, nature and the acids formed if any, when they react with water are given below.

Oxide Name	Formula	Nature	With water the oxide gives
Nitric oxide	NO	neutral	no reaction
Nitrous oxide (laughing gas)	N_2O	neutral	no reaction
Nitrogen dioxide (brown gas)	NO_2	acidic	Nitrous acid (HNO_2) and Nitric acid (HNO_3)
Nitrogen tetraoxide	N_2O_4	acidic	Nitrous acid (HNO_2) and Nitric acid (HNO_3)
Nitrogen trioxide	N_2O_3	acidic	Nitrous acid (HNO_2)
Nitrogen pentoxide	N_2O_5	acidic	Nitric acid (HNO_3)
Nitrogen hexoxide	N_2O_6	acidic	Nitric acid (HNO_3) and per nitric acid (HNO_4).

12.9 AMMONIA AND AMMONIUM HYDROXIDE

Occurrence

Free state: It is present in traces in air and natural water. It is present in larger quantities where organic matter decays in the absence of air.

Combined state: Destructive distillation of coal gives ammoniacal liquor. It is also found near volcanoes as NH_4SO_4.

Laboratory Preparation of ammonia

Ammonia can be prepared by two methods outlined below.

(a) Ammonia is liberated when ammonium salts react with alkalis.

$$NH_4Cl + NaOH \rightarrow NH_3\uparrow + NaCl + H_2O$$

$$(NH_4)_2SO_4 + Ca(OH)_2 \rightarrow 2NH_3\uparrow + CaSO_4 + 2H_2O$$

$$(NH_4)_3PO_4 + 3NaOH \rightarrow 3NH_3\uparrow + Na_3PO_4 + 3H_2O$$

(b) Ammonia is liberated when metallic nitrides react with water.

$$Na_3N + 3H_2O \rightarrow 3NaOH + NH_3\uparrow$$

$$Ca_3N_2 + 6H_2O \rightarrow 3Ca(OH)_2 + 2NH_3\uparrow$$

$$Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3\uparrow$$

$$AlN + 3H_2O \rightarrow Al(OH)_3 + NH_3\uparrow$$

These methods are discussed below.

(a) By heating ammonium chloride (NH_4Cl) with slaked lime:

A mixture of NH_4Cl and $Ca(OH)_2$ are heated gently. Ammonia (NH_3) is liberated as shown in the diagram. It is to be noted that only ammonium salts react with alkalis to give ammonia gas. All other salts do not give ammonia gas when they react with alkalis.

Ammonia is also called nitrogen hydride.
Formula: NH_3
Molecular weight = 17
Nature – Basic gas, non-poisonous.

$$2NH_4Cl + Ca(OH)_2 \rightarrow 2NH_3\uparrow + CaCl_2 + 2H_2O$$

Certain important precautions that have to be observed are as follows.

(i) Slaked lime and ammonium chloride (NH_4Cl) are taken in the ratio 2 : 3, to compensate for loss of ammonium chloride by sublimation on heating.

(ii) Slaked lime is used as it is economical.

(iii) Ammonium nitrate (NH_4NO_3) is not used as a reactant as it is an explosive besides being unstable.

(iv) The reactant chamber is kept in an inclined position to prevent water that is formed from trickling back. This may lead to cracking of the glass.

Collection and drying of ammonia

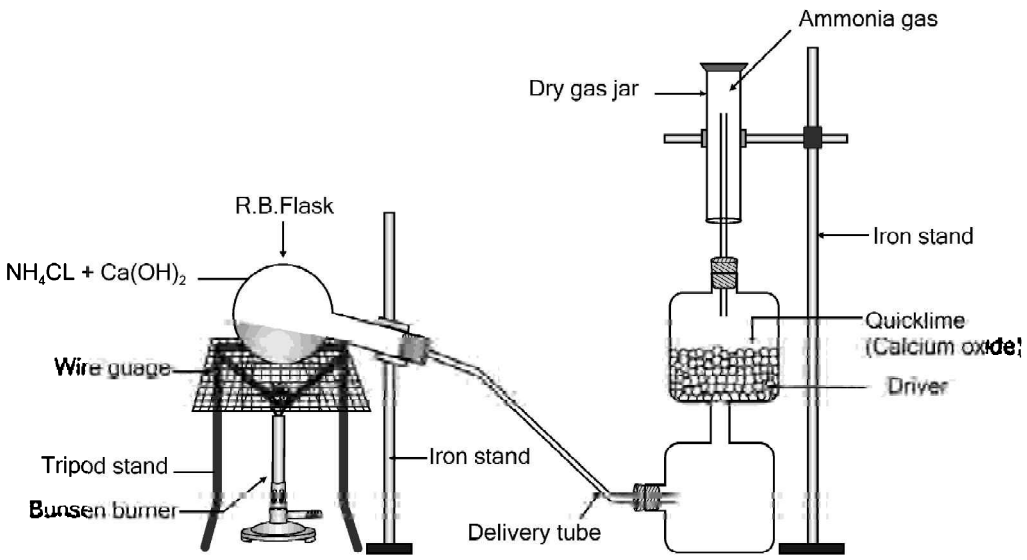
Method of collection: Ammonia gas is collected by downward displacement of air, as ammonia is extremely soluble in water and lighter than air.

Drying agent: The liberated gas is passed through quicklime (CaO) to absorb moisture. Other drying agents like concentrated sulphuric acid (H_2SO_4) or phosphorous pentoxide (P_2O_5) or calcium chloride ($CaCl_2$) cannot be used to dry ammonia as they react with it as follows.

$$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$$

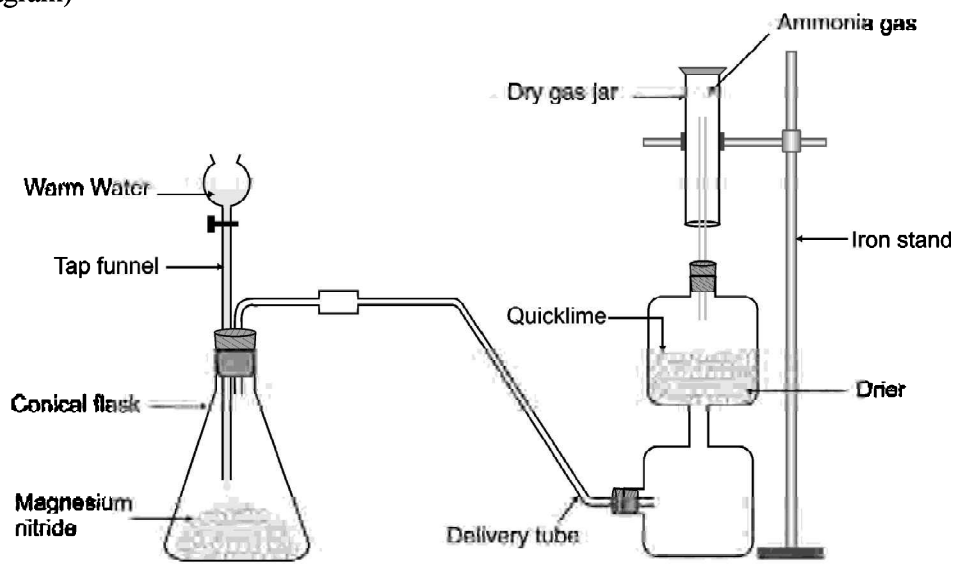
$$6NH_3 + P_2O_5 + 3H_2O \rightarrow 2(NH_4)_3PO_4$$

$$8NH_3 + CaCl_2 \rightarrow CaCl_2.8NH_3$$



Apparatus for laboratory preparation of ammonia gas.

(b) Ammonia can also be prepared by heating metallic nitrides like magnesium nitride (Mg_3N_2) with water. (See diagram)



Preparation of ammonia from magnesium nitride.

12.10 INDUSTRIAL PREPARATION OF AMMONIA BY HABER'S PROCESS

Reactants: The reactant – dry nitrogen (N_2) gas and dry hydrogen (H_2) gas are taken in the ratio 1 : 3. *Care must be exercised that the reactants are dry and pure to prevent poisoning of catalyst*. If the catalyst is spoilt the whole reacting tower and machinery has to be dismantled for replacement.

Conditions of the reaction:

Temperature and pressure: The reactants are heated to a temperature of 450-500°C and compressed to a pressure of 200-900 atm.

Catalyst: The catalyst used is either molybdenised iron or a mixture of $Fe_2O_3 + 1\%K_2O + 3\%Al_2O_3$.

Reaction: $N_2 + 3H_2 \xrightarrow{(Fe+Mo, 450°C, 200\text{-}900 atm} 2NH_3$

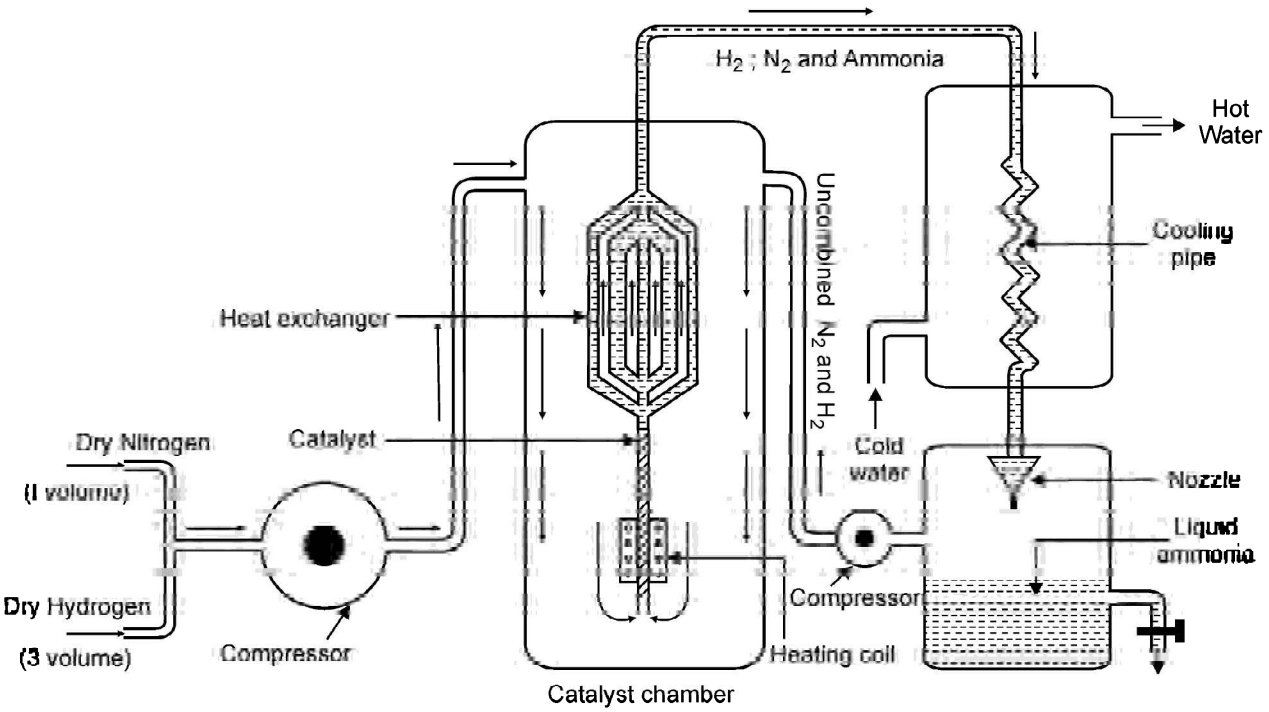
Reaction flow chart : H_2, N_2 → Compressor → Catalyst Chamber → Condenser → Condenser → NH_3 gas sent for liquefaction

Collection and storage of ammonia gas

Ammonia is collected and stored by the following two methods.

(a) Ammonia is easily liquefied on being compressed to 8 atm and cooling to –33°C.

(b) Ammonia gas is absorbed in water (1vol of water dissolves 702 vol of ammonia).



Haber's process for the synthesis of ammonia

12.11 PHYSICAL PROPERTIES OF AMMONIA GAS

The physical properties of ammonia are listed below.

(i) Ammonia is a colourless gas with a very strong characteristic smell that brings tears to the eyes and causes burning in the nose.

(ii) Ammonia has a bitter taste.

(iii) Ammonia is extremely soluble. 1 vol of water can dissolve 702 volumes of NH_3 at STP.

(iv) Ammonia is lighter than air. Under high pressure and low temperature it can be liquefied and then solidified on further increasing the pressure and lowering the temperature.

(v) Ammonia has a melting point of –78°C and boiling point of –33.4°C.

(vi) Ammonia is non-poisonous but due to its basic nature can damage the respiratory system.

12.12 CHEMICAL PROPERTIES OF AMMONIA GAS

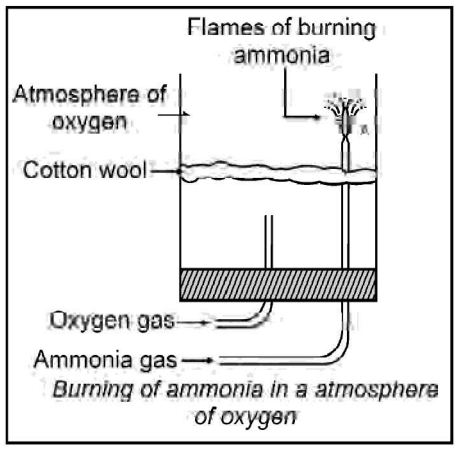
The chemical properties of ammonia are listed below.

(i) Ammonia is neither combustible nor supports combustion.

(ii) Ammonia fumes in air due to formation of NH_4OH.

(iii) *Simple burning*: Ammonia burns in pure oxygen with a pale greenish yellow flame. (See diagram).

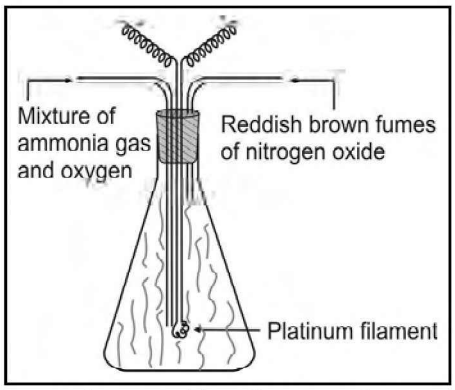
$$4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$$



Burning of ammonia in a atmosphere of oxygen

(iv) *Catalysed burning:* Catalysed burning of ammonia is called *Ostwald's reaction.*

If air or oxygen is present in limited quantities, nitric oxide and water are formed. If air or oxygen is present in excess, then nitric oxide is further oxidized to brown nitrogen dioxide (NO_2) gas (See diagram).



(v) Ammonia turns blue litmus red.

(vi) Ammonia reacts with acids to form salt.

Example: With HCl it forms dense white fumes of ammonium chloride.

$$NH_3 + HCl \rightarrow NH_4Cl$$

(vii) Action with chlorine (See box):

If chlorine is in excess

$$NH_3 + 3Cl_2 \rightarrow NCl_3 + 3HCl$$

If NH_3 is in excess

$$8NH_3 + 3Cl_2 \rightarrow 6NH_4Cl + N_2$$

An easy method to remember how ammonia reacts with chlorine

Chlorine is a powerful oxidizing agent.

(i) Chlorine in excess

Step 1: Chlorine oxidizes ammonia to nitrogen.

Step 2: As chlorine is in excess, it reacts further with nitrogen

$$2NH_3 + 3Cl_2 \rightarrow 6HCl + N_2$$
$$N_2 + 3Cl_2 \rightarrow 2NCl_3$$
$$NH_3 + 3Cl_2 \rightarrow NCl_3 + 3HCl$$

(ii) Ammonia in excess

Step 1: Chlorine oxidizes ammonia to nitrogen.

Step 2: As ammonia is in excess, it reacts further with HCl.

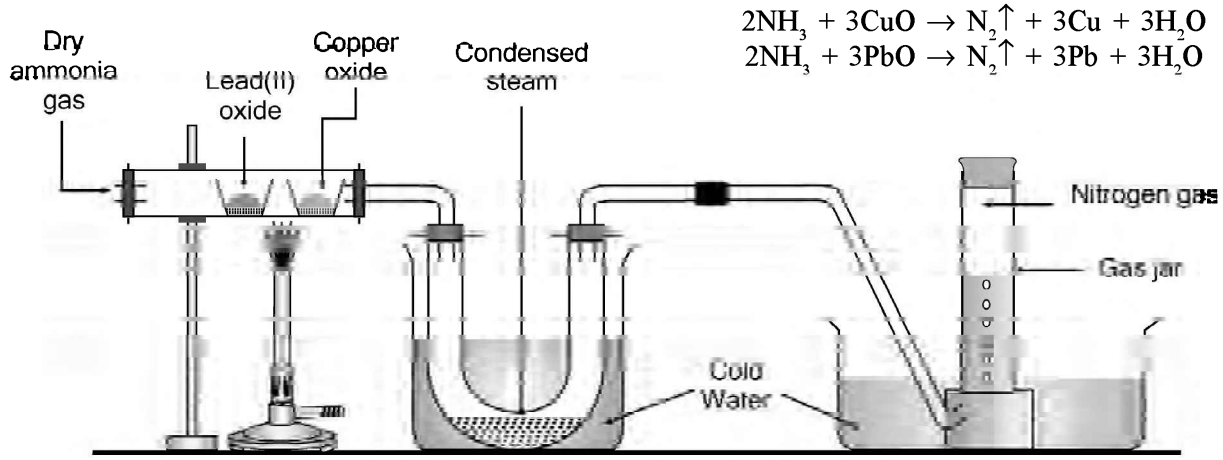
$$2NH_3 + 3Cl_2 \rightarrow 6HCl + N_2$$
$$6NH_3 + 6HCl \rightarrow 6NH_4Cl$$
$$8NH_3 + 3Cl_2 \rightarrow N_2 + 6NH_4Cl$$

(viii) With active metals, ammonia forms amides liberating hydrogen:

$$2Na + 2NH_3 \rightarrow 2NaNH_2 + H_2\uparrow$$

(ix) Ammonia has reducing properties: Ammonia reduces heated copper oxide (CuO) and lead oxide (PbO) to respective metals. (See diagram below)

$$2NH_3 + 3CuO \rightarrow N_2\uparrow + 3Cu + 3H_2O$$
$$2NH_3 + 3PbO \rightarrow N_2\uparrow + 3Pb + 3H_2O$$



Reducing action of ammonia on PbO and CuO

12.13 CHEMICAL PROPERTIES OF AMMONIUM HYDROXIDE (NH_4OH)

Ammonia reacts with water forming ammonium hydroxide. *A saturated solution of ammonia is known as ammonia liquor or ammonia fortis.*

Ammonium hydroxide has the following properties.

(i) Ammonium hydroxide is a weak base ionising weakly in aqueous solution.

$$NH_4OH \rightleftharpoons NH_4^+ + OH^-$$

(ii) Ammonium hydroxide turns red litmus blue.

(iii) Ammonium hydroxide reacts with acids forming ammonium salts.

(a) With hydrochloric acid, ammonium hydroxide forms one ammonium salt:

$$NH_4OH + HCl \rightarrow NH_4Cl + H_2O$$

(b) With sulphuric acid, ammonium

hydroxide forms two ammonium salts:

$NH_4OH + H_2SO_4 \rightarrow NH_4HSO_4 + H_2O$

(one H-atom of H_2SO_4 replaced forming ammonium bisulphate)

$2NH_4OH + H_2SO_4 \rightarrow (NH_4)_2SO_4 + 2H_2O$

(Both H-atoms of H_2SO_4 replaced forming ammonium sulphate)

(c) With phosphoric acid, ammonium hydroxide forms three ammonium salts:

$NH_4OH + H_3PO_4 \rightarrow NH_4H_2PO_4 + H_2O$

(one H-atom of H_3PO_4 replaced forming mono ammonium dihydrogen phosphate – MAP)

$2NH_4OH + H_3PO_4 \rightarrow (NH_4)_2HPO_4 + 2H_2O$

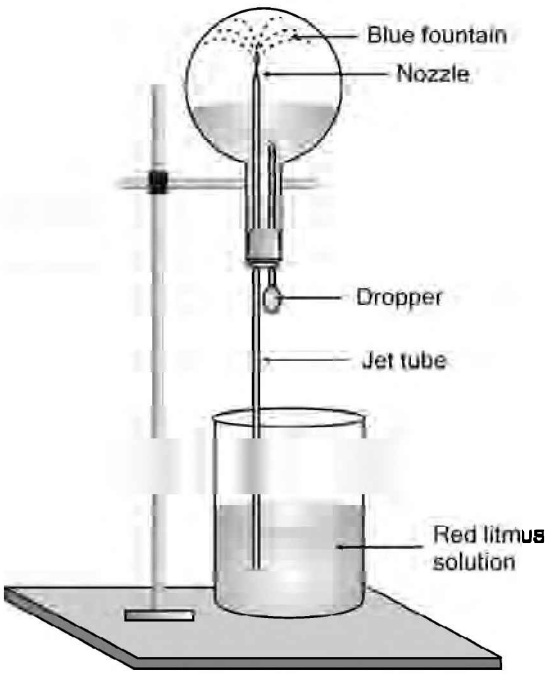
(two H-atoms of H_3PO_4 replaced forming di-ammonium hydrogen phosphate - DAP)

$3NH_4OH + H_3PO_4 \rightarrow (NH_4)_3PO_4 + 3H_2O$

(all three H-atoms of H_3PO_4 replaced forming ammonium phosphate)

(iv) All salts of metals react giving insoluble metallic hydroxides (The only exceptions are salts of sodium and potassium that give soluble hydroxides). This property is used in qualitative analysis to identify the metal in the metallic salts (See box).

(v) **Fountain experiment**: Solubility of ammonia can be demonstrated by the experiment shown. The round bottom flask is filled with ammonia gas and a dropper filled with water is fitted as shown. When the dropper is pressed water enters the flask. *The red litmus solution rises up the jet tube and creates a fountain at the top that is blue in colour.*



Fountain experiment

Explanation: The high solubility of ammonia creates a partial vacuum inside the flask. Atmospheric pressure acting on the surface of the red litmus in the beaker forces the red litmus up the jet tube. The force of red litmus entering the flask through the small nozzle creates a fountain like effect. The change in colour occurs because ammonia is a basic gas that changes red litmus spontaneously to blue.

12.14 TEST FOR AMMONIUM HYDROXIDE

The following properties of ammonium hydroxide are used to test for its presence.

(i) Ammonium hydroxide has a characteristic smell.

(ii) Ammonium hydroxide turns red litmus blue.

(iii) Ammonium hydroxide forms dense white fumes with HCl.

(iv) Ammonium hydroxide gives a brown precipitate with Nessler's reagent (K_2HgI_4).

(v) Ammonium hydroxide forms bluish precipitate with copper sulphate ($CuSO_4$) solution that dissolves if excess ammonium hydroxide is added to give a deep inky blue solution.

12.15 USES OF AMMONIUM HYDROXIDE

(i) Ammonium hydroxide is used to manufacture *urea*. *Urea* is an important fertilizer. Ammonia and carbon dioxide are heated to 100 – 150°C to form *urea*.

$$2NH_3 + CO_2 \xrightarrow{100\text{-}150^\circ C} H_2NCONH_2 \text{ or } (CON_2H_4)$$

(ii) Ammonium hydroxide is used for the manufacture of fertilisers such as ammonium sulphate, $(NH_4)_2SO_4$, calcium ammonium nitrate, ammonium hydrogen phosphate, urea, etc.

(iii) Ammonia is used for the manufacture of nitric acid by Ostwald's process.

(iv) Ammonium compounds are used in medicine:

(a) Ammonium carbonate, $(NH_4)_2CO_3$, mixed with lime is used as smelling salts for relief from bad colds and for reviving consciousness.

(b) Ammonium chloride (NH_4Cl) is used in cold syrups.

(v) Ammonium chloride, (NH_4Cl) is used in dry cells as an electrolyte.

(vi) Ammonium hydroxide is used in tinning industry as it acts as a flux, removing the fine coating of metallic oxides from metal surfaces.

(vii) Ammonium hydroxide is used for the manufacture of washing and baking soda.

(viii) Ammonium hydroxide is used for manufacture of synthetic products like dyes, drugs, explosives and plastics.

(ix) Ammonium hydroxide is used as a refrigerant in ice plants, as it has a high latent heat of vaporisation of 5700 cal/mole and a low boiling point.

(x) Ammonium hydroxide is used as a cleansing agent to remove grease stains from clothes by emulsification and cleaning windows panes and porcelain.

(xi) Ammonium hydroxide is used as a laboratory reagent for qualitative analysis (See box).

12.16 AMMONIUM SALTS

When ammonium hydroxide reacts with acids the corresponding ammonium salts are formed.

(i) Ammonium chloride is formed when ammonium hydroxide reacts with hydrochloric acid.

$$NH_4OH + HCl \rightarrow NH_4Cl + H_2O$$

(ii) Ammonium sulphate is formed when ammonium hydroxide reacts with sulphuric acid.

$$2NH_4OH + H_2SO_4 \rightarrow (NH_4)_2SO_4 + 2H_2O$$

(iii) Ammonium phosphate is formed when ammonium hydroxide reacts with phosphoric acid.

$$3NH_4OH + H_3PO_4 \rightarrow (NH_4)_3PO_4 + 3H_2O$$

(iv) Ammonium nitrate is formed when ammonium hydroxide reacts with nitric acid .

$$NH_4OH + HNO_3 \rightarrow NH_4NO_3 + H_2O$$

12.17 USES OF AMMONIUM SALTS

(i) *Ammonium chloride* (NH_4Cl) is used in soldering, in Leclanche battery as an electrolyte, in dyeing and calico painting, in medicines and as a laboratory reagent.

(ii) *Ammonium sulphate* $(NH_4)_2SO_4$, is used as a fertiliser and preparing other ammonium salts.

(iii) *Ammonium nitrate* (NH_4NO_3) is used as a fertiliser, in the manufacture of nitrous oxide and in making explosives. *Ammotal* is a mixture of ammonium nitrate (NH_4NO_3) and 20% Tri-Nitro Toluene (TNT). *Ammonal* is another explosive – a mixture of ammonium nitrate (NH_4NO_3) and aluminium powder

(iv) All ammonium salts of phosphoric acid are used as fertilisers.

(v) A mixture of $CaCO_3$ and NH_4NO_3 when dried, powdered and made into pellets is used as fertilizer.

Introduction to Salt Analysis

Ammonium hydroxide reacts with *metallic* salts giving *hydroxides*. The hydroxides exhibit one or more of the following properties.

(a) The hydroxides may be either soluble or insoluble.

(b) Some of the insoluble hydroxides dissolve when excess ammonium hydroxide is added.

(c) Many of the metal hydroxides have distinct colours.

The above properties are used to identify the metal in quantitative analysis.

Metallic salt + NH_4OH → Ammonium salt + metallic hydroxide

	Reaction	Colour of precipitate	Solubility in excess NH_4OH
(a)	$FeSO_4 + 2NH_4OH \rightarrow (NH_4)_2SO_4 + Fe(OH)_2\downarrow$	Dirty green	Insoluble
(b)	$FeCl_3 + 3NH_4OH \rightarrow 3NH_4Cl + Fe(OH)_3\downarrow$	Reddish brown	Insoluble
(c)	$Pb(NO_3)_2 + 2NH_4OH \rightarrow 2NH_4NO_3 + Pb(OH)_2\downarrow$	Chalky white	Insoluble
(d)	$ZnSO_4 + 2NH_4OH \rightarrow (NH_4)_2SO_4 + Zn(OH)_2\downarrow$	White gelatinous	Soluble
(e)	$CuSO_4 + 2NH_4OH \rightarrow (NH_4)_2SO_4 + Cu(OH)_2\downarrow$	Pale blue	Soluble

When NH_4OH is added drop by drop to $CuSO_4$ or $ZnSO_4$ it forms a precipitate of the respective hydroxide. If NH_4OH is added in excess the precipitate dissolves forming complex salts.

With Zinc sulphate

(a) **Limited ammonium hydroxide** $ZnSO_4 + 2NH_4OH \rightarrow (NH_4)_2SO_4 + Zn(OH)_2\downarrow$

(b) **Excess ammonium hydroxide**

$(NH_4)_2SO_4 + Zn(OH)_2\downarrow + NH_4OH \rightarrow Zn(NH_3)_4SO_4 + 4H_2O$

Zincamine sulphate (soluble colourless)

With copper sulphate

(a) **Limited ammonium hydroxide** $CuSO_4 + 2NH_4OH \rightarrow (NH_4)_2SO_4 + Cu(OH)_2\downarrow$

(b) **Excess Ammonium hydroxide**

$(NH_4)_2SO_4 + Cu(OH)_2\downarrow + NH_4OH \rightarrow Cu(NH_3)_4SO_4 + 4H_2O$

(Cupramine sulphate – soluble, deep inky blue solution)

12.18 NITRIC ACID

Introduction

Nitric acid is also known, as *aqua fortis*. Nitric acid is the only liquid that can dissolve silver metal.

Occurrence

Free state**:** Nitric acid occurs in ***acid rain.***

Combined state: Nitric acid occurs as salts.

Examples : *Chile salt petre* ($NaNO_3$), *Bengal salt petre* (KNO_3)

Nitric acid

Other names: *Aqua fortis*; *spirit of nitre*; *sal petre acid*; *hydrogen nitrate*

O[N+](=O)[O⁻]

Molecular Formula HNO_3

Mol. Wt. 63

Appearance clear, colourless liquid

Density: 1.51 g/cm³

Melting point: – 42 °C,

Boiling point of pure acid: 83° C

(A 68% solution boils at 120° C)

Solubility in water: high

Laboratory preparation of nitric acid

Principle: The basic principle in the laboratory preparation of nitric acid is,

A volatile acid can be displaced by a less volatile acid from the salt of the volatile acid.

Illustration: Nitric acid being a more volatile acid than sulphuric acid is displaced by sulphuric acid from metal nitrates.

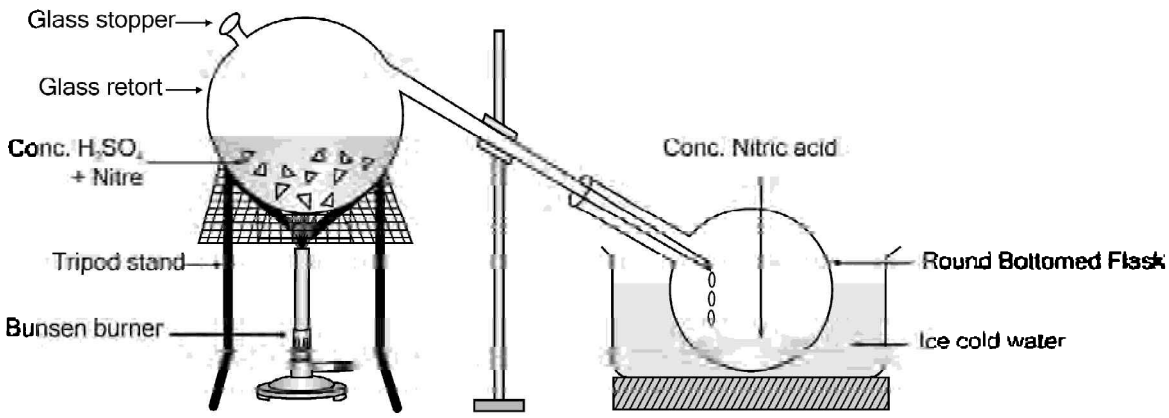
Reactants: 50 gm of potassium nitrate (KNO_3) + 25 ml of concentrated sulphuric acid (H_2SO_4) are taken in a round bottom flask. The reactants are heated to about 200°C taking care that the temperature does not cross 200°C.

The reaction:

$$KNO_3 + H_2SO_4 \xrightarrow{<200°C} KHSO_4 + HNO_3\uparrow$$

(Salt of more volatile acid + less volatile acid → displaces more volatile acid)

Method of collection: The vapours of nitric acid are cooled and condensed for collection. (See diagram)



Laboratory preparation of nitric acid.

Important precautions

(i) Concentrated hydrochloric acid (HCl) cannot be used as *it is volatile*.

(ii) Concentrated sulphuric acid is used as it is non-volatile.

(iii) The temperature is maintained below 200°C. Above 200°C, potassium sulphate (K_2SO_4) is formed instead of $KHSO_4$. Potassium sulphate fuses with glass and makes cleaning difficult. Besides this, at high temperatures nitric acid

(HNO_3) decomposes. Heat energy is also wasted.

(iv) The apparatus should be completely made of glass with no rubber stoppers, etc., used as nitric acid (HNO_3) is extremely corrosive.

(v) As nitric acid is a fuming volatile acid it is kept in an ice water bath.

12.19 INDUSTRIAL MANUFACTURE OF NITRIC ACID BY OSTWALD'S PROCESS

Reaction: The reactants are pure and dry ammonia and oxygen gases. They are taken in the ratio 1:2 by volume. The two gases are mixed and heated to a temperature of 900°C in the presence of platinum as catalyst, when nitric oxide (NO) is formed.

Catalytic chamber: The reactants are heated initially to 800 – 900°C.

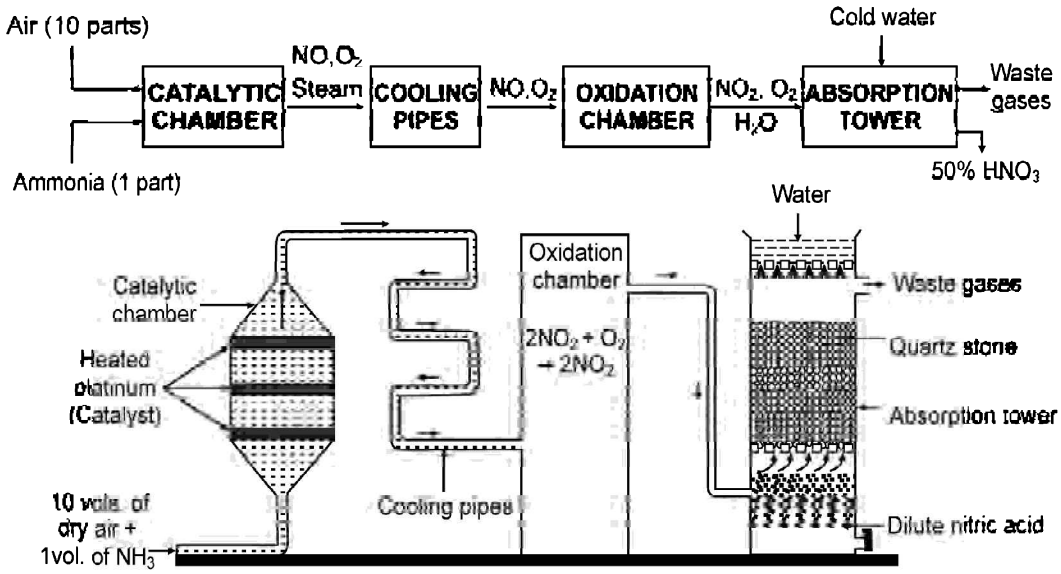
$$4NH_3 + 5O_2 \xrightarrow{Pt,\ 800-900°C} 4NO + 6H_2O + 21.5 \text{ kcal}$$

Oxidation chamber: The mixture of NO, O_2 and H_2O are next cooled to 50°C when NO is oxidized to NO_2.

$$2NO + O_2 \xrightarrow{50°C} 2NO_2$$

Absorption chamber: The absorption chamber is packed with quartz layers over which water is made to trickle down. The mixture of NO_2, O_2, H_2O is forced upwards into the absorption chamber where HNO_3 forms (See flow chart and diagram below).

$$4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$$



Industrial preparation of nitric acid by Ostwald's Process.

IMPORTANT PRECAUTIONS

(i) Atmospheric nitrogen (in the form of air) is not used as it prevents unnecessary heating of nitrogen.

(ii) As oxygen is required for reaction in all three chambers, the volume of oxygen taken is larger.

(iii) If atmospheric oxygen is to be used instead of pure oxygen, then the ratio of ammonia and air should be 1:10 as air contains only about 20% of oxygen.

(iv) Heating has to be done externally only initially as the reaction is exothermic. The heat evolved during the reaction maintains the temperature at the desired level.

(v) The gases leaving the catalytic chamber are at 800 – 900°C.

(vi) These gases are cooled to 50°C to facilitate the oxidation of nitric oxide (NO) and prevent simultaneous decomposition of nitrogen dioxide (NO_2) (At higher temperatures the reaction is reversible).

(vii) Quartz stone is used in the absorption chamber as it is non-corrosive.

(viii) The contact of nitrogen dioxide, oxygen and water is kept maximum in the absorption chamber to ensure more formation of acid.

(ix) The nitric acid (HNO_3) formed is of 50% concentration.

(x) The nitric acid is further concentrated to 68% by boiling or distillation.

(xi) Further concentration beyond 68% cannot be done as nitric acid forms a *Constant Boiling mixture* – both nitric acid and water boil off together at 121°C.

(xii) To concentrate to about 98% nitric acid is distilled with concentrated sulphuric acid (H_2SO_4).

12.20 PHYSICAL PROPERTIES

The physical properties of nitric acid are listed below.

(i) Pure nitric acid is an unstable fuming colourless acid with a choking smell and sour taste.

(ii) On standing and exposure to light nitric acid turns yellow over time due to dissolved NO_2. This is why nitric acid is stored in dark coloured bottles.

(iii) Nitric acid is hygroscopic in nature.

(iv) Nitric acid is very corrosive and chars the skin.

(v) Density of nitric acid is 1.54g/cc.

(vi) Nitric acid is completely soluble in water.

(vii) Nitric acid forms a constant boiling mixture of 68%.

(viii) Pure nitric acid boils at 86°C and freezes at –42°C forming white crystals.

12.21 CHEMICAL PROPERTIES

The chemical properties of nitric acid are listed below.

(i) Nitric acid is a very strong acid; turns blue litmus red.

(ii) Pure nitric acid is not very stable.

(iii) The acid anhydride of nitric acid is nitrogen dioxide (NO_2).

(iv) Nitric acid decomposes on standing forming brown nitrogen dioxide. This is the reason why though fresh nitric acid is colourless, it becomes brownish over time.

$$4HNO_3 \rightarrow 4NO_2 + O_2 + 2H_2O$$

(v) **Oxidising action of nitric acid**

Nitric acid decomposes, to form nascent (fresh) oxygen. The nascent oxygen acts as a good oxidizing agent. Nitric acid *is a powerful oxidizing agent*.

$$2HNO_3 \rightarrow H_2O + 2NO_2 + [O]$$

The oxidizing action can be seen in the following reactions:

With non-metals

$$C + 4HNO_3 \rightarrow CO_2 + 2H_2O + 4NO_2$$
$$S + 6HNO_3 \rightarrow H_2SO_4 + 2H_2O + 6NO_2$$
$$P + 5HNO_3 \rightarrow H_3PO_4 + H_2O + 5NO_2$$

With metals

In general, metals form a salt and hydrogen. Depending on the concentration, nitric acid reacts with metals as follows.

(a) **Metals react with very dilute nitric acid (1% concentration) to form the corresponding nitrates and liberate *hydrogen*.**

$$Mg + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2\uparrow$$
$$Mn + 2HNO_3 \rightarrow Mn(NO_3)_2 + H_2\uparrow$$

But, $4Zn + 10HNO_3 \rightarrow 4Zn(NO_3)_2 + NH_4NO_3 + 3H_2O$

(b) **Metals react with dilute nitric acid (20% concentration) to form the corresponding nitrates, *nitric oxide* and water**

$$3Mg + 8HNO_3 \rightarrow 3Mg(NO_3)_2 + 2NO + 4H_2O$$
$$3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$$

But, $4Zn + 10HNO_3 \rightarrow 4Zn(NO_3)_2 + N_2O + 5H_2O$

(c) **Metals react with concentrated nitric acid (68% concentration) to form the corresponding nitrates, *nitrogen dioxide* and water.**

$$Mg + 4HNO_3 \rightarrow Mg(NO_3)_2 + 2NO_2 + 2H_2O$$
$$Cu + HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$$

(Zinc also reacts similarly)

(d) Metals like iron, cobalt and nickel become *passive* when treated with concentrated nitric acid due to the formation of an oxide layer that *prevents further action*.

(e) **Action with *noble* metals**

Pure nitric acid does not react with noble metals like gold and platinum.

Aqua Regia: A mixture of nitric acid and hydrochloric acid in the ratio 1 : 3, forms a highly reactive mixture of acids called *Aqua Regia*.

Aqua Regia reacts with the noble metals, though *the individual acids do not react with the metals separately*. *Aqua Regia* forms *nascent* chlorine that reacts with the metals as follows.

$$3HCl + HNO_3 \rightarrow NOCl + 2H_2O + 2[Cl]$$

(nascent chlorine)

$$Au + 3[Cl] \rightarrow AuCl_3$$
$$Pt + 4[Cl] \rightarrow PtCl_4$$

(vi) **Action of nitric acid with inorganic compounds**

$3H_2S + 2HNO_3 \rightarrow 2NO + 3S + 4H_2O$

$3SO_2 + 2HNO_3 + 2H_2O \rightarrow 3H_2SO_4 + 2NO$

$6FeSO_4 + 3H_2SO_4 + 2HNO_3 \rightarrow 3Fe_2(SO_4)_3 + 2NO + 4H_2O$

(vii) **Action of nitric acid with organic compounds**

(a) With *cellulose*:

$C_6H_{10}O_5 + 24HNO_3 \rightarrow 6CO_2 + 17H_2O + 24NO_2$

(b) With *Toluene*

$C_7H_8 + 3HNO_3 \rightarrow C_7H_5(NO_2)_3 + 3H_2O$

(Tri Nitro Toluene or TNT is an explosive)

(viii) **Action of nitric acid with proteins**

When nitric acid comes in contact with any proteinous material, *xanthoproteic acid* is formed due to *nitration*. *Xanthoproteic acid* is yellow in colour. When treated with ammonia it forms an orange stain. *This property of nitric acid is used as a test for proteins*.

Test for nitric acid and nitrates

The presence of nitrate radical (NO_3^-) can be shown by using one of the following methods.

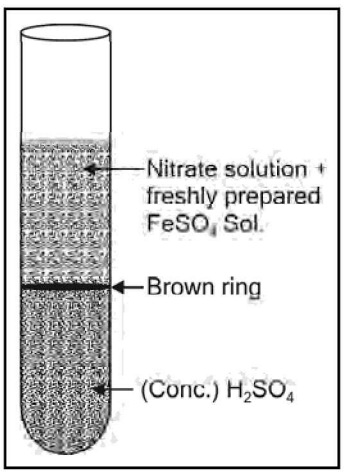
(i) All nitrates on strong heating liberate brown NO_2 gas.

(ii) All nitrates give Brown Ring Test.

Brown Ring Test

Freshly prepared $FeSO_4$ is mixed with the suspected nitrate salt (or HNO_3 acid) in a test tube.

Holding the test tube under running water, concentrated H_2SO_4 is slowly made to slide along the walls of the tube so that it slides below the solution in the tube.



If a nitrate salt or HNO_3 acid is present, a brown ring of $FeSO_4.NO$ is formed at the junction of the two liquids.

The reactions:

$NaNO_3 + H_2SO_4 \rightarrow NaHSO_4 + HNO_3$

$2HNO_3 \rightarrow H_2O + 2NO + 3[O]$

$FeSO_4 + NO \rightarrow FeSO_4.NO$

(Brown ring of Nitroso – ferrous sulphate)

The Brown ring is an unstable *addition product* of $FeSO_4$ and NO. The ring does not form if the test tube is not cooled as the ring is made of an unstable compound that decomposes and the reaction is exothermic. Even if the tube is shaken after the ring forms, the ring will disappear.

Uses of nitric acid

(a) In chemical industry nitric acid is used to manufacture nitrates. Sodium nitrate ($NaNO_3$) and potassium nitrate (KNO_3) are used in the manufacture of fertilisers and gunpowder. Calcium nitrate ($CaNO_3$) ammonium nitrate (NH_4NO_3) are used as fertilisers. Silver nitrate ($AgNO_3$) is used for preparing photographic material.

(b) Nitric acid is used in making dyes, drugs and perfumes.

(c) Nitric acid is used in the manufacture of explosives like tri-nitro-toluene (TNT), tri-nitro-glycerine (TNG) and nitro cellulose or gun cotton, blasting gelatin, amatol, etc.

(d) Nitric acid is used in metallurgy for purifying gold and silver.

(e) Nitric acid is used in etching designs on brass articles.

(f) Nitric acid in the form of *aqua Regia* is used in dissolving noble metals.

(g) Nitric acid is used in rockets as an oxidizing agent for fuels.

(h) Nitric acid is used in laboratories as a reagent.

12.22 NITROGEN CYCLE

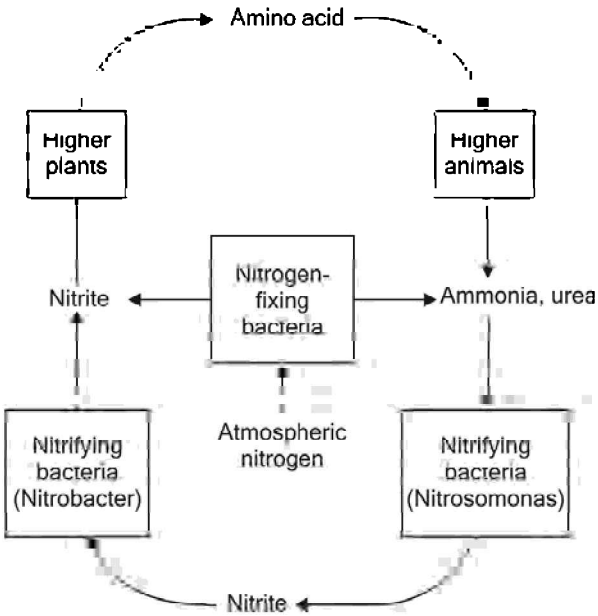
The process by which atmospheric nitrogen is converted into soluble nitrogen compounds for the growth and development of plants and animals and the liberation of free nitrogen on decomposition of these compounds into the atmosphere is called nitrogen cycle (See diagram).

Step 1: Fixation of atmospheric nitrogen

Step 2: Absorption of fixed nitrogen by plants and animals

Step 3: Conversion of fixed nitrogen from plants to nitrates in the soil

Step 4: Conversion of nitrates to atmospheric in nitrogen.



Fixation of nitrogen

Fixing nitrogen takes place either by lightning followed by rain (acid rain) or by direct absorption of atmospheric nitrogen by symbiotic bacteria present in the root nodules of leguminous plants like peas and beans. The absorbed nitrogen is converted by these plants into amino acids. Nitrogen can be supplied to plants through fertilizers also.

Formation and role of acid rain in nitrogen cycle

At normal temperatures, nitrogen and oxygen in the atmosphere do not combine. When lightning strikes, the two combine due to the extreme heat generated and form nitric oxide.

$$N_2 + O_2 \xrightarrow{\text{lightning}} 2NO$$

The nitric oxide combines with oxygen from the atmosphere and water from the rain to form nitric acid .

$$2NO + O_2 \rightarrow 2NO_2$$
$$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$$

The nitric acid in acid rain reacts with salts present in the soil undergoing double decomposition changing them to soluble nitrates.

Example:

$2HNO_3 + CaCO_3 \rightarrow Ca(NO_3)_2 + H_2O + CO_2$

(from acid rain) (present in the soil)

EXERCISE

I. Choose the correct answer

1. **Which is a physical property of Nitrogen?**
   (a) liquid at room temperature
   (b) A = 28
   (c) Z = 7
   (d) metal
2. **Nitrides are**
   (a) nitrous compounds
   (b) compounds of nitrogen with metals
   (c) ammonium compounds
   (d) nitrates
3. **Nitrogen is manufactured on a large scale from**
   (a) organic waste (b) proteins
   (c) air (d) all three (a, b, c)
4. **The compound which gives nitrogen on heating is**
   (a) ammonium nitrate (b) nitric acid
   (c) ammonium nitrite (d) ammonia
5. **Nitrogen is collected in the laboratory by**
   (a) by downward displacement of water
   (b) by downward displacement of mercury
   (c) by downward displacement of air
   (d) by upward displacement of air
6. **The approximate composition by volume of air is ____% of nitrogen and ____% oxygen.**
   (a) 78, 22 (b) 22, 78
   (c) 3, 78 (d) 78, 3
7. **Drying agents that can be used to dry nitrogen in the laboratory is**
   (a) anhydrous $CaCl_2$ (b) P_2O_5
   (c) conc. H_2SO_4 (d) all three (a, b, c)
8. **Nitrogen is**
   (a) combustible
   (b) supporter of combustion
   (c) combustible but supports combustion
   (d) incombustible and does not support combustion
9. **Heating ammonium nitrate produces the gas**
   (a) nitrous oxide (N_2O) (b) nitrogen (N_2)
   (c) ammonia (NH_3)
   (d) nitrogen dioxide (NO_2)
10. **Nitrogen boils off _____ on warming liquid air.**
    (a) last (b) first
    (c) together with oxygen
    (d) together with carbon dioxide
11. **The oxides of nitrogen that is/are not neutral is/are**
    (a) NO (b) N_2O
    (c) NO_2 (d) both (a) and (b)
12. **The single largest use of nitrogen is in the manufacture of**
    (a) ammonia (b) nitric acid
    (c) both (a) and (b) (d) proteins

13. Ammonia is collected by
(a) by downward displacement of water
(b) by downward displacement of mercury
(c) by downward displacement of air
(d) by upward displacement of air

14. Ammonia cannot be collected by the downward displacement of water as it is
(a) completely soluble (b) volatile
(c) insoluble (d) heavy gas

15. Ammonia is dried using
(a) concentrated sulphuric acid
(b) phosphorous pentoxide
(c) quick lime (calcium oxide)
(d) any of the three (a, b or c)

16. Ammonia cannot be dried using
(a) concentrated sulphuric acid
(b) phosphorous pentoxide
(c) anhydrous calcium chloride
(d) all the three (a, b and c)

17. Simple burning of ammonia produces _____gas while catalysed burning produces _____gas.
(a) Nitrogen, Nitric oxide
(b) Nitric oxide, Nitrogen
(c) Oxygen, Nitrous oxide
(d) Steam, Nitrous oxide

18. Ammonia reacts with excess chlorine forming
(a) NCl_3 and HCl (b) NH_4Cl and HCl
(c) NCl_3 and N_2 (d) NH_4Cl and N_2

19. Excess Ammonia reacts with chlorine forming
(a) NCl_3 and HCl (b) NH_4Cl and HCl
(c) NCl_3 and N_2 (d) NH_4Cl and N_2

20. Ammonia is used
(a) for manufacture of nitric acid
(b) for manufacture fertiliser
(c) for manufacture of explosives
(d) all three (a, b, c)

21. Nitric acid is a good
(a) reducing agent (b) dehydrating agent
(c) oxidising agent (d) none of these

22. Nitric acid turns metals like iron, cobalt and nickel
(a) active (b) passive
(c) non-reactive (d) porous

23. Noble metals react with
(a) concentrated hydrochloric acid
(b) concentrated nitric acid
(c) a 3 : 1 mixture of hydrochloric acid and nitric acid
(d) a 1 : 3 mixture of hydrochloric acid and nitric acid

24. Fresh nitric acid is
(a) colourless (b) pale brown
(c) pale blue (d) white

25. Freshly prepared nitric acid turns brown over a period of time due to
(a) dissolved rust (b) dissolved NO
(c) dissolved NO_2 (d) dissolved N_2O

26. NH_4Cl is used
(a) in cough syrups (b) for soldering
(c) in Leclanche batteries (d) all three (a, b, c)

27. The formula of the brown ring formed in the test for nitrates (Brown Ring test) is
(a) NO_2 (b) $FeSO_4.NO$
(c) $Fe_2O_3.xH_2O$ (rust) (d) $FeSO_4.NO_2$

28. Nitric acid cannot be concentrated beyond _____% by boiling.
(a) 68% (b) 22%
(c) 98% (d) 50%

29. The *constant boiling mixture of nitric acid* is concentrated up to 98%
(a) by boiling
(b) by treating with $CaCl_2$
(c) by distilling with concentrated sulphuric acid
(d) by treating with phosphorous pentoxide

30. The nitrates that give oxygen as the only gaseous product on heating
(a) Sodium nitrate (b) Potassium nitrate
(c) Calcium nitrate (d) both (a) and (b)

II. Read the paragraphs below and answer the questions that immediately follow each paragraph. Each question has four options out of which only one option is correct. Choose the correct one.

Paragraph A

When an acid looses water the remaining part is called acid anhydride. Conversely, an acid anhydride reacts with water giving acid. For example the acid anhydride of sulphuric acid is sulphur trioxide (SO_3). Conversely, sulphur trioxide (SO_3) reacts with water giving sulphuric acid and so is called acid anhydride of sulphuric acid. The acid anhydride of an acid can be found by writing the formula and removing the elements of water molecule (two H-atoms and one O-atom) from the formula.

1. The acid anhydride of sulphurous acid is
(a) Sulphur dioxide (b) Sulphur trioxide
(c) Sulphur (d) All three

2. The acid anhydride of oxalic acid – $(COOH)_2$ is
(a) Carbon monoxide (b) Carbon dioxide
(c) Carbon suboxide (d) both (a) and (b)

3. The acid anhydride of nitric acid is
(a) nitrogen dioxide (b) nitric oxide
(c) nitrous oxide (d) nitrogen

Paragraph B

Atmospheric oxygen enters the soil by two methods – acid rain and by direct absorption by certain plants. This phenomenon is called fixing of nitrogen. When lightning strikes nitrogen combines with oxygen forming nitric oxide that further gets oxidised to nitrogen dioxide. This combines with more oxygen and water forming nitric acid. This comes down as acid rain. The nitric acid reacts with salts present in the soil converting them into nitrates. These nitrates are good source of nitrogen for plants. Some plants like pea beans can absorb atmospheric nitrogen directly. Symbiotic bacteria present in the root nodules of leguminous plants have the capacity of absorbing nitrogen directly from the atmosphere. Nitrogen can also be supplied artificially to plants by using fertilizers.

4. **Free nitrogen in air is converted into nitrates in the soil by**
   (a) acid rain (b) leguminous plants
   (c) both (a) and (b) (d) carbon dioxide
5. **Amino acids contain**
   (a) nitric acid (b) nitrates
   (c) nitrogen (d) rain water
6. **Lightning converts atmospheric nitrogen and oxygen to**
   (a) nitric oxide (b) nitric acid
   (c) water (d) none of these

III. This section contains three questions. Each question contains statements given in two columns, which have to be matched. Statements (i, ii, iii, iv) in Set A have to be matched with Statements (a, b, c, d) in Set B. More than one match or no match between options in the Sets is possible.

1. **Match non-metal oxide in Set A with the acid formed in Set B, when it reacts with water.**

Set A	Set B
(i) Nitrogen dioxide	(a) nitrous acid
(ii) Nitrogen hexoxide	(b) nitric acid
(iii) Nitrogen trioxide	(c) per nitric acid
(iv) Nitrogen pentoxide	(d) hydrogen cyanide

2. **Match the reactants in Set A with the with the products formed in Set B**

Set A	Set B
(i) Ammonia and excess chlorine	(a) $6NH_4Cl + N_2$
(ii) Excess ammonia and chlorine	(b) $3Mg(OH)_2 + 2NH_3$
(iii) Magnesium nitride and water	(c) $Zn(NH_3)_4SO_4 + 4H_2O$
(iv) Excess ammonia with zinc sulphate	(d) $NCl_3 + 3HCl$

3. **Match the hydroxides in Set A with its colour in Set B**

Set A	Set B
(i) Ferrous hydroxide	(a) Reddish brown
(ii) Lead hydroxide	(b) Dirty green
(iii) Zinc hydroxide	(c) White gelatinous
(iv) Ferric hydroxide	(d) Chalky white

4. **Match the ion in Set A with the colour of solution or precipitate formed when it reacts with excess ammonium hydroxide in Set B**

Set A	Set B
(i) Pb^{2+}	(a) Reddish brown
(ii) Zn^{2+}	(b) deep inky blue
(iii) Cu^{2+}	(c) White gelatinous
(iv) Fe^{3+}	(d) Chalky white

IV. Each question contains two statements. Statement A is an Assertion Statement and Statement R is the Reason Statement. Each question has four choices out of which only one is correct. Choose the correct one.

Choose A if Statement A is true, Statement R is true; Statement R is correct explanation for statement A

Choose B if Statement A is true, statement R is true; Statement R is not correct explanation for statement A

Choose C if Statement A is true, Statement R is false

Choose D if Statement A is false, statement R is true

1. **Statement –A:** Simple burning of ammonia yields nitrogen and water.

   Statement –R: Catalytic oxidation of ammonia yields nitric oxide and water.
2. **Statement –A**: Nitrogen is used to provide an inert atmosphere in some reactions.

   Statement –R: Nitrogen is quite non reactive.
3. **Statement –A**: Ammonium nitrate is used to prepare nitrogen.

   Statement –R: Ammonium nitrite is unstable.
4. **Statement –A**: Noble metals do not react with nitric acid.

   Statement –R: Nitric acid loses its property of a strong acid in the presence of noble metals.
5. **Statement –A**: Freshly prepared nitric acid is colourless.

   Statement –R: On standing nitric acid decomposes to form brown nitrogen dioxide which remains dissolved giving the acid a yellowish tinge.
6. **Statement –A**: Noble metals react with *Aqua Regia* but do not react with concentrated nitric acid.

   Statement –R: *Aqua Regia* forms nascent chlorine

which is a stronger oxidising agent than nitric acid.

7. **Statement –A**: Nitrogen dioxide is a mixed oxide.

   Statement –R: Nitrogen dioxide reacts with water forming two acids – nitric acid and nitrous acid.

8. **Statement –A**: Ammonium hydroxide gives two salts with sulphuric acid.

   Statement –R: Sulphuric acid is diacidic.

9. **Statement –A**: In the Fountain Experiment, if blue litmus is used instead of red litmus no fountain will be seen at the top.

   Statement –R: Ammonia is a basic gas that turns moist red litmus blue.

10. **Statement –A**: Hydrochloric acid cannot be used instead of sulphuric acid in the laboratory preparation of nitric acid.

    Statement –R: Hydrochloric acid is weak acid.

V. Answer the following questions on Nitrogen

1. List the properties of nitrogen with respect to its physical state, atomic number, mass number, atomic weight, electronic configuration and position in the Periodic Table.
2. How does nitrogen occur in free state and combined state?
3. Give the equations of three chemical reactions that can be used to prepare Nitrogen.
4. How is nitrogen collected and why?
5. Nitrogen in the atmosphere controls combustion. Explain.
6. Give reasons why nitrogen is non-reactive.
7. Give reactions of Haber's Process and Birkeland Eyde Process.
8. What is acid rain? Write reactions to show how nitrogen enters the soil from the atmosphere.
9. Why is it difficult to test for nitrogen directly?
10. Three unlabelled jars A, B and C contain nitrogen, oxygen and hydrogen gases respectively. How would you identify the gases?
11. List the important compounds of nitrogen.
12. How can nitrogen be used as a food preservative?
13. Give an outline of the separation of nitrogen from air in the laboratory.
14. What are the three main types of artificial fertilizers?
15. What is superphosphate of lime?
16. Name two oxides of nitrogen each that are (a) acidic (b) neutral.
17. What weight of nitrogen and hydrogen will be required to form 170 gm of ammonia in its synthesis by Haber's Process?
18. Complete and balance these equations. Where required indicate the reaction conditions.

(a) $N_2 + H_2 \xrightarrow{?}$ (b) $N_2 + O_2 \xrightarrow{?}$

(c) $Na + N_2 \xrightarrow{\Delta}$ (d) $Mg + N_2 \xrightarrow{\Delta}$

(e) $Al + N_2 \xrightarrow{\Delta}$ (f) $Ca + N_2 \xrightarrow{\Delta}$

(g) $CaC_2 + N_2 \xrightarrow{?}$ (h) $NO + O_2 \rightarrow$

(i) $NO_2 + H_2O + O_2 \rightarrow$ (j) $HNO_3 + CaCO_3 \rightarrow$

(k) $NH_4NO_2 \rightarrow$ (l) $(NH_4)_2Cr_2O_7 \rightarrow$

(m) $NH_3 + O_2 \xrightarrow{\text{simple burning}}$

(n) $NH_3 + CuO \xrightarrow{\Delta}$ (o) $NH_4OH + CaOCl_2 \rightarrow$

(p) $8NH_3 + 3Cl_2 \rightarrow$ (q) $NaNO_2 + NH_4Cl \rightarrow$

(r) $NH_4NO_2 \rightarrow$

VI. Answer the following questions on Ammonia

1. Name a liquid source of ammonia.
2. How would you obtain ammonia from metal nitrides? Write the equation for one such reaction.
3. What precautions are to be observed in the laboratory preparation of ammonia using NH_4Cl and H_2SO_4?
4. What is the method of collection of ammonia? Why is this method used?
5. How is ammonia dried in the laboratory? Why are other drying agents like P_2O_5, conc. H_2SO_4 or $CaCl_2$ not used?
6. Why should the reactants be pure in the synthesis of ammonia?
7. How is nitrogen and hydrogen manufactured for large scale use in Haber's Process?
8. What are the reaction conditions for Haber's process?
9. How is ammonia stored after its manufacture?
10. Ammonia does not burn in air but burns in pure oxygen. Write the reaction.
11. Write the equations for the reaction of ammonium hydroxide with any three acids showing *all* the salts formed.
12. Give at least two reactions of ammonium hydroxide with metals salts that give soluble salts.
13. Name two insoluble metal hydroxides that dissolve in excess ammonium hydroxide.
14. Give one reaction that shows the reducing nature of ammonia gas.
15. How would you test for ammonia using Nessler's Reagent?
16. Give the reaction of ammonia and chlorine when both

are taken in equal volumes under same conditions. (Hint: Use the concept of limiting reagent and excess reagent. Excess ammonia reacts with chlorine in the ratio 8:1 of $NH_3 : Cl_2$ while excess chlorine reacts in the ratio of 1 : 3 of $NH_3 : Cl_2$. If equal volumes of both gases are taken then clearly the ratio is in favour of excess chlorine. At the end of the reaction, excess chlorine will be left along with the products of ammonia reacting with excess chlorine).

17. List any five uses of ammonia.
18. What are Ammotal and ammonal?
19. Which ammonium salt is used for soldering?
20. Which ammonium salt is used to prepare nitrous oxide? Give the equation for the reaction.
21. Which ammonium salt is used in Leclanche Cell?
22. Which ammonium salt is used in cough and cold syrups?
23. Give one use of liquid ammonia.
24. Catalytic oxidation of ammonia forms brown fumes if air or oxygen is in excess. Why?
25. Complete and balance the following equations. Write the reaction conditions where required.

(a) $N_2 + 3H_2 \xrightarrow{?}$

(b) $NH_4Cl + NaOH \rightarrow$

(c) $(NH_4)_2SO_4 + Ca(OH)_2 \rightarrow$

(d) $(NH_4)_3PO_4 + NaOH \rightarrow$

(e) $Na_3N + H_2O \rightarrow$

(f) $Ca_3N_2 + H_2O \rightarrow$

(g) $Mg_3N_2 + H_2O \rightarrow$

(h) $AlN + H_2O \rightarrow$

(i) $NH_4Cl + Ca(OH)_2 \rightarrow$

(j) $NH_3 + H_2SO_4 \rightarrow$

(k) $NH_3 + P_2O_5 + H_2O \rightarrow$

(l) $NH_3 + CaCl_2 \rightarrow$

(m) $NH_3 + O_2 \xrightarrow{\text{simple burning}}$

(n) $NH_3 + O_2 \xrightarrow{?}$

(o) $NH_3 + HCl \rightarrow$

(p) $Na + NH_3 \rightarrow$

(q) If chlorine is in excess $-NH_3 + Cl_2 \rightarrow$

If NH_3 is in excess $-NH_3 + Cl_2 \rightarrow$

(r) $NH_3 + CuO \rightarrow$

(s) $NH_4OH \rightleftharpoons$

(t) $NH_4OH + HCl \rightarrow$

(u) One H-atom of H_2SO_4 replaced:

$NH_4OH + H_2SO_4 \rightarrow$

Both H-atoms of H_2SO_4 replaced:

$NH_4OH + H_2SO_4 \rightarrow$

(v) $NH_4OH + H_3PO_4 \rightarrow NH_4H_2PO_4 + H_2O$

(w) $2NH_4OH + H_3PO_4 \rightarrow$

(x) $3NH_4OH + H_3PO_4 \rightarrow$

(y) Salts with ammonium hydroxide

$FeSO_4 + NH_4OH \rightarrow$

$FeCl_3 + NH_4OH \rightarrow$

$Pb(NO_3)_2 + NH_4OH \rightarrow$

$ZnSO_4 + 2NH_4OH \rightarrow$

$CuSO_4 + NH_4OH \rightarrow$

$(NH_4)_2SO_4 + Zn(OH)_2\downarrow + NH_4OH \rightarrow$

$(NH_4)_2SO_4 + Cu(OH)_2\downarrow + NH_4OH \rightarrow$

VII. Answer the following questions on Nitric acid

1. What is acid rain?
2. Give reasons for the following in the laboratory preparation of nitric acid.
   (a) Why should the reactants not be heated beyond 200°C?
   (b) Why should an all glass apparatus be used?
   (c) Why can HCl not be used instead of H_2SO_4 acid?
3. What is the advantage of using pure oxygen over air in Ostwald's Process?
4. In case air is used in Ostwald's process instead of oxygen, what ratio of ammonia and air is to be taken?
5. Why should the oxygen ratio be higher in Ostwald's process?
6. Why is only initial heating required in Ostwald's Process?
7. Write the equations with conditions of the reactions taking place in the Catalytic, Oxidation and Absorption Chambers in Ostwald's Process.
8. Why does nitric acid on standing acquire a brown colour?
9. How is nitric acid concentrated in Ostwald's Process?
10. What is the reason for some metals like iron, becoming passive (non-reactive to nitric acid though they react initially with it?
11. Give the equations for the reaction that take place in the Brown Ring Test.
12. Give a test for proteins.
13. What is the cation in nitric acid that turns red litmus blue?
14. List any five uses of nitric acid.
15. Nitric acid is a strong oxidising agent. Why? Give one equation to show this.
16. How can you convert nitric acid to sulphuric acid? Write the equation.

17. How can you obtain nascent chlorine from nitric acid?
18. What does solid nitric acid look like?
19. Nitric acid is different from other acids in the way it reacts with metals. Explain.
20. Complete and balance these equations.

a. $HNO_3 \rightarrow$ b. $H_2O + H^+ \rightarrow$ c. $Mg + HNO_3 \rightarrow$

d. $Mn + HNO_3 \rightarrow$ e. $ZnO + HNO_3 \rightarrow$ f. $PbO + HNO_3 \rightarrow$

g. $Ca(OH)_2 + HNO_3 \rightarrow$ h. $Cu(OH)_2 + HNO_3 \rightarrow$ i. $CaCO_3 + HNO_3 \rightarrow$

j. $PbCO_3 + HNO_3 \rightarrow$ k. $Ca(HCO_3)_2 + HNO_3 \rightarrow$ l. $NaHCO_3 + HNO_3 \rightarrow$

m. $HNO_3 \rightarrow$ n. $C + HNO_{-3} \rightarrow$ o. $S + HNO_3 \rightarrow$

p. $P + HNO_3 \rightarrow$

q. Metals with very dilute HNO_3 (1% conc.) $Mg + HNO_3 \rightarrow$

$Mn + HNO_3 \rightarrow$

and, $Zn + HNO_3 \rightarrow$

r. Metals with dilute HNO_3 (20% conc.) $Mg + HNO_3 \rightarrow$

$Cu + HNO_3 \rightarrow$

and, $Zn + HNO_3 \rightarrow$

s. Metals with concentrated HNO_3 (68% conc.) $Mg + 4HNO_3 \rightarrow$

$Cu + HNO_3 \rightarrow$

t. $H_2S + HNO_3 \rightarrow$ u. $SO_2 + HNO_3 + H_2O \rightarrow$

v. $FeSO_4 + H_2SO_4 + HNO_3 \rightarrow$ w. $C_6H_{10}O_5 + HNO_3 \rightarrow$ x. $C_7H_8 + HNO_3 \rightarrow$

21. Nitric oxide (NO) gas can be absorbed in which aqueous solution?

KEY TO EXERCISE

I. 1. c 2. b 3. c 4. c 5. a 6. a 7. d 8. d
9. a 10. b 11. c 12. a 13. c 14. a 15. c 16. d
17. a 18. a 19. d 20. d 21. c 22. b 23. c 24. a
25. c 26. d 27. b 28. a 29. c 30. d

II. 1. a 2. d 3. a 4. c 5. c 6. a

III. 1. i. a, b ii. b, c iii. a iv. b 2.i. d ii. a iii. b iv. c
3. i. b ii. d iii. c iv. a 4. i. d ii. d iii. b iv. a

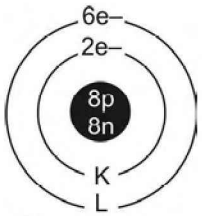
IV. 1. B 2. A 3. D 4. C 5. A 6. A 7. A
8. C 9. D

VII. 21. Aqueous solution of $FeSO_4$.

13 Study of Group VIA Element-Oxygen

13.0 INTRODUCTION TO OXYGEN

Priestley discovered oxygen in 1774 and called it *active air* as a glowing splinter when put in a jar of oxygen made it glow more brightly. It was Lavoisier who proved that oxygen is an element and named it oxygen – meaning acid former, as he believed it to be a part of all acids. It is vital for sustenance of life.



Atom of oxygen

13.1 OCCURRENCE

Oxygen is the most abundant element on earth.

Free state: Oxygen occurs in the free state in air. In air, about 23% by weight and 21% by volume is oxygen. Oxygen occurs as a dissolved gas in large water bodies like sea water and oceans.

Combined state: Oxygen occurs in the combined state in many compounds. In the combined state it occurs as oxides, carbonates, sulphates, nitrates and in many organic compounds like carbohydrates.

(i) In water, about 89% by weight is oxygen.

(ii) About 70% of living tissues by weight is oxygen.

(iii) About 50% by weight of the earth's crust is oxygen.

13.2 GENERAL METHODS OF PREPARATION OF OXYGEN

The following reactions can be used for the general methods of preparation of oxygen.

(a) On heating mercuric oxide, oxygen is liberated.

$$HgO \xrightarrow{\Delta} Hg + O_2 \uparrow - \Delta$$

(red)

(b) On heating silver oxide, oxygen is liberated.

$$Ag_2O \xrightarrow{\Delta} Ag + O_2\uparrow - \Delta$$

(brown)

Oxygen

Symbol of oxygen is O

Molecular formula is O_2.

Atomicity of oxygen is 2.

Atomic Number = 8

Mass Number = 16

Atomic Weight = 16

Molecular weight = 32

Group VIA and Period 2 element

Electronic configuration is (2, 6)

Valence electrons = 6

Valency is –2

Physical state – a gas at room temperature and pressure; non-metal

(c) On heating red lead, oxygen is liberated.

$$Pb_3O_4 \xrightarrow{\Delta} PbO + O_2\uparrow - \Delta$$

(red lead)

13.3 LABORATORY PREPARATION OF OXYGEN

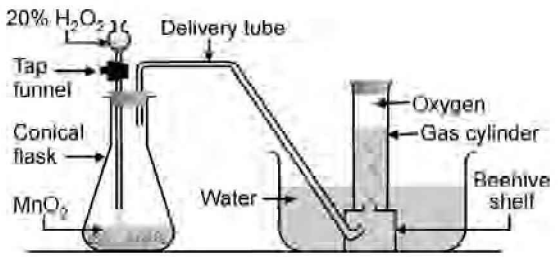
(a) Oxygen is produced from sodium peroxide (Na_2O_2) or hydrogen peroxide (H_2O_2). When H_2O_2 is trickled over water it forms water and oxygen.

$$2H_2O_2 \xrightarrow{MnO_2} 2H_2O + O_2$$

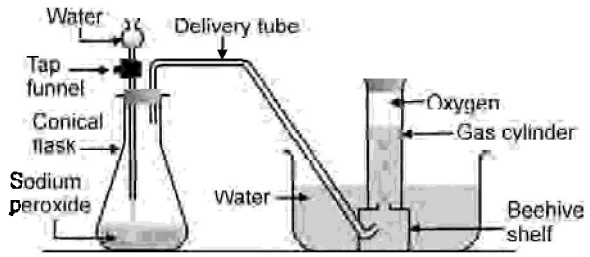
Similarly, when water is allowed to trickle into Na_2O_2 oxygen is liberated.

$$2Na_2O_2 + 2H_2O \rightarrow 4NaOH + O_2$$

The apparatus is set up as shown in the diagrams below.



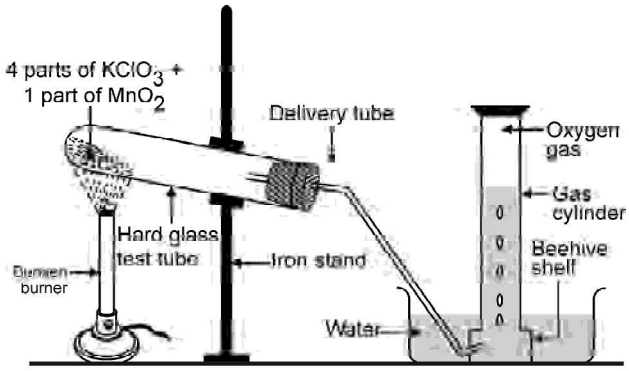
Laboratory preparation hydrogen peroxide



Laboratory preparation using sodium peroxide

(b) Heating oxidising agents like potassium chlorate ($KClO_3$) and potassium permanganate ($KMnO_4$) liberates oxygen in the presence of a catalyst like manganese dioxide (MnO_2). (See diagram)

$$2KClO_3 \xrightarrow[MnO_2]{\Delta} 2KCl + 3O_2$$



Preparation of oxygen using potassium chlorate

Method of collection of gas: As oxygen is almost as heavy as air and very slightly soluble in water, it is collected by downward displacement of water.

Drying of oxygen: Dry oxygen can be obtained by passing through drying agents like anhydrous calcium chloride ($CaCl_2$) or phosphorous pentoxide (P_2O_5) or passing through conc. H_2SO_4 and collecting over mercury.

Important points to note in the laboratory preparation of oxygen

(a) As oxygen is a supporter of combustion, care must be exercised in its collection.

(b) If $KClO_3$ is used, the ratio of $KClO_3$ and the catalyst (MnO_2) should be 4 : 1. This mixture is called ***oxygen mixture***. Also the mouth of the test tube should be slanting downwards so that if any water vapour form, it does not condense and flow backwards into the test tube and thus prevent breaking of the tube.

13.4 INDUSTRIAL PREPARATION OF OXYGEN

Oxygen is manufactured on a large scale for industrial use by two methods.

(a) Electrolysis of water

(b) From air

(a) Electrolysis of water

This method is used only if electricity is cheap. Electrolysis of acidulated water yields oxygen at the anode and hydrogen at the cathode in the ratio 1 : 2 by volume. (See Chapter 9 – Hydrogen)

(b) From air

Since about 20% of air is made up of oxygen, oxygen can be obtained by fractional distillation of *liquid air*.

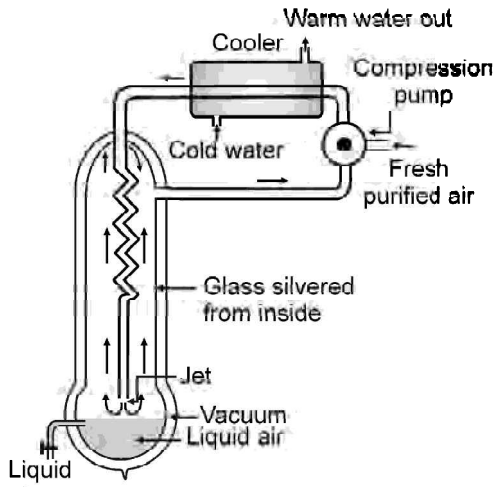
Air is prepared for liquefaction by removing dust particles, carbon dioxide, sulphur dioxide, pollutants and water vapour as follows.

1. Dust particles in air are removed by passing air through filters and water.
2. The air is passed through dilute caustic soda when gases like CO_2, SO_2 are removed.
3. The air is now cooled (–20°C) when moisture present in it solidifies to ice making the air dry.

   Now the air is free from dust particles, carbon dioxide, sulphur dioxide, water vapour and other pollutants. Air is now liquefied as follows.
4. The pure air is compressed to a pressure of 200 atm due to which its temperature increases. The hot compressed air is passed into a cooling tank.
5. Next the cooled air is *repeatedly* compressed and expanded into an evacuated chamber through a fine nozzle. Due to *Joule Thomson Effect*, every cycle of compression and expansion results in a drop in the temperature of air. As the compressed air comes out of the nozzle its pressure suddenly decreases and the temperature falls. The cooled air is then compressed again and sent through the nozzle into an evacuated chamber. Repeated compression and expansion of air leads to a fall in temperature for every cycle *till it liquefies.* (See figure and box).

Joule Thomson Effect

What is Joule-Thomson effect? When a gas kept at a very high pressure suddenly expands into a very low-pressure region (like an evacuated chamber) there is a temperature drop. As the molecules move apart work is done. The energy is drawn from kinetic energy of the molecules of the gas. The kinetic energy thus drops and this results in the drop in temperature. For example, if a cylinder of CO_2 is opened to the atmosphere, one can see a spray of fine dry ice particles emerging at about –78°C. Expansion of air from a very high pressure to atmospheric pressure can be used to cool the air to the point of liquefaction. Low-temperature distillation of the liquid thus produced is then used to make liquid nitrogen, liquid oxygen and even liquid argon (1% of natural air).



Liquefaction of air

6. The liquid air is at a temperature of –200° C. The liquid air consists mainly of nitrogen and oxygen and traces of inert gases. The BP of liquid nitrogen is –195.8° C while that of liquid oxygen is –183°C. The difference in boiling points is used to separate the two liquids by fractional distillation. The liquid air is very slowly warmed to –195° C, when nitrogen starts to boil off. Nitrogen is collected by compressing into steel cylinders. On further warming, oxygen boils off at –183° C. Oxygen gas is collected after compression in separate steel cylinders.

Oxygen prepared by fractional distillation of liquid air will be slightly denser than oxygen made by other methods as it contains traces of inert gases.

13.5 PHYSICAL PROPERTIES OF OXYGEN

Oxygen exhibits the following physical properties.

(i) Oxygen is colourless, odourless and tasteless gas.

(ii) Oxygen is heavier than air. This is the main reason why mountaineers carry oxygen cylinders. At high altitudes air has less oxygen.

(iii) Oxygen is almost insoluble in water. About 4ml dissolves in 100ml of water. This dissolved oxygen is responsible for sustenance of aquatic life near the poles where water freezes forming a layer of ice on the surface of oceans.

(iv) Liquid oxygen is pale blue.

(v) Oxygen is a non-poisonous gas which is a non-metal.

(vi) Other specific properties of oxygen are –
Vapour density = 16
Boiling point = –183°C
Melting point = –218.4°C

13.6 CHEMICAL PROPERTIES OF OXYGEN

1. **Oxygen is a neutral gaseous element**.
2. **Oxygen is quite reactive**. Proof of its high reactivity can be seen in the large number of naturally occurring compounds containing oxygen.
3. **Oxygen supports combustion.**
4. **Oxygen combines with almost all elements forming *oxides*.**

(a) Reaction of oxygen with non-metals

Some non-metals burn in oxygen producing flames with specific colour (see table below).

Non-Metal	Flame	Reaction	Comments
(i) Hydrogen	Blue flame	$2H_2 + O_2 \rightarrow 2H_2O$	If the supply of oxygen is uncontrolled, hydrogen burns with a loud blinding flash.If the supply of oxygen is controlled, the reaction produces a very hot, blue flame. This prop-

			erty is used in oxy-hydrogen torches.
(ii). Oxygen	–	$3O_2 \xrightarrow{\text{UV radiation}} 2O_3$	This reaction is known as self-oxidation. If a silent discharge of electricity is sent through oxygen gas it forms O_3. Ozone is present in the upper layer of the atmosphere and prevents skin cancer by blocking UV radiation from the sun.
(iii) Nitrogen	–	$N_2 + O_2 \rightleftharpoons 2NO$	Equal volumes of N_2 and O_2 form NO in a reversible reaction.
(iv) Carbon	–	$C + O_2 \rightarrow CO_2$	Carbon starts glowing as the reaction progresses.
(v) Sulphur	Beautiful blue flame	$S + O_2 \rightarrow SO_2$	SO_2 and traces of SO_3 are formed.
(vi) Phosphorous	Brilliant white flame	$P_4 + 5O_2 \rightarrow 2P_2O_5$	Forms dense white fumes of P_2O_5

(b) Reaction of oxygen with metals

Various metals burn in oxygen producing flames with specific colour (see box below).

Metal	Flame	Reaction	Comments
(i) Potassium, K	Lilac flame	$K + O_2 \rightarrow K_2O$	Reaction takes place at room temperature
(ii) Sodium, Na	Golden yellow	$Na + O_2 \rightarrow Na_2O$	Reaction takes place at room temperature
(iii) Calcium, Ca	Brick red	$2Ca + O_2 \rightarrow 2CaO$	Reaction takes place at room temperature
(iv) Magnesium, Mg	Dazzling white	$2Mg + O_2 \rightarrow 2MgO$	Burning magnesium inserted in a jar of oxygen
(v) Iron, Fe	No flame	$4Fe + 3O_2 \rightarrow 2Fe_2O_3$	Crackling sound with a reddish brown deposit
(vi) Copper, Cu	No flame	$2Cu + O_2 \rightarrow 2CuO$	Cu starts glowing forming black powder

(c) Oxides and their classification

Oxides are binary compounds of oxygen with other elements.

Depending on the property exhibited by the oxides they are classified as follows.

OXIDES			
Non-metal oxides		**Metal oxides**	
Acidic Oxides	**Neutral oxides**	**Basic Oxides**	**Amphoteric oxides**
React with bases giving salt and water	Do not react with acids or bases	React with acids giving salt and water	React with acids and bases giving salt and water
Examples: SO_2, NO_2	Examples: N_2O, CO	Examples: CaO, Na_2O	Examples: ZnO, Al_2O_3

(i) *With most metals, oxygen forms basic oxides*. Basic oxides react with acids forming salt and water.

Examples: sodium oxide and potassium oxide

$$Na_2O + 2HCl \rightarrow NaCl + H_2O$$
$$K_2O + 2HCl \rightarrow 2KCl + H_2O$$

(ii) *With some metals like zinc and aluminium, oxygen forms amphoteric oxides*. Amphoteric

oxides react with both acids and bases forming salt and water.

Examples: zinc oxide and aluminium oxide

Zinc oxide reacts with acids:

$ZnO + 2HCl \rightarrow ZnCl_2 + H_2O$

Zinc oxide reacts with bases:

$ZnO + 2NaOH \rightarrow Na_2ZnO_2 + H_2O$

Aluminium oxide reacts with acids:

$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$

Aluminium oxide reacts with bases:

$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$

(iii) *With many non-metals, oxygen forms acidic oxides*. Acidic oxides react with bases forming salt and water.

Examples: sulphur dioxide and carbon dioxide

$SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$

$CO_2 + 2NaOH \rightarrow Na_2CO_3 + 2H_2O$

(iv) *With some non-metals, oxygen forms neutral oxides*. Neutral oxides are neither acidic nor basic.

Examples: Nitrous oxide (N_2O) and carbon monoxide (CO).

5. Reaction of oxygen with other compounds (oxidation)

Action with carbon monoxide

CO burns with a pale blue luminous hot flame forming carbon dioxide.

$$2CO + O_2 \rightarrow 2CO_2$$

Action with sulphur dioxide

A mixture of 2 : 1 volumes of SO_2 and oxygen passed at 450°C over platinised asbestos or V_2O_5 forms SO_3.

$$2SO_2 + O_2 \xrightarrow{450^0\,C,\ Pt-Asbestos} 2SO_3$$

Action with nitric oxide

When NO (colourless gas) reacts with oxygen it forms brown nitrogen dioxide gas.

$$2NO + O_2 \rightarrow 2NO_2$$

Action with ammonia

Simple burning – Simple burning of ammonia in pure oxygen forms nitrogen gas and steam.

$$4NH_3 + 3O_2 \xrightarrow{\text{simple burning}} 2N_2 + 6H_2O$$

Catalysed burning – A mixture of ammonia and oxygen in the ratio 4 : 5 by volume, heated to 800°C and passed over platinum gauze catalyst forms nitric oxide and steam.

$$4NH_3 + 5O_2 \xrightarrow{Pt,\ 800^0\,C} 4NO + 6H_2O$$

6. Reaction with hydrocarbons

Hydrocarbons are compounds of carbon and hydrogen only. Hydrocarbons burn in oxygen, forming carbon dioxide and steam with the release of heat.

Methane :

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + Heat released

7. (a) Reaction with non-metallic sulphides: Hydrogen sulphide (H_2S) and carbon disulphide (CS_2) react with oxygen giving sulphur dioxide (SO_2) as common product.

Reaction with H_2S:

$2H_2S + 3O_2 \rightarrow 2H_2O + 2SO_2$

Reaction with CS_2:

$2CS_2 + 3O_2 \rightarrow 2H_2O + 2SO_2$

(b) Reaction with metallic sulphides: Sulphides of zinc, iron, lead, copper and mercury catch fire when heated in oxygen forming the respective oxides and sulphur dioxide.

$$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$$

Sulphides of potassium, sodium, calcium, magnesium, aluminium do not react with oxygen, as the affinity of these metals for sulphur is more than that for oxygen.

13.7 TEST FOR OXYGEN

The reactions of oxygen given below are unique and can be used as a test for detecting its presence.

(i) Oxygen rekindles a glowing wooden splinter.

(ii) Oxygen turns alkaline pyrogallol solution brown.

(iii) Oxygen forms brownish fumes of NO_2 with NO.

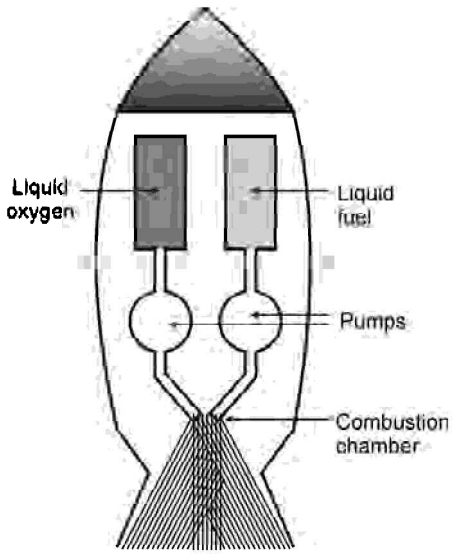
13.8 USES OF OXYGEN

Oxygen is a very important element and finds many uses as follows.

(i) **General uses** – 21% by volume of air is oxygen. It is essential for respiration of all living beings.

(ii) **Artificial respiration** – Oxygen filled cylinders are used by mountaineers, firemen, astronauts, divers, in submarines and in deep mining.

(iii) **Liquid oxygen** under the market name of LOX is used to burn rocket fuel in space. (See box below)

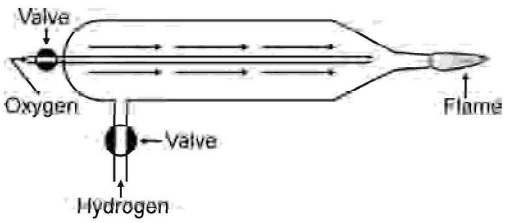


(iv) **In hospitals** – *Carbogen* is a mixture of 5% CO_2 and 95% O_2. It is used in hospitals to stimulate breathing in patients of asthma, drowning or poisoning.

(v) **Industrial uses** – Oxygen is used by industries for cutting and welding. Hydrogen and acetylene when burnt in a jet of pure oxygen achieve very high temperatures that are required for welding.

(a) Oxy-hydrogen torches can achieve temperatures of 2800°C. (See figure below)

(b) Oxy-acetylene torches can achieve temperatures of 3200°C.



(c) Oxygen is used in blast furnaces for the manufacture of iron in place of air to prevent loss of heat energy by unnecessary heating of nitrogen in air.

(d) Oxygen is used in mining for bursting explosives.

(e) A mixture of one part coal dust, one part petroleum jelly and 8 parts of liquid oxygen are used for blasting rocks in mining.

(f) Oxygen is used in chemical industry to manufacture sulphuric acid from sulphur and nitric acid from ammonia.

13.9 RUSTING OF IRON

Iron slowly converts into its hydrated oxide in the presence of moisture and air to form a brown dusty layer of *rust*.

$$4Fe + 3O_2 \rightarrow 2Fe_2O_3$$

$$Fe_2O_3 + xH_2O \rightarrow Fe_2O_3 . xH_2O$$

(rust)

Rusting is a slow oxidation process in the presence of moisture. Rusting of iron is a totally undesirable reaction and must be prevented.

Galvanisation

Galvanisation is a method of preventing rusting of iron by coating it with a layer of zinc. As zinc has more affinity for oxygen and is more reactive than iron, it gets preferentially oxidized thus preventing the rusting of iron.

Alloys

Alloys are mixtures of metals. Alloying of iron consists of mixing it with other elements like manganese, chromium or carbon. Alloying of iron prevents rusting as other elements in the alloy get preferentially oxidized.

Prevention of rusting

Absence of moisture or oxygen or both can prevent rusting of iron. Absence of moisture or oxygen can be ensured by several methods as given below.

(i) By painting the iron surface

(ii) By applying oil / grease

(iii) By electroplating the iron surface with another metal

(iv) By galvanisation with zinc (See box).

(v) By alloying (See box).

EXERCISE

I. Choose the correct answer

1. Which is a physical property of oxygen?

(a) gas, non-metal (b) A = 16

(c) Z = 8 (d) a, b and c

2. Oxygen is manufactured on a large scale from

(a) water

(b) oxidising agents

(c) liquid air

(d) hydrogen peroxide

3. **Drying agents that can be used in the laboratory preparation of oxygen is**
   (a) anhydrous calcium chloride
   (b) concentrated sulphuric acid
   (c) phosphorous pentoxide
   (d) all three (a, b, c)
4. **Oxygen is**
   (a) a supporter of combustion
   (b) does not support combustion
   (c) combustible
   (d) not combustible
5. **Liquid oxygen is**
   (a) yellow (b) pale green
   (c) pale blue (d) brown gas
6. **The gas that boils last during fractional distillation of liquid air is**
   (a) inert gases (b) nitrogen
   (c) oxygen
   (d) all gases boil off together
7. **Oxides are binary compounds of oxygen with**
   (a) other elements (b) itself
   (c) hydrocarbons (d) inert gases
8. **Basic oxides are formed when oxygen combines with**
   (a) metals (b) non-metals
   (c) inert gas elements
   (d) any of the four options
9. **Oxygen is collected in the laboratory by**
   (a) by downward displacement of water
   (b) by collecting in a cylinder
   (c) by downward displacement of air
   (d) by upward displacement of air
10. **Acidic oxides are formed when oxygen combines with**
   (a) metals (b) non-metals
   (c) inert gas elements
   (d) any of the four options
11. **The salt formed when zinc oxide reacts with NaOH is**
   (a) Na_2PbO_2 (b) $Pb(OH)_2$
   (c) Na_2ZnO_2 (d) $PbCl_2$
12. **The formula of rust is**
   (a) Fe_2O_3 (b) $FeO.2H_2O$
   (c) $Fe_2O_3.H_2O$ (d) $F_3O_4.H_2O$
13. **The sulphide that reacts with oxygen when heated in air is**
   (a) K_2S (b) ZnS
   (c) Li_2S (d) Na_2S
14. **Rusting of iron can be prevented by**
   (a) alloying (b) electroplating
   (c) galvanizing
   (d) all three options (a, b and c).
15. **A definite test for oxygen is**
   (a) rekindles glowing wooden splinter.
   (b) forms brownish fumes of NO_2 with NO.
   (c) turns alkaline pyrogallol solution brown.
   (d) all three options (a , b and c).
16. **Amphoteric oxides differ from basic oxides in that they react with**
   (a) bases only. (b) acids only.
   (c) neither acids nor bases.
   (d) both acids and bases.
17. **Which is not an amphoteric oxide?**
   (a) ZnO (b) PbO
   (c) Al_2O_3 (d) CaO
18. **Which is not a neutral oxide?**
   (a) CO_2 (b) H_2O
   (c) N_2O (d) NO
19. **Which is not an acidic oxide?**
   (a) ZnO (b) P_2O_5
   (c) SO_2 (d) SO_3
20. **Which is an acidic oxide?**
   (a) NO_2 (b) PbO
   (c) Li_2O (d) N_2O
21. **Neutral oxides differ from acidic oxides in that they do not react with**
   (a) bases only. (b) acids only.
   (c) both acids and bases.
   (d) none of these options (a, b and c) are correct.
22. **The acid formed when sulphur trioxide reacts with water is**
   (a) sulphuric acid (b) sulphurous acid
   (c) pyrosulphuric acid (d) none of the above
23. **When oxygen combines catalytically with ammonia the gas formed before cooling is**
   (a) nitric oxide (b) water vapour
   (c) both (a) and (b) (d) none of the above
24. **The sacrificial metal that prevents rusting when iron is galvanized is**
   (a) iron (b) zinc
   (c) mercury (d) oxygen
25. **The carbon dioxide in carbogen is**
   (a) to activate the bio-feedback mechanism for increased inhalation
   (b) to make the artificial respiration feel almost like normal air
   (c) to decrease the amount of oxygen going into the lungs
   (d) as a solvent for oxygen

II. Match the following

This section contains three questions. Each question contains statements given in two columns, which have to be matched. Statements (i, ii, iii, iv) in Set A have to be matched with Statements (a, b, c, d) in Set B. More

than one match or no match between some options in the Sets is possible.

1. Match the nature of oxide given in Set A with how it forms in Set B

	Set A	Set B
(i)	Acidic oxide	(a) certain compounds formed be tween non-metals + oxygen
(ii)	Basic oxide	(b) metal + oxygen
(iii)	Amphoteric oxide	(c) certain metals (like zinc + oxygen
(iv)	Neutral oxide	(d) non-metal + oxygen

2. Match the reaction of oxygen with reactant given in Set A with the products formed in Set B

	Set A	Set B
(i)	Carbon disulphide	(a) carbon dioxide and water
(ii)	Ammonia	(b) carbon dioxide and sulphur dioxide
(iii)	Methane	(c) nitrogen and water
(iv)	Hydrogen sulphide	(d) sulphur dioxide and water

3. Match the type of use of oxygen in Set A with the use in Set B

	Set A	Set B
(i)	Artificial respiration	(a) as a oxidant
(ii)	Liquid oxygen	(b) in oxy-hydrogen torches (used for welding)
(iii)	In hospitals	(c) as *Carbogen* (CO_2 mixed with O_2)
(iv)	Industrial use	(d) as fuels in rockets

III. State whether the following statements are True or False

1. Oxygen is the most abundant element on the earth.
2. In the industrial preparation of oxygen from liquid air, oxygen boils off first.
3. The amount of oxygen is less at high altitudes when compared to the ground.
4. All basic oxides are amphoteric in nature.
5. A neutral oxide reacts with bases.
6. Oxygen undergoes self-oxidation to form ozone in the presence of UV radiation from the sun.
7. Sulphides of potassium, sodium, calcium, magnesium and aluminium do not react with oxygen even on strong heating.
8. Carbogen is a mixture of 50% carbon dioxide with 50% oxygen.
9. Oxy-acetylene torches produce temperatures in the range of 3200°C.
10. Rusting of iron is slow oxidation of iron in the presence of moisture.

IV. This section contains two paragraphs on 'GLOBAL WARMING' and 'POLLUTION' both of which are related to the air we breathe. Based on the paragraph three multiple-choice questions have to be answered. Each question has 4 choices, out of which only one is correct. Choose the correct one.

Paragraph I

Ozone (O_3) is present in the uppermost layer of the earth's atmosphere. Ozone absorbs ultraviolet rays from the sun. These rays are very harmful to both plants and human beings if they reach the earth's surface. The amount of ozone in the upper layers of the atmosphere has been coming down over the last 15 years. Due to this the ultraviolet rays are not getting filtered. More ultraviolet radiation is now reaching the earth's surface. This is mainly due to the release of man made chemicals containing chlorine such as CFC's (chloro fluoro carbons) other halogen compounds and oxides of nitrogen into the atmosphere. CFC's are a common industrial product used in refrigeration systems, air conditioners, aerosols and solvents and in the manufacture of some plastic packing materials. Oxides of nitrogen are released into the atmosphere from exhausts of engines specially aircraft engines. The first global agreement between countries to restrict CFCs came in 1987. The aim is to reduce the usage of these chemicals by half by the year 2000. The CFC's will not be produced by any of the countries after the end of 1995, except for a limited amount for essential uses, such as for medical sprays. The 'Ozone Hole' in the atmosphere is the largest in the atmosphere that is above the most indutrialised countries. This shows the flip side of industrialization. Along with global warming the Ozone Hole is an issue that needs to be tackled on a war footing.

1. The gas which prevents ultra violet rays from reaching the surface of the earth is
   (a) Oxygen (b) Carbon dioxide
   (c) Ozone (d) All the above
2. Ozone in the atmosphere is depleted when it comes in contact with chemicals like
   (a) Oxides of nitrogen
   (b) CFC's
   (c) some halogen compounds
   (d) all three options (a, b and c)
3. Global warming is due to increased levels of _____, whereas formation of Ozone Hole is due to decrease in levels of _____
   (a) Carbon dioxide (CO_2), sulphur dioxide (SO_2)
   (b) Carbon dioxide (CO_2), ozone (O_3)
   (c) hot air, CFC's
   (d) none of these

Paragraph 2

Vehicle exhausts contain gases formed due to combustion of fuel. Vehicles that pollute, release toxins in the form of particulate matter as well as CO – a poisonous gas. When air polluted with CO is inhaled, the lungs absorb CO preferentially. Carbon monoxide is 300 times more soluble in blood than oxygen forming carboxyhemoglobin thus depriving the tissues of oxygen. Improper burning of fuel causes incomplete combustion resulting in the formation of CO. Proper burning of fuel causes complete combustion resulting in the formation of CO_2 that is not poisonous. However, large-scale use of fuels is causing the CO_2 levels in the atmosphere to rise. This causes global warming. Global warming has disastrous consequences for mankind.

1. The gas which is poisonous when inhaled is
   (a) CO (b) CO_2
   (c) both CO and CO_2 (d) neither CO or CO_2
2. Deprival of oxygen in the tissues on inhaling air polluted with CO is due to
   (a) formation of haemoglobin
   (b) formation of CO_2
   (c) formation of carboxyhaemoglobin
   (d) insufficient oxygen
3. Global warming is due to increased level of
   (a) CO (b) CO_2
   (c) hot air (d) none of these

V. Fill in the blanks

1. _____ oxides react with acids and bases to give salt and water.
2. The percentage of oxygen in the air by weight is _____ and by volume is _____.
3. Dry oxygen is prepared in the laboratory by bubbling through _____.
4. The boiling point of nitrogen is _____ and that of oxygen is _____.
5. Potassium when burnt in air gives the flame a _____ colour.
6. Catalytic oxidation of ammonia gives _____and _____ as products.
7. Alkaline pyrogallol solution turns _____in the presence of oxygen.
8. _____ are binary compounds of oxygen with other elements.
9. Oxygen mixture used in the laboratory preparation of oxygen is a mixture of _____ and _____.
10. The ratio by weight of MnO_2 and $KClO_3$ in the laboratory preparation using a mixture of these two compounds is _____.

VI. Each question contains two statements. Statement A is an Assertion Statement and Statement R is the Reason Statement. Each question has four choices out of which only one is correct. Choose the correct one.

Choose A if Statement A is true, Statement R is true; Statement R is correct explanation for statement A

Choose B if Statement A is true, statement R is true; Statement R is not correct explanation for statement A

Choose C if Statement A is true, Statement R is false

Choose D if Statement A is false, statement R is true

1. **Statement –A**: Oxygen reacts with almost every element.

   Statement –R: Oxygen is the most abundant element on earth.
2. **Statement –A**: About 89% of water by weight is oxygen.

   Statement –R: Water has two atoms of hydrogen and one atom of oxygen.
3. **Statement –A**: At normal temperature nitrogen does not react with oxygen.

   Statement –R: Nitrogen in the atmosphere helps in making all burning reactions more controlled.
4. **Statement –A**: Alloying of iron prevents the iron from rusting.

   Statement –R: Iron on alloying cannot react with oxygen easily.
5. **Statement –A**: Ozone is a product formed by *self-addition* of oxygen.

   Statement –R: Ozone prevents skin cancer by blocking UV radiation from the sun.

VII. Answer the following questions.

1. How does oxygen occur on earth?
2. What are the changes noticed when red lead is heated?
3. How would you prepare and collect dry oxygen in the laboratory? Give reason for the method of collection.
4. What is the principle involved in the manufacture of oxygen by fractional distillation of liquid air?
5. Why is the density of oxygen prepared from air slightly more dense than when prepared by electrolysis?
6. Why do mountaineers carry oxygen cylinders?
7. Water freezes in the oceans near the poles. The ice formed forms a thick crust on the surface of oceans. How does aquatic life survive inside the water in the absence of oxygen?
8. What are oxides? Differentiate between basic and

acidic oxides. Give an example each.

9. What are mixed oxides and peroxides? Give examples.
10. How would you distinguish between hydrogen and oxygen?
11. Bring out the differences between burning and rusting.
12. How can galvanizing and alloying prevent rusting?
13. What is carbogen?
14. Which gives a higher temperature – oxy-acetylene or oxy-hydrogen torch?
15. More reactive metal sulphides do not react with oxygen. Why?
16. How much oxygen would be needed to burn 18 gm of carbon? (C=12)
17. Complete and balance the following reactions:
    (a) $H_2O_2 \rightarrow$ (b) $Na_2O_2 + H_2O \rightarrow$
18. Complete and balance the following reactions if the reaction takes place.
    (a) $N_2O + HCl \rightarrow$ (b) $ZnO + NaOH \rightarrow$
    (c) $CO + KOH \rightarrow$ (d) $Na_2O + HNO_3 \rightarrow$

EQUATION PRACTICE

Complete and balance the equations given below. Where applicable, state the physical change seen. Write the reaction conditions where required.

1. $H_2 + O_2 \rightarrow$
2. $3O_2 \rightarrow$
3. $ZnS + O_2 \rightarrow$
4. $C + O_2$ (excess air supply) $\rightarrow$
5. $C + O_2$ (limited air supply) $\rightarrow$
6. $S + O_2 \rightarrow$
7. $P_4 + O_2 \rightarrow$
8. $K + O_2 \rightarrow$
9. $Na + O_2 \rightarrow$
10. $Ca + O_2 \rightarrow$
11. $CS_2 + O_2 \rightarrow$
12. $Mg + O_2 \rightarrow$
13. $Fe + O_2 \rightarrow$
14. $Cu + O_2 \rightarrow$
15. $CO + O_2 \rightarrow$
16. $SO_2 + O_2 \xrightarrow{?}$
17. $NO + O_2 \rightarrow$
18. $NH_3 + O_2 \xrightarrow{\text{simple burning}}$
19. $NH_3 + O_2 \xrightarrow{\text{Pt, 800°C}}$
20. $CH_4 + O_2 \rightarrow$
21. $C_xH_y + O_2 \rightarrow$
22. $H_2S + O_2 \rightarrow$
23. $N_2 + O_2 \rightleftharpoons$

KEY TO EXERCISE

I. 1. d 2. c 3. d 4. a 5. c 6. c 7. a
8. a 9. a 10. b 11. c 12. c 13. b 14. d
15. d 16. d 17. d 18. a 19. a 20. a 21. c
22. a 23. c 24. b 25. b

II. 1. i. d ii. b iii. c iv. a 2. i. b ii. c iii. a iv. d
3. i. c ii. a, d iii. c iv. b

III. 1. true 2. false 3. true 4. false 5. false 6. true 7. true
8. false 9. true 10. true

IV. Paragraph 1 1. c 2. d 3. b
Paragraph 2 1. a 2. c 3. b

V. 1. Amphoteric 2. 23%, 21% 3. conc. H_2SO_4 acid
4. –195.8° C, –183°C 5. lilac 6. nitric oxide and water
7. brown 8. Oxides 9. MnO_2 and $KClO_3$
10. 1 : 4

VI. 1. B 2. B 3. B 4. A 5. B

14 Study of Group VIA Element–Sulphur and its compounds

14.0 INTRODUCTION TO SULPHUR

Sulphur is a bright yellow, tasteless solid and a very reactive element. It is found in a wide range of minerals and is one of the products in a volcanic eruption. Perhaps this is why ancient people associated sulphur (also known as brimstone) with hell's fires.

The pure element sulphur has always been thought to have strange properties. A spinning ball of it was used in one of the world's first demonstrations of static electricity. It was found that when the ball was touched by a hand, the ball began to glow!

The element sulphur is a non-metal and will not dissolve in water.

The pure element sulphur has very little smell. The smell that one might associate with sulphur is the smell of bad eggs. But the smell is actually due to a compound of sulphur, known as hydrogen sulphide. Sulphur compounds are also responsible for the smell in garlic, mustard, onions and cabbage. Sulphur is a part of all living tissues. Sulphur is found in proteins in plants. The Sanskrit name of sulphur is 'sulvaari' meaning copper's enemy.

14.1 OCCURRENCE

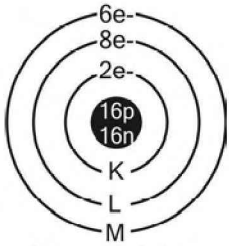
It is the 14th most abundant element in the earth's crust.

Free state – sulphur occurs in the free state near volcanoes in Sicily, Japan and Mexico and as underground deposits in Texas, USA (70% of world deposits are found here)

Combined state – Sulphur occurs as sulphates and sulphides in the earth's crust.

Some examples
- Salt cake (Na_2SO_4)
- Gypsum ($CaSO_4\ 2H_2O$)
- Zinc blende (ZnS)
- Iron pyrites (FeS_2)
- Silver glance (Ag_2S)
- Epsom salt ($MgSO_4,\ 7H_2O$)
- Baryte ($BaSO_4$)
- Galena (PbS)
- Copper pyrites ($CuFeS_2$)
- Cinnabar (HgS)

Sulphur is found in vegetables, animal proteins, garlic, fish, mustard, crude oil and coal.



Atom of sulphur

Sulphur

Symbol: S
Molecular formula S_8
Atomicity – 8
Z = 16, A = 32
Atomic Weight – 32
Electronic configuration (2, 8, 6)
Valence electrons 6
Valency 2
Yellow solid, non-metal
Sulphur belonging to Group VIA.

14.2 INDUSTRIAL EXTRACTION OF SULPHUR

Sulphur is extracted by two methods :
(a) The Sicilian Process
(b) The Frasch Process

Sicilian process

In this process, the crude sulphur is placed in a kiln called Calcaroni (see figure). The important features / steps involved in the Sicilian Process are as follows.

(i) These kilns have sloping floors to allow the molten sulphur to flow down.

(ii) The crude sulphur ore is stacked in such a way that air spaces are there in between the stacked sulphur.

(iii) The sulphur is ignited from the bottom of the kiln.

(iv) As the burning of sulphur is an exothermic reaction, the rest of the sulphur melts.

(v) The molten sulphur flows down the sloping floor and is collected in moist wooden crates.

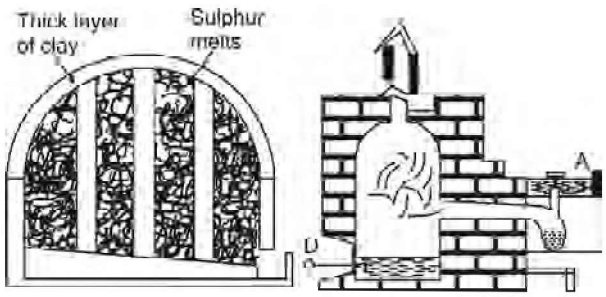
(vi) The sulphur extracted by this method is only 95% pure.

(vii) The impure sulphur is placed in iron retorts and heated so that it starts to boil.

(viii) The sulphur vapourises leaving the impurities behind.

(ix) The sulphur vapours are led into a big brick chamber where they condense into yellow powder on the walls.

(x) As the chamber starts getting hotter, the sulphur melts again and is collected at regular intervals by tapping.



Sicilian Process

FRASCH PROCESS

The Frasch process is used for underground deposits of sulphur. The important features / steps involved in the Sicilian Process are as follows:

(i) The deposits are found at a depth of about 300 – 400 m.

(ii) The underground deposits are 120 – 250 m thick.

(iii) A bore is drilled into the earth so that the lower end is embedded deep in the sulphur deposit.

(iv) Three concentric pipes are lowered into the bore (see figure).

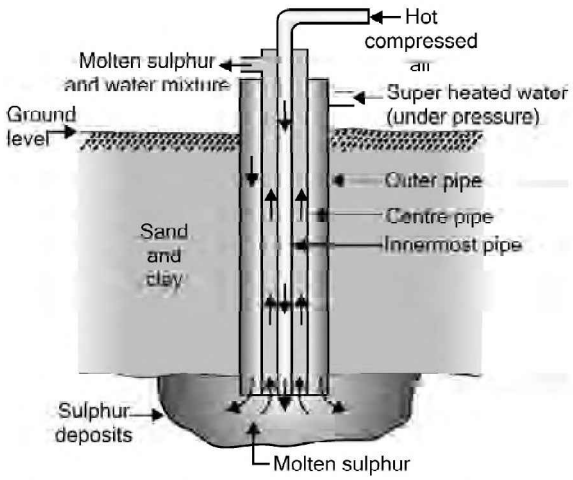
(v) Super heated steam at 170°C is passed through the outermost pipe to melt the underground sulphur.

(vi) Hot compressed air is passed simultaneously through the innermost pipe.

(vii) The compressed air and steam melts the sulphur. As the sulphur melts, the air and water churn the molten sulphur to form frothy foam.

(viii) The compressed air tries to come out through the middle pipe, carrying the frothy mass of sulphur with it.

(ix) The sulphur is collected in wooden vats and allowed to cool when 99.5% pure sulphur separates from water.

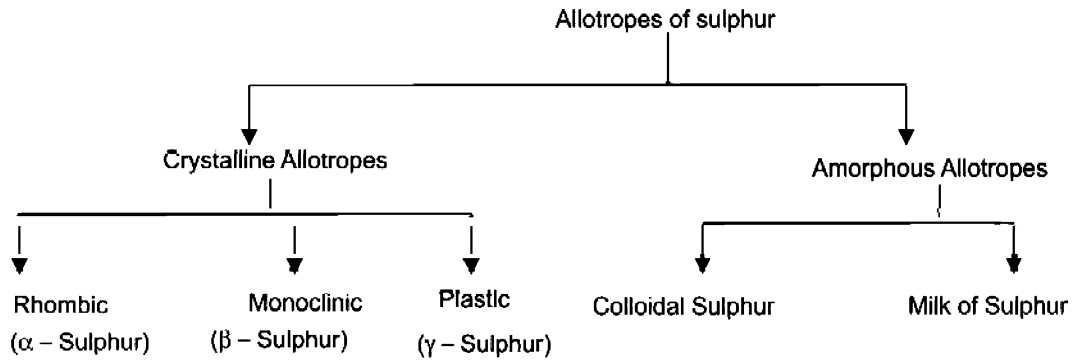


Frasch process for the extraction of sulphur

14.3 SULPHUR EXHIBITS ALLOTROPY

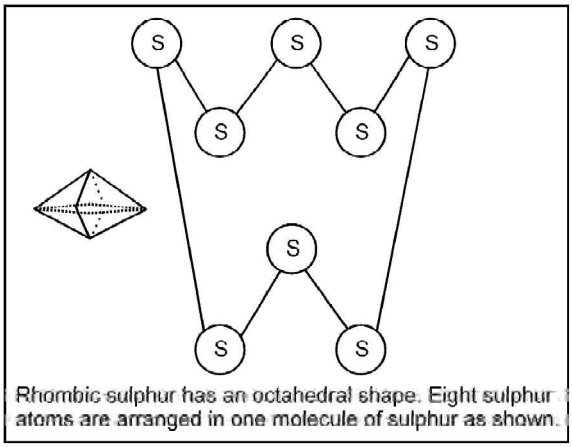
Allotropy is the phenomenon in which an element exists in two or more physical forms. Sulphur has many allotropic forms.

The allotropes of sulphur are broadly classified into two types: Crystalline allotropes and Amorphous allotropes.



A. Crystalline allotropes of sulphur

The crystalline allotropes of sulphur are further subdivided into Rhombic and Monoclinic sulphur.



Rhombic sulphur has an octahedral shape. Eight sulphur atoms are arranged in one molecule of sulphur as shown.

Rhombic sulphur

Rhombic sulphur is also known as octahedral or α-sulphur.

Preparation of Rhombic sulphur

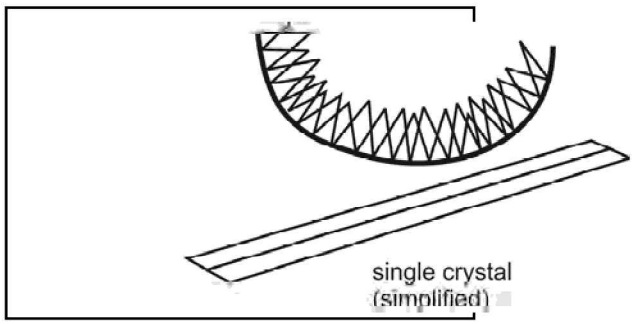
Rhombic sulphur is prepared by dissolving sulphur in a volatile organic solvent like carbon disulphide (CS_2) to form a saturated solution. This solution is placed in the sun, when the volatile solvent evaporates leaving behind pale yellow octahedral crystals of Rhombic sulphur.

Monoclinic sulphur

Monoclinic sulphur is also known as Prismatic or β-sulphur.

Preparation of Monoclinic sulphur

Monoclinic sulphur is prepared by melting sulphur in a dish on a low flame while stirring constantly. As it melts, more sulphur is added. The liquid sulphur is placed in lukewarm water. The surface of molten sulphur cools faster, and due to this a thin crust forms on the surface of liquid sulphur. The layer is pierced using a rod and the liquid sulphur inside is poured out. The crust is cut and peeled off. Needle like structures of monoclinic sulphur can be seen sticking to the sides of the dish.



B. Amorphous allotropes of sulphur

The amorphous allotropes of sulphur are further subdivided into Plastic sulphur, Colloidal sulphur and Milk of sulphur.

Plastic Sulphur

Plastic sulphur is also known as Rubber or γ (gamma) sulphur. Plastic sulphur is prepared by heating sulphur in a hard glass tube till it boils. The boiling sulphur is poured into cold water when it solidifies forming a dark brown mass of plastic sulphur

Colloidal Sulphur

Colloidal sulphur is prepared by passing hydrogen sulphide (H_2S) gas through concentrated nitric acid (HNO_3). Nitric acid being a strong oxidizing agent, oxidizes hydrogen sulphide to a white yellow suspension of colloidal sulphur

Milk of Sulphur

Milk of sulphur is prepared boiling sulphur with limewater solution [$Ca(OH)_2$] and then cooling it. Dilute hydrochloric acid (HCl) is added to neutralize the limewater, when a fine white deposit of milk of sulphur precipitates out which is then filtered.

14.4 PHYSICAL PROPERTIES OF SULPHUR

Property	Rhombic Sulphur	Monoclinic Sulphur	Plastic Sulphur
1. Colour	Pale yellow	Colourless, turning brown on exposure to air	Brown/amber
2. Physical appearance	Crystalline	Crystalline	Amorphous
3. Shape	Octahedral	Needle like	No shape
4. Stability	Stable below 95.6°C	Stable from 95.6–119°C	Slowly changes to α-S
5. Solubility in water	Insoluble	Insoluble	Insoluble
6. Solubility in CS_2	Soluble	Soluble	Insoluble
7. Relative density	2.06*	1.96*	1.95
8. MP	112.8°C	119°C	No clear MP

14.5 CHEMICAL PROPERTIES OF SULPHUR

1. Action of heat on Sulphur

(i) Absence of air

When sulphur is heated in the absence of air in a hard glass tube it melts at 114°C to form a pale straw liquid. On further heating, it becomes viscous and brown finally becoming black at 200°C. It is so viscous that it cannot be poured out. On further heating it starts softening at 250°C. It finally starts boiling at 444°C giving red vapours of sulphur that condense on the cooler parts of the tube forming a pale yellow and powdery deposits of Flowers of sulphur.

Action of Heat on sulphur

(In absence of air)

Sulphur

↓

(114°C, in absence of air)

↓

Pale straw-yellow liquid

(heat)

↓

Viscous brown liquid

(200°C)

↓

Jet-black liquid

(can not be poured out)

(250°C)

↓

Starts softening (444°C)

↓

Starts boiling forming red vapours that cool giving flowers of sulphur

Reasons for the changes observed on heating sulphur

Solid sulphur consists of a compact arrangement of sulphur molecules consisting of 8 atoms of S each.

(a) When it melts, the S_8 rings of sulphur slip over one another easily to form a pale yellow free flowing liquid.

(b) When molten sulphur is heated further, the S_8 rings of sulphur are broken to form open chains of eight sulphur atoms each. Due to this the colour darkens and molten sulphur starts becoming viscous.

(c) At about 114°C (387 K), the small open chains get entangled with one another due to which the colour changes to dark brown and the sulphur becomes extremely viscous.

Transition temperature

When rhombic, monoclinic and plastic sulphur are placed in three separate dishes, over a period of time, they all change to rhombic sulphur.

Rhombic sulphur is the most stable form of sulphur at room temperature. On heating, above 95.6°C it changes to monoclinic sulphur and on cooling below 95.6°C it changes back to rhombic sulphur.

This temperature of 95.6°C is known as Transition Temperature. Below the transition temperature only rhombic sulphur exists.

* Both rhombic sulphur and monoclinic sulphur are made up of S_8 molecules – the arrangement of these rings is tight in rhombic sulphur and loose in monoclinic sulphur due to which density of rhombic sulphur is higher than that of monoclinic sulphur.

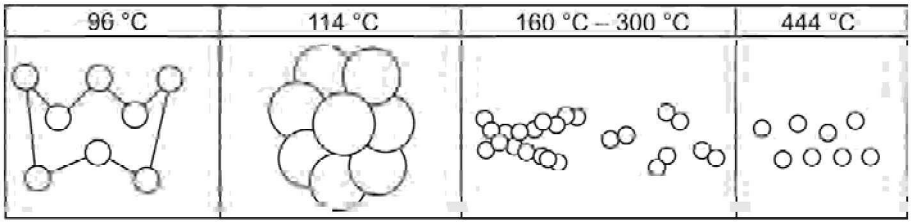
Enantiomerism : When rhombic S is heated to 95.6°C it slowly starts changing to monoclinic sulphur.

Thus at 95.6°C, both the allotropes of S are present such that they are in equilibrium.

$$S_{(Rh)} \overset{95.6^0C}{\Leftrightarrow} S_{(M)}$$

Such a condition where both allotropes are present at the same time in equilibrium is called enantiomerism.

As the temperature is raised further beyond 95.6°C concentration of $S_{(M)}$ will rise and if it falls below 95.6°C concentration of $S_{(Rh)}$ will rise. As the temperature touches > 112°C whatever $S_{(Rh)}$ is present starts melting and as the temperature reaches 119°C, $S_{(M)}$ starts melting.



(d) When the temperature is increased further beyond 160°C (433 K) the long chains of sulphur atoms break more and more to form smaller chains of S_6, S_4, S_2 (Above 600°C only S_2 molecules will be present). Due to this molten sulphur becomes lighter in colour, less viscous and free flowing again.

(e) At 444°C (717K), the heat energy breaks the small chains to form S_2 molecules thus forming yellowish-brown vapours of boiling sulphur.

(ii) Presence of air

When sulphur is heated in the presence of air all the above changes take place with the only difference that the red vapours of sulphur catch fire burning with a pale blue flame forming sulphur dioxide. Traces of sulphur trioxide react with moisture in the air forming sulphuric acid that fumes.

$$S + O_2 \xrightarrow{>300\,^0C} SO_2 \text{ (and traces of } SO_3)$$

2. Action with non-metals

Sulphur reacts with non-metal like hydrogen, chlorine, carbon and phosphorous as follows.

$$H_2 + S \rightarrow H_2S$$

$$Cl_2 + 2S \rightarrow S_2Cl_2$$

(sulphur monochloride)

$$C + 2S \rightarrow CS_2$$

$$4P + 8S \rightarrow P_2S_3 + P_2S_5$$

3. Action with metals

Sulphur reacts with active metals like sodium, potassium and calcium to form simple sulphides. The simple sulphides react further with excess sulphur forming polysulphides.

$$Ca + S \rightarrow CaS$$

$$CaS + 4S \rightarrow CaS_5$$

All other metals form respective sulphides.

Examples

Aluminium reacts with sulphur forming aluminium sulphide.

$$2\,Al + 3S \rightarrow Al_2S_3$$

However, reaction of copper with sulphur depends on the temperature as follows:

$$Cu + S \xrightarrow{<400°C} CuS$$

$$2Cu + S \xrightarrow{>400°C} Cu_2S$$

4. Action with oxy-acids like HNO_3, H_2SO_4

Sulphur reacts with oxy-acids as follows.

$$S + 6HNO_3 \rightarrow H_2SO_4 + 6NO_2 + 2H_2O$$

$$S + 2H_2SO_4 \xrightarrow{\text{Boiling acid}} 3SO_2 + 2H_2O$$

5. Action with alkalis

Limited quantities of sulphur react with alkalis forming thiosulphate and sulphides. Excess sulphur reacts with alkalis to form polysulphide.

(i) Action with NaOH or KOH

$$6NaOH + 12S \rightarrow 2Na_2S_5 + Na_2S_2O_3 + 3H_2O$$

(ii) Action with $Ca(OH)_2$

$$3Ca(OH)_2 + 12S \rightarrow 2CaS_5 + CaS_2O_3 + 3H_2O$$

14.6 USES OF SULPHUR

Sulphur has many uses in different categories as can be seen below:

(a) In the chemical industry

(i) For synthesis of sulphuric acid (H_2SO_4) by Contact process.

(ii) For manufacture of sodium thiosulphate ($Na_2S_2O_3$) – also called hypo, used in photography (as $Na_2S_2O_3$ dissolves silver halides).

(iii) For manufacture of sulphur dioxide (SO_2) – used for bleaching silk, wool and in sugar industry.

(iv) For manufacture of calcium sulphide (CaS) and calcium bisulphate – $Ca(HSO_3)_2$, (used in paper industry for bleaching).

(v) For manufacture of carbon disulphide (CS_2) – used as an industrial solvent and used extensively in manufacture of rayon.

(vi) For manufacture of phosphorous trisulphide (P_2S_3) and antimony sulphide (Sb_2S_3) – used in match industry.

(vii) For manufacture of explosives like gunpowder – a mixture of sulphur, charcoal and nitre. This mixture is also used in firecrackers.

(b) In agriculture

Sulphur is used as an insecticide and fungicide in grapes vines. It is used as sulphur dioxide for fumigation.

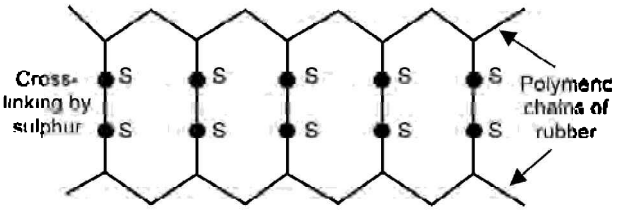
(c) In medicine

Sulphur is used in skin ointments and in other oral medicines for purification of blood. It is used in all forms of medicines including ayurveda.

(d) In tyre industry

Sulphur is used for vulcanizing rubber to make it hard for use in tyres and tubes.

Vulcanisation of rubber : Natural rubber is soft and wears off very fast. When natural rubber is warmed with sulphur, it becomes hard and tough due to the formation of cross chains with sulphur (see figure). This rubber is called vulcanized rubber and is used for making tyres and tubes.



Formation of vulcanised sulphur

14.7 INTRODUCTION SULPHUR DIOXIDE

Sulphur dioxide occurs in volcanic eruptions, wherever coal is burnt and near oil refineries.

SULPHUR DIOXIDE
Molecular Formula – SO_2
Molecular Weight – 64
Acidic gas and poisonous

14.8 GENERAL METHODS OF PREPARATION

(a) By burning sulphur in air or oxygen:

$$S + O_2 \rightarrow SO_2 (+ \text{ traces of } SO_3)$$

Sulphur burns with a pale blue flame in air. The gaseous products fume strongly as sulphur trioxide (SO_3) reacts with moisture in the air forming vapours of sulphuric acid (H_2SO_4)

(b) By roasting metallic sulphides in air

With the exception of sodium sulphide, potassium sulphide, magnesium sulphide and aluminium sulphide all metallic sulphides on heating in air form the respective oxides and sulphur dioxide

$$PbS + O_2 \rightarrow PbO + SO_2$$

$$ZnS + O_2 \rightarrow ZnO + SO_2$$

(c) By heating conc. sulphuric acid with metals

$$Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$$

$$Zn + 2H_2SO_4 \rightarrow ZnSO_4 + SO_2 + 2H_2O$$

(d) By action of dil. sulphuric acid with bisulphites or sulphites

$$2NaHSO_3 + H_2SO_4 \rightarrow Na_2SO_4 + 2SO_2 + 2H_2O$$

$$ZnSO_3 + H_2SO_4 \rightarrow ZnSO_4 + SO_2 + H_2O$$

14.9 LABORATORY PREPARATION OF SULPHUR DIOXIDE

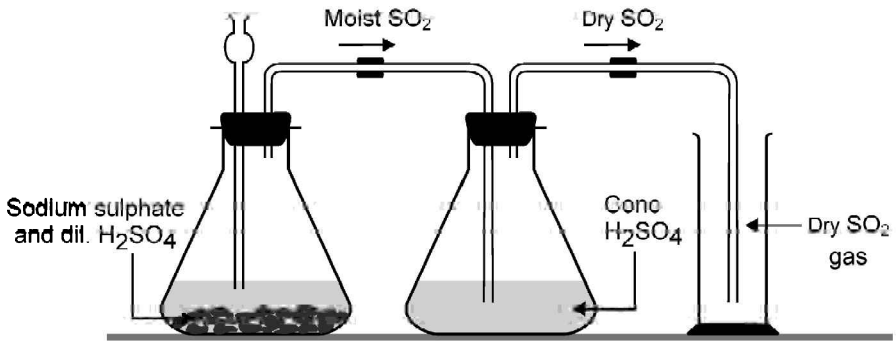
In the laboratory sulphur dioxide is prepared in two ways.

(i) By the action of copper turnings on conc. sulphuric acid

$$Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$$

(ii) By action of dil. sulphuric acid on most metal sulphites (usually sodium sulphite is used).

$$Na_2SO_3 + H_2SO_4 \rightarrow Na_2SO_4 + SO_2 + H_2O$$



Preparation of sulphur dioxide gas from sodium sulphite and dilute sulphuric acid

Drying of sulphur dioxide

The gas evolved is passed through conc. sulphuric acid to dry it.

Collection of sulphur dioxide

Sulphur dioxide is collected by the upward displacement of air, as it is heavier than air and extremely soluble in water.

14.10 PHYSICAL PROPERTIES OF SULPHUR DIOXIDE

The physical properties are listed below.

(i) Sulphur dioxide is a colourless, pungent suffocating gas with sour taste.

(ii) When inhaled sulphur dioxide causes coughing.

(iii) Sulphur dioxide is fairly soluble in water, 1 volume of water dissolving 80 volumes of the gas.

(iv) Sulphur dioxide is easily liquefied under pressure by placing it in a freezing mixture.

(v) Sulphur dioxide boils at 10^0C. Due to this property, sulphur dioxide can be used as a refrigerant.

(vi) Sulphur dioxide solidifies at – 76^0C to form colourless crystals.

14.11 CHEMICAL PROPERTIES OF SULPHUR DIOXIDE

The chemical properties are listed below.

(i) **Sulphur dioxide is neither combustible nor supports combustion**.

(ii) **Sulphur dioxide is an acid gas**. It turns blue litmus red and then *bleaches* it, making it colourless.

(iii) **Acidic properties of sulphur dioxide**

Sulphur dioxide exhibits acidic properties as follows.

– Sulphur dioxide dissolves in water giving hydronium ions (H^+) that are responsible for its acidic properties.

– Sulphur dioxide reacts with alkalis forming two types of salts – normal salt and acid salts.

$$2NaOH + SO_2 \rightarrow Na_2SO_3 + H_2O$$

($NaSO_3$ is a normal salt)

$$NaOH + SO_2 \rightarrow NaHSO_3$$

($NaHSO_3$ is an acid salt)

– On passing sulphur dioxide through an aqueous solution of sodium carbonate it forms bisulphites with the evolution of carbon dioxide.

$$Na_2CO_3 + H_2O + 2SO_2 \rightarrow 2NaHSO_3 + CO_2$$

$$K_2CO_3 + H_2O + 2SO_2 \rightarrow 2KHSO_3 + CO_2$$

(iv) **Sulphur dioxide forms sulphurous acid (H_2SO_3) with water**. Sulphur dioxide is the acid anhydride of sulphurous acid.

(v) **Bleaching action of sulphur dioxide**

Sulphur dioxide is a bleaching agent. When sulphur dioxide reacts with water nascent hydrogen is liberated. The bleaching is due to reduction with nascent hydrogen. However, the bleaching action is temporary as the colour is regained. The reactions are shown below :

(a) Sulphur dioxide forms nascent hydrogen in the presence of moisture.

$$SO_2 + 2H_2O \rightarrow H_2SO_4 + 2[H]$$

(b) The nascent hydrogen bleaches coloured substance by reduction.

Coloured substance + 2[H] → colourless substance

(c) The bleached substance however combines with atmospheric oxygen to form coloured substance.

Bleached substance + O_2 → coloured substance + H_2O
(oxygen from air)

(d) When compared to bleaching by chlorine, the bleaching action of sulphur dioxide is thus temporary.

Comparison of bleaching action of sulphur dioxide and chlorine

	Sulphur Dioxide	Chlorine
Bleaching	$SO_2 + 2H_2O \rightarrow H_2SO_4 + 2[H]$	$Cl_2 + H_2O \rightarrow HClO + HCl$ $HClO \rightarrow HCl + [O]$
Mechanism	Bleaching is due to nascent hydrogen	Bleaching is due to nascent oxyge
Bleaches by	Reduction Coloured substance + [H] → Colourless substance	Oxidation Coloured substance + [O] → Colourless substance
Bleaching Action	The bleaching is temporary as the bleached substance combines with oxygen from air over a period of time and regains its colour.	The bleaching is permanent as the bleaching itself is an oxidation process.

Bleaching Agent	Mild bleaching action with no damage to the bleached substance – used for bleaching woollen clothes.	Strong bleaching action
Similarity	Both sulphur dioxide and chlorine bleaches only in the presence of moisture	

(vi) **Additive properties**

Sulphur dioxide adds to certain compounds / elements. Three examples are given below.

(a) Action with oxygen

Sulphur dioxide reacts with oxygen forming sulphur trioxide.

$$2SO_2 + O_2 \xrightarrow{450°C,\ Pt\text{-}Asbestos} 2SO_3$$

(b) Action with chlorine

Dry sulphur dioxide reacts with dry chlorine in the presence of sunlight to form an unstable compound sulphuryl chloride.

$$SO_2 + Cl_2 \xrightarrow{sunlight} \underset{\text{Sulphuryl chloride}}{SO_2Cl_2}$$

(c) Action with PbO_2

Sulphur dioxide reacts with lead dioxide forming lead sulphite.

$$PbO_2 + SO_2 \xrightarrow{\Delta} PbSO_4$$

(vii) **Oxidising properties**

Sulphur dioxide exhibits oxidizing properties.

(a) Sulphur dioxide oxidizes hydrogen sulphide to sulphur.

$$2H_2S + SO_2 \rightarrow 3S + 2H_2O$$

(b) Sulphur dioxide oxidizes burning magnesium and burning sodium to the respective oxides.

$$2Mg + SO_2 \rightarrow 2MgO + S$$
$$4Na + SO_2 \rightarrow 2Na_2O + S$$

(viii) **Reducing properties**

Sulphur dioxide exhibits reducing properties due to the formation of nascent [H] when it reacts with water.

Some examples are given below.

(a) Sulphur dioxide reduces green chlorine water to HCl that is colourless.

$$Cl_2 + 2H_2O + SO_2 \rightarrow HCl + H_2SO_4$$

(b) Sulphur dioxide reduces red bromine water to HBr that is colourless.

$$Br_2 + 2H_2O + SO_2 \rightarrow 2HBr + H_2SO_4$$

(c) Sulphur dioxide reduces brown iodine solution to HI that is colourless

$$I_2 + 2H_2O + SO_2 \rightarrow 2HI + H_2SO_4$$

(d) Sulphur dioxide reduces brown ferric salt solution to green ferrous salt solution

$$Fe_2(SO_4)_3 + 2H_2O + SO_2 \rightarrow 2FeSO_4 + 2H_2SO_4$$

(e) Sulphur dioxide reduces pink acidified $KMnO_4$ to colourless sulphates

$$2KMnO_4 + 2H_2O + 5SO_2 \rightarrow K_2SO_4 + MnSO_4 + 2H_2SO_4$$

(f) Sulphur dioxide reduces orange red acidified $K_2Cr_2O_7$ to green chromic sulphite

$$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \rightarrow Cr_2(SO_4)_3 + K_2SO_4 + H_2O$$

14.12 TEST FOR SULPHUR DIOXIDE

(a) Sulphur dioxide causes coughing.

(b) Sulphur dioxide turns paper dipped in acidified potassium dichromate ($K_2Cr_2O_7$) green.

(c) Sulphur dioxide bleaches moist coloured petals. The colour is however regained after sometime.

(d) Sulphur dioxide decolourises bromine water.

14.13 USES OF SULPHUR DIOXIDE

Sulphur dioxide has the following uses.

(i) Sulphur dioxide is used for bleaching of woollen and silk fabrics, hair and sugar.

(ii) Sulphur dioxide is used as a disinfectant and for fumigation due to its germicidal properties.

(iii) Sulphur dioxide is used as a refrigerant.

(iv) Liquid sulphur dioxide is used in oil refineries.

(v) In chemical industry sulphur dioxide is used for manufacture of sodium sulphites and bisulphates. Calcium sulphite is used in paper industry.

(vi) Sulphur dioxide is used in the manufacture of sulphuric acid.

14.14 INTRODUCTION TO SULPHURIC ACID

Sulphuric acid is strong acid. It is soluble in water at all concentrations. It was named oil of vitriol by the 8th – century alchemist Jabir after his discovery of the chemical. Sulphuric acid has many

uses. Pure sulphuric acid (undiluted) does not occur on the earth, due to its great affinity for water. Sulphuric acid is a constituent of acid rain, which is formed by atmospheric oxidation of sulphur dioxide in the presence of water. Sulphur dioxide is the main byproduct produced when sulphur – containing fuels such as coal or oil are burned.

Sulphuric acid
Molecular Formula – H_2SO_4
Molecular weight – 98
Also known as Oil of Vitriol,
King of chemicals.

14.15 GENERAL METHODS OF PREPARATION

Sulphuric acid is prepared by the following general methods depending on convenience.

(a) By heating hydrated sulphates in the absence of air (Sulphuric acid was discovered for the first time using this reaction)

$$2FeSO_4.7H_2O \xrightarrow{\Delta} Fe_2O_3 + SO_2 + H_2SO_4 + 13H_2O$$

(b) By action of oxidizing agents on sulphurous acid

$$H_2SO_3 + [O] \rightarrow H_2SO_4$$

(c) By dissolving sulphur trioxide in water

$$SO_3 + H_2O \rightarrow H_2SO_4$$

(d) By hydrolyzing sulphuryl chloride

$$SO_2Cl_2 + H_2O \rightarrow H_2SO_4 + HCl$$

(e) By boiling sulphur with concentrated nitric acid

$$S + 6HNO_3 \rightarrow H_2SO_4 + 2H_2O + 6NO_2$$

14.16 CONTACT PROCESS – INDUSTRIAL MANUFACTURE OF SULPHURIC ACID

The stages in the Contact Process are as follows :

(a) Sulphur burners (conversion of sulphur to sulphur dioxide)

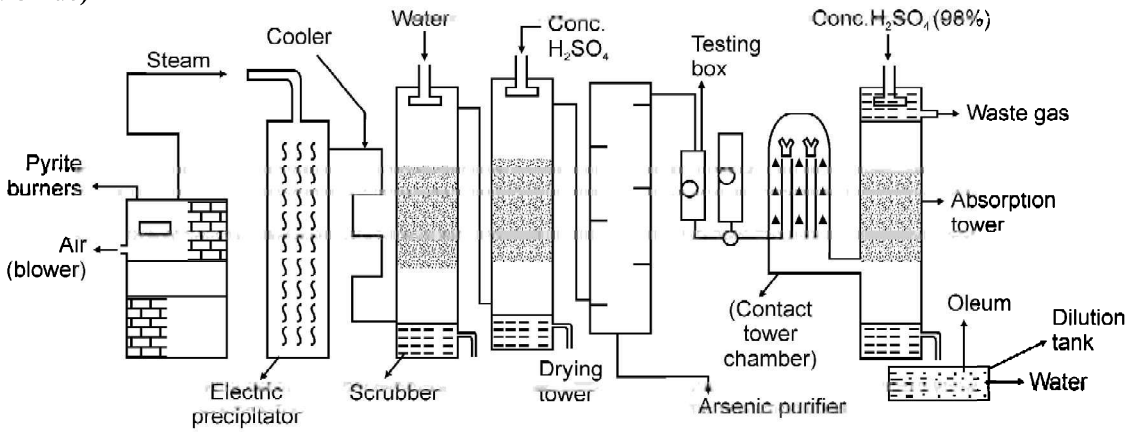
Pure sulphur is burnt with excess of pure oxygen (or dust free dry air) when sulphur dioxide is obtained. Instead of pure sulphur, iron pyrites can be used but this results in arsenic compounds forming as impurity. Pure oxygen is preferred over air as unnecessary heating of nitrogen in air can be avoided. Care is taken to ensure that oxygen is pure and dust free to avoid poisoning of the catalyst.

(b) Purification

The mixture of gases (SO_2 + air or pure oxygen) is purified to remove dust particles and arsenic compounds. This is done by passing through a dusting and scrubbing tower. In the tower the gases are mixed with steam and allowed to cool. The steam condenses on the dust particles. Next the mixture is passed through a drying tower where concentrated sulphuric acid is used to dry the mixture. The mixture is next passed through an Arsenic purifier which is a long tower containing shelves of heated ferric hydroxide. Arsenic compounds if present are absorbed. The mixture is now passed through a testing tower. The testing tower is a big black box with a glass window through which a beam of light is focused to test for dust particles. Any dust particles can be seen in the beam of light. If dust particles are present the whole dusting and scrubbing process is repeated.

(c) Converter or oxidation tower (conversion of sulphur dioxide to sulphur trioxide)

This tower consists of a huge cylindrical tower containing a series of vertical tubes lined with rows of catalyst. The catalyst is either platinised asbestos or vanadium pentoxide. The tubes with the catalyst are heated electrically to a temperature of 450°C. The heating is required only initially as the oxidation of sulphur dioxide is exothermic. As this reaction gets initiated the temperature is maintained at 450°C.



Contact Process for the manufacture of sulphuric acid

(d) Absorption of SO_3 (conversion of sulphur trioxide to oleum)

Sulphur trioxide is absorbed in concentrated sulphuric acid to form oleum.(Oleum is $H_2S_2O_7$. It is also known as fuming sulphuric acid or pyrosulphuric acid).

Note: The direct method of bubbling sulphur trioxide gas through water to form sulphuric acid is not used as the reaction is extremely exothermic and the heat generated cannot be controlled. The heat causes bubbles filled with steam to form. This causes spurting which may be dangerous. This can be avoided by converting sulphurous acid to oleum and then to sulphuric acid as this is a more controlled reaction.

(e) Conversion of oleum to sulphuric acid

Oleum is converted to sulphuric acid by mixing with calculated amount of soft water.

Flow chart of reactions of Contact Process

(i) Preparing sulphur dioxide:

$$S + O_2 \xrightarrow{\Delta} SO_2$$

or

$$4FeS_2 + 11O_2 \xrightarrow{\Delta} 8SO_2 + 2Fe_2O_3$$

(ii) Converting SO_2 to SO_3:

$$2SO_2 + O_2 \underset{}{\overset{V_2O_5,\ 450°C}{\rightleftharpoons}} 2SO_3$$

(iii) Converting SO_3 to oleum:

$$SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$$

(iv) Converting oleum to sulphuric acid:

$$H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$$

14.17 PHYSICAL PROPERTIES OF SULPHURIC ACID

Physical properties of sulphuric acid are listed below:

(i) Pure sulphuric acid is colourless, odourless fuming, dense, oily liquid sticking to the walls of the container.

(ii) Density of sulphuric acid is 1.84g/cc at 15^0C.

(iii) Sulphuric acid boils between 315^0 – 338^0C.

(iv) Sulphuric acid forms a Constant Boiling Mixture at 98%. It cannot be concentrated beyond 98% by boiling as both water and sulphuric acid boil off together at around 338^0C.

(v) Sulphuric acid is extremely hygroscopic, absorbing moisture easily from air (see box).

(vi) Sulphuric acid mixes easily with any volume of water releasing large amounts of heat called heat of hydration.

Diluting concentrated sulphuric acid

Concentrated sulphuric acid should be added slowly to water and not water to the concentrated acid. Each sulphuric acid molecule combines with water molecule in an extremely exothermic reaction.

$$H_2SO_4 + H_2O \rightarrow H_2SO_4.H_2O + \text{Heat of hydration}$$

If water is added to the acid, the first few drops of water coming in contact with the acid release so much heat that the water changes to steam causing spurting. However, if the acid is added to water the acid being heavier quickly reaches the bottom. Here the heat evolved disperses faster to the surrounding water and hence spurting is not there.

(vii) Concentrated sulphuric acid is a poor conductor. This is because concentrated sulphuric acid is poorly dissociated into ions. The number of charged particles available to conduct electricity is very less.

(viii) Dilute sulphuric acid is a good conductor. Dilute sulphuric acid is completely dissociated into ions.

$$H_2SO_4 \rightarrow 2H^+ + SO_4^{2-}$$

(Extent of dissociation is less in concentrated acid than in dilute acid)

The numbers of ions are more. The larger number of ions available to conduct electricity makes dilute sulphuric acid a better conductor than concentrated acid.

(ix) Sulphuric acid is highly corrosive and chars the skin and wood black.

14.18 CHEMICAL PROPERTIES OF SULPHURIC ACID

The Chemical Properties of sulphuric acid are listed below.

(i) Sulphuric acid shows acidic properties

The acidic properties are due to dissociation of the acid molecules in water to form hydronium ions.

$$H_2SO_4 \rightarrow 2H^+ + SO_4^{2-}$$

$$H^+ + H_2O \rightarrow H_3O^+$$

Hydronium ion

Hygroscopic nature of Sulphuric acid

Around 20 ml of sulphuric acid kept exposed to air, absorbs moisture from the air and increases to 40 ml at the end of three weeks.

It is the hydronium ion that is responsible for acidic property. The acidic properties shown are given :

(a) With metallic oxides dilute sulphuric acid forms the corresponding sulphates:

$MgO + H_2SO_4 \rightarrow MgSO_4 + H_2O$

$ZnO + H_2SO_4 \rightarrow ZnSO_4 + H_2O$

With lead oxide and calcium oxide the reaction is very slow and stops after sometime due to the formation of an insoluble layer of lead sulphate ($PbSO_4$) and calcium sulphate ($CaSO_4$).

(b) Neutralisation Reaction: Sulphuric acid is a dibasic acid. On neutralizing with a base, as each H-atom gets replaced an acid salt and then a normal salt is formed. So, when sulphuric acid is neutralized with reactive metallic hydroxides (caustic alkalis) dilute sulphuric acid forms two series of corresponding salts – acid salt and normal salt.

$NaOH + H_2SO_4 \rightarrow NaHSO_4 + H_2O$

Sodium hydrogen sulphate – an acid salt

$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

Sodium sulphate – a normal salt

With calcium hydroxide ($Ca(OH)_2$ and lead hydroxide ($Pb(OH)_2$), the reaction is very slow and stops after sometime due to the formation of an insoluble layer of lead sulphate ($PbSO_4$) and calcium sulphate ($CaSO_4$)

(c) With metallic carbonates and bicarbonates dilute sulphuric acid forms respective sulphates with the release of carbon dioxide.

With lead carbonate ($PbCO_3$) and calcium carbonate ($CaCO_3$) the reaction is very slow and stops after sometime due to the formation of an insoluble layer of lead sulphate ($PbSO_4$) and calcium sulphate ($CaSO_4$).

(ii) **Sulphuric acid shows oxidizing properties**

The oxidizing action of concentrated sulphuric acid is due to the reaction

$$H_2SO_4 \rightarrow H_2O + SO_2 + [O]$$

(a) Sulphuric acid oxidizes non-metals

Concentrated sulphuric acid when boiled with carbon, sulphur and phosphorous get oxidized to their respective oxides or corresponding acids.

$$C + 2H_2SO_4 \rightarrow 2H_2O + 2SO_2 + CO_2$$

$$S + 2H_2SO_4 \rightarrow 2H_2O + 3SO_2$$

$$2P + 5H_2SO_4 \rightarrow 2H_2O + 5SO_2 + 2H_3PO_4$$

(b) Sulphuric acid oxidizes metals

Dilute sulphuric acid reacts with more reactive metals like sodium, potassium and calcium explosively to liberate hydrogen gas.

$$2Na + 2H_2SO_4 \text{ (dil)} \rightarrow Na_2SO_4 + H_2$$

Hot concentrated sulphuric acid reacts with moderately reactive metals to form sulphates with the liberation of sulphur dioxide gas.

$$Zn + 2H_2SO_4 \xrightarrow{\Delta} ZnSO_4 + SO_2 + 2H_2O$$

(concentrated)

(iii) Sulphuric acid shows dehydrating properties (see box)

(a) Conc. sulphuric acid removes water of crystallization from hydrated salts:

$$CuSO_4.\ 5H_2O - 5H_2O \rightarrow CuSO_4$$

The crystal crumbles to form white powdery mass with the release of heat forming steam.

(b) Conc. sulphuric acid removes the elements of water from sugar, glucose:

The sugar turns brown initially with the formation of froth and steam finally leaving a black residue.

$$C_{12}H_{22}O_{11} \rightarrow 12C + 11H_2O$$

(c) Conc. sulphuric acid dehydrates oxalic acid to carbon monoxide and carbon dioxide.

$$(COOH)_2 \rightarrow CO + CO_2 + H_2O$$

(d) Conc. sulphuric acid dehydrates formic acid to CO:

$$HCOOH \rightarrow CO + H_2O$$

(e) Conc. sulphuric acid dehydrates ethanol to ethylene gas.

$$C_2H_5OH \rightarrow C_2H_4 + H_2O$$

(f) Conc. sulphuric acid dehydrates cellulose (skin/wood) and chars it black:

$$C_6H_{10}O_5 \rightarrow 6C + 5H_2O$$

Sulphuric acid is a dehydrating agent and drying agent

Drying agent removes water molecules that are present in a wet substance. There is no change in the chemical composition. Hence it is a physical process.

Dehydrating agent removes the elements of water molecule that are chemically combined with other elements in a given substance. Atoms of hydrogen and oxygen in the ratio 2 : 1 are removed from the substance.

Dehydration is a chemical change as there is a change in the composition of the substance.

(iv) **Sulphuric acid is a non-volatile acids when compared to other common acids.**

This property makes it suitable for use in preparing more volatile acids like hydrogen chloride and nitric acid by displacing them from their salts.

$$2NaNO_3 + H_2SO_4 \rightarrow Na_2SO_4 + HNO_3$$
$$2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$$

(v) **Reaction of sulphuric acid with some other compounds.**

(a) With sulphites and bisulphites, dilute sulphuric acid forms the corresponding sulphates, sulphur dioxide and water.

$$Na_2SO_3 + \text{dil. } H_2SO_4 \rightarrow Na_2SO_4 + SO_2 + H_2O$$
$$2NaHSO_3 + \text{dil. } H_2SO_4 \rightarrow Na_2SO_4 + 2SO_2 + 2H_2O$$

(b) With metallic sulphides, dilute sulphuric acid forms the corresponding sulphates and hydrogen sulphide.

$$FeS + H_2SO_4 \rightarrow FeSO_4 + H_2S$$

(c) With lead and barium salts, dil. sulphuric acid forms the corresponding precipitates of sulphates.

$$BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2\ HCl$$

(d) With metallic nitrates conc. sulphuric acid on heating forms the corresponding sulphates and nitric acid.

$$Mg(NO_3)_2 + H_2SO_4 \rightarrow MgSO_4 + 2HNO_3$$

(e) With metallic chlorides conc. H_2SO_4 on heating forms the corresponding sulphates and hydrochloric acid.

$$ZnCl_2 + H_2SO_4 \rightarrow ZnSO_4 + 2HCl$$

(f) With bleaching powder dilute H_2SO_4 liberates Cl_2 gas:

$$CaOCl_2 + H_2SO_4 \rightarrow CaSO_4 + Cl_2 + H_2O$$

(g) With oxidizing agents like $K_2Cr_2O_7$, $KMnO_4$ conc. H_2SO_4 liberates O_2 gas on warming:

$$2K_2Cr_2O_7 + 8H_2SO_4 \rightarrow 2K_2SO_4 + 2Cr_2(SO_4)_3 + 8H_2O + 3O_2$$
$$4KMnO_4 + 6H_2SO_4 \rightarrow 2K_2SO_4 + 4MnSO_4 + 6H_2O + 5O_2$$

14.19 TESTS FOR SULPHURIC ACID

(i) When conc. sulphuric acid is boiled with copper turnings it turns blue, liberating pungent suffocating sulphur dioxide gas which turns acidified potassium dichromate ($K_2Cr_2O_7$) paper green.

(ii) When conc. sulphuric acid is heated with sodium chloride, it gives off colourless hydrogen chloride, which fumes strongly in air. When a glass rod dipped in ammonia is brought near the acid, it gives off white dense fumes of ammonium chloride.

$$2NaCl + H_2SO_4 \rightarrow 2HCl + NaSO_4$$
$$HCl + NH_3 \rightarrow NH_4Cl$$

(iii) When dilute sulphuric acid is treated with barium chloride ($BaCl_2$) it forms a white precipitate of barium sulphite ($BaSO_4$)

$$BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2HCl$$

(iv) When dilute sulphuric acid is treated with lead acetate it forms a white precipitate of lead sulphate ($PbSO_4$).

$$(CH_3COO)_2Pb + H_2SO_4 \rightarrow PbSO_4 + 2CH_3COOH$$

14.20 USES OF SULPHURIC ACID

Sulphuric acid is known as the King of Chemicals and used almost in every industry.

(a) The single largest uses of sulphuric acid is in the manufacture of artificial fertilisers mainly $Ca(H_2PO_4)_2$ known as super phosphate of lime and other fertilizers like ammonium sulphates.

(b) Sulphuric acid is used in the manufacture of several other chemicals some of which are mentioned below.

(i) Explosives like tri-nitro-glycerine (TNG), tri-nitro-toluene (TNT) and picric acid.

(ii) Sulphuric acid is used in the manufacture of plastics, cellophane synthetic rubber, etc.

(iii) Nitric acid, hydrochloric acid and phosphoric acid

(iv) Dyes and drugs

(v) Paints and pigments

(vi) Laboratory reagents

(c) Sulphuric acid is used for pickling of metal surface. Pickling is the removal of the oxide or carbonate layer from metal surfaces before tin plating or electroplating.

(d) Sulphuric acid is used as an electrolyte in lead acid batteries.

(e) Sulphuric acid is used in leather tanning industry.

(f) Sulphuric acid is used in refining of petroleum.

(g) Sulphuric acid is used in the extraction of copper by electrochemical method.

14.21 HYDROGEN SULPHIDE

Hydrogen sulphide is an acidic gas, found near volcanoes, sewerages and furnaces.

Hydrogen sulphide is prepared in the laboratory as follows.

$$FeS + 2HCl \rightarrow FeCl_2 + H_2S.$$

The gas is collected by upward displacement of air.

Physical properties

Hydrogen sulphide is heavier than air, smells like rotten eggs, is soluble in water and acidic in nature.

Chemical properties

(a) Hydrogen sulphide is combustible – burns with a pale blue flame but does not support combustion.

(b) Hydrogen sulphide reacts with oxygen :

(i) limited air supply:

$$2H_2S + O_2 \rightarrow 2H_2O + 2S$$

(ii) excess air supply:

$$2H_2S + 3O_2 \rightarrow 2H_2O + 2SO_2$$

(c) Action with chlorine

$$H_2S + Cl_2 \rightarrow 2HCl + S$$

(d) Action with acidified $KMnO_4$

$$2KMnO_4 + 3H_2SO_4 + 5H_2S \rightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5S$$

(e) Reaction with salt solutions

$$Pb(NO_3)_2 + H_2S \rightarrow PbS\downarrow + 2HNO_3$$

$$CuSO_4 + H_2S \rightarrow CuS\downarrow + H_2SO_4$$

$$ZnSO_4 + H_2S \rightarrow ZnS\downarrow + H_2SO_4$$

Test for hydrogen sulphide

(a) Hydrogen sulphide has a smell of rotten eggs.

(b) Hydrogen sulphide turns a filter paper dipped in lead acetate $[Pb(CH_3COO)_2]$ black due to the formation of lead sulphide.

(c) Hydrogen sulphide decolourises acidified potassium permanganate but unlike sulphur dioxide gas, a deposit of yellow sulphur can also be seen. The reaction is

$$2KMnO_4 + 3H_2SO_4 + 5H_2S \rightarrow K_2SO_4 + MnSO_4 + 8H_2O + 5S$$

Uses of hydrogen sulphide in analysis

Hydrogen sulphide is used as an analytical gas. Insoluble salts of manganese, nickel, cobalt, zinc, silver, copper, mercury and cadmium are precipitated with specific colours. The colour of the precipitate is used to identify the metal cation present. The general reaction is :

Metal salt + H_2S → Metal sulphide precipitate + corresponding acid.

Reactions of hydrogen sulphide in acidic medium	**Colour of precipitate**
$2AgNO_3 + H_2S \xrightarrow{\text{acidic medium}} 2HNO_3 + Ag_2S\downarrow$	Black
$CuSO_4 + H_2S \xrightarrow{\text{acidic medium}} H_2SO_4 + CuS\downarrow$	Black
$HgSO_4 \xrightarrow{\text{acidic medium}} H_2SO_4 + HgS\downarrow$	Black
Reactions of hydrogen sulphide in basic medium	**Colour of precipitate**
$Zn(NO_3)_2 + H_2S \xrightarrow{\text{basic medium}} 2HNO_3 + ZnS\downarrow$	White
$MnSO_4 + H_2S \xrightarrow{\text{basic medium}} H_2SO_4 + MnS\downarrow$	Flesh Colour
$NiCl_2 + H_2S \xrightarrow{\text{basic medium}} 2HCl + NiS\downarrow$	Black

EXERCISE

I. Choose the correct answer

1. Which is not a physical property of sulphur
(a) solid (b) A = 32
(c) Z = 16 (d) green colour

2. For extracting sulphur, Sicilian Process is used for _____ deposits while Frasch Process is used for _____ deposits.
(a) surface, underground
(b) underground, underground
(c) surface, surface
(d) underground, surface

3. Plastic sulphur is _____ of sulphur
(a) also called gamma allotrope
(b) amorphous allotrope
(c) crystalline allotrope
(d) both (a) and (b)

4. Monoclinic sulphur is _____ of sulphur.
(a) also called beta allotrope

(b) crystalline allotrope
(c) amorphous allotrope
(d) both (a) and (b)

5. Rhombic sulphur is _____ of sulphur.
(a) also called alpha allotrope
(b) amorphous allotrope
(c) crystalline allotrope
(d) both (a) and (c)

6. At 95.6^{0}C, the allotropes of sulphur that exist are
(a) monoclinic and rhombic
(b) rhombic and colloidal
(c) all allotropes of sulphur
(d) rhombic and plastic

7. On standing _____ sulphur change(s) to _____ sulphur.
(a) all allotropes of, rhombic
(b) rhombic, monoclinic
(c) monoclinic, rhombic
(d) rhombic, plastic

8. Above 95.6^{0}C, the allotrope of sulphur that exists is
(a) all allotropes (b) rhombic
(c) monoclinic (d) plastic

9. The most stable allotrope of sulphur is _____sulphur.
(a) alpha (b) beta
(c) gamma (d) plastic

10. Enantiomerism is exhibited by sulphur at _____
(a) 0^{0}C (b) 96.5^{0}C
(c) 95.6^{0}C
(d) room temperature (25^{0}C)

11. Excess sulphur reacts with more reactive metals forming
(a) sulphides (b) sulphates
(c) polysulphides (d) sulphites

12. Burning sulphur fumes in air due to the formation
(a) sulphur dioxide (b) sulphuric acid
(c) sulphur trioxide (d) sulphurous acid

13. Atomicity of sulphur is
(a) two (b) eight
(c) five (d) four

14. The allotrope that is soluble in organic solvents is
(a) alpha-sulphur (b) beta-sulphur
(c) gamma-sulphur (d) both (a) and (b)

15. The allotrope that is insoluble in organic solvents is
(a) alpha-sulphur (b) beta-sulphur
(c) gamma-sulphur (d) both (a) and (b)

16. The allotrope that is soluble in water is
(a) alpha-sulphur (b) beta-sulphur
(c) gamma-sulphur (d) none

17. The drying agent used in the laboratory preparation of sulphur dioxide is
(a) hot air
(b) concentrated sulphuric acid
(c) sodium carbonate
(d) none of these

18. A drying agent brings about a chemical change in a substance
(a) true (b) false
(c) sometimes true (d) none

19. A dehydrating agent brings about a chemical change in a substance
(a) true (b) false
(c) sometimes true (d) none

20. A dehydrating agent removes _____ from substance
(a) dust
(b) moisture
(c) elements of water molecule
(d) none of these

21. The only residue on adding sulphuric acid to table sugar is carbon. This shows that table sugar is a compound made of
(a) carbon (b) hydrogen and oxygen
(c) both (a) and (b) (d) cannot say

22. The black circular mark that forms over time, beneath a bottle of concentrated sulphuric acid kept on a wooden table is due to
(a) paint
(b) wear and tear
(c) drying of cellulose
(d) dehydration of cellulose

23. Water can be added to concentrated sulphuric acid safely
(a) true
(b) false as the dissolution is endothermic
(c) false as the dissolution is exothermic
(d) none

24. Concentrated sulphuric acid should be added to water slowly to dilute it
(a) true
(b) false as the dissolution is endothermic
(c) false as the dissolution is exothermic
(d) none

25. Collection of sulphur dioxide in the laboratory is done by
(a) downward displacement of water
(b) downward displacement of mercury
(c) downward displacement of air
(d) upward displacement of air

26. The bleaching action of sulphur dioxide is
(i) permanent (ii) temporary
(iii) by reduction (iv) by oxidation

(a) i and ii (b) ii and iii
(c) i and iv (d) iii and iv

27. For bleaching of cotton, the bleaching agent that can be used is
(a) chlorine (b) bleaching powder
(c) sulphur dioxide (d) white colour

28. For sulphur dioxide to react with chlorine the essential conditions are
(i) both gases should be dry
(ii) light should be present
(iii) temperature should be high
(iv) pressure should be low
(a) (iii) and (iv) (b) (ii) and (iii)
(c) (i) and (iv) (d) (i) and (ii)

29. The gas evolved when concentrated sulphuric acid is added to oxalic acid
(a) carbon monoxide (b) carbon dioxide
(c) both (a) and (b) (d) neither (a) nor (b)

30. Sulphuric acid forms _____ salt(s) with alkalis.
(a) one (b) two
(c) three (d) none

II. Give one word answers

1. The temperature at which two allotropes of an element co-exist.
2. The process of making natural rubber hard for making tyres.
3. The property exhibited by sulphur dioxide when it decolourises coloured clothes.
4. The property exhibited by sulphur dioxide when it decomposes hydrogen sulphide to sulphur.
5. The property exhibited by sulphuric acid when it absorbs moisture from the atmosphere.
6. The mixture formed by sulphuric acid at 98% concentration when it cannot be concentrated any further by boiling.
7. The property exhibited by sulphuric acid when it removes the elements of water molecule from the chemical composition of a compound.
8. The property exhibited by sulphuric acid when it absorbs water molecules that are present in a wet gaseous substance.
9. The colour of paper dipped in lead acetate solution when exposed to hydrogen sulphide gas.

III. This section contains three questions. Each question contains statements given in two columns, which have to be matched. Statements (i, ii, iii, iv) in Set A have to be matched with statements (a, b, c, d) in Set B. More than one match or no match between options in the Sets is possible.

1. Match the reaction given in Set A with the product (s) formed in Set B

Set A	Set B
(i) $SO_2 + H_2O \rightarrow$	(a) $2H_2SO_4$
(ii) $2SO_2 + O_2 \rightarrow$	(b) $Na_2SO_4 + 2SO_2 + 2H_2O$
(iii) $2NaHSO_3 + H_2SO_4 \rightarrow$	(c) $2SO_3$
(iv) $H_2S_2O_7 + H_2O \rightarrow$	(d) H_2SO_3

2. Match the allotrope of sulphur in Set A with its structure in Set B

Set A	Set B
(i) Rhombic	(a) needle like
(ii) Monoclinic	(b) octahedral
(iii) Plastic	(c) fine white powdery
(iv) Colloidal	(d) no shape

3. Match the reactions in Set A with the property of sulphuric acid in Set B

Set A	Set B
(i) $C_{12}H_{22}O_{11} + nH_2SO_4 \rightarrow 12C + 11H_2O$	(a) acidic property
(ii) $S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$	(b) dehydrating property
(iii) $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$	(c) non-volatile acid
(iv) $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$	(d) oxidizing agent

4. Match the property in Set A with the substance in Set B

Set A	Set B
(i) dehydrating agent	(a) sulphuric acid
(ii) drying agent	(b) anhydrous calcium chloride
(iii) hygroscopic nature	(c) nitric acid
(iv) oxidizing property	(d) phosphorus pentoxide

IV. State whether the following statements are True or False

1. The naturally occurring sulphide of silver is called cinnabar.
2. Atomicity of sulphur is 2.
3. Colloidal sulphur is a crystalline allotrope
4. In Frasch Process of extraction of sulphur, molten sulphur comes out of the external pipe.
5. When heated in the absence of air at 114^{0}C, sulphur forms a viscous mass which cannot be poured out of the test tube.
6. Below 95.6^{0}C only rhombic sulphur exists.
7. Sulphur dioxide bleaches by reduction and the

bleaching is permanent.

8. Sulphur dioxide is a neutral oxide.
9. To dilute concentrated sulphuric acid, water should be added to the concentrated acid.
10. Hydrogen sulphide gas is used in analytical chemistry to identify metal cations present.

V. Fill in the blanks

1. The colour of the precipitate obtained when hydrogen sulphide is bubbled through a solution of zinc nitrate is _____
2. When a drop of concentrated sulphuric acid is put on copper sulphate crystals, the crystals form a white powdery mass due to _____
3. Arsenic compounds present as impurities are removed in the Contact process by heating with _____
4. If pure oxygen is used in the Contact Process instead of air, unnecessary heating of _____ can be avoided.
5. The acidic properties of sulphuric acid are due to dissociation of the acid molecules in water to form _____ ions.
6. The reaction of calcium oxide with dilute sulphuric acid stops after some time due to the formation of an _____ of _____ that surrounds the reactant.
7. Removing of the dull layer that forms on the surface of metals over time using sulphuric acid is called _____
8. In lead acid batteries the electrolyte that is used is _____
9. Both hydrogen sulphide and sulphur dioxide decolourise acidified potassium permanganate, with a difference that with hydrogen sulphide a deposit of _____ is also seen.
10. Firecrackers are a mixture of _____, _____ and _____

VI. Each question contains two statements. Statement A is an Assertion Statement and Statement R is the Reason Statement. Each question has four choices out of which only one is correct. Choose the correct one.

Choose A if Statement A is true, Statement R is true. Statement R is correct explanation for statement A

Choose B if statement A is true, Statement R is true; statement R is not correct explanation for statement A

Choose C if statement A is true, statement R is false.

Choose D statement A is false, statement R is true

1. **Statement – A :** The catalyst used in preference to platinised asbestos is vanadium pentoxide in Contact process.

   Statement – R : Asbestos is carcinogenic.
2. **Statement – A :** Sulphur trioxide is absorbed in concentrated sulphuric acid in its manufacture.

   Statement – R : Direct conversion of sulphur trioxide by reacting with water is highly exothermic reaction and may cause spurting.
3. **Statement – A :** Vulcanised rubber is stronger and more durable than rubber.

   Statement – R : Rubber is soft and is easily worn.
4. **Statement – A:** Concentrated sulphuric acid can turn blue vitriol ($CuSO_4$ $5H_2O$) colourless.

   Statement – R: Concentrated sulphuric acid bleaches the colour of the crystals.
5. **Statement – A:** Above 95.6°C only rhombic sulphur exists.

   Statement – R : Rhombic sulphur is the most stable allotrope of sulphur.
6. **Statement – A** : Hydrogen sulphide is used for qualitative analysis (identification of metal ions)

   Statement – R : Hydrogen sulphide has a smell of rotten eggs.
7. **Statement – A :** Plastic sulphur is insoluble in organic solvents.

   Statement – R: Rhombic and monoclinic sulphur are soluble in organic solvents.
8. **Statement – A:** Bleaching of sulphur dioxide is due to the formation of nascent oxygen with moisture.

   Statement – R: Bleaching of sulphur dioxide is by reduction.
9. **Statement – A :** Dilute sulphuric acid is a good conductor of electricity.

   Statement – R : Concentrated sulphuric acid is a bad conductor of electricity.
10. **Statement – A:** Heating sulphur in absence of air is a chemical change.

    Statement – R: As sulphur is heated in absence of air, the atoms get rearranged.

VII. Answer the following questions on sulphur.

1. Give a brief introduction to sulphur with respect to its physical state, atomic and mass numbers, electronic configuration, valency and position in the periodic table.
2. How does sulphur occur in the free state?
3. Give a brief outline of the extraction of sulphur from underground deposits.
4. Draw the structure of Rhombic, Monoclinic and Plastic sulphur.
5. Which isotope of sulphur is stable? What is transition temperature of sulphur?

6. Bring out the changes that take place as sulphur is heated from room temperature in the absence of air.
7. Apart from the physical form of the allotropes of sulphur how would you distinguish between Rhombic, Monoclinic and Plastic sulphur? (Hint: properties of solubility and stability of allotrope at room temperature can be used to distinguish the allotropes)
8. Burning sulphur fumes in air. Why?
9. How is sulphur used by the tyre industry?
10. Sulphur can be used to manufacture many chemicals. Make a list of the chemicals and their uses.
11. What is the atomicity of sulphur?
12. Outline the method of preparing the various allotropes of sulphur.
13. Draw a table classifying the various allotropes of sulphur.
14. How is sulphur extracted from surface deposits purified?
15. Complete and balance the following equations. Write the conditions where needed.

(a) $S + O_2 \xrightarrow{>300\,^\circ C}$ (b) $H_2 + S \rightarrow$

(c) $Cl_2 + S \rightarrow$ (d) $C + S \rightarrow$

(e) $P + S \rightarrow$

(f) $Ca + S \rightarrow$; if sulphur is in excess: $CaS + 4S \rightarrow$

(g) $Al + S$ (h) $Cu + S \xrightarrow{>400\,^\circ C}$

(i) $Cu + S \xrightarrow{>400\,^\circ C}$ (j) $S + HNO_3 \rightarrow$

(k) $S + H_2SO_4 \xrightarrow{?}$ (l) $NaOH + S \rightarrow$

VIII. Answer the following questions on sulphur dioxide

1. Give the reaction, method of drying and collection of sulphur dioxide in the laboratory.
2. Give two reactions to show the acidic property of sulphur dioxide.
3. Justify the given statement 'Bleaching action of sulphur dioxide is by reduction while that by chlorine is by oxidation'.
4. State the reason why the bleaching action of sulphur dioxide is temporary while that of chlorine is permanent.
5. Both dry sulphur dioxide and dry chlorine do not bleach dry petals of a flower. Why?
6. Both sulphur dioxide and carbon dioxide turn lime-water milky. How would you differentiate between the two gases? (Hint : See action with metals)
7. If four unlabeled jars contain hydrogen, oxygen, nitrogen and sulphur dioxide how would you differentiate between the four gases?
8. Name industrial uses of sulphur dioxide.
9. Write the equation to show the following conversions.

   Sulphur → sulphur dioxide → sulphur trioxide → sulphuric acid.
10. What is the acid anhydrides of sulphurous and sulphuric acid?
11. Sulphur dioxide forms two salts when it reacts with alkalis. What are these salts? (Hint: see normal and acid salt)
12. Sulphur dioxide is an acidic oxide. However it bleaches blue litmus colourless instead of turning it red. Why?
13. If 64 gm of sulphur is burnt in just sufficient air what is the weight of sulphur dioxide formed?
14. Give one reaction each that shows the additive property of sulphur dioxide where it gets added to an element and to a compound.
15. Complete and balance the following reactions.

(a) $S + O_2 \rightarrow$

(b) $PbS + O_2 \rightarrow$

(c) $ZnS + O_2 \rightarrow$

(d) $Cu + H_2SO_4(conc.) \rightarrow$

(e) $Zn + H_2SO_4(conc.) \rightarrow$

(f) $NaHSO_3 + H_2SO_4 \rightarrow$

(g) $ZnSO_3 + H_2SO_4 \rightarrow$

(h) $Na_2SO_3 + H_2SO_4 \rightarrow$

(i) NaOH limited quantities; $NaOH + SO_2 \rightarrow$

(j) NaOH in excess; $NaOH + SO_2 \rightarrow$

(k) $Na_2CO_3 + H_2O + SO_2 \rightarrow$

(l) $K_2CO_3 + H_2O + 2SO_2 \rightarrow$

(m) $SO_2 + O_2 \xrightarrow{450\,^\circ C \text{ pt-Asbestos}}$

(n) $SO_2 + Cl_2 \xrightarrow{\text{sunlight}}$

(o) $H_2S + SO_2 \rightarrow$

(p) $Mg + SO_2 \rightarrow$

(q) $Na + SO_2 \rightarrow$

(r) $Cl_2 + H_2O + SO_2 \rightarrow$

(s) $Br_2 + H_2O + SO_2 \rightarrow$

(t) $I_2 + H_2O + SO_2 \rightarrow$

(u) $Fe_2(SO_4)_3 + H_2O + SO_2 \rightarrow$

(v) $KMnO_4 + H_2O + SO_2 \rightarrow$

(w) $K_2Cr_2O_7 + H_2SO_4 + SO_2 \rightarrow$

IX. Answer the following questions on sulphuric acid.

1. Write the equation for the preparation of sulphuric acid from nitric acid.
2. What is the advantage of using pure oxygen over air in Contact Process?
3. How are arsenic impurities removed from the mixture of reactants in Contact Process?
4. Why is only initial heating required in Contact Process?
5. In the final step of Contact Process, instead of absorbing sulphur trioxide in water directly for conversion to acid, sulphur trioxide is made to react with conc. sulphuric acid. Why?
6. Why and how would you
   (a) dilute concentrated sulphuric acid
   (b) concentrate dilute sulphuric acid?
7. Give all possible reactions of a dibasic acid in aqueous solution.
8. Show the dissociation of sulphuric acid in aqueous solution.
9. Write the equations for the reactions of zinc with dilute and concentrated sulphuric acid.
10. Why does sulphuric acid form one acid salt and a normal salt when neutralized?
11. The reaction of lead and calcium salts with sulphuric acid stops after sometime. Why?
12. Concentrated sulphuric is a poor conductor of electricity when compared to dilute acid. Why?
13. Using an equation show why sulphuric acid behaves as an oxidizing agent?
14. Give two examples with reactions that show the dehydrating action of concentrated sulphuric acid.
15. Why is the skin charred black on contact with conc. sulphuric acid?
16. Over a period of time a black mark appears underneath a bottle of conc. sulphuric acid kept on a wooden shelf in a chemistry lab. Why?
17. List five important uses of sulphuric acid?
18. How would you distinguish between oxalic acid and formic acid using conc. sulphuric acid?
    (Hint – Identify the gases evolved on treating with conc. sulphuric acid)
19. Complete and balance the following reactions. Write conditions where needed.
    (a) $FeSO_4.7H_2O \xrightarrow{\Delta}$
    (b) $Al_2(SO_4)_3 + H_2O \xrightarrow{\Delta}$
    (c) $H_2SO_3 + [O] \rightarrow$ (d) $SO_3 + H_2O \rightarrow$
    (e) $SO_2.Cl_2 + H_2O \rightarrow$ (f) $S + HNO_3 \rightarrow$
    (g) $S+O_2 \rightarrow$
    (h) $SO_2 + O_2 \xrightarrow{\Delta}$
    (i) $SO_3 + H_2SO_4 \rightarrow$
    (j) $H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$
    (k) $H_2SO_4 + H_2O \rightarrow ?$ + Heat of Hydration
    (l) $H_2SO_4 \rightarrow$
    (m) $H^+ + H_2O \rightarrow$
    (n) $MgO + H_2SO_4 \rightarrow$
    (o) $ZnO + H_2SO_4 \rightarrow$
    (p) $NaOH + H_2SO_4 \rightarrow$
    (q) $2NaOH + H_2SO_4 \rightarrow$
    (r) $H_2SO_4 \rightarrow$
    (s) $C + H_2SO_4 \rightarrow$
    (t) $S + H_2SO_4 \rightarrow$
    (u) $P + H_2SO_4 \rightarrow$
    (v) $Na + H_2SO_4(dil) \rightarrow$
    (w) $Zn + H_2SO_4$ (conc. Heat) $\rightarrow$
    (x) $CuSO_4.5H_2O$(conc. $H_2SO_4 \rightarrow$
    (y) $C_{12}H_{12}O_6 - 7H_2O \xrightarrow{\Delta}$
    (z) $(COOH)_2 - H_2O \xrightarrow{\Delta}$
20. Complete and balance the following reactions. Write conditions where needed.
    (a) $HCOOH - H_2O \xrightarrow{\Delta}$
    (b) $C_2H_5OH - H_2O \xrightarrow{\Delta}$
    (c) $C_6H_{10}O_5 - ? H_2O \xrightarrow{?}$
    (d) $Na_2SO_3 +$ dil $H_2SO_4 \rightarrow$
    (e) $NaHSO_3 - ? H_2O \xrightarrow{?}$
    (f) $FeS + H_2SO_4 \rightarrow$
    (g) $BaCl_2 + H_2SO_4 \rightarrow$
    (h) $Mg(NO_3)_2 + H_2SO_4 \rightarrow$
    (i) $ZnCl_2 + H_2SO_4 \rightarrow$
    (j) $CaOCl_2 + H_2SO_4 \rightarrow$
    (k) $K_2Cr_2O_7 + H_2SO_4 \rightarrow$ (l) $KMnO_4 + H_2SO_4 \rightarrow$
    (m) $FeS + HCl \rightarrow$
    (n) Limited air supply : $H_2S + O_2 \rightarrow$
    (o) Excess air supply : $H_2S + O_2 \rightarrow$
    (p) $KMnO_4 + H_2SO_4 + H_2S \rightarrow$
    (q) $Pb(NO_3)_2 + H_2S \rightarrow$ (r) $CuSO_4 + H_2S \rightarrow$
    (s) $ZnSO_4 + H_2S \rightarrow$

KEY TO EXERCISE

I. 1. d 2. a 3. d 4. d 5. d 6. a 7. c 8. c
9. a 10. c 11. c 12. b 13. b 14. d 15. c 16. d
17. b 18. b 19. a 20. c 21. c 22. d 23. c 24. a
25. d 26. b 27. a 28. d 29. c 30. b

II. 1. Transition temperature 2. vulcanization 3. Bleaching action
4. Oxidising property 5. Hygroscopic nature 6. constant boiling mixture
7. Dehydrating action 8. drying agent 9. Black

III. 1. i. d ii. c iii. b iv. a 2. i. b ii. a iii. d iv. c
3. i. b ii. d iii. a, c iv. a 4. i. a ii. a, b, d iii. a iv. c

IV. 1. false 2. false 3. false 4. false 5. true
6. true 7. false 8. false 9. false 10. true

V. 1. white 2. dehydrating action 3. ferric hydroxide 4. nitrogen
5. hydronium 6. precipitate layer, calcium sulphate 7. pickling
8. sulphuric acid 9. yellow sulphur 10. sulphur, charcoal, nitre(potassium nitrate)

VI. 1. B 2. A 3. B 4. C 5. D 6. B 7. B
8. D 9. B 10. D

15 Study of Group VIIA Element-Chlorine

15.0 INTRODUCTION TO CHLORINE

Chlorine was first prepared and studied in 1774 by the Swedish chemist Scheele. However, he failed to establish chlorine as an element. He thought chlorine was an oxide as he obtained it from hydrochloric acid. He named the oxide as *muriaticum*. Regardless of what he thought, Scheele did isolate chlorine by reacting manganese dioxide with hydrochloric acid.

In 1809 Gay-Lussac tried to decompose the '*muriaticum*' by reacting it with charcoal to release the free 'element' as he thought wrongly then. They did not succeed and published a report in which they considered the possibility that *muriaticum* is an element, but were not convinced.

In 1810, Humphrey tried the same experiment again, and concluded that it was an element, and not a compound. He named this new element as chlorine, from the Greek word *chloros*, meaning green-yellow. *The name halogen, meaning salt producer, was originally defined for chlorine.* It was later applied to the rest of the elements in this family.

Occurrence

Free State: Chlorine does not occur free in nature. The reason for this is that chlorine is an extremely reactive element.

Combined state: Chlorine is present mainly as sodium chloride dissolved in seawater. Sea water has approximately 3% of sodium chloride. Other sources of chlorine in the combined state are rock salt (NaCl), carnalite ($KCl.MgCl_2.6H_2O$) and sylvine deposits (KCl).

15.1 LABORATORY PREPARATION OF CHLORINE

Chlorine can be prepared in the laboratory by oxidising hydrogen chloride using oxidising agents (See diagram of apparatus)

Reactions using different oxidising agents:

(a) Using Manganese dioxide:

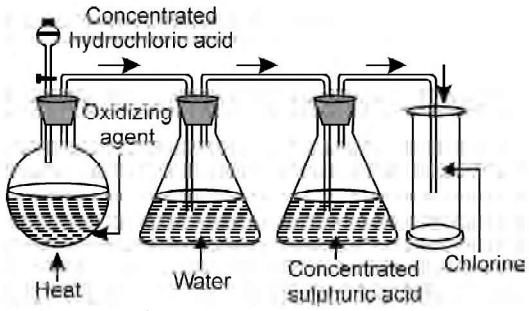
$$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$$

(b) Using Potassium permanganate:

$$2KMnO_4 + 16HCl \rightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2$$

Chlorine

Symbol - Cl
Molecular Formula - Cl_2
Atomicity = 2
Atomic Weight = 35.5
At. No: 17
Mass number 37 and 35 (Two isotopic forms)
At. Wt. 35.5
Electronic configuration: (2, 8, 7)
Number of valence electrons = 7
Valency: 1
Belongs to the halogen Group. Group VIIA and 3rd period in the Modern Periodic Table.
Very active, non-metal, gaseous element.



Preparation of chlorine from concentrated hydrochloric acid with the help of oxidising agents

(c) Using Bleaching powder:

$$CaOCl_2 + 2HCl \rightarrow CaCl_2 + H_2O + Cl_2$$

(d) Using Potassium dichromate:

$$K_2Cr_2O_7 + 14HCl \rightarrow 2KCl + 2CrCl_3 + 7H_2O + 3Cl_2$$

Drying of chlorine: The chlorine gas evolved is passed through water to absorb hydrogen chloride vapours that may be present. Chlorine is slightly soluble in water. The water gets saturated with chloride ions after which the excess chlorine comes out. Next the chlorine gas is bubbled through concentrated sulphuric acid (H_2SO_4).

Method of collection: Chlorine is heavier than air and is moderately soluble in water. Therefore, chlorine is collected by upward displacement of air.

Precautions to be taken in the laboratory preparation of Chlorine:

(i) The reactants are to be heated slowly to control the evolution of chlorine.

(ii) The vapours that are evolved have to be passed through water to absorb hydrogen chloride vapours that are also formed.

(iii) Chlorine is a highly poisonous gas. Hence, the apparatus should be placed in a fume chamber.

(iv) If chlorine is required dry, *it cannot be collected over mercury as it reacts easily with mercury* forming $HgCl_2$.

(v) A mixture of sodium chloride, sulphuric acid and manganese dioxide ($NaCl + H_2SO_4 + MnO_2$) can be used instead of HCl directly. Sodium chloride reacts with sulphuric acid giving hydrochloric acid.

$$NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$$

In the second step the HCl formed, reacts with MnO_2 to liberate chlorine gas.

$$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2\uparrow$$

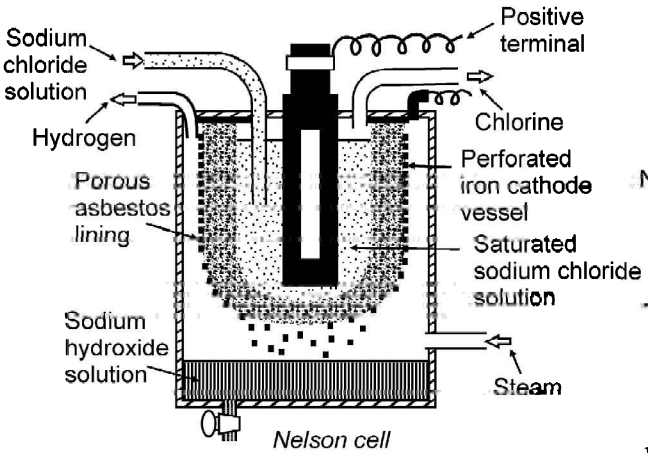
This method is to be preferred as there is relatively more production of chlorine. Besides sodium chloride is a more economical reactant. If hydrochloric acid is used directly, a part of HCl is used to form $MnCl_2$ and hence the amount of chlorine formed will be relatively less.

15.2 MANUFACTURE OF CHLORINE

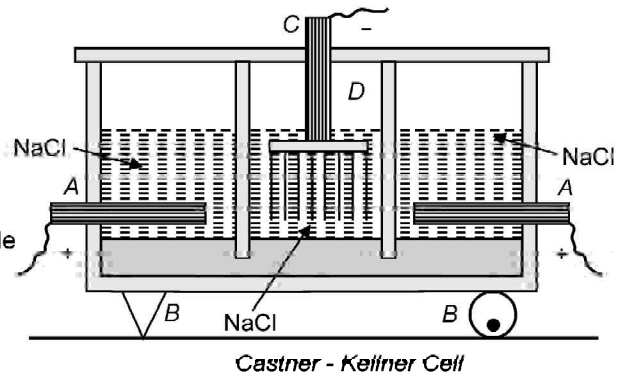
Chlorine is manufactured on a large scale by the *electrolysis of a saturated solution of sodium chloride.* A saturated solution of sodium chloride is called *brine.*

Electrolysis of brine

Electrolysis of brine can be done by two methods of electrolysis as given below.



Nelson cell



Castner - Kellner Cell

In both the methods chlorine *is formed at the anode*.

When NaCl is dissolved in water, it dissociates into ions as follows.

$$NaCl \rightarrow Na^+ + Cl^-$$

When electricity is passed through the solution, the positive ions are attracted to the cathode (negative plate) while the negative ions are attracted to the anode (positive plate).

Thus, Na^+ ions move to the cathode (negative plate) and Cl^- ions move to the anode (positive plate).

At the anode Cl^- ions lose electrons (they are reduced). The ions become atoms that combine to give molecules.

$$Cl^- - 1e^- \rightarrow [Cl]$$
$$2[Cl] \rightarrow Cl_2\uparrow$$

At the cathode Na^+ ions gain electrons (they are oxidized). The ions become atoms. *Sodium is a very reactive element. Hence sodium atoms react with the water present forming NaOH and hydrogen.*

$$2Na + 2H_2O \rightarrow 2NaOH + H_2\uparrow$$

Thus, Hydrogen gas is produced at the cathode.

The chlorine gas liberated at the anode is collected in steel cylinders. Hydrogen gas produced at the cathode is also collected separately.

15.3 PHYSICAL PROPERTIES OF CHLORINE

Physical properties of chlorine are listed below.

(a) Chlorine has a choking, suffocating odour and a sour taste. The gas has a greenish yellow colour.

(b) Chlorine is highly poisonous and fatal in large doses.

(c) Chlorine is heavier than air and is fairly soluble in water (1 volume of water dissolves 2.7 volumes of chlorine)

(d) Chlorine liquefies at 15°C and 6 atm pressure.

(e) The boiling point is –34.6°C and melting point is –202°C.

(f) Chlorine exists in two isotopic forms. *These*

two isotopic forms are present simultaneously in any given sample of chlorine. This fact is responsible for the fractional atomic weight of chlorine. (See box).

The two isotopic forms are	$_{17}Cl^{37}$ and	$_{17}Cl^{35}$
Number of protons, (Z)	17	17
Number of electrons,	17	17
Number of neutrons, (A–Z)	20	18

15.4 CHEMICAL PROPERTIES OF CHLORINE

Chemical properties are listed below.

(a) Chlorine gas is non-combustible, and does not support combustion.

(b) Chlorine undergoes *addition reactions* as follows.

With metals

Metals high in the Metal Activity Series (more reactive metals) like sodium (Na), potassium (K), magnesium (Mg) catch fire in chlorine forming the respective chlorides. Other metals like, iron (Fe), aluminium (Al), zinc (Zn) combine forming the respective chlorides when heated in chlorine.

As chlorine is a good oxidising agent, *it oxidises metals showing variable valency to the higher valency ion.*

Example: $2Fe + 3Cl_2 \rightarrow 2FeCl_3$
Ferric chloride

With non-metals

(i) With Sulphur:

$2S + Cl_2 \rightarrow S_2Cl_2$
Sulphur monochloride

Fractional Atomic weight of chlorine

Consider a sample of chlorine. The sample of chlorine contains both Cl^{37} and Cl^{35} isotopes. The percentage abundance of Cl^{37} is 25% while that of Cl^{35} is 75%. This means that out of 100 atoms of chlorine in the sample, 25 atoms are of Cl^{37} isotope and 75 atoms are of Cl^{35}.

The weight of each proton or neutron is almost 1amu. The mass number represents the sum of protons and neutrons present in the atom. Thus, each Cl^{37} atom weighs 37 amu and each Cl^{35} atom weighs 35 amu.

Thus in a sample of 100 atoms,

25 atoms of Cl^{37} weigh = 25 × 37 = 925 amu

75 atoms of Cl^{37} weight = 75 × 35 = 2625 amu

Total weight of 100 atoms = 3550 amu

Average weight of atom = 35.5 amu

Hence chlorine has fractional atomic weight.

Isotopes

When atoms of an element have the same atomic number but different mass numbers the atoms are said to be Isotopes of the element and the phenomenon is called Isotopy.

The isotopes have same chemical properties, as the number of electrons is the same in all isotopes of the element.

But the isotopes will have slightly different physical properties due to the difference in the atomic weights. The difference in atomic weights is due to the difference in number of neutrons present in the isotopes.

S_2Cl_2 is a red coloured liquid and an unstable compound in which both sulphur and chlorine are electronegative.

(ii) With Hydrogen:

$H_2 + Cl_2 \rightarrow 2HCl$

This reaction *does not take place in darkness, is slow and controlled in diffused light* and *explosive in bright light.*

(iii) With Phosphorous:

$P_4 + 6Cl_2 \rightarrow 4PCl_3$
Phosphorous Trichloride

$P_4 + 10Cl_2 \rightarrow 4PCl_5$
Phosphorous Pentachloride

Phosphorous burns with a pale green flame forming phosphorous trichloride and phosphorous pentachloride.

(c) Chlorine is a good oxidising agent.

The oxidising action of chlorine is due to the reasons given below.

(i) Chlorine accepts electrons easily.

(ii) Chlorine has high affinity for hydrogen.

(iii) Chlorine forms nascent oxygen in the presence of moisture which is responsible for the oxidizing action of chlorine.

$Cl_2 + H_2O \rightarrow HCl + HClO$
(Hydrochloric acid and hypochlorous acid)

$HClO \rightarrow HCl + [O]$
(Nascent oxygen)

Examples of oxidising action of chlorine:

(i) $H_2S + Cl_2 \rightarrow 2HCl + S$
(Hydrogen removed from H_2S by Cl_2)

(ii) $2FeCl_2 + Cl_2 \rightarrow 2FeCl_3$
[Ferrous (Fe^{2+}) oxidized to ferric ion (Fe^{3+})]

(iii) $Na_2SO_3 + H_2O + Cl_2 \rightarrow Na_2SO_4 + 2HCl$

(Affinity of chlorine for hydrogen).

(iv) $H_2SO_3 + [O] \rightarrow H_2SO_4$

(Oxidation of sulphurous acid).

(d) Chlorine is a good bleaching agent.

The bleaching action of chlorine is due to the formation of nascent oxygen in the presence of moisture.

$Cl_2 + H_2O \rightarrow HCl + HClO$

(Hydrochloric acid and Hypochlorous acid)

$HClO \rightarrow HCl + [O]$

(Nascent oxygen)

The nascent oxygen combines with coloured substance bleaching it. The bleaching is by oxidation.

Coloured substance + [O] → Colourless substance

Dry chlorine will not bleach dry coloured substances.

As the bleaching action by chlorine is permanent, chlorine is a more powerful bleaching agent than sulphur dioxide (See box).

Bleaching action of sulphur dioxide

Sulphur dioxide is also a bleaching agent. In the presence of moisture, sulphur dioxide forms nascent hydrogen.

$SO_2 + H_2O \rightarrow H_2SO_4 + [H]$

The nascent hydrogen combines with the coloured substance bleaching it. Thus, the *bleaching is by reduction.*

[H] + coloured substance → colourless substance

But, the bleached substance combines with atmospheric oxygen to give the coloured substance again. Thus, *the bleaching by sulphur dioxide is temporary.*

Bleached substance + atmospheric oxygen → coloured substance + water

As bleaching of chlorine is by combining with oxygen, there is no question of recombining with atmospheric oxygen. Hence *bleaching by chlorine is permanent.*

However both chlorine and sulphur dioxide bleach only in the presence of moisture

(e) Chlorine has good affinity for hydrogen.

The affinity of chlorine for hydrogen can be seen in the following reactions.

(i) Direct combination $H_2 + Cl_2 \rightarrow 2HCl$

(ii) *Free chlorine displaces other less reactive halogens from their salts.*

$2KBr + Cl_2 \rightarrow 2KCl + Br_2$

$2KI + Cl_2 \rightarrow 2KCl + I_2$

$KF + Cl_2 \rightarrow$ No reaction

With fluorides chlorine has no reaction, as fluorine is more reactive.

(iii) With ammonia (See chapter on Ammonia)

If chlorine is in excess :

$NH_3 + 3Cl_2 \rightarrow NCl_3 + 3HCl$

(NCl_3 is an oily explosive liquid)

If ammonia is in excess:

$8NH_3 + 3Cl_2 \rightarrow N_2 + 6NH_4Cl$

(iv) Hydrocarbons burn with a reddish yellow flame forming soot (carbon particles) and HCl.

$C_xH_y + ½\ yCl_2 \rightarrow xC + yHCl$

Example: With turpentine

$C_{10}H_{16} + 8Cl_2 \rightarrow 10C + 16HCl$

(f) Acidic nature of chlorine

(i) Moist chlorine is acidic as it forms hydrochloric acid.

(ii) Chlorine reacts with bases forming salt and water. The salts formed are simple chlorides, hypochlorites or chlorates depending on the concentration of the base and temperature.

$2NaOH + Cl_2 \rightarrow NaCl + NaClO + H_2O$

dilute Cold — sodium hypochlorite

$6NaOH + 3Cl_2 \rightarrow 5NaCl + NaClO_3 + 3H_2O$

hot, conc. — Sodium chlorate

Naming acids of chlorine

A simple method of remembering the names of hydracids of chlorine is given below.

HCl	Hydrogen chloride no oxygen
HClO	Hypochlorous acid one oxygen (less than chlorous so hypochlorous)
$HClO_2$	Chlorous acid two oxygen (suffix 'ous' as in Ferrous – Fe^{2+})
$HClO_3$	Chloric acid three oxygen (suffix 'ic' as in Ferric – Fe^{3+})
$HClO_4$	per chloric acid four oxygen (more than chloric so perchloric)

Test for chlorine

Apart from its greenish yellow colour and suffocating odour the following tests can be carried out to detect the presence of chlorine.

(i) Chlorine will turn moist blue litmus red and then bleach it colourless.

(ii) Chlorine will bleach moist coloured petals.

(iii) Chlorine will turn moist starch iodide paper (white in colour) to blue black. *Starch iodide* paper is a filter paper dipped in starch and potassium iodide (KI) solution. As chlorine is more reactive than iodine it displaces it from KI as follows

$$2KI + Cl_2 \rightarrow 2KCl + I_2$$

The Iodine reacts with starch (cellulose) turning it blue-black.

15.5 USES OF CHLORINE

Chlorine has many uses.

(a) As a germicide chlorine gas is used in drinking water and swimming pools.

(b) As a bleaching agent. For bleaching, chlorine is used in one of the following forms.

(i) Bleaching powder

(ii) Sodium hypochlorite (NaClO)

(iii) Liquid chlorine

(iv) Chlorine water

(c) Bleaching powder is used as a disinfectant in public toilets, etc.

(d) Sodium hypochlorite (NaClO) is used as a bleaching agent for whitening cotton clothes.

(e) Gammexene is $C_6H_6Cl_6$ used as a pesticide and disinfectant.

(f) Chlorine is used in the manufacture of the following important compounds due to the reasons given alongside.

- Potassium chlorate ($KClO_3$) – used in matchsticks and explosives
- Hydrochloric acid (HCl) – an important acid
- Carbon tetrachloride (CCl_4) – as an industrial solvent
- Antimony trichloride ($SbCl_3$) – used in match industry
- Poly Vinyl Chloride or PVC – a widely used plastic
- Carbonyl chloride ($COCl_2$) – also known as war gas used for chemical warfare but now banned
- Bleaching Powder ($CaOCl_2$) – used as a bleaching agent and
- Chloropierrin, $CCl_3.NO_2$ also known as tear gas used to disperse unruly crowds by the police
- Chloro Fluro Carbons (CFC) - used as a refrigerant in refrigerators. CFC is now being phased out of use by all countries as it destroys the ozone layer. If the ozone layer is destroyed, harmful UV radiation from the sun will enter the earth's atmosphere causing skin cancer and burns.

EXERCISE

I. Choose the correct answer

1. Common salt becomes damp in the rainy season though sodium chloride is neither hygroscopic nor deliquescent. The reason for this is
(a) sodium chloride is deliquescent.
(b) sodium chloride has affinity for water.
(c) impurities present in common salt are responsible for the dampness.
(d) sodium chloride dissolves in water.

2. Common salt melts the ice during snow fall as the freezing point of water _____ when sprinkled on ice.
(a) decreases (b) increases
(c) remains constant
(d) common salt dissolves in the ice forming solution

3. Common salt is used as a preservative as it removes water from microbes by
(a) reacting with the water
(b) dissolving in the water
(c) osmosis (d) none of these

4. Which is a physical property of chlorine?
(a) A liquid at room temperature and pressure
(b) A = 35 or A = 37
(c) Z = 16 (d) Atomicity = 3

5. Chlorine can be prepared
(a) by electrolysis of brine
(b) from seawater
(c) by oxidation of hydrochloric acid
(d) both (a) and (c)

6. The atomic weight of chlorine is taken as 35.5. This is closer to the mass number of Cl^{35} isotope. This suggests that
(a) this isotope (Cl^{35}) is more abundant in a sample of chlorine.
(b) this isotope (Cl^{35}) is less abundant in a sample of chlorine.
(c) the isotope (Cl^{37}) is more abundant in a sample of chlorine.
(d) cannot say anything.

7. Chlorine is collected in the laboratory preparation

by the upward displacement of air. This is because

(i) Chlorine is lighter than air.
(ii) Chlorine is heavier than air.
(iii) Chlorine is soluble in water.
(iv) Chlorine cannot be collected over mercury.

(a) all four reasons (b) (i, iii) and (iv)
(c) (ii, iii) and (iv) (d) (i) and (iv)

8. **To collect chlorine in its dry state mercury cannot be used because**
   (a) mercury is volatile (b) mercury is expensive
   (c) it reacts easily with mercury
   (d) none of these
9. **Dry chlorine is obtained in the laboratory by**
   (a) collecting over mercury
   (b) passing through concentrated sulphuric acid
   (c) heating it to remove water vapour
   (d) upward displacement of air
10. **In the laboratory preparation of chlorine, common salt is taken instead of hydrochloric acid as this is more**
    (a) economical (b) safer
    (c) part of the HCl reacts with the oxidizing agent
    (d) both (a) and (c)
11. **In the laboratory preparation of chlorine the apparatus is _____.**
    (a) made of glass only
    (b) kept in a fume chamber
    (c) made of inert materials
    (d) kept in water to dissolve it
12. **During electrolysis of brine, chlorine forms at the**
    (a) cathode (b) anode
    (c) bottom of the electrolyte
    (d) surface of the electrolyte
13. **One of the by-products of electrolysis of brine is**
    (a) sodium (b) oxygen
    (c) hydrogen (d) nitrogen
14. **_____ is a good oxidizing agent while _____is a good reducing agent.**
    (a) hydrogen, chlorine (b) chlorine, hydrogen
    (c) hydrogen, oxygen (d) oxygen, chlorine
15. **Chlorine dissolves in water forming initially**
    (a) HCl only (b) HCl and [O]
    (c) HCl and HClO (d) HClO only.
16. **The two isotopes of chlorine differ in terms of fundamental particles by**
    (a) 2 neutrons (b) 2 protons
    (c) 2 electrons
    (d) 2 neutrons and 2 protons
17. **The oxidising action of chlorine is due to**
    (a) ease with which it accepts electrons
    (b) affinity of chlorine for hydrogen
    (c) formation of nascent oxygen
    (d) all three (a, b and c)
18. **State whether the given statement is true: A more volatile acid can be displaced from its salt by a less volatile acid.**
    (a) sometimes true, sometimes false
    (b) false
    (c) true (d) cannot say
19. **Hydrogen chloride gas is collected by upward displacement of air as it**
    (a) is extremely soluble in water
    (b) is heavier than air (c) reacts with water
    (d) all three options (a, b and c)
20. **Nitric acid cannot be used instead of sulphuric acid in the laboratory preparation of hydrogen chloride as**
    (a) nitric acid is more volatile than HCl
    (b) nitric acid is more volatile than H_2SO_4
    (c) nitric acid decomposes easily
    (d) they both (HNO_3 and HCl) react with each other
21. **HCl will show acidic properties only if dissolved in which of the following solvent?**
    (a) Carbon disulphide (b) Water
    (c) Benzene (d) Carbon tetrachloride
22. **A definite test for presence of chlorine gas is**
    (a) turns moist red litmus blue
    (b) turns moist starch iodide paper to blue black
    (c) has green colour (d) suffocating odour
23. **A test for presence of HCl gas is**
    (a) turns moist red litmus blue
    (b) gives dense white fumes with ammonia
    (c) turns dry red litmus blue
    (d) very soluble in water
24. **Bleaching powder is actually a mixture of two compounds namely**
    (a) SO_2 and Cl_2
    (b) washing soda and chlorine
    (c) $Ca(OCl)_2$ and $CaCl_2$
    (d) none of these
25. **The bleaching action of bleaching powder is due to**
    (a) nascent oxygen (b) nascent chlorine
    (c) nascent hydrogen (d) reduction
26. **The atomic weight of chlorine is 35.5. This would suggest that Cl^{37} is _____abundant when compared to Cl^{35} isotope.**
    (a) more (b) less
    (c) as (equal)
    (d) cannot say from data given.
27. **The substance that reacts with both the metals gold and platinum is**
    (a) Nitric acid
    (b) Hydrochloric acid
    (c) 3 : 1 mixture of HNO_3 and HCl
    (d) 1 : 3 mixture of HNO_3 and HCl
28. **The action of Aqua Regia is primarily due to**
    (a) nascent oxygen (b) nascent hydrogen
    (c) nascent chlorine (d) carbon

29. Chlorine is dried in the laboratory using which drying agent?

(a) Concentrated sulphuric acid
(b) Calcium oxide
(c) Phosphorous pentoxide
(d) both (b) and (c)

30. The acid that is monobasic is

(a) Nitric acid (b) Hydrochloric acid
(c) Acetic acid (d) all three (a, b and c)

II. One of the very important uses of chlorines is as a disinfectant. Read the paragraph below and answer the questions that immediately follow the paragraph. Each question has four options out of which only one option is correct. Choose the correct one.

One of the important uses of chlorine is as a disinfectant. Chlorine based bleach is applied as a disinfectant on a large scale. Bleaching occurs as a result of oxidation of chlorine. About 20% of chlorine manufactured for industrial use is used to produce bleach and disinfectants. Chlorine is one of the most widely used disinfectants. Chlorine based bleach is formed when chlorine reacts with sodium hydroxide to form sodium hypochlorite (NaOCl). Chlorine kills pathogens such as bacteria and viruses by breaking the chemical bonds in their molecules. When enzymes come in contact with chlorine, one or more of the hydrogen atoms in the molecule are replaced by chlorine. This causes the entire molecule to change shape or fall apart.

1. Chlorine-based bleaches have
   (a) chloride ions (b) hypochlorite ions
   (c) nascent hydrogen (d) nascent oxygen
2. Chlorine acts as a disinfectant by
   (a) killing microbes on contact
   (b) poisoning the microbes
   (c) suppressing the microbes
   (d) none of these.
3. One of the major uses of chlorine is as a
   (a) bleaching agent (b) disinfectant
   (c) both (a) and (b)
   (d) none of the options is correct.

III. This section contains three questions. Each question contains statements given in two columns, which have to be matched. Statements (i, ii, iii, iv) in Set A have to be matched with Statements (a, b, c, d) in Set B. More than one match or no match between options in the Sets is possible.

1. Match the substance given in Set B with the percentage concentration given in Set A, at which the substance forms an Azeotropic Mixture (Constant Boiling Mixture).

Set A	Set B
(i) 68%	(a) Sulphuric acid
(ii) 22%	(b) Ethanol
(iii) 98.5%	(c) Nitric acid
(iv) 95.6%	(d) Hydrochloric acid

2. Match the element / compound in Set A with its property in Set B.

Set A	Set B
(i) Chlorine	(a) oxidising agent
(ii) Hydrogen Sulphide	(b) reducing agent
(iii) Hydrogen	(c. oxidised
(iv) Nitric acid	d. reduced

3. Match the reactants given in Set A with the products in Set B.

Set A	Set B
(i) $C_{10}H_{16} + 8Cl_2 \rightarrow$	(a) No reaction
(ii) $KCl + I_2 \rightarrow$	(b) $5NaCl + NaClO_3 + 3H_2O$
(iii) 2NaOH (cold, dilute. + $Cl_2 \rightarrow$	(c) 10C + 16HCl
(iv) 6NaOH (hot, conc.) + $3Cl_2 \rightarrow$	(d) $NaCl + NaClO + H_2O$

4. Match the acid in Set A with its name in Set B.

Set A	Set B
(i) $HClO_4$	(a) chloric acid
(ii) HClO	(b) chlorous acid
(iii) $HClO_2$	(c) per chloric aicd
(iv) $HClO_3$	(d) hypochlorous acid

IV. State whether the following statements are True or False

1. The largest source of sodium chloride is rock salt.
2. A saturated solution of any salt is called brine.
3. Microorganisms die in a salty environment as the water is drawn out of the cell by osmosis.
4. Dry chlorine can be obtained in the laboratory by collecting it over mercury instead of water.
5. The isotope of chlorine that is more abundant is Cl^{37}.
6. The salt formed when iron reacts with chlorine is ferrous chloride.
7. Hydrogen and chlorine react explosively in darkness.
8. Chlorine bleaches by oxidation and the bleaching is temporary.
9. Chlorine forms nitrogen gas and ammonium chloride with excess ammonia.
10. Hydrogen chloride is a less volatile acid when compared to sulphuric acid.

V. Fill in the blanks

1. When hydrogen chloride is dissolved in water it behaves as an _____.
2. When excess chlorine is reacted with ammonia the products are _____ and _____.
3. Aqua regia is a mixture of HNO_3 and HCl in the ratio _____.
4. When a rod dipped in ammonia is brought near a jar containing chlorine gas dense white fumes of _____ are formed.
5. Chlorine turns moist starch iodide paper from _____ to _____.

VI. Each question contains two statements. Statement A is an Assertion Statement and Statement R is the Reason Statement. Each question has four choices out of which only one is correct. Choose the correct one.

Choose A if Statement A is true, Statement R is true; Statement R is correct explanation for statement A

Choose B if Statement A is true, statement R is true; Statement R is not correct explanation for statement A

Choose C if Statement A is true, Statement R is false

Choose D if Statement A is false, statement R is true

1. **Statement A**: The atomicity of chlorine is 2.

   Statement R: The symbol of chlorine is Cl.

2. **Statement A**: The laboratory apparatus for preparing chlorine is kept in a fume chamber.

   Statement R: Fume chamber is used as chlorine fumes strongly in moist air.

3. **Statement A**: In the presence of moisture, hydrogen chloride turns blue litmus red.

   Statement R: Hydrogen chloride is a bleaching agent.

4. **Statement A**: Chlorine bleaches both moist red and moist blue litmus paper.

   Statement R: Chlorine behaves as an acid as well as a base.

5. **Statement A**: Both sulphur dioxide and chlorine bleach only in the presence of moisture.

   Statement R: Both are acidic gases.

6. **Statement A**: Chlorine is not a powerful bleaching agent as sulphur dioxide.

   Statement R: Sulphur dioxide bleaches by reduction.

7. **Statement A**: Only moist hydrogen chloride exhibits acidic properties.

   Statement R: In the presence of moisture hydrogen chloride ionizes forming hydronium ion.

8. **Statement A**: Bleaching action of sulphur dioxide is temporary.

   Statement R: Bleaching action of chlorine is permanent.

9. **Statement A**: Oxidation of hydrogen chloride gives chlorine.

   Statement R: Hydrogen chloride is an acidic gas.

10. **Statement A**: Hydrochloric acid forms only one salt on neutralization with a base.

    Statement R: Hydrochloric acid is a monobasic acid.

VII. Answer the following questions.

1. Write briefly about chlorine with respect to its atomic number, mass number, atomic weight, electronic configuration and position in the Modern periodic table.
2. What are the isotopes of chlorine and how are they responsible for its fractional weight?
3. In what way does Cl^{37} differ from Cl^{35}?
4. In the combined state what is the main source of chlorine?
5. What is the advantage of using a mixture of [NaCl + H_2SO_4 + MnO_2] over [HCl + MnO_2] in the laboratory preparation of chlorine?
6. What precautions would you take in preparing chlorine in the laboratory from NaCl?
7. Write and state the nature of the reactions taking place at cathode and anode in the electrolysis of brine.
8. Why should the reactants (NaCl + H_2SO_4) be heated to below 200°C in the laboratory?
9. What is the method of collecting chlorine and why? Why is chlorine not collected over mercury?
10. State the reasons for the oxidising nature of chlorine?
11. How does chlorine bleach? Why does dry chlorine not bleach?
12. How will you test for chlorine using starch iodide paper? Give the reaction.
13. State all the industrial uses of chlorine where it is used to manufacture other chemicals / compounds.
14. What are the different forms in which chlorine is used for its bleaching action?
15. Complete and balance the following equations.
    (a) $Fe + Cl_2 \rightarrow$
    (b) $S + Cl_2 \rightarrow$
    (c) $H_2 + Cl_2 \xrightarrow{\text{darkness}}$
    (d) $H_2 + Cl_2 \xrightarrow{\text{diffused light}}$
    (e) $H_2 + Cl_2 \xrightarrow{\text{Bright light}}$
    (f) chlorine is in limited quantity: $P_4 + Cl_2 \rightarrow$
    (g) chlorine is in excess quantity: $P_4 + Cl_2 \rightarrow$
    (h) $Cl_2 + H_2O \rightarrow$
    (i) $HClO \rightarrow$
    (j) $H_2S + Cl_2 \rightarrow$
    (k) $FeCl_2 + Cl_2 \rightarrow$
    (l) $Na_2SO_3 + H_2O + Cl_2 \rightarrow$
    (m) $H_2SO_3 + [O] \rightarrow H_2SO_4$
    (n) Coloured substance + [O] $\rightarrow$
    (o) $Cl_2 + H_2O \rightarrow$
    (p) $C_xH_y + ½\ yCl_2 \rightarrow$
    (q) $C_{10}H_{16} + Cl_2 \rightarrow$
    (r) If ammonia is in excess: $NH_3 + Cl_2 \rightarrow$
    (s) If chlorine is in excess: $NH_3 + Cl_2 \rightarrow$
    (t) $KBr + Cl_2 \rightarrow$
    (u) $KI + Cl_2 \rightarrow$
    (v) $KF + Cl_2 \rightarrow$

(w) Cold, dilute NaOH: $NaOH + Cl_2 \rightarrow$

(x) Hot, concentrated NaOH: $NaOH + Cl_2$

16. How would you prepare HCl in the lab?
17. What is the principle involved in the preparation of HCl from NaCl in the laboratory?
18. In preparing HCl using NaCl, why cannot HNO_3 be used instead of H_2SO_4?
19. How would you convert HCl gas to HCl acid? What precaution needs to be observed and why?
20. Dry HCl does not act as an acid. Why?
21. HCl dissolved in a non-polar solvent does not show acidic properties. Why?
22. What is the basicity of HCl? Does it produce acid salts when it reacts with alkalis?
23. Why can HCl not be concentrated beyond 22% by boiling?
24. Write an equation with conditions to show the thermal decomposition of HCl.
25. Give five equations that show the acidic properties of HCl acid.
26. Why cannot CaO or P_2O_5 be used as a drying agent for HCl?
27. How would you test for HCl?
28. What is Aqua Regia? How does it act?
29. Give two uses of HCl.
30. Bleaching powder is actually a mixture of two compounds. What are they? Derive its empirical formula from this information.
31. Give a reaction in which bleaching powder loses chlorine.
32. State how bleaching powder bleaches.
33. Name two acids other than HCl that form Constant Boiling Mixtures. State the percentages.
34. An element 'Y' with atomic weight 'x' exists in two isotopic forms. If the heavier isotope has mass number (x + 2) and the lighter isotope has mass number (x – 1), find the percentage abundance of the two isotopes. [Hint: Let the percentage abundance of the heavier isotope be 'a'; thus percentage abundance of lighter isotope is (100 – a). From this proceed to derive an expression to find the weight of 100 atoms and equate it to 100x as the At. Wt. of element is given to be x. Simplify to get your answer]
35. Complete and balance the following equations.

(a) $NaCl + H_2SO_4 \xrightarrow{< 200°C}$ (b) $HCl + CaO \rightarrow$

(c) $HCl + P_2O_5 \rightarrow$

(d) $HCl\ H_2 + Cl_2 \xrightarrow{> 500°C}$

(e) $Mg + HCl \rightarrow$ (f) $Fe + HCl \rightarrow$

(g) $Zn + HCl \rightarrow$ (h) $NaOH + HCl \rightarrow$

(i) $HCl \xrightarrow{\text{in aqueous solution}}$ (j) $CaCO_3 + HCl \rightarrow$

(k) $ZnO + HCl \rightarrow$ (l) $NH_4OH + HCl \rightarrow$

(m) $Na_2SO_3 + HCl \rightarrow$

(n) $Ca(HSO_3)_2 + HCl \rightarrow$ (o) $ZnS + HCl \rightarrow$

(p) $Na_2S_2O_3 + HCl \rightarrow$

(q) $MnO_2 + HCl \xrightarrow{\Delta}$

(r) $KMnO_4 + HCl \rightarrow$

(s) $3HCl + HNO_3 \rightarrow$

(t) $Au + 3[Cl] \rightarrow$

(u) $Pt + 4[Cl] \rightarrow$

(v) $NH_3 + HCl \rightarrow NH_4Cl$

(w) $AgNO_3 + HCl \rightarrow$

(x) $CaOCl_2 + CO_2 \rightarrow$

(y) $CaOCl_2 + HCl \rightarrow$

(z) $CaOCl_2 + H_2SO_4 \rightarrow$

KEY TO EXERCISE

I. 1. c 2. a 3. c 4. b 5. d 6. a 7. c 8. c
9. b 10. d 11. b 12. b 13. c 14. b 15. c 16. a
17. d 18. c 19. d 20. a 21. b 22. b 23. b 24. c
25. b 26. b 27. d 28. c 29. a 30. d

II. 1. b 2. a 3. c

III. 1. i. c ii. d iii. a iv. b 2. i. a, d ii. b, c iii. b, c iv. a, d
3. i. c ii. a iii. d iv. b 4. i. c ii. d iii. b iv. a

IV. 1. false 2. false 3. true 4. false 5. false 6. false 7. false 8. false
9. true 10. false

V. 1. acid 2. HCl, NCl_3 3. 1 : 3 4. ammonium chloride 5. white, blue black

VI. 1. B 2. C 3. C 4. C 5. B 6. D 7. A 8. B
9. B 10. A